Mathematics for Economists

A Graduate Reference Volume

Prepared for the Swiss Program for Beginning Doctoral Students in Economics

Study Center Gerzensee

Abstract

These notes develop the mathematics required for the first year of an economics doctoral program, and somewhat beyond it. They follow the topics of the Mathematics Review Course taught at the Study Center Gerzensee within the Swiss Program for Beginning Doctoral Students in Economics — probability and conditional expectation, single- and multi-variable calculus, static optimisation, real analysis, and discrete-time dynamic programming — and extend each of them to the level of rigor at which the results are actually used in microeconomic theory, macroeconomics, and econometrics.

The treatment is organised around hypotheses: what a theorem requires of the space it is stated on, which of its conclusions survive when a requirement is dropped, and which counterexample shows that the requirement cannot simply be dropped. The economic applications run from the environment and its primitives through the derivation to the comparative statics, and close with the failure the hypotheses are there to exclude.

Preface

A first-year economics doctoral student meets three kinds of mathematical difficulty. The first is technical: an argument uses the implicit function theorem, or the Banach fixed point theorem, or the fact that a continuous function on a compact set attains its maximum, and the student has either never seen the result or has seen it without its hypotheses. The second is translational: the mathematics is familiar but the economic object it stands for is not, so that a Lagrange multiplier is manipulated correctly without being recognised as a shadow price and a fixed point is computed without being recognised as an equilibrium. The third is architectural: the results are known individually but not as a system, so that the student does not see that the existence of a Walrasian equilibrium, the existence of a value function solving a Bellman equation, and the existence of a least-squares projection are three instances of the same handful of theorems about compactness, completeness, and convexity.

These notes address all three. They follow the topics of the Mathematics Review Course of the Swiss Program for Beginning Doctoral Students in Economics, taught at the Study Center Gerzensee: probability and conditional expectation, calculus in one and several variables, static optimisation, real analysis, and discrete-time dynamic programming (Fukuda 2026e). That course compresses these subjects into four days, and its slides are deliberately economical — they state what is needed, illustrate it, and move on. A student who wants to know why a hypothesis is there, what fails without it, or how a result generalises must look elsewhere. The purpose of this volume is to be that elsewhere, while keeping the course's sequence, its notation, and above all its economic motivation.

The mathematical standard aimed at is that of a graduate text rather than that of a review course: hypotheses are attached to the space on which a statement is made, necessary conditions are distinguished from sufficient ones, existence from uniqueness, and local statements from global ones. Where a teaching-level simplification appears in the course slides, the simplification is retained for its intuitive value and the exact statement is given alongside it, with the relation between the two made explicit; the Kuhn–Tucker theorem in Chapter 12 and the envelope formula in Chapter 15 are the clearest instances. A result that is quoted rather than proved is marked as such, with the source and the reason for the omission given where it is used.

The economic applications are worked rather than named. Each one specifies the environment, the preferences or technology, the constraint set, the correspondence between mathematical and economic objects, the derivation, and what goes wrong when a hypothesis fails. The applications are drawn from the course itself: profit maximisation by price-taking and price-setting firms, the risk premium and the Arrow–Pratt coefficient, ordinary least squares as an orthogonal projection, the consumer-choice problem, intertemporal consumption and the Euler equation, Nash bargaining, Bayesian updating in a market-microstructure setting, normal-normal learning, and the cake-eating, Ramsey growth, and Lucas asset-pricing problems.

Sources

Three layers of source material stand behind the volume.

The first is the set of six slide decks of the Gerzensee course (Fukuda 2026a,b,c,e,f,g). These fix the scope: the topics are the course's topics, and the results the slides state are carried over here with their hypotheses supplied.

The second is the accompanying reference volume (Fukuda 2026d), a 768-page set of lecture notes of which the slides are a distillation. It supplies definitions the slides state informally, proofs the slides sketch, exercises the slides mark optional, and the technical conditions the slides suppress. Where a result appears in both, the numbering of the reference volume is given so that the two can be read together.

The third consists of the mathematical texts that carry each topic beyond what a review course can reach: Rudin (1976) and Amann and Escher (2005, 2008, 2009) for real analysis; Axler (2024) for linear algebra; Boyd and Vandenberghe (2004) and Rockafellar (1970) for convexity and optimisation; Le Gall (2022) and Le Gall (2016) for measure-theoretic probability; Corbae et al. (2009) and Ok (2007) for analysis written with economic applications in view; and Ljungqvist and Sargent (2018) and Stokey et al. (1989) for recursive methods. Each is cited, with the relevant chapter or section, at the points where its content is used. Le Gall (2016) is listed because it is the natural continuation for readers who go on to continuous-time finance; nothing in this volume requires stochastic calculus, and it is therefore not cited again.

How to Use These Notes

Order of reading

The chapters are ordered by logical dependence, not by the order in which the four-day course presents them. The course opens with probability because Bayes' rule is immediately useful; but the sharpest statement about conditional expectation — that it is the orthogonal projection of a square-integrable random variable onto the space of functions of the conditioning variable — needs completeness, convexity, and a projection theorem. Those are developed in Chapters 6 and 9, so probability appears here as Chapter 14. The dependence structure is as follows.

- **Foundations.** Chapter 1 (sets, relations, logic, the real number system) and Chapter 2 (vector spaces, linear maps, matrices, spectra) presuppose nothing.
- **Analysis.** Chapter 3 (inner products, norms, metrics) requires Chapter 2. Chapter 4 (open and closed sets, sequences, compactness) requires Chapter 3. Chapter 5 (continuity, semicontinuity, the Weierstrass theorem) requires Chapter 4. Chapter 6 (completeness, the contraction mapping theorem, Blackwell's conditions) requires Chapter 5.
- **Calculus.** Chapter 7 (one variable) requires only Chapter 1 and the limit notions of Chapter 4. Chapter 8 (several variables, the implicit function theorem) requires Chapter 7, Chapter 2 and, for the proof of the implicit function theorem, Chapter 6. Chapter 10 (integration) requires Chapter 7.
- **Convexity and optimisation.** Chapter 9 (convex sets and functions, separation) requires Chapter 4 and Chapter 8. Chapter 11 (projection, orthogonality, least squares) requires Chapter 6 and Chapter 9. Chapter 12 (Lagrange and Kuhn–Tucker) requires Chapter 9 and Chapter 8. Chapter 13 (correspondences, the maximum theorem, fixed points) requires Chapter 5 and Chapter 9.
- **Probability and dynamics.** Chapter 14 requires Chapter 10 and Chapter 11. Chapter 15 (discrete-time dynamic programming) requires Chapter 6, Chapter 12, Chapter 13, and, for the stochastic case, Chapter 14. Chapter 16 (continuous-time optimisation) requires Chapter 8 and Chapter 15.

A reader following the four-day course can read Chapter 14, then Chapters 7, 8 and 10, then Chapter 12, then Chapters 3–6, then Chapter 15, consulting earlier chapters as the cross-references demand.

Conventions used throughout

Definitions are boxed in green, formal statements in blue, and remarks and examples are set plainly. A statement labelled **Assumption** is a standing hypothesis that remains in force until it is explicitly withdrawn; a statement labelled **Theorem**, **Proposition**, **Lemma**, or **Corollary** carries all of its hypotheses within it and depends on no standing assumption except those stated in its own text or in an assumption it cites by number.

Every result is numbered within its section, so that Theorem 7.3.2 is the second numbered statement of Section 7.3. Cross-references are hyperlinked. The two indices at the end of the volume list, respectively, defined terms and named results, and the symbols with the page on which each is introduced.

Mathematical Prerequisites

The volume is self-contained in the sense that every object it uses is defined before it is used. It is not self-contained in the sense of assuming no prior exposure: a reader meeting the derivative or the determinant for the first time here will find the treatment too fast. What is assumed is the content of a solid undergraduate program in mathematics for economics, namely

(i) elementary set notation and the manipulation of quantifiers;

(ii) single-variable differential and integral calculus at the computational level — differentiating and integrating elementary functions, and the statement of the fundamental theorem of calculus;

(iii) matrix algebra: multiplication, inversion, determinants, and the solution of linear systems;

(iv) elementary probability: random variables, expectation, variance, and the common families of distributions.

Each of these is revisited from a higher standpoint rather than assumed silently. Chapter 1 restates set theory in the language of relations and mappings and constructs the real numbers axiomatically; Chapter 2 replaces matrix algebra by the theory of linear maps between vector spaces, recovering matrices as coordinate representations; Chapter 7 rebuilds the derivative as a local linear approximation; and Chapter 14 builds probability on a measure space. A reader who is comfortable with (i)–(iv) can read the corresponding chapters quickly, but should not skip them: the higher standpoint is what the later chapters use.

No prior exposure to metric-space topology, convex analysis, measure theory, or dynamic programming is assumed.

Notation and Conventions

The notation follows the Gerzensee course materials wherever they fix a convention, and standard mathematical usage elsewhere. Conflicts between sources have been resolved in favour of the version that makes hypotheses most visible; where a competing convention is common in economics, it is noted at the point of first use.

Sets and numbers

Symbol	Meaning
$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	natural numbers (excluding 0), integers, rationals, reals, complex numbers
$\mathbb{N}_0$	$\mathbb{N} \cup \{0\}$
$\mathbb{R}_+$, $\mathbb{R}_{++}$	$[0, \infty)$ and $(0, \infty)$
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$
$\mathbb{R}^n$	n-fold Cartesian product of $\mathbb{R}$; elements are column vectors
$\mathbb{R}^n_+$, $\mathbb{R}^n_{++}$	vectors with all coordinates ≥ 0, respectively > 0
A^{c}, $A \setminus B$	complement of A, set difference
$\mathcal{P}(X)$	power set of X
$\operatorname{Int} A$, $\overline{A}$, ∂A	interior, closure, boundary of A
$\operatorname{ri} A$, $\operatorname{co} A$, $\operatorname{aff} A$	relative interior, convex hull, affine hull of A
$\mathbb{1}_A$	indicator function of the set A
$:=$	equality that defines the left-hand side

Vectors, matrices and maps

Vectors in $\mathbb{R}^n$ are columns; x^T denotes the transpose, so that $x^\mathsf{T} y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$. Matrices are written A, B, X, with A^T the transpose, A^{-1} the inverse when it exists, $\operatorname{rank} A$ the rank, $\operatorname{tr} A$ the trace, $\det A$ the determinant, $\operatorname{Ker} A$ the kernel and $\operatorname{Im} A$ the image. The i-th standard basis vector of $\mathbb{R}^n$ is e^i. For a matrix A, the notation $A \succeq 0$ ($A \succ 0$) means that A is symmetric positive semidefinite (positive definite).

For $f \colon A \to B$ the notation $f(S)$ denotes the image of $S \subseteq A$ and $f^{-1}(T)$ the preimage of $T \subseteq B$; f^{-1} is used for the inverse map only where f is stated to be a bijection. A correspondence, that is a set-valued map, is written $\Gamma \colon A \rightrightarrows B$ and its graph is $\operatorname{gr} \Gamma$.

Derivatives use $f'(x)$ in one variable, $\nabla f(x) \in \mathbb{R}^n$ for the gradient of a real-valued f on $\mathbb{R}^n$, $Df(x)$ for the $m \times n$ Jacobian matrix of $f \colon \mathbb{R}^n \to \mathbb{R}^m$, and $\mathrm{D}^2 f(x)$ for the $n \times n$ Hessian. The gradient is a column vector and $Df(x) = \nabla f(x)^\mathsf{T}$ when $m = 1$.

Analysis

A metric space is a pair (X, d) and a normed space a pair $(X, \|\cdot\|)$; the pair is abbreviated to X when the metric or norm is clear. Open balls are $B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$. Sequences are $(x_n)_{n \in \mathbb{N}}$ and subsequences $(x_{n_k})_{k \in \mathbb{N}}$. The spaces of continuous and of bounded functions from X to Y are $\mathcal{C}(X, Y)$ and $\mathcal{B}(X, Y)$; both carry the supremum norm $\|f\|_\infty = \sup_{x \in X} |f(x)|$ unless stated otherwise.

Optimisation

A constrained problem is written

$$\max_{x \in \mathbb{R}^m} f(x) \quad \text{subject to} \quad g_j(x) \geq 0,\ j = 1, \ldots, n,$$

with the constraints in the ≥ 0 orientation used throughout the Gerzensee course (Fukuda 2026c, slide 3). The feasible set is D, the Lagrangian is $\mathcal{L}$, and multipliers are $y = (y_1, \ldots, y_n) \in \mathbb{R}^n_+$. Writing constraints as $g_j \geq 0$ rather than $g_j \leq 0$ fixes the sign of the multiplier at a maximum to be non-negative, which is what makes the shadow-price reading correct without a case distinction; the opposite convention, common in the optimisation literature, is noted in Chapter 12.

Probability

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, random variables are X, Y, Z, and realisations are x, y, z. Expectation, variance and covariance are $\mathbb{E}$, Var and Cov; $X \perp\!\!\!\perp Y$ denotes independence. Conditional expectation given an event H is $\mathbb{E}[X \mid H]$ and given a random variable Y is $\mathbb{E}[X \mid Y]$, the latter being a random variable, that is, a function of Y. Precision is $\tau = 1/\sigma^2$, following the treatment of normal-normal updating in (Fukuda 2026f, slides 27–28).

Dynamics

Time is $t \in \mathbb{N}_0$ in discrete time and $t \in [0, \infty)$ in continuous time. State variables are x and controls u; the discount factor is $\beta \in (0, 1)$; the transition is $x' = g(x, u)$, with a prime denoting the next period, as in (Fukuda 2026b, slide 8). The value function is V, the policy function is h, and the Bellman operator is T.

Contents

Chapter 1

Sets, Relations, Logic, and the Real Number System

1.1 Motivation: Why an Economist Starts with Sets

An economic model is a statement about a set of alternatives, a rule for comparing them, and a subset of them that is feasible. A consumer facing prices $p \in \mathbb{R}^n_{++}$ and wealth $m > 0$ chooses from the budget set $\mathcal{B}(p, m) = \{x \in \mathbb{R}^n_+ : \langle p, x \rangle \leq m\}$; a firm chooses from a production set; a player in a game chooses from a strategy set. The comparison rule is a binary relation on the set of alternatives, and the object economists call a utility function is a numerical representation of that relation. Every subsequent chapter presupposes that these three objects — a set, a relation, a distinguished subset — have been given a precise meaning.

This chapter fixes set-theoretic language (Section 1.2–Section 1.3); it records the forms of argument used throughout the volume (Section 1.4); and it constructs the real number system axiomatically (Section 1.5), because the single property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$ — the least upper bound property — is what makes every existence theorem in later chapters true. The chapter closes with the first substantive economic application, the representation of a preference relation by a utility function (Section 1.6).

1.2 Sets, Operations, and Mappings

1.2.1 Sets and their specification

We take the notion of a set as primitive: a *set* is a collection of objects, called its *elements* or members. We write $a \in A$ when a is an element of A and $a \notin A$ otherwise. The set with no elements is the *empty set*, written $\emptyset$.

A set may be specified by enumeration, as in $\{a_1, \ldots, a_n\}$, or by a defining property relative to an ambient set, as in $\{x \in X : P(x)\}$. The ambient set matters: the collection of all x with $P(x)$, taken without an ambient set, is not in general a set, and the restriction to subsets of a given set is what avoids the classical paradoxes. All constructions in this volume take place inside an ambient set that is either named or clear from context.

Definition 1.2.1 (Inclusion and equality). Let A and B be sets. We say A is a *subset* of B, written $A \subseteq B$, if $a \in A$ implies $a \in B$. We say $A = B$ if $A \subseteq B$ and $B \subseteq A$. If $A \subseteq B$ and $A \neq B$, then A is a *proper subset* of B, written $A \subsetneq B$.

The definition of equality dictates the standard proof technique: to establish $A = B$ one establishes two inclusions. This is used so often below that it is worth naming once.

Definition 1.2.2 (Operations on sets). Let A, B be subsets of an ambient set X, and let $(A_i)_{i \in I}$ be a family of subsets of X indexed by a set I.

(i) The *complement* of A in X is $A^{\mathsf{c}} := \{x \in X : x \notin A\}$.

(ii) The *union* and *intersection* of the family are

$$\bigcup_{i\in I} A_i := \{x \in X : x \in A_i \text{ for some } i \in I\}, \qquad \bigcap_{i\in I} A_i := \{x \in X : x \in A_i \text{ for all } i \in I\}.$$

(iii) The *set difference* is $A \setminus B := A \cap B^{\mathsf{c}}$.

(iv) The *Cartesian product* of sets $A_1, \dots, A_n$ is $\prod_{i=1}^n A_i := \{(a_1, \dots, a_n) : a_i \in A_i \text{ for each } i\}$, the set of ordered n-tuples.

(v) The *power set* of X is $\mathcal{P}(X) := \{A : A \subseteq X\}$.

The index set I in (ii) is arbitrary: it may be finite, countably infinite, or uncountable. This generality is not idle. In Chapter 4 the collection of open sets is required to be closed under arbitrary unions but only under finite intersections, and the whole of point-set topology turns on that asymmetry.

Proposition 1.2.1 (De Morgan's laws)**.** Let $(A_i)_{i\in I}$ be a family of subsets of X with $I \neq \emptyset$. Then

$$\Big(\bigcup_{i\in I} A_i\Big)^{\mathsf{c}} = \bigcap_{i\in I} A_i^{\mathsf{c}}, \qquad \Big(\bigcap_{i\in I} A_i\Big)^{\mathsf{c}} = \bigcup_{i\in I} A_i^{\mathsf{c}}.$$

Proof. For the first identity, $x \in (\bigcup_i A_i)^{\mathsf{c}}$ holds if and only if $x \in X$ and it is not the case that $x \in A_i$ for some $i \in I$; that is, if and only if $x \in X$ and $x \notin A_i$ for every $i \in I$; that is, if and only if $x \in A_i^{\mathsf{c}}$ for every i, which is $x \in \bigcap_i A_i^{\mathsf{c}}$.

The second identity follows from the first applied to the family $(A_i^{\mathsf{c}})_{i\in I}$ together with $(A^{\mathsf{c}})^{\mathsf{c}} = A$: indeed $(\bigcup_i A_i^{\mathsf{c}})^{\mathsf{c}} = \bigcap_i A_i$, and taking complements of both sides gives the claim. ∎

Remark 1.2.1. The hypothesis $I \neq \emptyset$ is needed. With $I = \emptyset$ the convention $\bigcup_{i\in\emptyset} A_i = \emptyset$ and $\bigcap_{i\in\emptyset} A_i = X$ makes both identities hold, but only because the intersection over the empty family is defined to be the ambient set. Outside an ambient set that convention has no meaning, which is a second reason for insisting on one.

1.2.2 Mappings

Definition 1.2.3 (Mapping)**.** Let A and B be sets. A *mapping* (or function) $f\colon A \to B$ assigns to each $a \in A$ exactly one element $f(a) \in B$. The set A is the *domain* and B the *codomain*. For $S \subseteq A$ and $T \subseteq B$,

$$f(S) := \{f(a) : a \in S\} \subseteq B, \qquad f^{-1}(T) := \{a \in A : f(a) \in T\} \subseteq A$$

are the *image* of S and the *preimage* of T. The set $f(A)$ is the *range*.

The preimage is better behaved than the image, and the difference is the reason that continuity is defined in Chapter 5 through preimages rather than images.

Proposition 1.2.2 (Images and preimages under set operations)**.** Let $f\colon A \to B$, let $(S_i)_{i\in I}$ be subsets of A, and let $(T_i)_{i\in I}$ be subsets of B with $I \neq \emptyset$. Then

(i) $f^{-1}(\bigcup_i T_i) = \bigcup_i f^{-1}(T_i)$ and $f^{-1}(\bigcap_i T_i) = \bigcap_i f^{-1}(T_i)$;

(ii) $f^{-1}(T^{\mathsf{c}}) = \big(f^{-1}(T)\big)^{\mathsf{c}}$ for every $T \subseteq B$;

(iii) $f(\bigcup_i S_i) = \bigcup_i f(S_i)$, whereas in general only $f(\bigcap_i S_i) \subseteq \bigcap_i f(S_i)$.

Proof. (i) We have $a \in f^{-1}(\bigcap_i T_i)$ if and only if $f(a) \in T_i$ for every i, if and only if $a \in f^{-1}(T_i)$ for every i. The union case is identical with "some" in place of "every".

(ii) $a \in f^{-1}(T^{\mathsf{c}})$ if and only if $f(a) \notin T$, if and only if $a \notin f^{-1}(T)$.

(iii) If $b \in f(\bigcup_i S_i)$ then $b = f(a)$ for some $a \in S_{i_0}$, so $b \in f(S_{i_0})$; conversely each $f(S_i) \subseteq f\big(\bigcup_j S_j\big)$. For intersections, if $a \in \bigcap_i S_i$ then $f(a) \in f(S_i)$ for every i, giving the stated inclusion. ∎

Example 1.2.1 (Failure of equality for images of intersections)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$, $S_1 = (-\infty, 0)$ and $S_2 = (0, \infty)$. Then $S_1 \cap S_2 = \emptyset$, so $f(S_1 \cap S_2) = \emptyset$, while $f(S_1) = f(S_2) = (0, \infty)$, so $\bigcap_i f(S_i) = (0, \infty)$. The inclusion in Proposition 1.2.2(iii) is therefore strict in general, and it is an equality for all families if and only if f is injective.

Definition 1.2.4 (Injection, surjection, bijection)**.** A mapping $f\colon A \to B$ is *injective* if $f(a) = f(a')$ implies $a = a'$; *surjective* if $f(A) = B$; and *bijective* if it is both. A bijection f has a unique inverse $f^{-1}\colon B \to A$ satisfying $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$.

Definition 1.2.5 (Finite, countable, uncountable)**.** A set A is *finite* if $A = \emptyset$ or there is a bijection $\{1, \ldots, n\} \to A$ for some $n \in \mathbb{N}$; *countably infinite* if there is a bijection $\mathbb{N} \to A$; *countable* if it is finite or countably infinite; and *uncountable* otherwise.

Proposition 1.2.3. A countable union of countable sets is countable. The set $\mathbb{Q}$ is countably infinite; the set $\mathbb{R}$ is uncountable.

Proof. The first two assertions are standard diagonal enumerations; see Rudin (1976, Thm. 2.12 and its corollary). For the third, Cantor's diagonal argument shows that no map $\mathbb{N} \to (0, 1)$ is surjective: given $f\colon \mathbb{N} \to (0, 1)$ with decimal expansions $f(n) = 0.d_{n1}d_{n2}\ldots$ (choosing the expansion that does not terminate in all nines), the number $x = 0.c_1c_2\ldots$ with $c_n = 5$ if $d_{nn} \neq 5$ and $c_n = 4$ otherwise lies in $(0, 1)$ and differs from every $f(n)$ in its n-th digit. Since $(0, 1) \subseteq \mathbb{R}$, the set $\mathbb{R}$ is uncountable. ∎

Remark 1.2.2 (Where countability matters in economics)**.** Countability decides what can be measured. In Chapter 14 a probability measure can be assigned to every subset of a countable sample space, but on an uncountable sample space such as $[0, 1]$ no countably additive probability measure exists on the full power set that extends length; one must restrict attention to a σ-algebra of measurable sets (Le Gall 2022, Ch. 1). The Gerzensee treatment makes this point when it moves from "an at most countable set Ω" (Fukuda 2026f, slide 3) to the remark that a continuum sample space requires a σ-algebra (Fukuda 2026g, slide 37).

1.3 Relations, Orders, and Preferences

Definition 1.3.1 (Binary relation)**.** A *binary relation* R on a set X is a subset $R \subseteq X \times X$; one writes xRy for $(x, y) \in R$. The relation is

(i) *reflexive* if xRx for all $x \in X$;

(ii) *complete* if for all $x, y \in X$, xRy or yRx (or both);

(iii) *transitive* if xRy and yRz imply xRz;

(iv) *symmetric* if xRy implies yRx;

(v) *antisymmetric* if xRy and yRx imply $x = y$.

Completeness implies reflexivity (take $y = x$), so the two are not independent hypotheses. Economics texts frequently list both; nothing is gained by it, and we list completeness alone where it is available.

Definition 1.3.2 (Equivalence relation and partition)**.** An *equivalence relation* on X is a reflexive, symmetric and transitive relation, usually written $\sim$. For $x \in X$ the *equivalence class* of x is $[x] := \{y \in X : y \sim x\}$. A *partition* of X is a family of non-empty, pairwise disjoint subsets whose union is X.

Proposition 1.3.1. The equivalence classes of an equivalence relation on X form a partition of X; conversely, every partition of X arises in this way from exactly one equivalence relation.

Proof. Reflexivity gives $x \in [x]$, so the classes are non-empty and cover X. Suppose $[x] \cap [y] \neq \emptyset$ and take z in the intersection. Then $z \sim x$ and $z \sim y$, so by symmetry and transitivity $x \sim y$; hence for any $w \in [x]$ we get $w \sim x \sim y$ and $w \in [y]$, and symmetrically, so $[x] = [y]$. Thus distinct classes are disjoint.

Conversely, given a partition $\mathcal{P}$, define $x \sim y$ if x and y lie in the same cell. This is reflexive, symmetric and transitive because the cells are non-empty, pairwise disjoint, and cover X; and its classes are exactly the cells. Uniqueness holds because the relation is determined by its classes. ∎

This proposition is used twice in later chapters: in Chapter 14, a finite partition $(H_j)_{j=1}^m$ of the sample space is precisely the information structure underlying the law of iterated expectations (Fukuda 2026f, slide 18); and in the theory of choice, the indifference relation derived from a preference partitions the consumption set into indifference curves.

1.3.1 Preference relations

Definition 1.3.3 (Preference relation). Let X be a non-empty set of alternatives. A *preference relation* $\succsim$ on X is a complete and transitive binary relation. Its *strict part* $\succ$ and *indifference part* $\sim$ are defined by

$$x \succ y \iff (x \succsim y \text{ and not } y \succsim x), \qquad x \sim y \iff (x \succsim y \text{ and } y \succsim x).$$

Proposition 1.3.2. Let $\succsim$ be a preference relation on X. Then $\sim$ is an equivalence relation, $\succ$ is transitive and irreflexive, and for all $x, y \in X$ exactly one of $x \succ y$, $y \succ x$, $x \sim y$ holds.

Proof. Reflexivity of $\sim$ follows from completeness. Symmetry is immediate from the definition. For transitivity of $\sim$, if $x \sim y$ and $y \sim z$ then $x \succsim y \succsim z$ gives $x \succsim z$, and $z \succsim y \succsim x$ gives $z \succsim x$.

For $\succ$: irreflexivity is immediate since $x \succsim x$. For transitivity, suppose $x \succ y$ and $y \succ z$. Transitivity of $\succsim$ gives $x \succsim z$. If also $z \succsim x$, then $z \succsim x \succsim y$ gives $z \succsim y$, contradicting $y \succ z$. Hence $x \succ z$.

For the trichotomy, completeness gives $x \succsim y$ or $y \succsim x$. If both, then $x \sim y$; if only the first, $x \succ y$; if only the second, $y \succ x$. These three cases are mutually exclusive by construction. ∎

Definition 1.3.4 (Utility representation). A function $u\colon X \to \mathbb{R}$ *represents* the preference relation $\succsim$ if

$$x \succsim y \iff u(x) \geq u(y) \qquad \text{for all } x, y \in X.$$

Theorem 1.3.1 (Representation on a countable set). Let X be a countable non-empty set and let $\succsim$ be a preference relation on X. Then there exists $u\colon X \to \mathbb{R}$ representing $\succsim$, and u may be chosen with $u(X) \subseteq (0,1)$.

Proof. Enumerate $X = \{x_1, x_2, \dots\}$ (finite or infinite) and choose weights $w_n > 0$ with $W := \sum_n w_n < \infty$, for instance $w_n = 2^{-n}$. Define

$$u(x) := \frac{1}{W} \sum_{n:\, x \succ x_n} w_n.$$

Suppose $x \succsim y$. If $y \succ x_n$ then $x \succ x_n$: indeed $x \succsim y \succsim x_n$ gives $x \succsim x_n$, and were $x_n \succsim x$ we would get $x_n \succsim x \succsim y$, contradicting $y \succ x_n$. Hence $\{n : y \succ x_n\} \subseteq \{n : x \succ x_n\}$ and $u(x) \geq u(y)$.

Conversely, suppose $u(x) \geq u(y)$ and, for a contradiction, that $y \succ x$. Then as above $\{n : x \succ x_n\} \subseteq \{n : y \succ x_n\}$. The inclusion is strict: since $y \succ x$ and $x = x_{n_0}$ for some n_0, we have $n_0 \in \{n : y \succ x_n\}$ while $n_0 \notin \{n : x \succ x_n\}$ by irreflexivity of $\succ$. As $w_{n_0} > 0$, this gives $u(y) > u(x)$, a contradiction. Hence $x \succsim y$.

Finally $0 \leq u \leq 1$; replacing u by $(1+u)/3$ places the range in $(0,1)$ without disturbing the representation, since the transformation is strictly increasing. ∎

Remark 1.3.1 (Why countability is not innocuous). Theorem 1.3.1 fails on uncountable X without further structure. Let $X = \mathbb{R}^2$ with the *lexicographic preference*

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1 > y_1, \text{ or } \big(x_1 = y_1 \text{ and } x_2 \geq y_2\big).$$

This relation is complete and transitive but admits no utility representation. Indeed, suppose u represented it. For each $t \in \mathbb{R}$ we have $(t, 1) \succ (t, 0)$, so $u(t, 1) > u(t, 0)$, and we may pick a rational $q(t) \in \big(u(t, 0), u(t, 1)\big)$ by Proposition 1.5.2(ii). If $t < t'$ then $(t', 0) \succ (t, 1)$, so $q(t) < u(t, 1) < u(t', 0) < q(t')$; hence $t \mapsto q(t)$ is a strictly increasing injection $\mathbb{R} \to \mathbb{Q}$, contradicting Proposition 1.2.3.

The lesson carries into Chapter 5: a continuous utility representation requires the preference to be continuous, in the sense that the sets $\{y : y \succsim x\}$ and $\{y : x \succsim y\}$ are closed, and lexicographic preference violates this. The general statement is Debreu's theorem; see Ok (2007, Ch. 2) and Corbae et al. (2009, Ch. 4).

1.4 Logic and Methods of Proof

1.4.1 Connectives and quantifiers

For propositions P, Q, the implication $P \Rightarrow Q$ is by definition the proposition $(\neg P) \vee Q$. This is the source of the "vacuous truth" convention: $P \Rightarrow Q$ is true whenever P is false, regardless of Q. The truth table is

P	Q	$\neg P$	$(\neg P) \vee Q$	$P \Rightarrow Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

which reproduces the table given in (Fukuda 2026g, slide 40). The convention is used there to show that $\emptyset$ is open: the statement "if $x \in \emptyset$ then $B_\varepsilon(x) \subseteq \emptyset$" is vacuously true, so $\emptyset$ satisfies the definition of an open set. That argument recurs in Chapter 4.

Quantifier order is not commutative and is a frequent source of error. The statement

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A : \; d(x, x_0) < \delta \Rightarrow d(f(x), f(x_0)) < \varepsilon$$

defines continuity of f at x_0, where δ may depend on both ε and x_0. Moving the quantifier $\forall x_0 \in A$ inside, to

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x_0 \in A \; \forall x \in A : \; d(x, x_0) < \delta \Rightarrow d(f(x), f(x_0)) < \varepsilon,$$

gives uniform continuity, a strictly stronger property (Chapter 5). The negation of a quantified statement exchanges $\forall$ and $\exists$ and negates the matrix: $\neg\big(\forall x \, P(x)\big)$ is $\exists x \, \neg P(x)$.

1.4.2 Three methods of proof

Proposition 1.4.1 (Principle of mathematical induction). Let $S \subseteq \mathbb{N}$ satisfy $1 \in S$ and $n \in S \Rightarrow n + 1 \in S$. Then $S = \mathbb{N}$.

The principle is equivalent to the well-ordering of $\mathbb{N}$ — every non-empty subset of $\mathbb{N}$ has a least element — and either may be taken as an axiom. Strong induction, from "$\{1, \ldots, n\} \subseteq S$ implies $n + 1 \in S$" conclude $S = \mathbb{N}$, follows by applying Proposition 1.4.1 to $S' = \{n : \{1, \ldots, n\} \subseteq S\}$.

Example 1.4.1 (Induction in a dynamic argument). The error bound in the contraction mapping theorem (Theorem 6.4.1) rests on the estimate $d(x_n, x_{n+1}) \leq \beta^n d(x_0, x_1)$ for all n, which is an induction: the case $n = 0$ is trivial, and $d(x_{n+1}, x_{n+2}) = d(Tx_n, Tx_{n+1}) \leq \beta d(x_n, x_{n+1}) \leq \beta^{n+1} d(x_0, x_1)$.

Proof by contradiction establishes P by assuming $\neg P$ and deriving a false statement; *proof by contraposition* establishes $P \Rightarrow Q$ by establishing $\neg Q \Rightarrow \neg P$, the two being equivalent by the truth table above. The uniqueness half of the contraction mapping theorem is a proof by contradiction; Proposition 1.3.2 used contraposition twice.

> **Remark 1.4.1** (A common error). "Suppose not; then . . . ; hence P" is not a proof by contradiction; it is a proof of P with an unused hypothesis. A genuine argument by contradiction must terminate in a statement known to be false. Confusing the two conceals which hypotheses are actually used, which matters when one later asks whether a hypothesis can be weakened — the recurring question of Chapter 12.

1.5 The Real Number System

The real numbers are characterised, up to isomorphism, by three groups of axioms. We state them because the third is used in essentially every existence theorem in this volume.

Definition 1.5.1 (Ordered field). A *field* is a set F with operations $+$ and $\cdot$ satisfying the usual commutative, associative and distributive laws, with additive identity 0, multiplicative identity $1 \neq 0$, additive inverses, and multiplicative inverses for every element other than 0. An *ordered field* is a field with a complete transitive antisymmetric relation $\leq$ such that $x \leq y$ implies $x + z \leq y + z$ for all z, and $0 \leq x$, $0 \leq y$ imply $0 \leq xy$.

Definition 1.5.2 (Bounds, supremum, infimum). Let $A \subseteq \mathbb{R}$ be non-empty. An *upper bound* for A is $b \in \mathbb{R}$ with $a \leq b$ for all $a \in A$; A is *bounded above* if such a b exists. The *supremum* $\sup A$ is the least upper bound: an upper bound s such that $s \leq b$ for every upper bound b. *Lower bound*, *bounded below* and *infimum* $\inf A$ are defined symmetrically. A set bounded above and below is *bounded*.

Assumption 1.5.1 (Completeness axiom). $\mathbb{R}$ is an ordered field in which every non-empty subset that is bounded above has a supremum in $\mathbb{R}$.

This is the *least upper bound property*. It fails in $\mathbb{Q}$: the set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound there. The whole difference between $\mathbb{Q}$ and $\mathbb{R}$, and with it every convergence theorem below, rests on this one axiom.

Proposition 1.5.1 (Characterisation of the supremum). Let $A \subseteq \mathbb{R}$ be non-empty and bounded above, and let $s \in \mathbb{R}$. Then $s = \sup A$ if and only if

(i) $a \leq s$ for all $a \in A$; and

(ii) for every $\varepsilon > 0$ there exists $a \in A$ with $a > s - \varepsilon$.

Proof. Suppose $s = \sup A$. Then (i) holds since s is an upper bound. For (ii), if some $\varepsilon > 0$ had $a \leq s - \varepsilon$ for all $a \in A$, then $s - \varepsilon$ would be an upper bound smaller than s, contradicting the minimality of s among upper bounds.

Conversely, assume (i) and (ii). By (i), s is an upper bound. Let b be any upper bound and suppose $b < s$. Applying (ii) with $\varepsilon = s - b > 0$ produces $a \in A$ with $a > s - (s - b) = b$, contradicting that b is an upper bound. Hence $s \leq b$ for every upper bound b, so s is least. ■

Part (ii) is the form in which the supremum is actually used: it converts the abstract existence of a least upper bound into the ability to produce elements of A arbitrarily close to it. Every "choose $a_n \in A$ with $a_n > \sup A - 1/n$" argument in later chapters is an application of it.

Proposition 1.5.2 (Archimedean property and density of $\mathbb{Q}$). (i) For all $x, y \in \mathbb{R}$ with $x > 0$ there exists $n \in \mathbb{N}$ with $nx > y$.

(ii) For all $x < y$ in $\mathbb{R}$ there exists $q \in \mathbb{Q}$ with $x < q < y$.

Proof. (i) Suppose not: then $A = \{nx : n \in \mathbb{N}\}$ is bounded above by y. By Assumption 1.5.1 it has a supremum s. Since $x > 0$, the number $s - x$ is not an upper bound, so $mx > s - x$ for some $m \in \mathbb{N}$, whence $(m + 1)x > s$, contradicting that s is an upper bound.

(ii) Since $y - x > 0$, part (i) gives $n \in \mathbb{N}$ with $n(y-x) > 1$, that is $ny > nx + 1$. Again by (i) there are integers $m_1 > nx$ and $m_2 > -nx$, so $-m_2 < nx < m_1$; let m be the least integer in $\{k \in \mathbb{Z} : -m_2 \le k \le m_1\}$ with $k > nx$, which exists by well-ordering. Then $m - 1 \le nx$, so $nx < m \le nx + 1 < ny$, and $q = m/n$ satisfies $x < q < y$. ∎

> **Remark 1.5.1** (The extended real line). It is convenient to write $\sup A = +\infty$ when A is unbounded above and $\inf A = -\infty$ when A is unbounded below, and to set $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$. The resulting object $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ is not a field — $\infty - \infty$ is undefined — and no algebraic identity may be applied to it without checking that the terms are finite. The convention separates two failures in optimisation: the value of an infeasible maximisation problem is $-\infty$ by the convention $\sup \emptyset = -\infty$, whereas the value of a feasible but unbounded problem is $+\infty$, and the two failures call for different remedies.

1.6 Economic Application: Choice, Feasibility, and the Existence Question

We can now state, in the language of this chapter, the problem that occupies the rest of the volume.

Example 1.6.1 (The consumer's problem, stated set-theoretically). Let $X = \mathbb{R}^n_+$ be the *consumption set*, let $\succsim$ be a preference relation on X, and let prices $p \in \mathbb{R}^n_{++}$ and wealth $m > 0$ be given. The *budget set* is

$$\mathcal{B}(p, m) := \{x \in \mathbb{R}^n_+ : \langle p, x\rangle \le m\}.$$

The consumer's problem is to find

$$x^* \in \mathcal{B}(p, m) \quad \text{such that} \quad x^* \succsim x \text{ for all } x \in \mathcal{B}(p, m),$$

and the *demand correspondence* $x^*(p, m)$ is the set of all such x^*.

Four questions arise immediately, and each is answered by a theorem in a later chapter.

(a) *Does a maximal element exist?* If $\succsim$ is represented by a continuous u and $\mathcal{B}(p, m)$ is compact, the Weierstrass theorem (Theorem 5.4.1) gives existence. Compactness of $\mathcal{B}(p, m)$ requires $p \in \mathbb{R}^n_{++}$: if some $p_i = 0$ the set is unbounded and existence can fail. This is why the strict positivity of prices is a hypothesis and not a normalisation.

(b) *Is the maximiser unique?* Uniqueness requires strict quasi-concavity of u, equivalently strict convexity of preferences (Chapter 9).

(c) *Can it be characterised by first-order conditions?* This requires differentiability of u and a constraint qualification (Chapter 12); the Cobb–Douglas case worked in (Fukuda 2026c, slides 21–32) is developed there in full.

(d) *How does it move with (p, m)?* Continuity of $x^*(\cdot, \cdot)$ as a correspondence is Berge's maximum theorem (Theorem 13.5.1); its differentiability, where it holds, follows from the implicit function theorem (Theorem 8.6.1).

These are the same four questions — existence, uniqueness, characterisation, comparative statics — asked of every optimisation problem in economics, including the dynamic ones of Chapter 15, where the object being chosen is an infinite sequence and the "budget set" is a set of feasible paths.

1.7 Chapter Summary

Sets are specified relative to an ambient set; equality of sets is proved by double inclusion; preimages commute with all set operations while images commute only with unions. A preference relation is a complete, transitive binary relation; its indifference part partitions the alternatives; and it admits a utility representation on any countable set, but not in general on an uncountable one, as lexicographic preference shows. Implication is $(\neg P) \vee Q$, which makes statements with false hypotheses vacuously true — the device by which $\emptyset$ is open. The real numbers form an ordered field with the least upper bound property; the working form of that property is Proposition 1.5.1(ii), and the Archimedean property and the density of the rationals are its first two consequences.

Exercise 1.7.1. Show that $f\colon A \to B$ is injective if and only if $f\left(\bigcap_{i\in I} S_i\right) = \bigcap_{i\in I} f(S_i)$ for every non-empty family $(S_i)_{i\in I}$ of subsets of A. (One direction is Example 1.2.1; for the other, consider two-element families.)

Exercise 1.7.2. Let $\succsim$ be a preference relation on a finite set X. Show directly, by induction on the number of elements of X, that a $\succsim$-maximal element exists. Where does the argument break down when X is infinite, and which hypothesis of the Weierstrass theorem repairs it?

Exercise 1.7.3. Let $A, B \subseteq \mathbb{R}$ be non-empty and bounded above. Prove that $\sup(A+B) = \sup A + \sup B$, where $A + B = \{a + b : a \in A,\ b \in B\}$. Show by example that $\sup(A \cdot B) = \sup A \cdot \sup B$ can fail, and identify a sign condition under which it holds.

Chapter 2

Vector Spaces, Linear Maps, and Matrices

2.1 Motivation: Linearity as the First Approximation

Linear algebra enters economics along two routes, which draw on different parts of the theory.

The first route is exact. Some economic models are linear by construction: a Leontief input–output system, a linear-quadratic control problem, a system of linear difference equations describing the dynamics of a linearised macroeconomic model, the normal equations of ordinary least squares. Here one needs solvability of $Ax = b$, invertibility, rank, and — for dynamics — the spectrum of A.

The second route is approximate, and it is the more pervasive. The central idea of differential calculus is that a smooth function is, near a point, well approximated by a linear one; the derivative *is* that linear map. The Gerzensee slides make this explicit when they present the total differential $\mathrm{d}y = \langle \nabla f, \mathrm{d}x \rangle$ as an inner product and the Jacobian as the matrix representing df (Fukuda 2026g, slides 10–15). Every comparative-statics computation in economics is the solution of a linear system obtained in this way, and the invertibility condition in the implicit function theorem (Theorem 8.6.1) is exactly the requirement that the relevant linear system be uniquely solvable.

This chapter develops the theory in the coordinate-free language of vector spaces and linear maps, with matrices appearing as coordinate representations. The payoff is that the same theorems apply verbatim in the infinite-dimensional function spaces of Chapter 6 and Chapter 15, where no matrix exists.

2.2 Vector Spaces

Definition 2.2.1 (Vector space). A *vector space* over $\mathbb{R}$ is a set V with an addition $+\colon V \times V \to V$ and a scalar multiplication $\cdot\colon \mathbb{R} \times V \to V$ such that $(V, +)$ is an abelian group with neutral element 0, and for all $\lambda, \mu \in \mathbb{R}$ and $v, w \in V$,

$$\lambda(v + w) = \lambda v + \lambda w, \quad (\lambda + \mu)v = \lambda v + \mu v, \quad (\lambda\mu)v = \lambda(\mu v), \quad 1 \cdot v = v.$$

Elements of V are *vectors*.

Example 2.2.1 (Vector spaces used in this volume).

(i) $\mathbb{R}^n$ with coordinatewise operations. This is the space of commodity bundles, price vectors, and parameter vectors.

(ii) The set $M_{m\times n}(\mathbb{R})$ of $m \times n$ real matrices.

(iii) The set $\mathbb{R}^S$ of all functions $f\colon S \to \mathbb{R}$ on a set S, with pointwise operations. When $S = \{1, \dots, n\}$ this is $\mathbb{R}^n$; when $S = \mathbb{N}$ it is the space of real sequences, the natural home of an infinite consumption path $(c_t)_{t\in\mathbb{N}_0}$; when $S = \Omega$ is a sample space it is the space of random variables.

(iv) The set $\mathcal{C}([a, b], \mathbb{R})$ of continuous real functions on $[a, b]$, and the set $\mathcal{B}(X, \mathbb{R})$ of bounded real functions on a set X. These are the spaces on which the Bellman operator of Chapter 15 acts.

Items (iii) and (iv) are infinite-dimensional, and no choice of coordinates reaches them; the coordinate-free definitions do.

Definition 2.2.2 (Subspace, span, linear independence, basis). Let V be a vector space over $\mathbb{R}$.

(i) A *subspace* of V is a non-empty $W \subseteq V$ closed under addition and scalar multiplication. Every subspace contains 0.

(ii) The *span* of $S \subseteq V$ is $\operatorname{span} S := \{\sum_{i=1}^{k} \lambda_i v_i : k \in \mathbb{N},\ \lambda_i \in \mathbb{R},\ v_i \in S\}$, with $\operatorname{span} \emptyset := \{0\}$. It is the smallest subspace containing S.

(iii) A finite list $v_1, \ldots, v_k$ is *linearly independent* if $\sum_i \lambda_i v_i = 0$ implies $\lambda_1 = \cdots = \lambda_k = 0$, and *linearly dependent* otherwise.

(iv) A *basis* of V is a linearly independent list spanning V. If V has a finite basis it is *finite-dimensional* and its *dimension* $\dim V$ is the common length of all its bases.

Theorem 2.2.1 (Well-definedness of dimension). In a finite-dimensional vector space, every two bases have the same length, and every linearly independent list can be extended to a basis.

Proof. The engine is the exchange lemma: in a vector space spanned by a list of length m, every linearly independent list has length at most m. Given that, if B_1 and B_2 are bases of lengths m_1 and m_2, then B_1 is independent and B_2 spans, giving $m_1 \le m_2$; exchanging roles gives $m_2 \le m_1$.

For the exchange lemma itself, let $w_1, \ldots, w_m$ span V and let $v_1, \ldots, v_k$ be independent with $k > m$. Adjoin v_1 to the spanning list: the list $v_1, w_1, \ldots, w_m$ is dependent (since $v_1 \in \operatorname{span}\{w_j\}$), so some w_j lies in the span of its predecessors and can be deleted, leaving a spanning list of length m containing v_1. Repeating with $v_2, \ldots, v_k$ — at each step the deleted vector must be some w_j rather than a previously inserted v_i, because the v_i are independent — exhausts the w_j after m steps while v_{m+1} still remains, a contradiction. The extension statement follows by adjoining vectors outside the current span until the span is all of V, which terminates by the same length bound. A complete treatment is Axler (2024, Ch. 2). ■

2.3 Linear Maps and their Matrix Representation

Definition 2.3.1 (Linear map). Let V, W be vector spaces over $\mathbb{R}$. A map $T\colon V \to W$ is *linear* if $T(\lambda v + \mu v') = \lambda T(v) + \mu T(v')$ for all $\lambda, \mu \in \mathbb{R}$ and $v, v' \in V$. Its *kernel* and *image* are

$$\operatorname{Ker} T := \{v \in V : T(v) = 0\}, \qquad \operatorname{Im} T := \{T(v) : v \in V\},$$

subspaces of V and W respectively. The *rank* of T is $\operatorname{rank} T := \dim \operatorname{Im} T$.

Theorem 2.3.1 (Matrix representation theorem). A map $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if there exists a unique $m \times n$ matrix A with $T(x) = Ax$ for every $x \in \mathbb{R}^n$. Moreover

$$A = \begin{bmatrix} T(e^1) & T(e^2) & \cdots & T(e^n) \end{bmatrix},$$

where e^i is the i-th standard basis vector, so that the i-th column of A is $Ae^i = T(e^i)$.

Proof. If $T(x) = Ax$ then linearity is the distributive law for matrix multiplication.

Conversely, let T be linear and define A by the displayed formula. Writing $x = \sum_{i=1}^{n} x_i e^i$ and using linearity $n - 1$ times,

$$T(x) = T\Bigl(\sum_{i=1}^{n} x_i e^i\Bigr) = \sum_{i=1}^{n} x_i\, T(e^i) = \begin{bmatrix} T(e^1) & \cdots & T(e^n) \end{bmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = Ax.$$

For uniqueness, suppose $Ax = Bx$ for all x. Taking $x = e^i$ gives $Ae^i = Be^i$, that is, the i-th columns of A and B agree; as this holds for every i, $A = B$. ■

This is Theorem 5.1 of Fukuda (2026d) and is presented on slides 12–14 of (Fukuda 2026g). Two readings of Ax are useful and both appear later. Row by row, $(Ax)_i = \langle a^i, x\rangle$ where a^i is the i-th row of A; this is the reading under which each row is a linear functional, used in Chapter 3 for the Riesz representation theorem. Column by column, $Ax = \sum_i x_i Ae^i$ is a linear combination of the columns of A with weights x_i; this is the reading under which $\operatorname{Im} A$ is the column space, used in Chapter 11 for least squares.

Theorem 2.3.2 (Rank–nullity)**.** Let V be finite-dimensional and $T\colon V \to W$ linear. Then

$$\dim V = \dim \operatorname{Ker} T + \operatorname{rank} T.$$

Proof. Let $u_1, \dots, u_k$ be a basis of $\operatorname{Ker} T$ and extend it, by Theorem 2.2.1, to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V, so $\dim V = k + r$. We claim $T(v_1), \dots, T(v_r)$ is a basis of $\operatorname{Im} T$.

They span: any $T(v)$ with $v = \sum_i \alpha_i u_i + \sum_j \beta_j v_j$ equals $\sum_j \beta_j T(v_j)$ because $T(u_i) = 0$. They are independent: if $\sum_j \beta_j T(v_j) = 0$ then $T\big(\sum_j \beta_j v_j\big) = 0$, so $\sum_j \beta_j v_j \in \operatorname{Ker} T$ and hence $\sum_j \beta_j v_j = \sum_i \alpha_i u_i$ for some α_i; independence of the full basis forces all β_j (and α_i) to vanish. Therefore $\operatorname{rank} T = r$ and $\dim V = k + r$. ∎

Corollary 2.3.1 (Solvability of $Ax = b$)**.** Let A be $m \times n$ and $b \in \mathbb{R}^m$.

(i) $Ax = b$ has a solution if and only if $b \in \operatorname{Im} A$, equivalently $\operatorname{rank}[A\ b] = \operatorname{rank} A$.

(ii) If a solution exists, the solution set is $x_0 + \operatorname{Ker} A$ for any particular solution x_0; it is a singleton if and only if $\operatorname{rank} A = n$.

(iii) For $m = n$: A is invertible $\iff$ $\operatorname{rank} A = n$ $\iff$ $\operatorname{Ker} A = \{0\}$ $\iff$ $\det A \neq 0$ $\iff$ $Ax = b$ has exactly one solution for every b.

Proof. (i) is the definition of the image. (ii) If $Ax_0 = b$ then $Ax = b$ iff $A(x - x_0) = 0$. The solution is unique iff $\operatorname{Ker} A = \{0\}$, which by Theorem 2.3.2 is $\operatorname{rank} A = n$. (iii) For $m = n$, Theorem 2.3.2 makes $\operatorname{Ker} A = \{0\}$ and $\operatorname{Im} A = \mathbb{R}^n$ equivalent; the equivalence with $\det A \neq 0$ is the standard determinant criterion, and invertibility is then existence and uniqueness for every b. ∎

Remark 2.3.1 (Square matters)**.** Part (iii) is a statement about square matrices only. For $m \neq n$ injectivity and surjectivity are genuinely different, and the distinction is economically substantive: in the least-squares problem of Chapter 11 the design matrix X is $n \times k$ with $n \gg k$, so $Xb = y$ typically has no solution — which is why one projects instead of solving. In econometrics the condition $\operatorname{rank} X = k$ is the no-perfect-multicollinearity assumption, and by Corollary 2.3.1(ii) it is precisely what makes the estimator unique.

2.4 Determinants and Volume

Definition 2.4.1 (Determinant)**.** The *determinant* is the unique function $\det\colon M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ that is linear in each column separately, vanishes when two columns coincide, and satisfies $\det I_n = 1$. Explicitly

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{\sigma(i)i},$$

the sum running over permutations of $\{1, \dots, n\}$.

Proposition 2.4.1 (Properties)**.** For $A, B \in M_{n\times n}(\mathbb{R})$: $\det(AB) = \det A \det B$; $\det A^\mathsf{T} = \det A$; A is invertible if and only if $\det A \neq 0$, in which case $\det(A^{-1}) = (\det A)^{-1}$; and for $\lambda \in \mathbb{R}$, $\det(\lambda A) = \lambda^n \det A$.

The geometric content is what the Gerzensee slides emphasise: $|\det A|$ is the factor by which the linear map $x \mapsto Ax$ magnifies n-dimensional volume, and the sign of $\det A$ records whether orientation is preserved (Fukuda 2026g, slide 15). Two consequences are used later.

Example 2.4.1 (Change of variables). In the change-of-variables formula for multiple integrals (Theorem 10.4.2), the factor $|\det J|$ appears exactly because J is the linear approximation of the substitution and $|\det J|$ is its volume magnification. The polar-coordinate computation of the Gaussian integral in (Fukuda 2026a, slides 91–93), where $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$, is this statement with

$$J = \begin{pmatrix} \partial x/\partial r & \partial x/\partial\theta \\ \partial y/\partial r & \partial y/\partial\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det J = r(\cos^2\theta + \sin^2\theta) = r > 0.$$

The full derivation is given in subsection 10.7.2.

Example 2.4.2 (Cramer's rule and comparative statics). If $\det A \neq 0$, the unique solution of $Ax = b$ has i-th coordinate $x_i = \det A_i / \det A$, where A_i is A with its i-th column replaced by b. In comparative statics the system $Ax = b$ arises by differentiating equilibrium conditions, x collects the responses of endogenous variables, and b the direct effect of the parameter; Cramer's rule then delivers the sign of an individual response from the signs of two determinants. This is how sign restrictions on a Hessian or a Jacobian translate into comparative-statics predictions in Chapter 8.

2.5 Eigenvalues, Diagonalisation, and Symmetric Matrices

Definition 2.5.1 (Eigenvalue and eigenvector). Let $A \in M_{n\times n}(\mathbb{R})$. A scalar $\lambda \in \mathbb{C}$ is an *eigenvalue* of A if $Av = \lambda v$ for some $v \neq 0$; such a v is an *eigenvector*. The eigenvalues are the roots of the *characteristic polynomial* $p_A(\lambda) = \det(A - \lambda I)$. The *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \text{ an eigenvalue of } A\}$.

Theorem 2.5.1 (Spectral theorem for symmetric matrices). Let $A \in M_{n\times n}(\mathbb{R})$ be symmetric. Then all eigenvalues of A are real, and there is an orthogonal matrix Q (that is, $Q^\mathsf{T} Q = I$) and a diagonal matrix $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with $A = Q\Lambda Q^\mathsf{T}$. Equivalently, $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A.

Proof sketch. Reality of the eigenvalues: if $Av = \lambda v$ with $v \in \mathbb{C}^n \setminus \{0\}$, then $\bar{v}^\mathsf{T} Av = \lambda \bar{v}^\mathsf{T} v$; taking conjugate transposes and using $A = A^\mathsf{T} = \bar{A}$ gives $\bar{v}^\mathsf{T} Av = \bar{\lambda}\bar{v}^\mathsf{T} v$; since $\bar{v}^\mathsf{T} v = \|v\|^2 > 0$ we get $\lambda = \bar{\lambda}$.

Diagonalisation proceeds by induction on n. Let λ_1 be an eigenvalue with unit eigenvector q_1; the subspace $q_1^\perp$ is A-invariant, because $\langle q_1, Aw\rangle = \langle Aq_1, w\rangle = \lambda_1 \langle q_1, w\rangle = 0$ for $w \perp q_1$. Applying the induction hypothesis to the restriction of A to $q_1^\perp$, which is symmetric with respect to the induced inner product, and adjoining q_1 produces the orthonormal eigenbasis. The full argument, in the form of the real spectral theorem, is Axler (2024, Ch. 7). ■

Definition 2.5.2 (Quadratic form and definiteness). Let A be symmetric. The *quadratic form* associated with A is $q_A(x) = x^\mathsf{T} Ax$. The matrix A is *positive definite* ($A \succ 0$) if $q_A(x) > 0$ for all $x \neq 0$; *positive semidefinite* ($A \succeq 0$) if $q_A(x) \geq 0$ for all x; and negative (semi)definite if $-A$ is positive (semi)definite. Otherwise A is *indefinite*.

Proposition 2.5.1 (Definiteness via the spectrum). A symmetric A is positive definite if and only if all its eigenvalues are strictly positive, and positive semidefinite if and only if all are non-negative. Moreover

$$\lambda_{\min} \|x\|^2 \leq x^\mathsf{T} Ax \leq \lambda_{\max} \|x\|^2 \qquad \text{for all } x \in \mathbb{R}^n,$$

where $\lambda_{\min}, \lambda_{\max}$ are the smallest and largest eigenvalues.

Proof. Write $A = Q\Lambda Q^\mathsf{T}$ as in Theorem 2.5.1 and set $y = Q^\mathsf{T} x$. Since Q is orthogonal, $\|y\| = \|x\|$ and $x \neq 0 \iff y \neq 0$, and

$$x^\mathsf{T} Ax = y^\mathsf{T} \Lambda y = \sum_{i=1}^n \lambda_i y_i^2.$$

If every $\lambda_i > 0$ this is strictly positive for $y \neq 0$. Conversely, if $\lambda_j \leq 0$ for some j, taking $y = e^j$, that is $x = Qe^j$, gives $x^\mathsf{T} Ax = \lambda_j \leq 0$ with $x \neq 0$. The semidefinite case is identical with weak inequalities. The two-sided bound follows from $\lambda_{\min} \sum_i y_i^2 \leq \sum_i \lambda_i y_i^2 \leq \lambda_{\max} \sum_i y_i^2$. ■

Remark 2.5.1 (Only the symmetric part matters). For any square A, $x^\top Ax = x^\top\big(\frac{1}{2}(A + A^\top)\big)x$, because $x^\top Ax$ is a scalar and hence equals its own transpose $x^\top A^\top x$. A quadratic form therefore determines and is determined by the symmetric part of A alone, and one may always assume A symmetric. This is the content of the gradient computation in (Fukuda 2026a, slide 49): for $f(x) = x^\top Ax$,

$$\nabla f(x) = (A + A^\top)x,$$

which reduces to $2Ax$ exactly when A is symmetric. Forgetting the factor $A + A^\top$ is one of the most common errors in matrix differentiation.

Proposition 2.5.2 (Leading principal minors). Let A be symmetric $n \times n$ and let D_k denote the determinant of its upper-left $k \times k$ submatrix. Then $A \succ 0$ if and only if $D_k > 0$ for $k = 1, \dots, n$; and $A \prec 0$ if and only if $(-1)^k D_k > 0$ for $k = 1, \dots, n$, that is, the leading principal minors alternate in sign starting negative.

Remark 2.5.2 (The semidefinite case is different). The analogous criterion for *semi*definiteness requires all principal minors, not merely the leading ones, to be non-negative. The matrix $A = \left(\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ has $D_1 = D_2 = 0$, so all leading principal minors are non-negative, yet A is negative semidefinite and not positive semidefinite. This trap matters when checking second-order conditions at a boundary optimum in Chapter 12, where semidefiniteness rather than definiteness is what the theory delivers.

2.6 Singular Values, the Pseudoinverse, and Conditioning

Theorem 2.5.1 applies only to symmetric square matrices. The data matrices of econometrics are rectangular and are almost never symmetric, and the object one needs from them — how much a linear map can stretch, how nearly it fails to be injective, how much a solution moves when the data move — is supplied by the singular value decomposition. It is the tool behind the numerical behaviour of least squares in Chapter 11 and behind the diagnosis of collinearity.

Definition 2.6.1 (Singular values). Let $A \in M_{m\times n}(\mathbb{R})$. The *singular values* of A are the non-negative square roots $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_{\min(m,n)} \ge 0$ of the eigenvalues of the symmetric positive semidefinite matrix $A^\top A$.

The definition is legitimate: $A^\top A$ is symmetric, and $x^\top A^\top Ax = \|Ax\|^2 \ge 0$, so by Proposition 2.5.1 its eigenvalues are real and non-negative.

Lemma 2.6.1. $\operatorname{Ker}(A^\top A) = \operatorname{Ker} A$, and hence $\operatorname{rank}(A^\top A) = \operatorname{rank} A$.

Proof. If $Ax = 0$ then $A^\top Ax = 0$. Conversely, if $A^\top Ax = 0$ then $\|Ax\|^2 = x^\top A^\top Ax = 0$, so $Ax = 0$. The rank statement is Theorem 2.3.2 applied to both matrices, which have the same domain $\mathbb{R}^n$. ■

Theorem 2.6.1 (Singular value decomposition). Let $A \in M_{m\times n}(\mathbb{R})$ have rank r. There exist orthogonal matrices $U \in M_{m\times m}(\mathbb{R})$ and $V \in M_{n\times n}(\mathbb{R})$ such that

$$A = U\Sigma V^\top, \tag{2.1}$$

where $\Sigma \in M_{m\times n}(\mathbb{R})$ has (i,i) entry σ_i for $i \le r$ and all other entries zero, and $\sigma_1 \ge \cdots \ge \sigma_r > 0$ are the non-zero singular values of A.

Proof. Apply Theorem 2.5.1 to $A^\top A$: there is an orthogonal $V = [v^1 \ \cdots \ v^n]$ with $A^\top Av^i = \lambda_i v^i$ and $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$. By Lemma 2.6.1 the rank of $A^\top A$ is r, so $\lambda_i > 0$ for $i \le r$ and $\lambda_i = 0$ for $i > r$; set $\sigma_i = \sqrt{\lambda_i}$.

For $i \le r$ define $u^i := Av^i/\sigma_i$. These are orthonormal:

$$\langle u^i, u^j \rangle = \frac{(v^i)^\top A^\top Av^j}{\sigma_i\sigma_j} = \frac{\lambda_j \langle v^i, v^j \rangle}{\sigma_i\sigma_j} = \delta_{ij},$$

using $\lambda_j = \sigma_j^2$. Extend $u^1, \dots, u^r$ to an orthonormal basis $u^1, \dots, u^m$ of $\mathbb{R}^m$ and let $U = [u^1 \ \cdots \ u^m]$.

For $i > r$ we have $\|Av^i\|^2 = (v^i)^\mathsf{T} A^\mathsf{T} A v^i = \lambda_i = 0$, so $Av^i = 0$. Hence $Av^i = \sigma_i u^i$ for $i \le r$ and $Av^i = 0$ for $i > r$, which is precisely $AV = U\Sigma$. Right-multiplying by V^T gives (2.1). ■

Corollary 2.6.1 (What the decomposition displays)**.** With the notation of Theorem 2.6.1:

(i) $\operatorname{rank} A = r$, $\operatorname{Im} A = \operatorname{span}\{u^1, \dots, u^r\}$ and $\operatorname{Ker} A = \operatorname{span}\{v^{r+1}, \dots, v^n\}$;

(ii) the operator norm satisfies $\|A\|_2 := \max_{\|x\|=1} \|Ax\| = \sigma_1$, and $\min_{\|x\|=1} \|Ax\| = \sigma_n$ when $m \ge n$;

(iii) $A = \sum_{i=1}^r \sigma_i u^i (v^i)^\mathsf{T}$, a sum of r rank-one matrices.

Proof. (i) is read off the proof of Theorem 2.6.1. For (ii), put $y = V^\mathsf{T} x$; orthogonality gives $\|y\| = \|x\|$, and $\|Ax\|^2 = \|U\Sigma y\|^2 = \|\Sigma y\|^2 = \sum_i \sigma_i^2 y_i^2$, which over $\|y\| = 1$ is maximised at $y = e^1$ and, when $m \ge n$ so that every coordinate of y carries a σ_i, minimised at $y = e^n$. For (iii), expand $U\Sigma V^\mathsf{T}$ column by column. ■

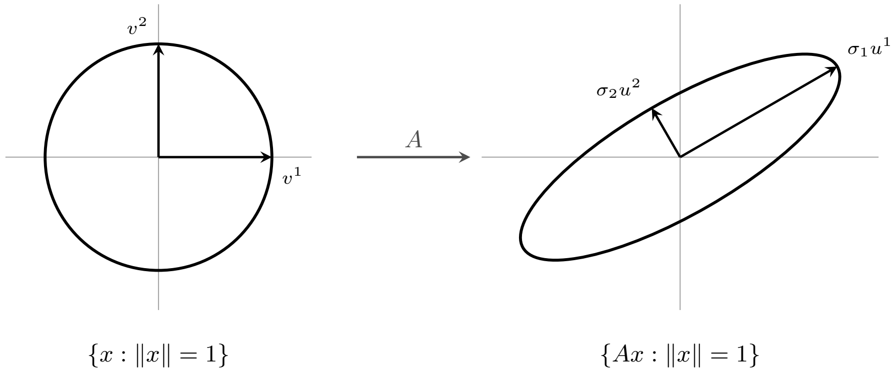


Figure 2.1: Corollary 2.6.1 read geometrically. A linear map carries the unit sphere onto an ellipsoid; the right singular vectors v^i are the directions in the domain that map to the axes of the ellipsoid, the left singular vectors u^i are those axes, and the singular values σ_i are their half-lengths. The largest, σ_1, is the operator norm, and the smallest is the distance to the nearest singular matrix; their ratio is the condition number of Definition 2.6.3.

Definition 2.6.2 (Moore–Penrose pseudoinverse)**.** With $A = U\Sigma V^\mathsf{T}$ as in Theorem 2.6.1, the *Moore–Penrose pseudoinverse* of A is

$$A^+ := V\Sigma^+ U^\mathsf{T} \in M_{n\times m}(\mathbb{R}),$$

where $\Sigma^+ \in M_{n\times m}(\mathbb{R})$ has (i,i) entry $1/\sigma_i$ for $i \le r$ and all other entries zero.

Theorem 2.6.2 (The pseudoinverse solves least squares)**.** Let $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$. The set of minimisers of $x \mapsto \|Ax - b\|$ over $\mathbb{R}^n$ is the affine set $A^+ b + \operatorname{Ker} A$, and $A^+ b$ is its element of smallest norm. If $\operatorname{rank} A = n$, the minimiser is unique and

$$A^+ = (A^\mathsf{T} A)^{-1} A^\mathsf{T}.$$

Proof. Write $y = V^\mathsf{T} x$ and $c = U^\mathsf{T} b$. Since U and V are orthogonal,

$$\|Ax - b\| = \|U\Sigma V^\mathsf{T} x - b\| = \|\Sigma y - c\| \quad \text{and} \quad \|x\| = \|y\|,$$

and $\|\Sigma y - c\|^2 = \sum_{i=1}^r (\sigma_i y_i - c_i)^2 + \sum_{i=r+1}^m c_i^2$. The coordinates $y_{r+1}, \dots, y_n$ do not appear, so the minimisers are exactly the y with $y_i = c_i/\sigma_i$ for $i \le r$ and $y_{r+1}, \dots, y_n$ arbitrary. Among these, $\|y\|^2$ is smallest when the free coordinates vanish, which is $y = \Sigma^+ c$, that is $x = A^+ b$. The free directions are $\operatorname{span}\{v^{r+1}, \dots, v^n\} = \operatorname{Ker} A$ by Corollary 2.6.1(i).

If rank $A = n$ then $r = n$, Ker $A = \{0\}$ and the minimiser is unique. In that case $A^\top A = V\Sigma^\top\Sigma V^\top = V\,\mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2)V^\top$ is invertible, and

$$(A^\top A)^{-1}A^\top = V\,\mathrm{diag}(\sigma_i^{-2})V^\top \cdot V\Sigma^\top U^\top = V\,\mathrm{diag}(\sigma_i^{-2})\Sigma^\top U^\top = V\Sigma^+U^\top = A^+. \qquad \blacksquare$$

Theorem 2.6.2 is the algebraic half of the least-squares problem; Chapter 11 gives the geometric half, and Proposition 11.5.1 identifies A^+b with the ordinary least squares estimator when A is the design matrix. What the singular values add is a quantitative statement about how badly conditioned that estimator can be.

Definition 2.6.3 (Condition number)**.** The *condition number* of an invertible $A \in M_{n\times n}(\mathbb{R})$ is

$$\kappa(A) := \|A\|_2 \left\|A^{-1}\right\|_2 = \frac{\sigma_1}{\sigma_n},$$

and $\kappa(A) := +\infty$ for a singular square matrix. For a general $A \in M_{m\times n}(\mathbb{R})$ of rank $r \geq 1$, the ratio

$$\kappa_{\mathrm{eff}}(A) := \frac{\sigma_1}{\sigma_r} = \|A\|_2 \left\|A^+\right\|_2$$

is the *effective condition number*, the condition number of A restricted to $(\mathrm{Ker}\,A)^\perp \to \mathrm{Im}\,A$, on which A is a bijection.

The two agree when A is square and invertible, and only then. A singular square matrix has $\kappa(A) = +\infty$, because arbitrarily small perturbations of b move the solution set of $Ax = b$ from a coset of Ker A to the empty set, while $\kappa_{\mathrm{eff}}(A)$ is finite and measures the sensitivity of the least-squares solution A^+b of Theorem 2.6.2. Reporting κ_{eff} as "the condition number" of a rank-deficient matrix would say that a singular system is well conditioned.

Proposition 2.6.1 (Sensitivity of a linear solve)**.** Let $A \in M_{n\times n}(\mathbb{R})$ be invertible, $b \neq 0$, and let x and $x + \delta x$ solve $Ax = b$ and $A(x + \delta x) = b + \delta b$. Then

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A)\,\frac{\|\delta b\|}{\|b\|}. \tag{2.2}$$

The bound is attained for some pair $(b, \delta b)$.

Proof. Subtracting the two systems, $\delta x = A^{-1}\delta b$, so $\|\delta x\| \leq \left\|A^{-1}\right\|_2 \|\delta b\| = \|\delta b\|/\sigma_n$ by Corollary 2.6.1(ii) applied to A^{-1}, whose singular values are $\sigma_n^{-1} \geq \cdots \geq \sigma_1^{-1}$. Also $\|b\| = \|Ax\| \leq \sigma_1 \|x\|$, so $1/\|x\| \leq \sigma_1/\|b\|$. Multiplying the two inequalities gives (2.2). Taking $b = \sigma_1 u^1$, so that $x = v^1$, and $\delta b = \varepsilon u^n$, so that $\delta x = (\varepsilon/\sigma_n)v^n$, turns both into equalities. $\blacksquare$

Corollary 2.6.2 (Normal equations square the conditioning)**.** For any $A \in M_{m\times n}(\mathbb{R})$ of rank $r \geq 1$ the non-zero singular values of $A^\top A$ are $\sigma_1^2 \geq \cdots \geq \sigma_r^2$, so $\kappa_{\mathrm{eff}}(A^\top A) = \kappa_{\mathrm{eff}}(A)^2$. If A has full column rank $r = n$, then $A^\top A$ is invertible and the identity reads $\kappa(A^\top A) = \kappa_{\mathrm{eff}}(A)^2$.

Proof. $A^\top A = V\Sigma^\top\Sigma V^\top$ is symmetric positive semidefinite with eigenvalues σ_i^2, and for a symmetric positive semidefinite matrix the singular values coincide with the eigenvalues. $\blacksquare$

Remark 2.6.1 (Collinearity, and why one does not invert $X^\top X$)**.** Let the $T \times n$ design matrix X have full column rank n, so that $\sigma_n > 0$. Then σ_n is the distance in operator norm from X to the nearest rank-deficient matrix, and $\kappa_{\mathrm{eff}}(X) = \sigma_1/\sigma_n$ is large when the columns of X are close to linearly dependent. Two consequences follow, one statistical and one numerical.

Statistically, Theorem 2.6.2 and Corollary 2.6.1 give $\mathrm{Var}(\hat\beta) = \sigma^2(X^\top X)^{-1} = \sigma^2 \sum_i \sigma_i^{-2} v^i (v^i)^\top$ under homoskedastic errors of variance σ^2, so the variance of $\left\langle v^n, \hat\beta \right\rangle$, the coefficient along the least-excited direction of the data, is σ^2/σ_n^2. Collinearity does not bias the estimator; it makes one linear combination of its coordinates imprecise, and (2.2) says the same thing about the deterministic problem.

Numerically, Corollary 2.6.2 says that assembling $X^\top X$ and inverting it, as the expression inv(X'*X)*X'*y in the accompanying programs does, works with a matrix whose condition number is the square of the one carried by X. If $\kappa_{\text{eff}}(X) = 10^8$, then $\kappa(X^\mathsf{T} X) = 10^{16}$, which in double precision — where the relative spacing of representable numbers is about 2.2×10^{-16} — leaves no correct digit. The QR route of subsection 11.4.1 and the pseudoinverse of Definition 2.6.2 both operate on X itself and inherit $\kappa_{\text{eff}}(X)$ rather than its square, which is why library least-squares routines use them.

2.7 Non-negative Matrices

Transition matrices of Markov chains, input–output coefficient matrices, and migration or trade-share matrices share a feature that the spectral theorem does not see: their entries are non-negative. That single sign restriction forces the spectral radius itself to be an eigenvalue, with an eigenvector that can be normalised to a probability distribution.

Definition 2.7.1 (Non-negative, irreducible, primitive)**.** A matrix $A \in M_{n\times n}(\mathbb{R})$ is *non-negative*, written $A \geq 0$, if every entry is non-negative, and *positive*, written $A \gg 0$, if every entry is strictly positive. A non-negative A is *irreducible* if for every pair (i, j) there is $k \geq 1$ with $(A^k)_{ij} > 0$, and *primitive* if there is a single $k \geq 1$ with $A^k \gg 0$. A non-negative A is *row-stochastic* if $A\mathbb{1} = \mathbb{1}$, where $\mathbb{1} = (1, \dots, 1)^\mathsf{T}$.

Theorem 2.7.1 (Perron–Frobenius)**.** Let $A \in M_{n\times n}(\mathbb{R})$ be non-negative and irreducible. Then

(i) $\rho(A) > 0$ is an eigenvalue of A, and it is a simple root of the characteristic polynomial;

(ii) there is an eigenvector $x \gg 0$ with $Ax = \rho(A)x$, and up to a positive scalar it is the only non-negative eigenvector of A;

(iii) if A is primitive, every other eigenvalue λ satisfies $|\lambda| < \rho(A)$, and $\rho(A)^{-k}A^k \to xy^\mathsf{T}/\langle y, x\rangle$ as $k \to \infty$, where $y \gg 0$ is the corresponding eigenvector of A^T.

The proof of (i) and (ii) rests on a fixed-point argument that the present chapter cannot yet supply: one maximises $\min_{i:x_i>0}(Ax)_i/x_i$ over the unit simplex and shows the maximiser is an eigenvector, which requires the Weierstrass theorem of Chapter 5 on a set whose compactness is established in Chapter 4, and part (iii) requires a further argument on the modulus of the subdominant eigenvalue. Both are omitted here and are given in full in Meyer (2000, Ch. 8); the statement is used below and in Chapter 14 only through the corollaries, which are proved.

Corollary 2.7.1 (Stochastic matrices)**.** Let $P \geq 0$ be row-stochastic. Then $\rho(P) = 1$ and $P\mathbb{1} = \mathbb{1}$. If P is in addition primitive, there is a unique $\pi \in \mathbb{R}^n$ with $\pi \geq 0$, $\sum_i \pi_i = 1$ and $\pi^\mathsf{T} P = \pi^\mathsf{T}$, it satisfies $\pi \gg 0$, and $P^k \to \mathbb{1}\pi^\mathsf{T}$ entrywise: every row of P^k converges to π whatever the initial state.

Proof. $P\mathbb{1} = \mathbb{1}$ is the definition, so 1 is an eigenvalue and $\rho(P) \geq 1$. For the reverse inequality let $Pv = \lambda v$ with $v \neq 0$ and pick i with $|v_i| = \max_j |v_j| > 0$; then $|\lambda|\,|v_i| = \left|\sum_j P_{ij}v_j\right| \leq \sum_j P_{ij}\,|v_j| \leq |v_i|$, so $|\lambda| \leq 1$. Hence $\rho(P) = 1$. Applying Theorem 2.7.1(ii) to P^T, which is irreducible whenever P is, produces a strictly positive left eigenvector for the eigenvalue 1; normalising its coordinates to sum to 1 gives π, and simplicity in (i) gives uniqueness. The limit in (iii) reads $P^k \to \mathbb{1}\pi^\mathsf{T}/\langle \pi, \mathbb{1}\rangle = \mathbb{1}\pi^\mathsf{T}$, since $\langle \pi, \mathbb{1}\rangle = 1$. ■

Corollary 2.7.2 (Hawkins–Simon)**.** Let $A \geq 0$ be the input–output coefficient matrix of Example 2.8.1. The following are equivalent: (a) $\rho(A) < 1$; (b) $(I - A)^{-1}$ exists and is non-negative; (c) there is $x \geq 0$ with $x \gg Ax$. Under any of them, $(I - A)^{-1} = \sum_{k\geq 0} A^k$, and the gross output $x = (I - A)^{-1}d$ required by a non-negative final demand d is non-negative.

Proof. (a)⇒(b). The Neumann series $\sum_k A^k$ converges when $\rho(A) < 1$ and sums to $(I - A)^{-1}$, which is then a sum of non-negative matrices.

(b)⇒(c). Take any $d \gg 0$ and put $x = (I - A)^{-1}d \geq 0$. Then $x - Ax = d \gg 0$.

(c)⇒(a). From $x \gg Ax \geq 0$ we get $x \gg 0$, so $\theta := \max_i (Ax)_i/x_i$ is well defined and lies in $[0,1)$, and $Ax \leq \theta x$ coordinatewise. Since $A \geq 0$ preserves the coordinatewise order, induction gives $A^k x \leq \theta^k x$, hence $A^k x \to 0$. For any $z \in \mathbb{C}^n$ choose $c > 0$ with $|z_i| \leq c x_i$ for every i, possible because $x \gg 0$; then $\left|(A^k z)_i\right| \leq (A^k |z|)_i \leq c(A^k x)_i \to 0$, so $A^k z \to 0$ for every z. If $Av = \lambda v$ with $v \neq 0$, then $\lambda^k v = A^k v \to 0$ forces $|\lambda| < 1$; hence $\rho(A) < 1$.

The final assertions are immediate from (b). ■

The economic content of Corollary 2.7.2 is that a productive technology — one in which some activity level leaves a surplus in every good — is exactly one whose Leontief inverse is well defined and non-negative, so that every non-negative bill of final demand is met by a non-negative vector of gross outputs. subsection 6.6.4 revisits the same system as a contraction and obtains a computable error bound for the iteration $x_{k+1} = Ax_k + d$.

Corollary 2.7.1 is what makes the stationary distribution of a finite Markov chain well defined and the long-run behaviour of the chain independent of its starting point. It is used in Chapter 14 when the transition of a state variable is specified as a Markov matrix, and in Chapter 15, where the ergodic distribution of the state under the optimal policy is what stationary equilibrium computations report. Primitivity is what rules out periodic chains: the two-state matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is irreducible but not primitive, has eigenvalues ± 1, and its powers alternate between I and itself without converging, although the stationary distribution $\pi = (\frac{1}{2}, \frac{1}{2})$ still exists.

2.8 Economic Application: Linear Dynamics and Stability

Consider the linear difference system

$$x_{t+1} = Ax_t, \qquad x_0 \in \mathbb{R}^n \text{ given,} \tag{2.3}$$

which arises as the linearisation of a non-linear dynamic model around a steady state — for instance the Ramsey model of Section 15.6 linearised about (k^*, c^*). Iterating, $x_t = A^t x_0$.

> **Proposition 2.8.1** (Stability)**.** For system (2.3): $x_t \to 0$ for every $x_0 \in \mathbb{R}^n$ if and only if $\rho(A) < 1$. If $\rho(A) > 1$, then x_t diverges for every x_0 outside a proper subspace of $\mathbb{R}^n$.

Proof for the diagonalisable case. Suppose $A = P\Lambda P^{-1}$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $A^t = P\Lambda^t P^{-1}$ with $\Lambda^t = \operatorname{diag}(\lambda_1^t, \ldots, \lambda_n^t)$. Writing $z_t = P^{-1}x_t$ decouples the system into $z_{t,i} = \lambda_i^t z_{0,i}$. Each coordinate tends to 0 for every initial condition if and only if $|\lambda_i| < 1$, and grows without bound whenever $|\lambda_i| > 1$ and $z_{0,i} \neq 0$. The set of x_0 with $z_{0,i} = 0$ for all i with $|\lambda_i| > 1$ — the stable subspace — is a proper subspace when some $|\lambda_i| > 1$. The non-diagonalisable case is handled by the Jordan form, where a λ-block of size k contributes terms $t^j \lambda^t$ with $j < k$; these still tend to 0 when $|\lambda| < 1$, so the criterion is unchanged. ■

> **Remark 2.8.1** (Saddle-path stability)**.** In macroeconomic models the interesting case is neither $\rho(A) < 1$ nor $\rho(A) > 1$ but a mixture: some eigenvalues inside the unit circle and some outside. The state vector then splits into predetermined variables, whose initial values are given, and jump variables, whose initial values are chosen. A unique bounded equilibrium path exists when the number of eigenvalues outside the unit circle equals the number of jump variables — the Blanchard–Kahn count. Economically, the jump variables must be placed on the stable subspace, and the counting condition says that the placement problem has exactly one solution, which by Corollary 2.3.1(iii) is a statement about a square invertible system. The recursive treatment is Ljungqvist and Sargent (2018, Ch. 2).

Example 2.8.1 (Leontief input–output)**.** Let $A \in M_{n\times n}(\mathbb{R}_+)$ have a_{ij} equal to the units of good i required per unit of good j, and let $d \in \mathbb{R}^n_+$ be final demand. Gross output x must satisfy $x = Ax + d$, that is $(I - A)x = d$. A non-negative solution exists for every $d \in \mathbb{R}^n_+$ if and only if $(I - A)^{-1}$ exists and is non-negative, which holds if and only if $\rho(A) < 1$; in that case

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k,$$

the Neumann series, whose k-th term is the input requirement generated at the k-th round of production. The condition $\rho(A) < 1$ is thus the productivity condition that the economy generates a surplus. Convergence of the series is a fixed-point statement: $x \mapsto Ax + d$ is a contraction in a suitable norm, and Theorem 6.4.1 applies. The same series reappears in Chapter 15 as the present-value expansion of a value function.

2.9 Chapter Summary

A linear map between finite-dimensional spaces is completely described, once bases are fixed, by a matrix whose columns are the images of the basis vectors (Theorem 2.3.1). Rank–nullity converts questions about solvability into questions about dimension (Theorem 2.3.2, Corollary 2.3.1); for square matrices, and only for square matrices, injectivity, surjectivity, invertibility, and non-vanishing determinant coincide. The determinant measures oriented volume magnification, which is why it appears in the change-of-variables formula. Symmetric matrices are orthogonally diagonalisable with real eigenvalues (Theorem 2.5.1), so definiteness of a quadratic form is a statement about the sign of the spectrum (Proposition 2.5.1) — the fact used for second-order conditions throughout Chapter 12. Only the symmetric part of a matrix affects its quadratic form, and $\nabla(x^\top Ax) = (A + A^\top)x$. Stability of a linear dynamic system is governed by the spectral radius.

> **Exercise 2.9.1.** Verify directly, without invoking Remark 2.5.1, that for $f(x_1, x_2) = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2$ one has $\nabla f(x) = (A + A^\top)x$, and conclude that two distinct matrices can induce the same quadratic form. (This is Exercise 3.3 of Fukuda (2026d); see (Fukuda 2026a, slide 49).)

> **Exercise 2.9.2.** Let X be $n \times k$ with $\operatorname{rank} X = k \leq n$. Show that $X^\top X$ is symmetric positive definite, hence invertible. Where is the rank hypothesis used, and what fails without it? (The result is what makes the OLS formula $\hat{\beta} = (X^\top X)^{-1}X^\top y$ of (Fukuda 2026a, slide 60) well posed.)

> **Exercise 2.9.3.** Let $A \in M_{n\times n}(\mathbb{R}_+)$ with all row sums strictly less than 1. Show $\rho(A) < 1$, and deduce from Example 2.8.1 that the Leontief system is productive. Interpret the row-sum condition economically.

Chapter 3

Inner Products, Norms, and Metric Spaces

3.1 Motivation: Geometry Without Coordinates

The Gerzensee course opens its real-analysis block with the inner product and immediately lists what it buys: length, angle, and orthogonality in $\mathbb{R}^n$ and in function spaces; the projection theorem, which delivers both ordinary least squares and conditional expectation; and the separation theorems, which deliver the second welfare theorem and the existence of state prices in a no-arbitrage economy (Fukuda 2026g, slide 3). All of these are geometric statements, and all of them survive the passage from $\mathbb{R}^n$ to spaces of functions and of random variables — provided the geometry is set up abstractly.

The structures form a strict hierarchy:

$$\text{inner product spaces} \subsetneq \text{normed spaces} \subsetneq \text{metric spaces},$$

reproduced on slide 25 of (Fukuda 2026g). Each step forgets something. A metric gives distance but no notion of adding points; a norm gives length and a vector structure but no notion of angle; an inner product gives angle as well. Knowing which structure a theorem needs is what tells us how far it generalises, and this chapter states each result at the weakest level at which it is true.

3.2 Inner Product Spaces

Definition 3.2.1 (Inner product). Let V be a real vector space. An *inner product* on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ such that for all $x, y, z \in V$ and $\lambda \in \mathbb{R}$:

(i) $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0$ if and only if $x = 0$ (positive definiteness);

(ii) $\langle x, y \rangle = \langle y, x \rangle$ (symmetry);

(iii) $\langle x, \lambda y \rangle = \lambda \langle x, y \rangle$ and $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ (linearity in the second argument; with (ii) this gives linearity in each argument separately).

The pair $(V, \langle \cdot, \cdot \rangle)$ is an *inner product space*.

These are exactly the five properties verified in (Fukuda 2026g, slides 5–6) for the Euclidean inner product $\langle a, x \rangle = \sum_{i=1}^{n} a_i x_i$ on $\mathbb{R}^n$; the abstraction consists in taking them as the definition.

Example 3.2.1 (Inner product spaces used in this volume). (i) $\mathbb{R}^n$ with $\langle x, y \rangle = x^\mathsf{T} y = \sum_i x_i y_i$.

(ii) $\mathbb{R}^n$ with $\langle x, y \rangle_W = x^\mathsf{T} W y$ for a symmetric $W \succ 0$. This is an inner product because W is positive definite (Proposition 2.5.1), and it is the metric under which generalised least squares is an orthogonal projection.

(iii) $\mathcal{L}^2(\mathbb{R}) = \left\{ f : \mathbb{R} \to \mathbb{R} \text{ measurable} : \int_{-\infty}^{\infty} |f(x)|^2 \, \mathrm{d}x < \infty \right\}$ with $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) g(x) \, \mathrm{d}x$.

(iv) The space of square-integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with $\langle X, Y\rangle = \mathbb{E}[XY]$; or, on the subspace of centred variables, $\langle X, Y\rangle = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathrm{Cov}(X, Y)$.

Items (iii) and (iv) are stated in (Fukuda 2026g, slide 7).

Remark 3.2.1 (The qualification the slides leave out). On $\mathcal{L}^2$ the map $\langle f, g\rangle = \int fg$ satisfies (ii) and (iii) but only half of (i): $\langle f, f\rangle = 0$ forces $f = 0$ almost everywhere, not everywhere. The slide's hedge that "roughly, the inner product satisfies the properties" (Fukuda 2026g, slide 7) refers to this. The remedy is standard: work with the quotient of $\mathcal{L}^2$ by the subspace of functions vanishing almost everywhere, writing L^2 for the quotient, whose elements are equivalence classes under the relation of Proposition 1.3.1. On L^2, positive definiteness holds. In the random-variable case (iv) the same convention identifies random variables that agree $\mathbb{P}$-almost surely, and it is what makes conditional expectation in Chapter 14 a well-defined element rather than a function defined up to null sets.

Lemma 3.2.1 (Cauchy–Schwarz inequality). Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space and set $\|x\| = \sqrt{\langle x, x\rangle}$. Then

$$|\langle x, y\rangle| \leq \|x\| \, \|y\| \qquad \text{for all } x, y \in V,$$

with equality if and only if x and y are linearly dependent.

Proof. If $y = 0$ both sides vanish and x, y are dependent. Assume $y \neq 0$ and set $\lambda := \langle x, y\rangle \,/\, \langle y, y\rangle$. Then

$$0 \leq \|x - \lambda y\|^2 = \langle x - \lambda y, x - \lambda y\rangle = \|x\|^2 - 2\lambda \langle x, y\rangle + \lambda^2 \|y\|^2 = \|x\|^2 - \frac{\langle x, y\rangle^2}{\|y\|^2},$$

which rearranges to $\langle x, y\rangle^2 \leq \|x\|^2 \|y\|^2$.

Equality holds if and only if $\|x - \lambda y\| = 0$, that is $x = \lambda y = \frac{\langle x,y\rangle}{\langle y,y\rangle} y$, which is linear dependence; and conversely if $x = \mu y$ then both sides equal $|\mu| \, \|y\|^2$. ∎

The choice of λ is the one that makes $x - \lambda y$ orthogonal to y, so the proof is the projection argument of Chapter 11 in miniature. This is the content of the geometric derivation on (Fukuda 2026g, slide 17), where the condition $\langle \lambda x, y - \lambda x\rangle = 0$ is solved for λ.

Definition 3.2.2 (Orthogonality and angle). Vectors x, y in an inner product space are *orthogonal*, written $x \perp y$, if $\langle x, y\rangle = 0$. For non-zero x, y the *angle* between them is the unique $\theta \in [0, \pi]$ with $\cos\theta = \langle x, y\rangle / (\|x\| \, \|y\|)$, which is well defined precisely because Lemma 3.2.1 places the ratio in $[-1, 1]$.

Example 3.2.2 (Correlation is a cosine). Let $x, y \in \mathbb{R}^n$ with sample means μ_x, μ_y, and put $\tilde{x} = x - \mu_x \mathbf{1}$, $\tilde{y} = y - \mu_y \mathbf{1}$ where $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$. With $\sigma_{x,y} = \frac{1}{n} \langle \tilde{x}, \tilde{y}\rangle$, $\sigma_x^2 = \frac{1}{n} \|\tilde{x}\|^2$ and $\sigma_y^2 = \frac{1}{n} \|\tilde{y}\|^2$, the Pearson correlation coefficient is

$$\rho = \frac{\sigma_{x,y}}{\sigma_x \sigma_y} = \frac{\langle \tilde{x}, \tilde{y}\rangle}{\|\tilde{x}\| \, \|\tilde{y}\|} = \cos\theta.$$

Lemma 3.2.1 therefore gives $|\rho| \leq 1$ immediately, and the equality case gives $|\rho| = 1$ if and only if $\tilde{y} = c\tilde{x}$ for some $c \in \mathbb{R}$, that is

$$y_i - \mu_y = \frac{\sigma_{x,y}}{\sigma_x^2} (x_i - \mu_x) \qquad \text{for all } i,$$

since c must equal $\langle \tilde{x}, \tilde{y}\rangle / \|\tilde{x}\|^2$. This settles Exercise 7.6 of (Fukuda 2026g, slide 18) and identifies the constant as the population regression slope of Chapter 11. The same computation with $\langle X, Y\rangle = \mathrm{Cov}(X, Y)$ shows that the correlation of two random variables is the cosine of the angle between them in L^2.

3.2.1 Riesz representation in $\mathbb{R}^n$

Theorem 3.2.1 (Riesz representation theorem for $\mathbb{R}^n$). (i) For each $a \in \mathbb{R}^n$ the map $f_a \colon \mathbb{R}^n \to \mathbb{R}$, $f_a(x) = \langle a, x\rangle$, is linear.

(ii) Conversely, for every linear $f\colon \mathbb{R}^n \to \mathbb{R}$ there is a unique $a \in \mathbb{R}^n$ with $f(x) = \langle a, x\rangle$ for all $x \in \mathbb{R}^n$; explicitly $a = (f(e^1), \dots, f(e^n))^\mathsf{T}$.

Proof. (i) is linearity of the inner product in its second argument.

(ii) Existence: writing $x = \sum_i x_i e^i$ and using linearity, $f(x) = \sum_i x_i f(e^i) = \langle a, x\rangle$ with a as displayed. Uniqueness: if $\langle a, x\rangle = \langle a', x\rangle$ for all x, then $\langle a - a', x\rangle = 0$ for all x; taking $x = a - a'$ gives $\|a - a'\|^2 = 0$, hence $a = a'$ by positive definiteness. ∎

This is Theorem 7.1 of Fukuda (2026d); the two-dimensional picture is (Fukuda 2026g, slide 9). It is the special case $m = 1$ of Theorem 2.3.1, but stated this way it says something the matrix version does not: *a linear functional is the same thing as a vector.* Three economic readings follow.

Example 3.2.3 (Expected utility as an inner product)**.** Let $\Omega = \{\omega_1, \dots, \omega_n\}$ and let $p \in \mathbb{R}^n$ be a probability vector. Expected utility $\mathbb{E}[u] = \sum_i p_i u_i = \langle u, p\rangle$ is linear in the probability vector p for fixed utility index u, and linear in u for fixed p (Fukuda 2026g, slide 8). Read in the first way, Theorem 3.2.1 says that *any* functional on lotteries that is linear in probabilities has an expected-utility representation, with the representing vector a being the utility index. The von Neumann–Morgenstern theorem is the statement that the behavioural axioms force linearity; the Riesz theorem then supplies the index.

Example 3.2.4 (State prices and the stochastic discount factor)**.** Let there be n states and let an asset be a payoff vector $z \in \mathbb{R}^n$. Suppose the pricing functional $\pi\colon \mathbb{R}^n \to \mathbb{R}$ is linear — which is exactly the law of one price plus portfolio additivity. By Theorem 3.2.1 there is a unique $q \in \mathbb{R}^n$ with $\pi(z) = \langle q, z\rangle = \sum_i q_i z_i$; the q_i are *state prices*. If moreover $\pi(z) > 0$ whenever $z \geq 0$, $z \neq 0$ — absence of arbitrage — then $q \in \mathbb{R}^n_{++}$, and writing $q_i = p_i m_i$ for a probability p gives $\pi(z) = \mathbb{E}[mz]$, exhibiting m as the *stochastic discount factor*. The step from "no arbitrage" to "a strictly positive linear functional exists" is not Riesz but a separation theorem, proved in Chapter 9; Riesz converts the functional into a vector. See LeRoy and Werner (2000, Ch. 2).

Example 3.2.5 (The total differential)**.** For $f\colon \mathbb{R}^n \to \mathbb{R}$ differentiable at x, the differential

$$\mathrm{d}y = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x)\,\mathrm{d}x_i$$

is a linear functional of the displacement $\mathrm{d}x$. By Theorem 3.2.1 it is represented by a unique vector, and that vector is by definition the gradient: $\mathrm{d}y = \langle \nabla f(x), \mathrm{d}x\rangle$ (Fukuda 2026g, slide 10). This is the cleanest statement of what a gradient is, and it explains why the gradient is a column vector while the derivative $Df(x)$ is a row: the derivative is the functional, the gradient is its Riesz representative.

3.3 Normed Spaces

Definition 3.3.1 (Norm)**.** Let V be a real vector space. A *norm* is a map $\|\cdot\| : V \to \mathbb{R}$ such that for all $x, y \in V$, $\lambda \in \mathbb{R}$:

(i) $\|x\| \geq 0$;

(ii) $\|x\| = 0$ if and only if $x = 0$;

(iii) $\|\lambda x\| = |\lambda|\,\|x\|$ (absolute homogeneity);

(iv) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

The pair $(V, \|\cdot\|)$ is a *normed space*.

Proposition 3.3.1 (An inner product induces a norm)**.** If $\langle\cdot,\cdot\rangle$ is an inner product on V, then $\|x\| := \sqrt{\langle x, x\rangle}$ is a norm.

Proof. (i) and (ii) are positive definiteness; (iii) follows from bilinearity, since $\|\lambda x\|^2 = \lambda^2 \|x\|^2$. For (iv),

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

using Lemma 3.2.1; taking square roots gives the claim. ∎

Proposition 3.3.2 (Reverse triangle inequality). In any normed space, $\big|\, \|x\| - \|x_0\| \,\big| \le \|x - x_0\|$.

Proof. $\|x\| = \|(x - x_0) + x_0\| \le \|x - x_0\| + \|x_0\|$ gives $\|x\| - \|x_0\| \le \|x - x_0\|$. Exchanging x and x_0 and using $\|x_0 - x\| = \|x - x_0\|$ gives $\|x_0\| - \|x\| \le \|x - x_0\|$. Together these bound the absolute value. ∎

This is Exercise 7.4 of (Fukuda 2026g, slide 20). It is used at once in Chapter 5 to show that the norm is a continuous function — indeed Proposition 3.3.2 says it is Lipschitz with constant 1.

Example 3.3.1 (Norms used in this volume). On $\mathbb{R}^n$: the Euclidean norm $\|x\|_2 = \left(\sum_i x_i^2\right)^{1/2}$, induced by the Euclidean inner product; the ℓ^1 norm $\|x\|_1 = \sum_i |x_i|$; the supremum norm $\|x\|_\infty = \max_i |x_i|$. On $\mathcal{C}([a,b],\mathbb{R})$ and on $\mathcal{B}(X,\mathbb{R})$: the *supremum norm*

$$\|f\|_\infty := \sup_x |f(x)|\,,$$

which is the norm used for value functions in Chapter 15 (Fukuda 2026g, slides 21–22, 73).

Proposition 3.3.3 (The supremum norm is a norm). Let X be a non-empty set and $\mathcal{B}(X,\mathbb{R})$ the set of bounded functions $f\colon X \to \mathbb{R}$. Then $\mathcal{B}(X,\mathbb{R})$ is a vector space under pointwise operations and $\|f\|_\infty = \sup_{x\in X} |f(x)|$ is a norm on it.

Proof. Boundedness is preserved by sums and scalar multiples, so $\mathcal{B}(X,\mathbb{R})$ is a subspace of $\mathbb{R}^X$, and $\|f\|_\infty < \infty$ for every f in it. Property (i) is clear. For (ii), $\|f\|_\infty = 0$ forces $|f(x)| = 0$ for every x, that is $f = 0$ — note that here, unlike in Remark 3.2.1, no almost-everywhere qualification arises. For (iii), $\sup_x |\lambda f(x)| = |\lambda| \sup_x |f(x)|$. For (iv), for each x, $|f(x) + g(x)| \le |f(x)| + |g(x)| \le \|f\|_\infty + \|g\|_\infty$; as the right-hand side does not depend on x, it is an upper bound for $\{|(f+g)(x)|\}$, so by Proposition 1.5.1 it dominates the supremum. ∎

Remark 3.3.1 (Not every norm comes from an inner product). A norm is induced by an inner product if and only if it satisfies the *parallelogram law*

$$\|x+y\|^2 + \|x-y\|^2 = 2\,\|x\|^2 + 2\,\|y\|^2\,.$$

Necessity is immediate by expanding both sides. The supremum norm fails it: in $\mathbb{R}^2$ with $x = (1,0)$, $y = (0,1)$ we get $\|x+y\|_\infty = \|x-y\|_\infty = 1$, so the left side is 2 while the right side is 4. Hence $\mathcal{B}(X,\mathbb{R})$ with the supremum norm carries no compatible inner product, and no notion of angle. This is why the value-function arguments of Chapter 15 use the contraction mapping theorem, which needs only a metric, rather than a projection argument, which needs an inner product.

3.4 Metric Spaces

Definition 3.4.1 (Metric). Let X be a non-empty set. A *metric* (or distance) on X is a map $d\colon X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

(i) $d(x,y) \ge 0$;

(ii) $d(x,y) = 0$ if and only if $x = y$;

(iii) $d(x,y) = d(y,x)$ (symmetry);

(iv) $d(x,z) \le d(x,y) + d(y,z)$ (triangle inequality).

The pair (X,d) is a *metric space*.

Proposition 3.4.1 (A norm induces a metric). If $(V, \|\cdot\|)$ is a normed space, then $d(x,y) := \|x - y\|$ is a metric on V.

Proof. (i)–(iii) are immediate from Definition 3.3.1(i)–(iii), using $\|y - x\| = \|-(x-y)\| = \|x - y\|$. For (iv), $d(x,z) = \|(x-y) + (y-z)\| \le \|x-y\| + \|y-z\| = d(x,y) + d(y,z)$. ∎

Combining Proposition 3.3.1 and Proposition 3.4.1 gives the hierarchy of (Fukuda 2026g, slide 25): every inner product space is a normed space and every normed space is a metric space. Both inclusions are strict — Remark 3.3.1 shows the first, and the next example the second.

Example 3.4.1 (A metric that comes from no norm)**.** On any non-empty set X the *discrete metric*

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

is a metric. If X is a vector space with more than one point, d is induced by no norm: a norm satisfies $\|2x\| = 2\|x\|$, so $d(2x, 0) = 2d(x, 0)$, but the discrete metric gives $d(2x, 0) = d(x, 0) = 1$ for any $x \neq 0$. More generally, a metric on a vector space comes from a norm if and only if it is translation invariant and absolutely homogeneous.

The discrete metric is not a curiosity: under it every subset is open, every sequence that converges is eventually constant, and the compact sets are exactly the finite ones. It is the standard source of counterexamples in Chapter 4.

Example 3.4.2 (Metrics on economic objects)**.** (i) Consumption bundles in $\mathbb{R}^n$: Euclidean metric.

(ii) Consumption *plans* $c\colon [0, \infty) \to \mathbb{R}$ or $(c_t)_{t\in\mathbb{N}_0}$: supremum metric on $\mathcal{B}(\cdot, \mathbb{R})$. The slide's remark that one chooses "a consumption function $c\colon [0, \infty) \to \mathbb{R}$ instead of $(c_1, c_2) \in \mathbb{R}^2$" (Fukuda 2026g, slide 4) is precisely the passage from (i) to (ii), and it is the passage that makes the infinite-dimensional theory necessary.

(iii) Value functions $V : \mathbb{R}^n \to \mathbb{R}$: supremum metric, under which the Bellman operator is a contraction (Theorem 6.5.1).

(iv) Probability distributions: many metrics are in use (total variation, Kolmogorov–Smirnov, Wasserstein), and they are not equivalent in infinite dimensions; which one is meant must always be stated.

Definition 3.4.2 (Open ball, bounded set)**.** Let (X, d) be a metric space, $x_0 \in X$ and $\lambda > 0$. The *open ball* with centre x_0 and radius λ is

$$B_\lambda(x_0) := \{x \in X : d(x_0, x) < \lambda\}.$$

A set $A \subseteq X$ is *bounded* if $A = \emptyset$ or $A \subseteq B_r(x_0)$ for some $x_0 \in X$ and $r > 0$.

Proposition 3.4.2. A set $A \subseteq \mathbb{R}^n$ is bounded in the sense of Definition 3.4.2 if and only if there exists $\alpha > 0$ with $\|x\| \leq \alpha$ for all $x \in A$.

Proof. If $\|x\| \leq \alpha$ on A, then $A \subseteq B_{\alpha+1}(0)$. Conversely if $A \subseteq B_r(x_0)$ then for $x \in A$, $\|x\| \leq \|x - x_0\| + \|x_0\| < r + \|x_0\|$, so $\alpha = r + \|x_0\|$ works. ∎

This settles Exercise 7.20 of (Fukuda 2026g, slide 28). The economic instance given there is the budget set: for $p \in \mathbb{R}^n_{++}$ and $m > 0$, every $x \in \mathcal{B}(p, m)$ satisfies $x_i \leq m/p_i$, so $\|x\|_\infty \leq m/\min_i p_i$ and $\mathcal{B}(p, m)$ is bounded. If some $p_i = 0$, the bound fails and so does the conclusion — the substantive reason for requiring strictly positive prices, flagged already in Example 1.6.1.

3.5 Complete Normed and Inner Product Spaces

The definitions below are stated here for completeness of the hierarchy; the notion of a Cauchy sequence on which they rest is developed in Chapter 4 and their consequences in Chapter 6.

Definition 3.5.1 (Banach and Hilbert spaces)**.** A normed space that is complete as a metric space (Definition 6.3.1) is a *Banach space*. An inner product space that is complete in the induced norm is a *Hilbert space*.

These are Definitions 7.12 and 7.13 of Fukuda (2026d), given on slide 77 of (Fukuda 2026g). The two examples that matter here are L^2, which is a Hilbert space and is where Chapter 11 operates, and $\mathcal{B}(X, \mathbb{R})$ with the supremum norm, which is a Banach space but not a Hilbert space by Remark 3.3.1, and is where Chapter 15 operates.

Remark 3.5.1 (Why completeness is stated at the level of the metric). Completeness is a metric notion: it makes sense in (X, d) without any vector structure. Banach and Hilbert spaces are therefore not new kinds of object but conjunctions — "normed and complete", "inner product and complete". This matters when one restricts attention to a subset: a closed subset of a complete metric space is complete (Proposition 6.3.2), and a closed subset of a Banach space need not be a subspace, but is still a complete metric space. The contraction mapping theorem applies to it regardless, which is what allows Chapter 15 to work on the set of bounded, continuous, *increasing and concave* functions — a closed subset, not a subspace.

3.6 Chapter Summary

An inner product supplies length, angle and orthogonality; the Cauchy–Schwarz inequality is the statement that the resulting cosine lies in $[-1, 1]$, and its equality case characterises linear dependence, which in the statistical reading is perfect correlation. Every linear functional on $\mathbb{R}^n$ is an inner product with a unique vector (Theorem 3.2.1); this identification is what turns linearity in probabilities into an expected-utility index, linearity of a pricing functional into a vector of state prices, and the differential into the gradient. A norm forgets angle, a metric forgets the vector structure as well; both inclusions in the hierarchy are strict, as the supremum norm and the discrete metric show. Boundedness of the budget set requires strictly positive prices. Completeness, defined in Chapter 4, upgrades normed and inner product spaces to Banach and Hilbert spaces, the two settings in which Chapters 11 and 15 respectively work.

Exercise 3.6.1. Prove the polarisation identity $\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right)$ in any inner product space, and deduce that an inner product is determined by its norm. Combine this with Remark 3.3.1 to give a complete criterion for when a norm comes from an inner product.

Exercise 3.6.2. Show that on $\mathbb{R}^n$ the norms $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ satisfy $\|x\|_\infty \le \|x\|_2 \le \|x\|_1 \le n\|x\|_\infty$, so that all three induce the same bounded sets and, by Chapter 4, the same open sets. Explain why the analogous statement fails on $\mathcal{C}([0,1], \mathbb{R})$ for $\|f\|_1 = \int_0^1 |f|$ versus $\|f\|_\infty$, and give a sequence witnessing the failure.

Exercise 3.6.3. Let $W \in M_{n\times n}(\mathbb{R})$ be symmetric. Show that $\langle x, y\rangle_W = x^\mathsf{T} W y$ is an inner product on $\mathbb{R}^n$ if and only if $W \succ 0$, and that positive *semi*definiteness gives only a seminorm. Identify the subspace on which the seminorm vanishes in terms of $\operatorname{Ker} W$.

Chapter 4

Topology of Metric Spaces: Open Sets, Sequences, and Compactness

4.1 Motivation: What "Nearby" Has to Mean

Three of the most-used theorems in economic theory — that a continuous function on a compact set attains its maximum, that a bounded sequence has a convergent subsequence, and that a contraction on a complete space has a unique fixed point — are statements about which sets contain their limit points and which sequences converge. This chapter builds that vocabulary.

The order of business follows the slides (Fukuda 2026g, slides 26–48): open balls, then interior and closure, then the three axioms that isolate what a topology is, then sequences and their limits, then the two results that make compactness usable — Bolzano–Weierstrass and Heine–Borel. Throughout, (X, d) is a metric space; the specialisation to $\mathbb{R}^n$ is stated whenever it is genuinely used, because several of the results below are false in general metric spaces and knowing which is the point of stating them abstractly.

4.2 Open and Closed Sets

Definition 4.2.1 (Interior, closure, boundary). Let (X, d) be a metric space and $A \subseteq X$.

(i) A point $a \in X$ is *interior to* A if $B_\varepsilon(a) \subseteq A$ for some $\varepsilon > 0$. The *interior* $\operatorname{Int} A$ is the set of interior points of A; by definition $\operatorname{Int} A \subseteq A$. The set A is *open* if $A = \operatorname{Int} A$.

(ii) A point $a \in X$ is a *point of closure* of A if $A \cap B_\varepsilon(a) \neq \emptyset$ for every $\varepsilon > 0$. The *closure* $\overline{A}$ is the set of points of closure; by definition $A \subseteq \overline{A}$. The set A is *closed* if $A = \overline{A}$.

(iii) The *boundary* is $\partial A := \overline{A} \setminus \operatorname{Int} A$.

Example 4.2.1 (Closure can add points that are "reached but not attained"). In $\mathbb{R}$ with $d(x, y) = |x - y|$ let $A = \{1/n : n \in \mathbb{N}\}$. Then $0 \in \overline{A}$: for any $\varepsilon > 0$, Proposition 1.5.2 supplies n with $1/n < \varepsilon$, so $1/n \in A \cap B_\varepsilon(0)$. Since $0 \notin A$, A is not closed, and $\overline{A} = A \cup \{0\}$. Also $\operatorname{Int} A = \emptyset$, since every ball around $1/n$ contains irrational numbers. This is Example 7.1 of Fukuda (2026d), illustrated on (Fukuda 2026g, slide 33).

Proposition 4.2.1 (Duality of interior and closure). For every $A \subseteq X$,

$$\left(\overline{A}\right)^{\mathsf{c}} = \operatorname{Int}\left(A^{\mathsf{c}}\right), \qquad \text{equivalently} \qquad \left(\operatorname{Int} A\right)^{\mathsf{c}} = \overline{\left(A^{\mathsf{c}}\right)}.$$

Consequently A is open if and only if A^{c} is closed, and A is closed if and only if A^{c} is open.

Proof. A point a fails to be a point of closure of A if and only if some $\varepsilon > 0$ has $A \cap B_\varepsilon(a) = \emptyset$, that is $B_\varepsilon(a) \subseteq A^{\mathsf{c}}$, that is $a \in \operatorname{Int}(A^{\mathsf{c}})$. This is the first identity; the second follows by applying the first to A^{c} and taking complements.

For the consequence: A open means $A = \operatorname{Int} A$, hence $A^{\mathsf{c}} = (\operatorname{Int} A)^{\mathsf{c}} = \overline{(A^{\mathsf{c}})}$, which is A^{c} closed; and conversely. ■

This is Proposition 7.5 and Corollary 7.1 of Fukuda (2026d) (Fukuda 2026g, slide 35). Almost every proof about closed sets below is obtained by complementing a proof about open sets.

Proposition 4.2.2 (Open balls are open). In any metric space, $B_\lambda(a)$ is open.

Proof. Let $x \in B_\lambda(a)$ and put $\rho := d(a,x) < \lambda$, so $\lambda - \rho > 0$. For $y \in B_{\lambda-\rho}(x)$ the triangle inequality gives

$$d(a,y) \leq d(a,x) + d(x,y) < \rho + (\lambda - \rho) = \lambda,$$

so $y \in B_\lambda(a)$. Hence $B_{\lambda-\rho}(x) \subseteq B_\lambda(a)$ and $x \in \operatorname{Int} B_\lambda(a)$. As x was arbitrary, $B_\lambda(a) = \operatorname{Int} B_\lambda(a)$. ■

Proposition 4.2.3 ($\operatorname{Int} A$ is open and $\overline{A}$ is closed). For every $A \subseteq X$: $\operatorname{Int} A$ is open, $\overline{A}$ is closed, $\operatorname{Int} A$ is the largest open set contained in A, and $\overline{A}$ is the smallest closed set containing A.

Proof. Let $a \in \operatorname{Int} A$, so $B_\varepsilon(a) \subseteq A$ for some $\varepsilon > 0$. By Proposition 4.2.2, every $x \in B_\varepsilon(a)$ has a ball $B_\delta(x) \subseteq B_\varepsilon(a) \subseteq A$, so $x \in \operatorname{Int} A$; hence $B_\varepsilon(a) \subseteq \operatorname{Int} A$ and $\operatorname{Int} A$ is open. Applying this to A^{c} and using Proposition 4.2.1 shows $\overline{A}$ is closed.

If $G \subseteq A$ is open then every $a \in G$ has a ball inside $G \subseteq A$, so $G \subseteq \operatorname{Int} A$; this is maximality. The statement for $\overline{A}$ follows by complementation. ■

Theorem 4.2.1 (The three axioms of a topology). Let $\mathcal{O}$ denote the collection of open subsets of a metric space (X,d). Then

(i) the union of any subfamily of $\mathcal{O}$ belongs to $\mathcal{O}$;

(ii) the intersection of any *finite* subfamily of $\mathcal{O}$ belongs to $\mathcal{O}$;

(iii) $\emptyset \in \mathcal{O}$ and $X \in \mathcal{O}$.

Proof. (i) Let $(G_i)_{i\in I} \subseteq \mathcal{O}$ and $x \in G = \bigcup_i G_i$. Then $x \in G_{i_0}$ for some i_0, and openness of G_{i_0} supplies $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq G_{i_0} \subseteq G$.

(ii) Let $G_1, \ldots, G_n \in \mathcal{O}$ and $x \in G = \bigcap_{i=1}^n G_i$. For each i pick $\varepsilon_i > 0$ with $B_{\varepsilon_i}(x) \subseteq G_i$ and set $\varepsilon = \min\{\varepsilon_1, \ldots, \varepsilon_n\}$. Then $\varepsilon > 0$ — this is where finiteness is used — and $B_\varepsilon(x) \subseteq B_{\varepsilon_i}(x) \subseteq G_i$ for every i, so $B_\varepsilon(x) \subseteq G$.

(iii) For X: any ε works since $B_\varepsilon(x) \subseteq X$ always. For $\emptyset$: the statement "$x \in \emptyset$ implies $B_\varepsilon(x) \subseteq \emptyset$" is vacuously true, by the truth table of Section 1.4. ■

Example 4.2.2 (Infinitely many open sets can intersect in a closed set). In $\mathbb{R}$, $\bigcap_{n\in\mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$, an intersection of open sets that is not open. The minimum in the proof of Theorem 4.2.1(ii) becomes an infimum equal to 0, and the argument collapses. Dually, $\bigcup_{n\in\mathbb{N}} \left[\frac{1}{n}, 1\right] = (0,1]$ is a union of closed sets that is not closed.

Definition 4.2.2 (Topological space). Let X be a set. A collection $\mathcal{O} \subseteq \mathcal{P}(X)$ satisfying (i)–(iii) of Theorem 4.2.1 is a *topology* on X, the pair $(X, \mathcal{O})$ is a *topological space*, and the members of $\mathcal{O}$ are the open sets. A topology arising from a metric as above is *metrisable*.

Remark 4.2.1 (Two abstractions of the same list of axioms). Compare Definition 4.2.2 with the definition of a σ-algebra: a collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ containing $\emptyset$ and Ω, closed under complementation, and closed under *countable* unions. The two are different closure requirements on a collection of sets, and the difference is exactly where the two theories diverge: topology tolerates arbitrary unions but only finite intersections; measure theory tolerates countable unions and countable intersections but requires complements. On $\mathbb{R}$ the Borel σ-algebra is generated by the open sets, which is how the two are linked (Fukuda 2026g, slide 37). The parallel between the two definitions of "measurable/continuous via preimages" drawn there — f is continuous if $f^{-1}(B)$ is open whenever B is open; X is a random variable if $X^{-1}(B)$ is measurable whenever B is measurable — is made precise in Theorem 5.2.1 and Definition 14.2.1.

4.3 Sequences and Limits

Definition 4.3.1 (Sequence, subsequence, limit). A *sequence* in X is a map $\mathbb{N} \to X$, written $(x_n)_{n\in\mathbb{N}}$. If $\tau\colon \mathbb{N} \to \mathbb{N}$ is strictly increasing, then $(x_{\tau(n)})_{n\in\mathbb{N}}$ is a *subsequence*, usually written $(x_{n_k})_{k\in\mathbb{N}}$.

The sequence *converges* to $x^* \in X$, written $x_n \to x^*$ or $x^* = \lim_{n\to\infty} x_n$, if for every $\varepsilon > 0$ there is $n_0 \in \mathbb{N}$ such that $n \geq n_0$ implies $d(x^*, x_n) < \varepsilon$.

Proposition 4.3.1 (Basic facts). Let (x_n) be a sequence in a metric space X.

(i) A limit, if it exists, is unique.

(ii) If $x_n \to x^*$ then every subsequence converges to x^*.

(iii) A convergent sequence is bounded.

Proof. (i) If $x_n \to x$ and $x_n \to y$, then for every $\varepsilon > 0$ we may choose n with $d(x, x_n) < \varepsilon/2$ and $d(y, x_n) < \varepsilon/2$, so $d(x, y) \leq d(x, x_n) + d(x_n, y) < \varepsilon$. As $\varepsilon > 0$ was arbitrary, $d(x, y) = 0$, hence $x = y$.

(ii) Since τ is strictly increasing, $\tau(k) \geq k$ by induction. Given $\varepsilon > 0$ take n_0 from the definition; for $k \geq n_0$ we have $\tau(k) \geq k \geq n_0$, so $d(x^*, x_{\tau(k)}) < \varepsilon$.

(iii) Take $\varepsilon = 1$ and the corresponding n_0. Then $d(x^*, x_n) < 1$ for $n \geq n_0$, and setting $M := \max\{1, d(x^*, x_1), \ldots, d(x^*, x_{n_0-1})\}$ gives $(x_n) \subseteq B_{M+1}(x^*)$; the maximum is over a finite set and hence finite. ∎

Parts (ii) and (iii) are Exercises 7.30 and 7.31 of (Fukuda 2026g, slide 43).

Theorem 4.3.1 (Sequential characterisation of closedness). A set $A \subseteq X$ is closed if and only if the limit of every convergent sequence of elements of A lies in A.

Proof. ($\Rightarrow$) Let A be closed, $(x_n) \subseteq A$ and $x_n \to x^*$. For every $\varepsilon > 0$ there is n with $x_n \in B_\varepsilon(x^*)$, and $x_n \in A$, so $A \cap B_\varepsilon(x^*) \neq \emptyset$. Hence $x^* \in \overline{A} = A$.

($\Leftarrow$) Suppose the stated condition holds and let $x^* \in \overline{A}$. For each $n \in \mathbb{N}$ the set $A \cap B_{1/n}(x^*)$ is non-empty; pick x_n in it. Then $d(x^*, x_n) < 1/n \to 0$, so $x_n \to x^*$ with $(x_n) \subseteq A$, and the hypothesis gives $x^* \in A$. Hence $\overline{A} \subseteq A$, and the reverse inclusion always holds. ∎

This is Proposition 7.11 of Fukuda (2026d) (Fukuda 2026g, slide 43). It converts a statement about balls into a statement about sequences, and it is the form in which closedness is checked in practice.

4.3.1 Monotone subsequences and Bolzano–Weierstrass

Proposition 4.3.2 (Every real sequence has a monotone subsequence). Every sequence $(x_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ has a subsequence that is increasing or a subsequence that is decreasing (possibly both).

Proof. Call $m \in \mathbb{N}$ a *peak* if $x_m > x_n$ for all $n > m$, and let $S := \{m \in \mathbb{N} : m \text{ is a peak}\}$.

If S is infinite, enumerate it increasingly as $n_1 < n_2 < \cdots$. For each k, since n_k is a peak and $n_{k+1} > n_k$, we get $x_{n_k} > x_{n_{k+1}}$; hence (x_{n_k}) is strictly decreasing.

If S is finite, let N exceed every element of S and set $n_1 := N$. Since n_1 is not a peak, there is $n_2 > n_1$ with $x_{n_2} \geq x_{n_1}$; since $n_2 > N$ is not a peak either, there is $n_3 > n_2$ with $x_{n_3} \geq x_{n_2}$, and so on. This produces an increasing subsequence. ∎

This is Proposition 7.12 of Fukuda (2026d); the argument is the one sketched on (Fukuda 2026g, slides 45–46), made explicit.

Proposition 4.3.3 (Monotone convergence). An increasing sequence in $\mathbb{R}$ that is bounded above converges to $\sup_n x_n$; a decreasing sequence bounded below converges to $\inf_n x_n$.

Proof. Let (x_n) be increasing and bounded above and put $s := \sup_n x_n$, which exists by Assumption 1.5.1. Given $\varepsilon > 0$, Proposition 1.5.1(ii) supplies n_0 with $x_{n_0} > s - \varepsilon$. For $n \geq n_0$, monotonicity and the upper-bound property give $s - \varepsilon < x_{n_0} \leq x_n \leq s$, so $|s - x_n| < \varepsilon$. The decreasing case follows by applying this to $(-x_n)$. ∎

Theorem 4.3.2 (Bolzano–Weierstrass)**.** Every bounded sequence in $\mathbb{R}$ has a convergent subsequence. More generally, every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. In $\mathbb{R}$: by Proposition 4.3.2 extract a monotone subsequence, which inherits boundedness, and apply Proposition 4.3.3.

In $\mathbb{R}^n$: let $(x_m)_{m\in\mathbb{N}}$ be bounded, with coordinates $x_m = (x_m^1, \ldots, x_m^n)$. Each coordinate sequence is bounded in $\mathbb{R}$, since $\left|x_m^i\right| \le \|x_m\|$. Apply the one-dimensional case to $(x_m^1)_m$ to get a subsequence along which the first coordinate converges; apply it again, along that subsequence, to the second coordinate; and so on. As n is finite, after n extractions we obtain a single subsequence along which every coordinate converges. By Proposition 4.3.4 the subsequence converges in $\mathbb{R}^n$. ■

Proposition 4.3.4 (Coordinatewise convergence)**.** A sequence (x_m) in $\mathbb{R}^n$ converges if and only if each coordinate sequence $(x_m^i)_m$ converges in $\mathbb{R}$, and then $\lim_m x_m = (\lim_m x_m^1, \ldots, \lim_m x_m^n)$.

Proof. If $x_m \to x^*$, then $\left|x_m^i - x^{i*}\right| \le \|x_m - x^*\| \to 0$ for each i. Conversely, if $x_m^i \to x^{i*}$ for each i, then given $\varepsilon > 0$ choose n_i with $\left|x_m^i - x^{i*}\right| < \varepsilon/\sqrt{n}$ for $m \ge n_i$; for $m \ge \max_i n_i$, $\|x_m - x^*\| = \left(\sum_i \left|x_m^i - x^{i*}\right|^2\right)^{1/2} < \varepsilon$. ■

This is Proposition 7.14 of Fukuda (2026d) (Fukuda 2026g, slide 48). The finiteness of n is essential: the argument in Theorem 4.3.2 performs n successive extractions, and in an infinite-dimensional space it cannot be completed. Indeed Bolzano–Weierstrass *fails* there, as Example 4.3.1 shows.

Example 4.3.1 (Bolzano–Weierstrass fails in infinite dimensions)**.** In the space of square-summable sequences with $\|x\|^2 = \sum_i x_i^2$, consider the orthonormal vectors $e^1, e^2, \ldots$, where e^k has a 1 in position k and zeros elsewhere. Then $\left\|e^k\right\| = 1$ for all k, so the sequence is bounded, while $\left\|e^j - e^k\right\| = \sqrt{2}$ for $j \ne k$, so no subsequence is Cauchy and hence none converges. Bounded and closed therefore does not imply compact outside finite dimensions, which is why Theorem 4.4.2 is stated for $\mathbb{R}^n$ and why the fixed-point arguments of Chapter 15 use completeness and contraction rather than compactness.

4.4 Compactness

Definition 4.4.1 (Compactness)**.** Let (X, d) be a metric space and $K \subseteq X$.

(i) An *open cover* of K is a family $(G_i)_{i\in I}$ of open sets with $K \subseteq \bigcup_{i\in I} G_i$. The set K is *compact* if every open cover of K admits a finite subcover, that is, a finite $J \subseteq I$ with $K \subseteq \bigcup_{i\in J} G_i$.

(ii) K is *sequentially compact* if every sequence in K has a subsequence converging to a point of K.

Theorem 4.4.1 (Compactness and sequential compactness agree in metric spaces)**.** In a metric space, a set is compact if and only if it is sequentially compact.

The proof is standard and somewhat long; see Rudin (1976, Thm. 3.28 and its converse) or Fukuda (2026d, Sec. 8.1). In general *topological* spaces the two notions differ, which is one reason for staying within metric spaces throughout this volume.

Proposition 4.4.1 (Compact implies closed and bounded)**.** In any metric space, a compact set is closed and bounded.

Proof. Boundedness: fix $x_0 \in X$; the balls $B_n(x_0)$, $n \in \mathbb{N}$, cover K, and a finite subcover is contained in the largest of them.

Closedness: let $x^* \in \overline{K}$ and take a sequence $(x_n) \subseteq K$ with $x_n \to x^*$, as in the proof of Theorem 4.3.1. By sequential compactness (Theorem 4.4.1) some subsequence converges to a point of K; by Proposition 4.3.1(ii) that limit is x^*. Hence $x^* \in K$ and K is closed by Theorem 4.3.1. ■

Theorem 4.4.2 (Heine–Borel)**.** A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. Necessity is Proposition 4.4.1. For sufficiency, let $K \subseteq \mathbb{R}^n$ be closed and bounded and let $(x_m) \subseteq K$. Boundedness and Theorem 4.3.2 give a convergent subsequence $x_{m_k} \to x^*$; closedness and Theorem 4.3.1 give $x^* \in K$. So K is sequentially compact, hence compact by Theorem 4.4.1. ■

Remark 4.4.1 (The equivalence is special to $\mathbb{R}^n$). Example 4.3.1 exhibits a closed bounded set — the closed unit ball of an infinite-dimensional space — that is not compact. In fact the closed unit ball of a normed space is compact if and only if the space is finite-dimensional (Riesz's lemma; see Aliprantis and Border (2006, Ch. 5)). Whenever a proof in economics invokes "closed and bounded, hence compact", the underlying space must be checked to be finite-dimensional.

Proposition 4.4.2 (Stability properties). (i) A closed subset of a compact set is compact.

(ii) A finite union of compact sets is compact; an arbitrary intersection of compact sets is compact.

(iii) A finite product of compact sets is compact in the product metric.

Proof. (i) Let $F \subseteq K$ be closed with K compact, and let $(x_n) \subseteq F$. Compactness of K gives a subsequence converging to some $x^* \in K$; closedness of F and Theorem 4.3.1 give $x^* \in F$.

(ii) For a finite union, a sequence in $\bigcup_{j=1}^{m} K_j$ has infinitely many terms in some K_{j_0}, and that subsequence has a further subsequence converging in K_{j_0}. An intersection of compact sets is a closed subset (by Proposition 4.4.1 and Proposition 4.2.1) of any one of them, so (i) applies.

(iii) Extract successively in each coordinate, as in the proof of Theorem 4.3.2; finiteness of the number of factors is what makes the extraction terminate. ■

Example 4.4.1 (The budget set is compact). For $p \in \mathbb{R}^n_{++}$ and $m > 0$, $\mathcal{B}(p, m) = \{x \in \mathbb{R}^n_+ : \langle p, x\rangle \le m\}$ is compact. It is bounded, as shown after Proposition 3.4.2. It is closed: if $(x_k) \subseteq \mathcal{B}(p, m)$ and $x_k \to x^*$, then $x^* \in \mathbb{R}^n_+$ because each coordinate is a limit of non-negative numbers, and $\langle p, x^*\rangle = \lim_k \langle p, x_k\rangle \le m$ by continuity of the inner product (Proposition 5.2.1). Theorem 4.4.2 applies.

Every hypothesis is doing work. If $p_i = 0$ for some i the set is unbounded. If the constraint were $\langle p, x\rangle < m$ the set would not be closed and the supremum of a strictly increasing utility would not be attained. If consumption were not restricted to $\mathbb{R}^n_+$, the set would again be unbounded whenever some $p_i < 0$. The existence half of Example 1.6.1 rests on this proposition, via the Weierstrass theorem of Chapter 5.

4.5 Connectedness

Definition 4.5.1 (Connected set). A metric space X is *connected* if the only subsets of X that are both open and closed are $\emptyset$ and X; equivalently, X cannot be written as the union of two disjoint non-empty open sets. A subset $A \subseteq X$ is connected if it is connected as a metric space in the induced metric.

Proposition 4.5.1. A subset of $\mathbb{R}$ is connected if and only if it is an interval, that is, if $x, y \in A$ and $x < z < y$ imply $z \in A$. Every convex subset of $\mathbb{R}^n$ is connected.

Proof. Connected ⇒ interval. If A is not an interval, take $x < z < y$ with $x, y \in A$ and $z \notin A$; then $A \cap (-\infty, z)$ and $A \cap (z, \infty)$ are disjoint, non-empty, relatively open, and cover A, so A is not connected.

Interval ⇒ connected. This direction is proved from the supremum property of $\mathbb{R}$ (Assumption 1.5.1), not from the connectedness of $[0, 1]$, which is the same statement. Let A be an interval and suppose $A = U \cup W$ with U, W relatively open, disjoint and non-empty. Pick $x \in U$ and $y \in W$; without loss of generality $x < y$, and $[x, y] \subseteq A$ because A is an interval. Put

$$s := \sup\{t \in [x, y] : [x, t] \subseteq U\},$$

the set being non-empty, since x belongs to it, and bounded above by y; so s is a real number lying in $[x, y] \subseteq A$. Since $A = U \cup W$, either $s \in W$ or $s \in U$.

Suppose $s \in W$. Relative openness gives $\varepsilon > 0$ with $A \cap (s - \varepsilon, s + \varepsilon) \subseteq W$, and the definition of the supremum gives $t \in [x, y]$ with $s - \varepsilon < t \le s$ and $[x, t] \subseteq U$. Then $t \in A \cap (s - \varepsilon, s + \varepsilon) \subseteq W$ and $t \in U$, so $t \in U \cap W = \emptyset$.

Suppose instead $s \in U$, and take $\varepsilon > 0$ with $A \cap (s-\varepsilon, s+\varepsilon) \subseteq U$ and t as above. Put $s' := \min\{s+\varepsilon/2,\ y\}$. Every point of $[t, s']$ lies in $[x, y] \subseteq A$ and in $(s-\varepsilon, s+\varepsilon)$, hence in U; together with $[x, t] \subseteq U$ this gives $[x, s'] \subseteq U$ and therefore $s' \leq s$. If $s < y$ then $s' > s$, a contradiction; so $s = y$ and $y \in U$, contradicting $y \in W$.

Both cases are impossible, so no such decomposition exists and A is connected.

Convex sets in $\mathbb{R}^n$. Let $C \subseteq \mathbb{R}^n$ be convex and non-empty and suppose $C = U \cup W$ with U, W relatively open, disjoint and non-empty. Pick $x \in U$, $y \in W$ and consider $\gamma(t) := (1-t)x + ty$, which is continuous and maps $[0,1]$ into C. Then $\gamma^{-1}(U)$ and $\gamma^{-1}(W)$ are relatively open in $[0,1]$ by Theorem 5.2.1, disjoint, non-empty and cover $[0,1]$, contradicting connectedness of the interval $[0,1]$ just established. ■

Connectedness is what makes the intermediate value theorem true, and hence what underlies both the existence of an interior solution between two boundary regimes and Debreu's continuous utility representation theorem alluded to in Remark 1.3.1.

4.6 Chapter Summary

Open sets are those equal to their interior, closed sets those equal to their closure, and the two notions are exchanged by complementation. The open sets are closed under arbitrary unions and finite intersections and contain $\emptyset$ and X; abstracting these three properties gives a topology, and the countable-union-plus-complement variant gives a σ-algebra. A set is closed if and only if it contains the limits of its convergent sequences. Every bounded real sequence has a monotone, hence convergent, subsequence (Theorem 4.3.2); by coordinatewise extraction the same holds in $\mathbb{R}^n$, and the argument requires n to be finite. Compactness is equivalent to sequential compactness in metric spaces, implies closed and bounded always, and is equivalent to closed and bounded exactly in $\mathbb{R}^n$ (Theorem 4.4.2). The budget set with strictly positive prices is compact, which is the hypothesis under which Chapter 5 produces an optimal consumption bundle.

Exercise 4.6.1. Show that $\partial A = \overline{A} \cap \overline{A^c}$, that ∂A is always closed, and that A is closed if and only if $\partial A \subseteq A$. Compute $\operatorname{Int} A$, $\overline{A}$ and ∂A for $A = \mathbb{Q} \cap [0,1]$ in $\mathbb{R}$, and comment on which of the three is empty.

Exercise 4.6.2. Prove that in the discrete metric of Example 3.4.1 every subset is both open and closed, that a sequence converges if and only if it is eventually constant, and that a set is compact if and only if it is finite. Which hypothesis of Theorem 4.4.2 fails for an infinite bounded set in this metric?

Exercise 4.6.3. Let $K \subseteq \mathbb{R}^n$ be compact and non-empty. Show that $\sup K$ and $\inf K$ are attained coordinatewise, that is, for each i there exist $x, y \in K$ with $x_i = \max\{z_i : z \in K\}$ and $y_i = \min\{z_i : z \in K\}$. Give an example of a closed but unbounded $K \subseteq \mathbb{R}$ for which the conclusion fails.

Chapter 5

Continuity, Semicontinuity, and the Weierstrass Theorem

5.1 Motivation: Why Optima Exist

An economic model that begins "the consumer chooses x to maximise $u(x)$ subject to $\langle p, x\rangle \le m$" has presupposed that a maximiser exists. Nothing guarantees this. The supremum of a function over a set need not be attained, and when it is not, the demand function is undefined, comparative statics are vacuous, and equilibrium existence arguments have no starting point.

The Weierstrass theorem gives the standard sufficient conditions: a compact feasible set and enough continuity of the objective. Chapter 4 supplied compactness. This chapter supplies continuity, and then weakens it as far as the existence conclusion permits — to upper semicontinuity for maxima and lower semicontinuity for minima. Indirect utility functions, value functions of dynamic programs and profit functions arise as suprema of families of continuous functions, and Proposition 5.3.3 shows that such suprema are in general only semicontinuous.

5.2 Continuity

Throughout, $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}^m$ unless stated otherwise. Two metrics are in play — the Euclidean metric on the domain and on the codomain — and both are written d.

> **Definition 5.2.1** (Relative topology)**.** Let $A \subseteq X$ with (X, d) a metric space. A set $U \subseteq A$ is *open relative to* A if $U = G \cap A$ for some open $G \subseteq X$; equivalently, if for every $x \in U$ there is $\varepsilon > 0$ with $B_\varepsilon(x) \cap A \subseteq U$. Relative closedness is defined analogously. This is the topology induced by restricting d to $A \times A$, so "relatively open in A" and "open in the metric space $(A, d|_{A\times A})$" are the same statement.

The distinction matters because $[0, 1)$ is not open in $\mathbb{R}$ but is open relative to $[0, 1]$, and hypotheses about feasible sets are almost always relative hypotheses.

> **Definition 5.2.2** (Continuity)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}^m$. The mapping f is *continuous at* $x_0 \in A$ if for every $\varepsilon > 0$ there is $\delta > 0$ such that
>
> $$x \in A \text{ and } d(x_0, x) < \delta \quad \Longrightarrow \quad d\big(f(x_0), f(x)\big) < \varepsilon.$$
>
> It is *continuous on* A if it is continuous at every point of A.

Two features of the definition are easy to pass over. First, δ may depend on x_0 as well as on ε; requiring that it not is uniform continuity (Definition 5.2.3). Second, the implication is restricted to $x \in A$, so continuity is a statement relative to the domain. A function defined only on $\mathbb{Z}$ is continuous at every point of its domain, for trivial reasons.

Example 5.2.1 (Removable and essential discontinuities)**.** Define $f\colon \mathbb{R} \to \mathbb{R}$ by $f(x) = \sin(x)/x$ for $x \ne 0$ and $f(0) = 1$. Then f is continuous at 0: since $\cos x \le \sin(x)/x \le 1$ for $0 < |x| < \pi/2$ and cos is continuous with $\cos 0 = 1$, the value 1 is the only value at 0 that makes f continuous, and it has been assigned. The discontinuity of the function $x \mapsto \sin(x)/x$ on $\mathbb{R} \setminus \{0\}$ is *removable*.

Define $g\colon \mathbb{R} \to \mathbb{R}$ by $g(x) = \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$. No assignment at 0 makes g continuous. Taking $x_k = \big(\pi/2 + 2k\pi\big)^{-1}$ and $y_k = \big(-\pi/2 + 2k\pi\big)^{-1}$ gives $x_k \to 0$, $y_k \to 0$ but $g(x_k) = 1$, $g(y_k) = -1$, so no value of $g(0)$ is compatible with Proposition 5.2.3. The discontinuity is *essential* (Fukuda 2026g, slides 50–51).

Proposition 5.2.1 (Continuity of linear functionals and of the norm)**.** Let $a \in \mathbb{R}^n$.

(i) The functional $f(x) = \langle a, x\rangle$ is continuous on $\mathbb{R}^n$.

(ii) Every linear map $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is continuous.

(iii) The norm $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ is continuous.

Proof. (i) If $a = 0$ then $f \equiv 0$ is constant, hence continuous. Let $a \neq 0$, fix $x_0 \in \mathbb{R}^n$ and $\varepsilon > 0$, and set $\delta := \varepsilon / \|a\| > 0$. If $\|x - x_0\| < \delta$ then by Cauchy–Schwarz (Lemma 3.2.1)

$$|f(x) - f(x_0)| = |\langle a, x - x_0\rangle| \leq \|a\| \, \|x - x_0\| < \|a\| \, \delta = \varepsilon.$$

(ii) Write T in the canonical bases as in Theorem 2.3.1, so that the ith component of Tx is $\langle a_i, x\rangle$ with a_i the ith row of the representing matrix. Given $\varepsilon > 0$, apply (i) to each component with tolerance $\varepsilon/\sqrt{m}$ and take the smallest of the resulting δ_i.

(iii) By the reverse triangle inequality (Proposition 3.3.2), $|\|x\| - \|x_0\|| \leq \|x - x_0\|$, so $\delta = \varepsilon$ works. ∎

Part (i) is Proposition 7.17 of Fukuda (2026d), proved on (Fukuda 2026g, slide 52); part (iii) is Proposition 7.18 there, whose hint is Exercise 7.4, recorded here as Proposition 3.3.2. The δ produced in (i) does not depend on x_0, so linear functionals are in fact uniformly continuous; this is a consequence of linearity, not of the specific bound.

Proposition 5.2.2 (Algebra of continuous functions)**.** Let $A \subseteq \mathbb{R}^n$ and let $f, g\colon A \to \mathbb{R}$ be continuous at x_0.

(i) $f + g$, fg and αf for $\alpha \in \mathbb{R}$ are continuous at x_0; f/g is continuous at x_0 provided $g(x_0) \neq 0$.

(ii) If $h\colon B \to \mathbb{R}^m$ is continuous at $f(x_0)$ and $f(A) \subseteq B$, then $h \circ f$ is continuous at x_0.

(iii) If $f_1, \ldots, f_m\colon A \to \mathbb{R}$ are continuous at x_0, then so are $\max_{1\leq i\leq m} f_i$ and $\min_{1\leq i\leq m} f_i$.

Proof. (i) and (ii) are the standard ε–δ arguments; see Rudin (1976, Thms. 4.4 and 4.7).

(iii) For $m = 2$ use the identity $\max\{s, t\} = \frac{1}{2}\big(s + t + |s - t|\big)$ and combine (i), (ii) and the continuity of $|\cdot|$; the general case follows by induction on m, since $\max_{i\leq m} f_i = \max\{\max_{i\leq m-1} f_i, f_m\}$. The corresponding identity for the minimum is $\min\{s, t\} = \frac{1}{2}(s + t - |s - t|)$. ∎

Part (iii) is Exercise 7.34 of (Fukuda 2026g, slide 56). The restriction to *finitely* many functions is essential and is what fails in Example 5.3.1; the induction consumes one step per function and cannot be run to infinity.

Proposition 5.2.3 (Sequential characterisation)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}^m$. Then f is continuous at $x_0 \in A$ if and only if for every sequence $(x_k) \subseteq A$ with $x_k \to x_0$ one has $f(x_k) \to f(x_0)$; that is, $f\big(\lim_k x_k\big) = \lim_k f(x_k)$.

Proof. ($\Rightarrow$) Given $\varepsilon > 0$, take δ from Definition 5.2.2 and then k_0 with $d(x_0, x_k) < \delta$ for $k \geq k_0$; then $d(f(x_0), f(x_k)) < \varepsilon$ for $k \geq k_0$.

($\Leftarrow$) Suppose f is not continuous at x_0. Then there is $\varepsilon_0 > 0$ such that for every $\delta > 0$ some $x \in A$ has $d(x_0, x) < \delta$ and $d(f(x_0), f(x)) \geq \varepsilon_0$. Taking $\delta = 1/k$ produces $x_k \in A$ with $x_k \to x_0$ and $d(f(x_0), f(x_k)) \geq \varepsilon_0$ for every k, so $f(x_k) \not\to f(x_0)$. ∎

This is Proposition 7.22 of Fukuda (2026d) (Fukuda 2026g, slide 57). The contrapositive is the practical tool: to refute continuity, exhibit two sequences approaching the same point along which f has different limits, as in Example 5.2.2.

Theorem 5.2.1 (Topological characterisation)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}^m$. The following are equivalent.

(i) f is continuous on A.

(ii) $f^{-1}(V)$ is open relative to A for every open $V \subseteq \mathbb{R}^m$.

(iii) $f^{-1}(F)$ is closed relative to A for every closed $F \subseteq \mathbb{R}^m$.

Proof. (i)⇒(ii). Let V be open and $x_0 \in f^{-1}(V)$. Since V is open there is $\varepsilon > 0$ with $B_\varepsilon(f(x_0)) \subseteq V$; continuity supplies $\delta > 0$ with $f\big(B_\delta(x_0) \cap A\big) \subseteq B_\varepsilon(f(x_0)) \subseteq V$, that is $B_\delta(x_0) \cap A \subseteq f^{-1}(V)$.

(ii)⇒(i). Fix $x_0 \in A$ and $\varepsilon > 0$. The set $V = B_\varepsilon(f(x_0))$ is open, so $f^{-1}(V)$ is relatively open and contains x_0; hence some $\delta > 0$ has $B_\delta(x_0) \cap A \subseteq f^{-1}(V)$, which is the required implication.

(ii)⇔(iii). Preimages commute with complements: $f^{-1}(F^{\mathsf{c}}) = \big(f^{-1}(F)\big)^{\mathsf{c}}$ relative to A (Proposition 1.2.2). Apply Proposition 4.2.1. ∎

Remark 5.2.1 (Continuity, measurability, and the shape of the definition)**.** Theorem 5.2.1 states continuity without reference to any metric: only the two collections of open sets are used. This is the definition of continuity in a general topological space, and it is what makes the parallel of Remark 4.2.1 exact. A map is continuous when preimages of open sets are open; a map is measurable when preimages of measurable sets are measurable (Definition 14.2.1). The same asymmetry appears in both: the condition constrains preimages, never images. Images of open sets under continuous maps need not be open — $x \mapsto x^2$ sends the open set $(-1, 1)$ to $[0, 1)$ — and this is why Proposition 5.2.4, which does concern images, requires compactness rather than openness.

Example 5.2.2 (Separate continuity does not imply joint continuity)**.** Let $f\colon \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}^m$ be continuous. For fixed y_0 the section $x \mapsto f(x, y_0)$ is continuous, since $x \mapsto (x, y_0)$ is continuous and Proposition 5.2.2(ii) applies; symmetrically in the other argument. The converse fails. Define $f\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1 x_2}{x_1^2 + x_2^2}, & (x_1, x_2) \neq (0, 0), \\ 0, & (x_1, x_2) = (0, 0). \end{cases}$$

Both sections through the origin are identically zero, hence continuous. But along the diagonal $f(\rho, \rho) = \frac{1}{2}$ and along the anti-diagonal $f(\rho, -\rho) = -\frac{1}{2}$ for every $\rho \neq 0$, so by Proposition 5.2.3 f is not continuous at the origin. In fact f is constant on every ray through the origin, and takes every value in $[-\frac{1}{2}, \frac{1}{2}]$ in every ball around it (Fukuda 2026g, slides 54–55).

The example is a standing warning in comparative statics. Verifying that a function of prices and income responds continuously to each argument separately establishes nothing about its joint behaviour, and it is joint continuity that the maximum theorem of Chapter 13 requires.

5.2.1 Continuity, compactness, and connectedness

Proposition 5.2.4 (Continuous images of compact sets)**.** Let $K \subseteq \mathbb{R}^n$ be compact and $f\colon K \to \mathbb{R}^m$ continuous. Then $f(K)$ is compact.

Proof. Let $(y_k) \subseteq f(K)$ and choose $x_k \in K$ with $f(x_k) = y_k$. By compactness of K some subsequence $x_{k_j} \to x^* \in K$, and by Proposition 5.2.3 $y_{k_j} = f(x_{k_j}) \to f(x^*) \in f(K)$. So $f(K)$ is sequentially compact, hence compact by Theorem 4.4.1. ∎

Proposition 5.2.5 (Continuous images of connected sets)**.** Let $C \subseteq \mathbb{R}^n$ be connected and $f\colon C \to \mathbb{R}^m$ continuous. Then $f(C)$ is connected.

Proof. Suppose $f(C) = U_1 \cup U_2$ with U_1, U_2 non-empty, disjoint and open relative to $f(C)$. By Theorem 5.2.1 the sets $f^{-1}(U_1)$ and $f^{-1}(U_2)$ are open relative to C, non-empty, disjoint, and their union is C — contradicting connectedness of C. ∎

Corollary 5.2.1 (Intermediate value theorem)**.** Let $f\colon [a,b] \to \mathbb{R}$ be continuous and let y lie between $f(a)$ and $f(b)$. Then $f(c) = y$ for some $c \in [a,b]$.

Proof. $[a,b]$ is connected by Proposition 4.5.1, so $f([a,b])$ is connected by Proposition 5.2.5, hence an interval by Proposition 4.5.1 again. An interval containing $f(a)$ and $f(b)$ contains everything between them. ■

5.2.2 Uniform and Lipschitz continuity

Definition 5.2.3 (Uniform, Lipschitz and Hölder continuity)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}^m$.

(i) f is *uniformly continuous* on A if for every $\varepsilon > 0$ there is $\delta > 0$, *depending only on* ε, such that $x, y \in A$ and $d(x,y) < \delta$ imply $d(f(x), f(y)) < \varepsilon$.

(ii) f is *Lipschitz continuous* with constant $L \geq 0$ if $d(f(x), f(y)) \leq L\, d(x,y)$ for all $x, y \in A$. If $L < 1$ the map is a contraction (Definition 6.4.1).

(iii) f is *Hölder continuous* of order $\alpha \in (0,1]$ if $d(f(x), f(y)) \leq L\, d(x,y)^\alpha$ for all $x, y \in A$.

Lipschitz continuity is Hölder continuity of order 1; Hölder continuity of any order implies uniform continuity, with $\delta = (\varepsilon/L)^{1/\alpha}$; and uniform continuity implies continuity. None of the three implications reverses.

Example 5.2.3 (The implications are strict)**.** On $(0,1]$, $x \mapsto 1/x$ is continuous but not uniformly continuous: taking $x_k = 1/k$ and $y_k = 1/(k+1)$ gives $|x_k - y_k| \to 0$ while $|f(x_k) - f(y_k)| = 1$ for every k. On $[0,1]$, $x \mapsto \sqrt{x}$ is uniformly continuous — it is Hölder of order $\frac{1}{2}$ — but not Lipschitz, since $\big(\sqrt{x} - \sqrt{0}\big)/(x-0) = x^{-1/2} \to \infty$ as $x \downarrow 0$.

Theorem 5.2.2 (Heine–Cantor)**.** A continuous function on a compact set is uniformly continuous.

Proof. Let K be compact, f continuous on K, and suppose f is not uniformly continuous. Then some $\varepsilon_0 > 0$ admits, for each $k \in \mathbb{N}$, points $x_k, y_k \in K$ with $d(x_k, y_k) < 1/k$ and $d(f(x_k), f(y_k)) \geq \varepsilon_0$. By compactness pass to a subsequence with $x_{k_j} \to x^* \in K$; then $y_{k_j} \to x^*$ as well, because $d(y_{k_j}, x^*) \leq d(y_{k_j}, x_{k_j}) + d(x_{k_j}, x^*) \to 0$. Continuity and Proposition 5.2.3 give $f(x_{k_j}) \to f(x^*)$ and $f(y_{k_j}) \to f(x^*)$, so $d(f(x_{k_j}), f(y_{k_j})) \to 0$, contradicting the lower bound ε_0. ■

Uniform continuity is what allows a single δ to be quoted for an entire domain, and it is the hypothesis under which Riemann sums converge in Chapter 10 and under which numerical approximation error can be bounded uniformly over a state space in Chapter 15.

5.3 Semicontinuity

Continuity is more than existence of optima requires. The following weakening isolates what is needed, and it is the form in which the hypothesis is verifiable for the value functions that arise in economics.

Definition 5.3.1 (Semicontinuity)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}$.

(i) f is *lower semicontinuous* (lsc) if $f^{-1}\big((\lambda, \infty)\big) = \{x \in A : f(x) > \lambda\}$ is open relative to A for every $\lambda \in \mathbb{R}$; equivalently, if every lower section $\{x \in A : f(x) \leq \lambda\}$ is closed relative to A.

(ii) f is *upper semicontinuous* (usc) if $-f$ is lower semicontinuous; equivalently, if every upper section $\{x \in A : f(x) \geq \lambda\}$ is closed relative to A.

Proposition 5.3.1 (Pointwise form)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}$.

(i) f is lsc if and only if for every $x_0 \in A$ and every $\lambda < f(x_0)$ there is $\delta > 0$ such that $x \in B_\delta(x_0) \cap A$ implies $\lambda < f(x)$.

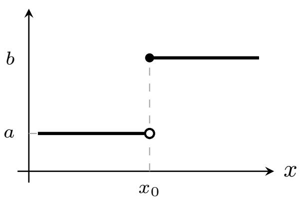
(ii) f is usc if and only if for every $x_0 \in A$ and every $\lambda > f(x_0)$ there is $\delta > 0$ such that $x \in B_\delta(x_0) \cap A$ implies $f(x) < \lambda$.

Equivalently, in sequential form: f is lsc at x_0 if $x_k \to x_0$ implies $f(x_0) \le \liminf_k f(x_k)$, and usc at x_0 if $x_k \to x_0$ implies $\limsup_k f(x_k) \le f(x_0)$.

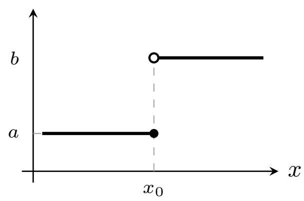
Proof. (i) If f is lsc and $\lambda < f(x_0)$, then x_0 belongs to the relatively open set $\{f > \lambda\}$, which therefore contains $B_\delta(x_0) \cap A$ for some $\delta > 0$. Conversely, given λ and $x_0 \in \{f > \lambda\}$, the hypothesis produces such a δ, so $\{f > \lambda\}$ is relatively open. Part (ii) follows by replacing f with $-f$.

For the sequential form, suppose f is lsc at x_0 and $x_k \to x_0$. For any $\lambda < f(x_0)$ the ball $B_\delta(x_0)$ eventually contains every x_k, so $f(x_k) > \lambda$ eventually and $\liminf_k f(x_k) \ge \lambda$; letting $\lambda \uparrow f(x_0)$ gives the claim. The converse is the same argument run backwards. ■

This is Proposition 8.10 of Fukuda (2026d, p. 271). Verbally: at a point of lower semicontinuity the function cannot *drop* suddenly as one moves away from x_0, and at a point of upper semicontinuity it cannot *jump up*. A jump downward is compatible with upper semicontinuity, which is why the maximum of a usc function can sit exactly at the top of a jump. Figure 5.1 draws the two cases.



(a) upper semicontinuous at x_0



(b) lower semicontinuous at x_0

Figure 5.1: The two ways a jump can be closed, for a function equal to a on one side of x_0 and to $b > a$ on the other. In (a) the value at x_0 is b, so $\limsup_{x\to x_0} f(x) = b = f(x_0)$ and the function is upper semicontinuous at x_0; it is not lower semicontinuous, since $\liminf_{x\to x_0} f(x) = a < b$. In (b) the value at x_0 is a and the two conclusions are exchanged. Filled circles mark the value taken at x_0.

Corollary 5.3.1. f is continuous if and only if it is both lower and upper semicontinuous.

Proof. If f is continuous, then f^{-1} of the open sets (λ, ∞) and $(-\infty, \lambda)$ are relatively open by Theorem 5.2.1, giving both properties. Conversely, if both hold then $f^{-1}\big((\lambda,\mu)\big) = f^{-1}((\lambda,\infty)) \cap f^{-1}((-\infty,\mu))$ is relatively open, and every open subset of $\mathbb{R}$ is a countable union of such intervals, so Theorem 4.2.1(i) and Theorem 5.2.1 apply. ■

Proposition 5.3.2 (Epigraph and hypograph)**.** Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}$. Then f is lsc if and only if its *epigraph*

$$\operatorname{epi} f := \{(x,z) \in A \times \mathbb{R} : f(x) \le z\}$$

is closed in $A \times \mathbb{R}$, and f is usc if and only if its *hypograph* $\operatorname{hyp} f := \{(x,z) \in A \times \mathbb{R} : f(x) \ge z\}$ is closed.

Proof. Suppose f is lsc and $(x_k, z_k) \to (x, z)$ in $A \times \mathbb{R}$ with $f(x_k) \le z_k$. Proposition 5.3.1 gives $f(x) \le \liminf_k f(x_k) \le \liminf_k z_k = z$, so that $(x,z) \in \operatorname{epi} f$; now apply Theorem 4.3.1.

Conversely, let $\operatorname{epi} f$ be closed and $x_k \to x$ in A; we must show $f(x) \le \ell$, where $\ell := \liminf_k f(x_k) \in [-\infty, +\infty]$. If $\ell = +\infty$ there is nothing to prove. If $\ell = -\infty$, choose a subsequence with $f(x_{k_j}) \to -\infty$; then for each fixed $z \in \mathbb{R}$ one has $f(x_{k_j}) \le z$ for all large j, so $(x_{k_j}, z) \in \operatorname{epi} f$, and $(x_{k_j}, z) \to (x, z)$ forces $(x,z) \in \operatorname{epi} f$, that is $f(x) \le z$. As $z \in \mathbb{R}$ was arbitrary this contradicts $f(x) \in \mathbb{R}$, so the case $\ell = -\infty$ does not arise. There remains $\ell \in \mathbb{R}$: pass to a subsequence along which $f(x_{k_j}) \to \ell$, so that $(x_{k_j}, f(x_{k_j})) \in \operatorname{epi} f$ converges to $(x, \ell) \in A \times \mathbb{R}$; closedness gives $(x, \ell) \in \operatorname{epi} f$, that is $f(x) \le \ell$. The hypograph statement follows by applying this to $-f$. ■

This is Proposition 8.12 of Fukuda (2026d, p. 273). The epigraph characterisation is the bridge to Chapter 9: convexity of f is convexity of $\operatorname{epi} f$, and lower semicontinuity is closedness of $\operatorname{epi} f$, so "closed convex function" means precisely that the epigraph is a closed convex set.

Proposition 5.3.3 (Suprema and infima)**.** Let $A \subseteq \mathbb{R}^n$ and let $(f_i)_{i\in I}$ be a family of functions $A \to \mathbb{R}$ indexed by an arbitrary set I.

(i) If every f_i is lsc and $f := \sup_{i\in I} f_i$ is finite on A, then f is lsc.

(ii) If every f_i is usc and $f := \inf_{i\in I} f_i$ is finite on A, then f is usc.

Proof. (i) For any λ, $\{x : \sup_i f_i(x) > \lambda\} = \bigcup_{i\in I}\{x : f_i(x) > \lambda\}$: the supremum exceeds λ exactly when some member does. Each set on the right is relatively open, and Theorem 4.2.1(i) permits arbitrary unions. Part (ii) follows by negation. ■

Example 5.3.1 (A supremum of continuous functions that is not usc)**.** For each $a \in (0,1]$ define the continuous function $f_a\colon [0,1] \to \mathbb{R}$ by $f_a(x) = x/a$ for $0 \le x \le a$ and $f_a(x) = 1$ for $a \le x \le 1$. Then

$$f(x) := \sup_{a\in(0,1]} f_a(x) = \begin{cases} 0, & x = 0, \\ 1, & 0 < x \le 1. \end{cases}$$

Indeed $f_a(0) = 0$ for every a, while for $x > 0$ the choice $a = x$ gives $f_x(x) = 1$. The limit function is lsc — consistent with Proposition 5.3.3(i) — and it is not usc, since $\{f \ge 1\} = (0,1]$ is not closed in $[0,1]$. This is Example 8.4 of Fukuda (2026d, p. 272).

The pattern recurs throughout economics. Indirect utility, the profit function and the value function of a dynamic program are all defined as suprema over a feasible set indexed by a parameter, and the general theory delivers only one of the two semicontinuity properties for free. Establishing the other requires additional structure — compactness of the feasible correspondence and continuity of the objective, which is the content of Berge's theorem in Theorem 13.5.1.

5.4 The Weierstrass Theorem

Theorem 5.4.1 (Weierstrass)**.** Let $A \subseteq \mathbb{R}^n$ be compact and non-empty.

(i) A lower semicontinuous $f\colon A \to \mathbb{R}$ attains its minimum: there is $x_* \in A$ with $f(x_*) = \inf_{x\in A} f(x)$, and this infimum is finite.

(ii) An upper semicontinuous $f\colon A \to \mathbb{R}$ attains its maximum: there is $x^* \in A$ with $f(x^*) = \sup_{x\in A} f(x)$, and this supremum is finite.

Proof. It suffices to prove (i); (ii) follows by applying (i) to $-f$.

Write $\alpha := \inf_{x\in A} f(x) \in [-\infty, \infty)$ and consider the lower sections $L_\lambda := \{x \in A : f(x) \le \lambda\}$, each closed relative to A by Definition 5.3.1, hence compact by Proposition 4.4.2(i).

First, $\alpha > -\infty$. If not, $L_\lambda \neq \emptyset$ for every $\lambda \in \mathbb{R}$, and the family $(L_\lambda)_{\lambda\in\mathbb{R}}$ is nested decreasing as λ decreases, so every finite subfamily has non-empty intersection. A nested family of non-empty compact sets has non-empty intersection (Lemma 5.4.1), so some $x \in \bigcap_{\lambda\in\mathbb{R}} L_\lambda$, giving $f(x) \le \lambda$ for every real λ, which is impossible for a real-valued f. Hence α is finite.

Second, the infimum is attained. The sets L_λ for $\lambda > \alpha$ are non-empty by definition of the infimum, compact, and nested. By Lemma 5.4.1 again, $\bigcap_{\lambda>\alpha} L_\lambda \neq \emptyset$; any x_* in this intersection satisfies $f(x_*) \le \lambda$ for every $\lambda > \alpha$, hence $f(x_*) \le \alpha$, and $f(x_*) \ge \alpha$ because α is a lower bound. So $f(x_*) = \alpha = \min_{x\in A} f(x)$. ■

Lemma 5.4.1 (Finite intersection property)**.** Let $(K_i)_{i\in I}$ be a family of non-empty compact subsets of a metric space such that every finite subfamily has non-empty intersection. Then $\bigcap_{i\in I} K_i \neq \emptyset$.

Proof. Fix $i_0 \in I$ and suppose $\bigcap_i K_i = \emptyset$. Then the relatively open sets $K_i^{\mathrm{c}} \cap K_{i_0}$, $i \in I$, cover K_{i_0}: a point of K_{i_0} lying in no K_i^{c} would lie in every K_i. By compactness finitely many suffice, say for $i_1, \ldots, i_m$, which says $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$ — contradicting the hypothesis on finite subfamilies. ■

Corollary 5.4.1 (Weierstrass, continuous case)**.** A continuous real-valued function on a non-empty compact set attains both its minimum and its maximum.

Proof. By Corollary 5.3.1 the function is both lsc and usc; apply both parts of Theorem 5.4.1.

A second proof avoids semicontinuity altogether and generalises differently. By Proposition 5.2.4 the image $f(A)$ is a compact subset of $\mathbb{R}$, hence closed and bounded by Theorem 4.4.2. Boundedness gives finite $M := \sup f(A)$ and $m := \inf f(A)$, and by Proposition 1.5.1(ii) both are points of closure of $f(A)$; closedness then gives $M, m \in f(A)$, that is, both are attained. ■

This is Corollary 8.6 of Fukuda (2026d, p. 275), with both proofs as given there.

Example 5.4.1 (Each hypothesis is needed). (i) *Semicontinuity of the wrong type.* Let $f\colon [0,1] \to \mathbb{R}$ be $f(0) = \frac{1}{2}$ and $f(x) = 1 - x$ for $x \in (0,1]$. This function is lsc, so by Theorem 5.4.1(i) it attains its minimum, at $x = 1$. It has no maximum: $\sup f = 1$, approached as $x \downarrow 0$, but $f(0) = \frac{1}{2}$. Symmetrically, $g(0) = \frac{1}{2}$ and $g(x) = x$ on $(0,1]$ is usc, attains its maximum at $x = 1$, and has no minimum (Fukuda 2026d, Ex. 8.5, p. 274).

(ii) *Non-compact domain, unbounded.* $f(x) = x$ on $[0,\infty)$ is continuous and attains no maximum.

(iii) *Non-compact domain, not closed.* $f(x) = x$ on $[0,1)$ is continuous and bounded, and attains no maximum: the supremum 1 is not in the image because the point at which it would be attained is not in the domain.

(iv) *Infinite dimensions.* Let B be the closed unit ball of the space ℓ^2 of Example 4.3.1 and define $f\colon B \to \mathbb{R}$ by $f(x) = \sum_{k\ge 1}(1-\frac{1}{k})x_k^2$. The series converges because $\sum_k x_k^2 \le 1$, and f is continuous on B: for $x, y \in B$, $|f(x) - f(y)| \le \sum_k |x_k - y_k|\,|x_k + y_k| \le \|x-y\|\,\|x+y\| \le 2\,\|x-y\|$ by Cauchy–Schwarz. Its supremum is 1, since $f(e^k) = 1 - 1/k$, and the supremum is not attained: $f(x) = 1$ would force $\sum_k x_k^2/k \le \sum_k x_k^2 - f(x) \le 0$, hence $x = 0$ and $f(x) = 0$. The domain is closed and bounded but, by Example 4.3.1, not compact, and Theorem 5.4.1 does not apply. No *linear* functional would serve as the example: every continuous linear functional on ℓ^2 is $x \mapsto \langle a, x\rangle$ for some $a \in \ell^2$ (Theorem 3.2.1 is the finite-dimensional case) and attains its supremum over B at $a/\|a\|$.

Proposition 5.4.1 (The solution set is compact). Let $A \subseteq \mathbb{R}^n$ be compact and non-empty.

(i) If $f\colon A \to \mathbb{R}$ is usc, then $\arg\max_{x\in A} f(x)$ is a non-empty compact subset of A.

(ii) If $f\colon A \to \mathbb{R}$ is lsc, then $\arg\min_{x\in A} f(x)$ is a non-empty compact subset of A.

Proof. (i) Non-emptiness is Theorem 5.4.1(ii). Writing $M := \max_{x\in A} f(x)$, the solution set is $\{x \in A : f(x) \ge M\}$, which is closed relative to A by upper semicontinuity, hence compact by Proposition 4.4.2(i). Part (ii) is the same argument for $-f$. ■

This is Proposition 8.13 of Fukuda (2026d, p. 275). It is the first appearance of the object that Chapter 13 studies systematically: the solution of an optimisation problem is in general a *set*, and the statement that it varies continuously with the parameters is a statement about correspondences, not functions. Uniqueness — which reduces the set to a singleton and the correspondence to a function — requires strict quasiconcavity of f and convexity of A, and is supplied in Proposition 9.3.5.

5.5 Economic Application: Existence of Consumer Demand

The four questions raised in Example 1.6.1 can now be answered in part.

Environment. There are n goods. A consumption bundle is a vector $x \in \mathbb{R}^n_+$. Prices are $p \in \mathbb{R}^n_{++}$ and wealth is $m > 0$. The consumer's feasible set is the budget set $\mathcal{B}(p,m) = \{x \in \mathbb{R}^n_+ : \langle p, x\rangle \le m\}$, and preferences are represented by a utility function $u\colon \mathbb{R}^n_+ \to \mathbb{R}$.

Problem. The consumer solves

$$\max_x \; u(x) \qquad \text{subject to} \quad \langle p, x\rangle \le m, \quad x \ge 0. \tag{5.1}$$

Proposition 5.5.1 (Existence of a solution to the consumer's problem). Let $p \in \mathbb{R}^n_{++}$, $m > 0$, and let $u\colon \mathbb{R}^n_+ \to \mathbb{R}$ be upper semicontinuous. Then (5.1) has a solution, the set of solutions

$$x(p,m) := \operatorname*{arg\,max}_{x \in \mathcal{B}(p,m)} u(x)$$

is non-empty and compact, and the indirect utility $v(p,m) := \max_{x \in \mathcal{B}(p,m)} u(x)$ is finite.

Proof. $\mathcal{B}(p,m)$ is non-empty, since 0 belongs to it, and compact by Example 4.4.1. Apply Theorem 5.4.1(ii) and Proposition 5.4.1(i). ∎

Economic content of the hypotheses. Strict positivity of every price is what bounds the budget set: a free good can be consumed without limit, and if utility is strictly increasing in it, no optimum exists. Positive wealth guarantees a non-degenerate problem, though $m = 0$ is admissible and yields $x = 0$. Upper semicontinuity of u is weaker than continuity and permits preferences with an upward jump at a threshold — a consumer who strictly prefers reaching a subsistence level to being marginally below it, with the preferred value attained *at* the threshold. A *downward* jump at the threshold violates upper semicontinuity, and then the supremum of utility over the budget set may fail to be attained even though the budget set is compact, exactly as in Example 5.4.1(i).

Uniqueness, interiority and continuity. First, uniqueness. Proposition 5.5.1 produces a non-empty compact solution *set*. With linear utility $u(x) = \langle a, x\rangle$ and prices proportional to a, every point of the budget hyperplane is optimal. Uniqueness needs strict quasiconcavity (Chapter 9).

Second, an interior solution. Nothing in the argument prevents $x(p,m)$ from lying on the boundary of $\mathbb{R}^n_+$, and indeed corner solutions are generic when preferences are not strictly monotone in every good. First-order conditions therefore take the Kuhn–Tucker form of Chapter 12 rather than the gradient-equals-zero form.

Third, continuity of $x(\cdot,\cdot)$ and of $v(\cdot,\cdot)$ in (p,m). Proposition 5.5.1 is a statement for each fixed (p,m) separately, and Example 5.2.2 is a reminder that pointwise statements do not aggregate into joint continuity. The budget set itself moves with (p,m); establishing that the value moves continuously and that the solution set moves upper hemicontinuously is Berge's maximum theorem (Theorem 13.5.1), which requires the budget correspondence to be both upper and lower hemicontinuous. Both hold on $\mathbb{R}^n_{++} \times \mathbb{R}_+$, zero wealth included, by Proposition 13.4.1 and Example 13.4.1; what breaks the argument is a price falling to zero, and what breaks lower hemicontinuity is the absence of a point strictly inside the budget set.

5.5.1 Preferences without a utility function

Preferences, not utility functions, are the primitive. Three statements about them are often run together, and they are distinct.

Let $\succsim$ be a complete and transitive preference relation on a set $X \subseteq \mathbb{R}^n$. Say that $\succsim$ has *closed upper contour sets* if $U(x) := \{y \in X : y \succsim x\}$ is closed relative to X for every $x \in X$. This is a property of $\succsim$ alone. A function $u\colon X \to \mathbb{R}$ *represents* $\succsim$ if $u(y) \ge u(x)$ exactly when $y \succsim x$.

Proposition 5.5.2 (Semicontinuity of a representation and of the relation). Let $u\colon X \to \mathbb{R}$ represent $\succsim$.

(i) If u is upper semicontinuous, then $\succsim$ has closed upper contour sets.

(ii) The converse fails: $\succsim$ may have closed upper contour sets while u is not upper semicontinuous.

(iii) If u is upper semicontinuous and $\varphi\colon I \to \mathbb{R}$ is *strictly* increasing and continuous on an interval $I \supseteq u(X)$, then $\varphi \circ u$ is upper semicontinuous and represents $\succsim$. Weak monotonicity is not enough: a constant φ merges all utility levels and represents the indifference relation instead.

Proof. (i) $U(x) = \{y : u(y) \ge u(x)\}$, an upper section of u, closed by Definition 5.3.1(ii).

(ii) Take $X = [0,1]$ and let $\succsim$ be the usual order, represented by the continuous function $u(x) = x$, so that $U(x) = [x,1]$ is closed for every x. Let $\varphi(t) = t$ for $t \le \frac{1}{2}$ and $\varphi(t) = t + 1$ for $t > \frac{1}{2}$, which is increasing on $[0,1]$, and put $\tilde{u} := \varphi \circ u$. Then $\tilde{u}$ represents the same $\succsim$, but $\{\tilde{u} \geq 1\} = (\frac{1}{2}, 1]$ is not closed, so $\tilde{u}$ is not upper semicontinuous.

(iii) That $\varphi \circ u$ represents $\succsim$ is immediate from strict monotonicity of φ. For upper semicontinuity, let $x_k \to x_0$ in X and fix $\varepsilon > 0$ with $u(x_0) + \varepsilon \in I$. Upper semicontinuity of u gives $u(x_k) < u(x_0) + \varepsilon$ for all large k, whence $\varphi(u(x_k)) \leq \varphi(u(x_0) + \varepsilon)$ and $\limsup_k \varphi(u(x_k)) \leq \varphi(u(x_0) + \varepsilon)$. Letting $\varepsilon \downarrow 0$ and using continuity of φ at $u(x_0)$ gives $\limsup_k \varphi(u(x_k)) \leq \varphi(u(x_0))$. If instead $u(x_0)$ is the right endpoint of I, then $u(x_k) \leq u(x_0)$ for every k and the same conclusion is immediate. Apply Proposition 5.3.1. ∎

The distinction matters because upper semicontinuity of a *particular* representation is not an ordinal property, whereas closedness of the upper contour sets is. The gap does not, however, affect the existence question, which can be settled for $\succsim$ directly.

Proposition 5.5.3 (Existence of a greatest element)**.** Let $K \subseteq \mathbb{R}^n$ be non-empty and compact and let $\succsim$ be a complete and transitive relation on K with closed upper contour sets. Then the set

$$G := \{x^* \in K : x^* \succsim y \text{ for all } y \in K\}$$

of $\succsim$-greatest elements of K is non-empty and compact.

Proof. For $y \in K$ the set $U(y) = \{x \in K : x \succsim y\}$ is closed in K, hence compact by Proposition 4.4.2(i), and non-empty because $y \in U(y)$ by completeness. Given finitely many $y_1, \ldots, y_m \in K$, completeness and transitivity produce an index j with $y_j \succsim y_i$ for all i — take a maximal element of a finite set under a complete transitive relation, obtained by induction on m — and then $y_j \in \bigcap_{i \leq m} U(y_i)$, so every finite subfamily of $(U(y))_{y \in K}$ has non-empty intersection. Lemma 5.4.1 gives $G = \bigcap_{y \in K} U(y) \neq \emptyset$, and G is an intersection of closed subsets of the compact set K, hence compact. ∎

Proposition 5.5.1 is the special case $K = \mathcal{B}(p, m)$ with $\succsim$ represented by an upper semicontinuous u; the argument above shows that the utility function is a convenience rather than a necessity, and that the substantive hypotheses are compactness of the feasible set and closedness of the upper contour sets. What a utility representation buys is the machinery of Chapter 8 and Chapter 12, which needs a real-valued objective to differentiate.

5.6 Chapter Summary

Continuity at a point admits three equivalent formulations: the ε–δ condition, preservation of limits of sequences (Proposition 5.2.3), and relative openness of preimages of open sets (Theorem 5.2.1). The third is metric-free and is the definition that transfers to general topological spaces and, with "open" replaced by "measurable", to random variables. Linear maps and norms are continuous; sums, products, quotients with non-vanishing denominator, compositions and finite maxima of continuous functions are continuous; separate continuity in each argument is strictly weaker than joint continuity (Example 5.2.2). Continuous images of compact sets are compact and of connected sets are connected, which gives the intermediate value theorem and the boundedness half of the Weierstrass theorem. Continuity on a compact set upgrades to uniform continuity (Theorem 5.2.2).

Splitting continuity into its two halves isolates what existence of an optimum actually needs: upper semicontinuity for a maximum, lower semicontinuity for a minimum, in both cases on a compact domain (Theorem 5.4.1). The solution set is then non-empty and compact (Proposition 5.4.1). An arbitrary supremum of lsc functions is lsc, but an arbitrary supremum of continuous functions need not be usc (Example 5.3.1), which is why the continuity of value functions in their parameters is a separate theorem rather than a corollary. For the consumer's problem with strictly positive prices, upper semicontinuity of utility delivers existence of demand and finiteness of indirect utility, and nothing more.

Exercise 5.6.1. Prove Proposition 5.2.2(i) directly from Definition 5.2.2, and identify at which step the hypothesis $g(x_0) \neq 0$ is used in the quotient case. Then show by example that continuity of f and g at x_0 is not enough for f/g to have a *removable* discontinuity at a zero of g.

Exercise 5.6.2. Let $A \subseteq \mathbb{R}^n$ and $f\colon A \to \mathbb{R}$. Show that f is lsc if and only if $f = \sup_{i \in I} f_i$ for some family of continuous functions f_i when A is compact and f is bounded below. (Hint: for each x_0 and each $\lambda < f(x_0)$, construct a continuous function below f that takes the value λ at x_0.) Deduce that Proposition 5.3.3(i) is sharp.

Exercise 5.6.3. A firm with technology $Y \subseteq \mathbb{R}^n$ closed and prices $p \in \mathbb{R}^n$ solves $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$. Show that π is convex and lsc on $\mathbb{R}^n$ as a supremum of linear functions, that $\pi(p) = +\infty$ is possible when Y is unbounded, and that if Y is compact then π is finite, continuous, and the supremum is attained. Relate the failure at unbounded Y to Example 5.4.1(ii).

Chapter 6

Completeness, Contractions, and Fixed Points

6.1 Motivation: Proving a Limit Exists Without Knowing It

The definition of convergence names the limit. To use it one must already have a candidate. In most of the arguments that matter — iterating a Bellman operator, solving a differential equation, constructing an equilibrium price — the object whose existence is at issue is precisely the limit, and there is no candidate to name.

Cauchy's condition removes the limit from the statement: it asks only that the terms of the sequence eventually be close *to one another*. In a complete space that internal condition is enough. Completeness is thus the hypothesis that converts a self-referential existence claim into a verifiable one, and it is the hypothesis on which the contraction mapping theorem, and with it the entire recursive method in macroeconomics, rests.

6.2 Cauchy Sequences

Definition 6.2.1 (Cauchy sequence). A sequence $(x_n)_{n\in\mathbb{N}}$ in a metric space (X, d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there is $n_0 \in \mathbb{N}$ such that $m, n \geq n_0$ implies $d(x_m, x_n) < \varepsilon$.

Proposition 6.2.1 (Convergent implies Cauchy). Every convergent sequence in a metric space is Cauchy.

Proof. Let $x_n \to x^*$ and $\varepsilon > 0$. Choose n_0 with $d(x^*, x_n) < \varepsilon/2$ for $n \geq n_0$. For $m, n \geq n_0$, $d(x_m, x_n) \leq d(x_m, x^*) + d(x^*, x_n) < \varepsilon$. ■

Proposition 6.2.2 (Cauchy implies bounded). Every Cauchy sequence in a metric space is bounded.

Proof. Take $\varepsilon = 1$ and the corresponding n_0, so $d(x_{n_0}, x_n) < 1$ for $n \geq n_0$. Put

$$M := \max\{1,\ d(x_{n_0}, x_1),\ \ldots,\ d(x_{n_0}, x_{n_0-1})\},$$

a maximum over finitely many numbers. Then $(x_n) \subseteq B_{M+1}(x_{n_0})$. ■

These are Propositions 8.14 and 8.15 of Fukuda (2026d) (Fukuda 2026g, slides 58–59).

Lemma 6.2.1 (Cauchy with a convergent subsequence). If a Cauchy sequence has a subsequence converging to x^*, then the whole sequence converges to x^*.

Proof. Let $\varepsilon > 0$. Choose n_0 with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq n_0$, and choose an index $n_k \geq n_0$ of the subsequence with $d(x^*, x_{n_k}) < \varepsilon/2$. For $n \geq n_0$, $d(x^*, x_n) \leq d(x^*, x_{n_k}) + d(x_{n_k}, x_n) < \varepsilon$. ■

The converse of Proposition 6.2.1 is false in general: the implication "Cauchy $\Rightarrow$ convergent" is not a theorem about metric spaces but a property that some metric spaces have and others lack. That property is completeness.

6.3 Complete Metric Spaces

Definition 6.3.1 (Completeness)**.** A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a limit belonging to X. A subset $A \subseteq X$ is complete if $(A, d|_{A\times A})$ is complete.

In a complete space the two conditions coincide, and the practical consequence is the one stated on (Fukuda 2026g, slide 60): to prove that (x_n) has a limit it suffices to prove that (x_n) is Cauchy, a condition involving only the terms of the sequence.

Example 6.3.1 (An incomplete space)**.** The space $X = (0, 1]$ with the metric inherited from $\mathbb{R}$ is not complete. The sequence $x_n = 1/n$ is Cauchy — given $\varepsilon > 0$, Proposition 1.5.2 supplies n_0 with $1/n_0 < \varepsilon$, and then $|1/m - 1/n| \leq 1/n_0 < \varepsilon$ for $m, n \geq n_0$ — but its limit 0 does not belong to X (Fukuda 2026g, slide 61).

Nothing about the sequence has changed relative to $\mathbb{R}$; what changed is the ambient set. Completeness is not a property of a sequence but of a space, and it is destroyed by deleting a single point.

Theorem 6.3.1 ($\mathbb{R}^n$ is complete)**.** $\mathbb{R}$ is a complete metric space, and so is $\mathbb{R}^n$ for every finite n.

Proof. Let (x_k) be Cauchy in $\mathbb{R}$. By Proposition 6.2.2 it is bounded, so by Theorem 4.3.2 it has a convergent subsequence, and by Lemma 6.2.1 the whole sequence converges. The limit is a real number, so $\mathbb{R}$ is complete. The case of $\mathbb{R}^n$ is identical, using the $\mathbb{R}^n$ form of Theorem 4.3.2. ■

The chain of implications locates completeness within the logical structure of the real numbers: the least upper bound property (Assumption 1.5.1) gives monotone convergence (Proposition 4.3.3), which with the monotone subsequence lemma gives Bolzano–Weierstrass, which with Lemma 6.2.1 gives metric completeness. Completeness of $\mathbb{R}$ as a metric space is therefore a theorem, not an axiom, once the order-completeness axiom is in place.

Proposition 6.3.1 (Products)**.** If X and Y are complete metric spaces, then $X \times Y$ with the metric $d\big((x, y), (x', y')\big) = \big(d_X(x, x')^2 + d_Y(y, y')^2\big)^{1/2}$ is complete.

Proof. If (x_k, y_k) is Cauchy in the product, then $d_X(x_j, x_k) \leq d((x_j, y_j), (x_k, y_k))$ shows (x_k) is Cauchy in X, and symmetrically for (y_k). Their limits x^*, y^* exist by completeness, and $d((x_k, y_k), (x^*, y^*)) \to 0$. ■

Proposition 6.3.2 (Complete subsets are exactly the closed ones)**.** Let X be a complete metric space and $A \subseteq X$. Then A is complete if and only if A is closed in X.

Proof. ($\Leftarrow$) Let A be closed and $(x_n) \subseteq A$ Cauchy. By completeness of X it converges to some $x^* \in X$, and by Theorem 4.3.1 $x^* \in A$. So the sequence converges in A.

($\Rightarrow$) Let A be complete and $x^* \in \overline{A}$. By the construction in the proof of Theorem 4.3.1 there is $(x_n) \subseteq A$ with $x_n \to x^*$; this sequence is Cauchy by Proposition 6.2.1, hence converges to a point of A by completeness of A, and by uniqueness of limits (Proposition 4.3.1(i)) that point is x^*. So $\overline{A} \subseteq A$. ■

This is Proposition 8.19 of Fukuda (2026d) (Fukuda 2026g, slide 62). Note the asymmetry of the hypotheses: completeness of the ambient space is used only in one direction, and the equivalence is a statement about subsets of a complete space, not about metric spaces in general.

Corollary 6.3.1. Every vector subspace of $\mathbb{R}^n$ is complete, and hence closed.

Proof. Let $V \subseteq \mathbb{R}^n$ be a subspace with basis $v_1, \ldots, v_k$ and let (x_j) be Cauchy in V. Completing to a basis of $\mathbb{R}^n$ and using Theorem 6.3.1, $x_j \to x^* \in \mathbb{R}^n$. Writing $x_j = \sum_{i=1}^{k} c_j^i v_i$, the coordinate functionals are linear, hence continuous by Proposition 5.2.1(ii), so the coefficient vectors (c_j) converge in $\mathbb{R}^k$ to some c^* and $x^* = \sum_i c^{i*} v_i \in V$. Closedness follows from Proposition 6.3.2. ■

This is Proposition 8.20 of Fukuda (2026d, p. 279). In infinite dimensions the conclusion fails — the subspace of finitely supported sequences is not closed in ℓ^2 — and this is one of the practical differences that Chapter 11 must accommodate: the projection theorem is stated for closed subspaces because in infinite dimensions "subspace" does not imply "closed".

6.3.1 Function spaces

The space in which dynamic programming operates is not $\mathbb{R}^n$. It is a space of functions, and its completeness is what the whole method depends on.

Theorem 6.3.2 (The bounded functions are a Banach space)**.** Let S be any non-empty set and let $\mathcal{B}(S,\mathbb{R})$ be the set of bounded functions $f\colon S \to \mathbb{R}$, equipped with the supremum norm $\|f\|_\infty = \sup_{s\in S} |f(s)|$. Then $\mathcal{B}(S,\mathbb{R})$ is a Banach space.

Proof. That $\|\cdot\|_\infty$ is a norm is Proposition 3.3.3. For completeness, let (f_k) be Cauchy in $\mathcal{B}(S,\mathbb{R})$.

Step 1: pointwise limit. For each fixed $s \in S$, $|f_j(s) - f_k(s)| \le \|f_j - f_k\|_\infty$, so $(f_k(s))_k$ is Cauchy in $\mathbb{R}$ and converges by Theorem 6.3.1. Define $f(s) := \lim_k f_k(s)$.

Step 2: the convergence is uniform. Given $\varepsilon > 0$, take k_0 with $\|f_j - f_k\|_\infty < \varepsilon/2$ for $j, k \ge k_0$. Fix s and $k \ge k_0$, and let $j \to \infty$ in $|f_j(s) - f_k(s)| < \varepsilon/2$; the absolute value is continuous, so $|f(s) - f_k(s)| \le \varepsilon/2$. The bound is uniform in s, so $\|f - f_k\|_\infty \le \varepsilon/2 < \varepsilon$ for every $k \ge k_0$.

Step 3: the limit is bounded. With $\varepsilon = 1$ and the corresponding k_0, $\|f\|_\infty \le \|f - f_{k_0}\|_\infty + \|f_{k_0}\|_\infty \le 1 + \|f_{k_0}\|_\infty < \infty$. Hence $f \in \mathcal{B}(S,\mathbb{R})$ and $f_k \to f$ in the supremum norm. ∎

Remark 6.3.1 (Supremum-norm convergence is uniform convergence)**.** Step 2 is the content of the identification: $\|f - f_k\|_\infty \to 0$ says exactly that $f_k \to f$ uniformly on S, that is, with a single k_0 serving every s simultaneously. Pointwise convergence alone is a strictly weaker statement and does not preserve continuity: $f_k(s) = s^k$ on $[0,1]$ converges pointwise to the discontinuous limit that is 0 on $[0,1)$ and 1 at 1, and $\|f - f_k\|_\infty = 1$ for every k. The supremum norm is chosen for dynamic programming because convergence in it is strong enough to carry continuity, monotonicity and concavity to the limit, as Corollary 6.4.2 exploits.

Proposition 6.3.3 (Bounded continuous functions)**.** Let $S \subseteq \mathbb{R}^n$. The set $\mathcal{C}(S,\mathbb{R}) \cap \mathcal{B}(S,\mathbb{R})$ of bounded continuous functions is a closed subset of $\mathcal{B}(S,\mathbb{R})$, hence a Banach space. If S is compact, every continuous function on S is bounded by Corollary 5.4.1, so $\mathcal{C}(S,\mathbb{R})$ itself is a Banach space in the supremum norm.

Proof. By Proposition 6.3.2 it suffices to show closedness. Let $f_k \to f$ uniformly with each f_k continuous, fix $s_0 \in S$ and $\varepsilon > 0$. Choose k with $\|f - f_k\|_\infty < \varepsilon/3$ and then $\delta > 0$ with $|f_k(s) - f_k(s_0)| < \varepsilon/3$ for $s \in B_\delta(s_0) \cap S$. For such s,

$$|f(s) - f(s_0)| \le |f(s) - f_k(s)| + |f_k(s) - f_k(s_0)| + |f_k(s_0) - f(s_0)| < \varepsilon.$$

∎

6.3.2 Compactness in a function space

Completeness of $\mathcal{C}(K,\mathbb{R})$ delivers limits of Cauchy sequences. It does not deliver convergent subsequences of bounded sequences, and in an infinite-dimensional space the Heine–Borel theorem is false: the closed unit ball of $\mathcal{C}([0,1],\mathbb{R})$ is closed and bounded and not compact. Take $f_k(x) = x^k$, so that $\|f_k\|_\infty = 1$ for every k. Every subsequence still converges pointwise to the function equal to 0 on $[0,1)$ and to 1 at $x = 1$, which is discontinuous; a uniform limit of continuous functions is continuous by Proposition 6.3.3, so no subsequence converges uniformly. This is the same failure that Example 5.4.1 exhibits for the Weierstrass theorem, and the repair is the same: replace boundedness by a condition that controls the whole family at once.

Definition 6.3.2 (Equicontinuity)**.** Let (X,d) be a metric space. A family $\mathcal{F} \subseteq \mathcal{C}(X,\mathbb{R})$ is *equicontinuous* if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$d(x,y) < \delta \quad\implies\quad |f(x) - f(y)| < \varepsilon \qquad \text{for every } f \in \mathcal{F}.$$

It is *uniformly bounded* if $\sup_{f\in\mathcal{F}} \|f\|_\infty < \infty$.

The quantifier order carries the content: δ depends on ε but not on f, whereas continuity of each member separately would allow δ to shrink as f ranges over the family. The family $\{x \mapsto x^k\}_{k\in\mathbb{N}}$ fails the condition: for any $\delta \in (0,1)$ the pair $x = 1$, $y = 1-\delta$ gives $|f_k(x) - f_k(y)| = 1-(1-\delta)^k \to 1$, so no δ works for $\varepsilon = \frac{1}{2}$.

Theorem 6.3.3 (Arzelà–Ascoli)**.** Let $K \subseteq \mathbb{R}^n$ be compact and $\mathcal{F} \subseteq \mathcal{C}(K,\mathbb{R})$. Then $\mathcal{F}$ has compact closure in $(\mathcal{C}(K,\mathbb{R}), \|\cdot\|_\infty)$ if and only if $\mathcal{F}$ is uniformly bounded and equicontinuous.

Proof. Sufficiency. Let (f_k) be a sequence in $\mathcal{F}$. A compact subset of $\mathbb{R}^n$ has a countable dense subset $\{x^1, x^2, \dots\}$: for each m cover K by finitely many balls of radius $1/m$ and collect the centres. The sequence $(f_k(x^1))_k$ is bounded in $\mathbb{R}$, so by Theorem 4.3.2 it has a convergent subsequence (f_k^1); from it extract (f_k^2) converging at x^2, and so on. The diagonal sequence $g_j := f_j^j$ is, from index m onwards, a subsequence of (f_k^m), so $(g_j(x^m))_j$ converges for every m.

Fix $\varepsilon > 0$ and take δ from equicontinuity for $\varepsilon/3$. Compactness of K gives finitely many points $x^{i_1}, \dots, x^{i_p}$ of the dense set with $K \subseteq \bigcup_l B_\delta(x^{i_l})$. Choose J so that $\left|g_j(x^{i_l}) - g_{j'}(x^{i_l})\right| < \varepsilon/3$ for all $j, j' \geq J$ and all $l \leq p$. Given $x \in K$, pick l with $d(x, x^{i_l}) < \delta$; then for $j, j' \geq J$,

$$|g_j(x) - g_{j'}(x)| \leq \left|g_j(x) - g_j(x^{i_l})\right| + \left|g_j(x^{i_l}) - g_{j'}(x^{i_l})\right| + \left|g_{j'}(x^{i_l}) - g_{j'}(x)\right| < \varepsilon,$$

the outer terms by equicontinuity. Hence $\|g_j - g_{j'}\|_\infty \leq \varepsilon$ for $j, j' \geq J$, so (g_j) is Cauchy and converges in $\mathcal{C}(K,\mathbb{R})$ by Proposition 6.3.3. Every sequence in $\mathcal{F}$ therefore has a uniformly convergent subsequence. Passing to the closure costs one approximation: given (h_j) in $\overline{\mathcal{F}}$, choose $f_j \in \mathcal{F}$ with $\|h_j - f_j\|_\infty < 1/j$, extract $f_{j_i} \to f$ in $\mathcal{C}(K,\mathbb{R})$, and note that $\|h_{j_i} - f\|_\infty \leq 1/j_i + \|f_{j_i} - f\|_\infty \to 0$ with $f \in \overline{\mathcal{F}}$. So $\overline{\mathcal{F}}$ is sequentially compact, hence compact by Theorem 4.4.1.

Necessity. A compact set in a metric space is bounded, which gives uniform boundedness, and totally bounded, so for $\varepsilon > 0$ there are $h_1, \dots, h_N \in \mathcal{C}(K,\mathbb{R})$ with every $f \in \mathcal{F}$ within $\varepsilon/3$ of some h_i in supremum norm. Each h_i is uniformly continuous on the compact K by Theorem 5.2.2; let δ be the smallest of the finitely many moduli for $\varepsilon/3$. For $d(x,y) < \delta$ and f within $\varepsilon/3$ of h_i,

$$|f(x) - f(y)| \leq |f(x) - h_i(x)| + |h_i(x) - h_i(y)| + |h_i(y) - f(y)| < \varepsilon.$$ ∎

Corollary 6.3.2 (Uniformly Lipschitz families)**.** Let $K \subseteq \mathbb{R}^n$ be compact and let $\mathcal{F} \subseteq \mathcal{C}(K,\mathbb{R})$ satisfy $|f(x) - f(y)| \leq L\,\|x - y\|$ for every $f \in \mathcal{F}$ and all $x, y \in K$, with a single constant L, and $|f(x_0)| \leq M$ for some fixed $x_0 \in K$ and every f. Then $\mathcal{F}$ has compact closure in $\mathcal{C}(K,\mathbb{R})$.

Proof. Equicontinuity holds with $\delta = \varepsilon/L$. Uniform boundedness follows from $|f(x)| \leq |f(x_0)| + L\,\|x - x_0\| \leq M + L \operatorname{diam} K$. Apply Theorem 6.3.3. ∎

Remark 6.3.2 (Where this is used). Corollary 6.3.2 is the compactness input to existence proofs in continuous time. In Chapter 16 the admissible state trajectories of $\dot{x} = f(x,u)$ with controls taking values in a compact set and f bounded on the relevant region are all Lipschitz on $[0,T]$ with the same constant $\sup\|f\|$ and share the initial value $x(0) = x_0$. The family of admissible trajectories is therefore relatively compact in $\mathcal{C}([0,T],\mathbb{R}^n)$, and a maximising sequence has a uniformly convergent subsequence; Remark 16.3.2 states the conclusion this yields and the extra convexity hypothesis needed to keep the limit admissible.

The theorem is also what makes a family of value functions or policy functions generated by varying a parameter relatively compact, which is the standard route to continuity of an equilibrium in a parameter when no explicit formula is available. Boundedness alone never suffices, as $\sin(k\cdot)$ shows.

6.4 The Contraction Mapping Theorem

Definition 6.4.1 (Contraction)**.** Let (X,d) be a metric space. A map $T\colon X \to X$ is a *contraction* with *modulus* β if there is $\beta \in [0,1)$ with

$$d(Tx, Ty) \leq \beta\, d(x,y) \qquad \text{for all } x, y \in X.$$

A point $x^* \in X$ with $Tx^* = x^*$ is a *fixed point* of T.

Two features of the definition deserve emphasis. The modulus must be strictly less than 1 and must be uniform over all pairs x, y. The weaker condition $d(Tx, Ty) < d(x, y)$ for $x \neq y$ — a *contractive* map — does not suffice: on $X = [1, \infty)$ the map $Tx = x + 1/x$ satisfies it, is a self-map of a complete space, and has no fixed point, because the implicit modulus approaches 1 as $x \to \infty$.

Theorem 6.4.1 (Banach contraction mapping theorem)**.** Let (X, d) be a non-empty complete metric space and let $T\colon X \to X$ be a contraction with modulus $\beta \in [0, 1)$. Then T has exactly one fixed point $x^* \in X$. Moreover, for any $x_0 \in X$ the sequence of iterates $x_n := Tx_{n-1} = T^n x_0$ converges to x^*, and

$$d(x_n, x^*) \leq \frac{\beta^n}{1-\beta}\, d(x_0, Tx_0) \qquad \text{for every } n \in \mathbb{N}_0. \tag{6.1}$$

Proof. Step 1: the iterates are Cauchy. Fix $x_0 \in X$ and define $x_n = Tx_{n-1}$. Applying the contraction property k times, $d(x_k, x_{k+1}) = d(Tx_{k-1}, Tx_k) \leq \beta\, d(x_{k-1}, x_k) \leq \cdots \leq \beta^k d(x_0, x_1)$. For $m > n$, the triangle inequality applied $m - n - 1$ times gives

$$d(x_n, x_m) \leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) \leq \sum_{k=n}^{m-1} \beta^k d(x_0, x_1) = \beta^n \frac{1-\beta^{m-n}}{1-\beta}\, d(x_0, x_1) \leq \frac{\beta^n}{1-\beta}\, d(x_0, x_1),$$

where the geometric sum is finite because $\beta < 1$. Given $\varepsilon > 0$, since $\beta^n \to 0$ there is n_0 with $\beta^n d(x_0, x_1)/(1-\beta) < \varepsilon$ for $n \geq n_0$, and then $d(x_n, x_m) < \varepsilon$ for all $m, n \geq n_0$. So (x_n) is Cauchy.

Step 2: the limit exists. X is complete, so $x^* := \lim_n x_n$ exists in X.

Step 3: the limit is a fixed point. T is continuous: given $\varepsilon > 0$, take $\delta = \varepsilon$ (if $\beta = 0$, any δ); then $d(x, y) < \delta$ implies $d(Tx, Ty) \leq \beta d(x, y) < \varepsilon$. Hence, by Proposition 5.2.3,

$$Tx^* = T\Big(\lim_{n\to\infty} x_n\Big) = \lim_{n\to\infty} Tx_n = \lim_{n\to\infty} x_{n+1} = x^*.$$

Step 4: uniqueness. If $Tx^* = x^*$ and $Ty^* = y^*$, then $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta\, d(x^*, y^*)$. If $d(x^*, y^*) > 0$ this yields $1 \leq \beta$, contradicting $\beta < 1$. Hence $d(x^*, y^*) = 0$ and $x^* = y^*$.

Step 5: the error bound. Let $m \to \infty$ in the Step 1 estimate $d(x_n, x_m) \leq \beta^n (1-\beta)^{-1} d(x_0, x_1)$. The metric is continuous in each argument — an immediate consequence of the triangle inequality — so the left-hand side tends to $d(x_n, x^*)$, giving (6.1) with $x_1 = Tx_0$. ■

This is Theorem 8.3 of Fukuda (2026d, p. 284), proved over (Fukuda 2026g, slides 64–72); the proof above follows those steps, with the geometric sum and the passage to the limit in Step 5 written out.

Remark 6.4.1 (What each hypothesis buys)**.** Completeness is used once, in Step 2, and only to produce the limit; it is responsible for *existence*. The strict inequality $\beta < 1$ is used twice: in Step 1 to sum the geometric series, and in Step 4, where it is responsible for *uniqueness*. The self-map property $T(X) \subseteq X$ is used implicitly throughout, in defining the iterates. Dropping any one of the three destroys a different part of the conclusion: on the complete space $\mathbb{R}$ the isometry $Tx = x + 1$ ($\beta = 1$) has no fixed point; on the incomplete space $(0, 1]$ the contraction $Tx = x/2$ has none either, though its iterates are Cauchy.

Remark 6.4.2 (The error bound is what makes the theorem computational)**.** Inequality (6.1) is an *a priori* bound: it is computable from the first iteration alone, before the iteration is run. It converts the theorem from an existence statement into an algorithm with a stopping rule, since $n \geq \log\big(\varepsilon(1-\beta)/d(x_0, Tx_0)\big)/\log\beta$ guarantees accuracy ε. Convergence is geometric at rate β, so the discount factor of a dynamic program is simultaneously an economic parameter and the convergence rate of the algorithm that solves it: a model with $\beta = 0.99$ needs roughly $\log(0.5)/\log(0.99) \approx 69$ times as many iterations per digit as one with $\beta = 0.5$.

6.4.1 Three refinements

Corollary 6.4.1 (Iterates of a contraction). Let X be a non-empty complete metric space and $T: X \to X$ a map such that T^N is a contraction for some $N \in \mathbb{N}$. Then T has a unique fixed point.

Proof. By Theorem 6.4.1, T^N has a unique fixed point x^*. Then $T^N(Tx^*) = T(T^N x^*) = Tx^*$, so Tx^* is also a fixed point of T^N; by uniqueness $Tx^* = x^*$. Conversely any fixed point of T is a fixed point of T^N, hence equals x^*. ■

This is Exercise 8.30 of Fukuda (2026d, p. 290). It matters because operators that are not contractions can have contracting powers: in a dynamic program with periodic structure, or in a Markov chain whose transition matrix has zero entries, the one-step operator may fail the modulus condition while the N-step operator satisfies it.

Corollary 6.4.2 (Properties inherited by the fixed point). Let X be a complete metric space, $T: X \to X$ a contraction with fixed point x^*, and let $Y \subseteq X$ be non-empty and closed with $T(Y) \subseteq Y$. Then $x^* \in T(Y) \subseteq Y$.

Proof. Y is closed in a complete space, hence complete by Proposition 6.3.2, and T restricted to Y is a contraction of Y into itself. By Theorem 6.4.1 applied to $T|_Y$ there is a fixed point in Y; by uniqueness in X it equals x^*. Finally $x^* = Tx^* \in T(Y)$. ■

Remark 6.4.3 (How to prove that a value function is concave). Corollary 6.4.2 is the standard route to qualitative properties of an object defined only as a fixed point. Let P be a property, and let $Y = \{x \in X : x \text{ has } P\}$. If Y is closed and T preserves P, in the sense that $T(Y) \subseteq Y$, then the fixed point has P. In Chapter 15 this is applied three times over, with $X = \mathcal{C}(S, \mathbb{R})$ and P successively "increasing", "concave" and "strictly concave". The closedness requirement is why the supremum norm is the right one: the pointwise limit of increasing functions is increasing and of concave functions is concave, and uniform convergence implies pointwise convergence, so both sets are closed in $\mathcal{B}(S, \mathbb{R})$. Strict concavity is *not* a closed property — $f_k(s) = -s^2/k$ is strictly concave with uniform limit 0 — and must be obtained by a separate argument (Fukuda 2026d, Remark 8.11, p. 290).

Proposition 6.4.1 (Comparison of fixed points). Let X be a complete metric space and let $S, T: X \to X$ be contractions with a common modulus $\beta \in [0, 1)$ and fixed points x_S, x_T. If $d(Sx, Tx) \leq a$ for all $x \in X$, then

$$d(x_S, x_T) \leq \frac{a}{1 - \beta}.$$

Proof. $d(x_S, x_T) = d(Sx_S, Tx_T) \leq d(Sx_S, Tx_S) + d(Tx_S, Tx_T) \leq a + \beta\, d(x_S, x_T)$. Rearranging and dividing by $1 - \beta > 0$ gives the bound. ■

This is Exercise 8.29 of Fukuda (2026d, p. 290). It is the stability result behind numerical dynamic programming: if the true Bellman operator and its discretised approximation differ by at most a uniformly, the computed value function differs from the true one by at most $a/(1-\beta)$. The factor $(1-\beta)^{-1}$ is an amplification, and it is large exactly when the discount factor is close to 1.

6.5 Blackwell's Sufficient Conditions

Verifying the contraction inequality directly for a Bellman operator requires comparing two suprema, which is awkward. Blackwell's conditions replace that comparison with two structural properties that can be read off the problem.

For $f, g \in \mathcal{B}(S, \mathbb{R})$ write $f \leq g$ to mean $f(s) \leq g(s)$ for every $s \in S$, and for $\alpha \in \mathbb{R}$ let $f + \alpha$ denote the function $s \mapsto f(s) + \alpha$.

Theorem 6.5.1 (Blackwell). Let S be a non-empty set and let $T: \mathcal{B}(S, \mathbb{R}) \to \mathcal{B}(S, \mathbb{R})$ satisfy

(i) *monotonicity*: $f \leq g$ implies $Tf \leq Tg$; and

(ii) *discounting*: there is $\beta \in [0, 1)$ such that $T(f + \alpha)(s) \leq (Tf)(s) + \beta\alpha$ for every $f \in \mathcal{B}(S, \mathbb{R})$, every $\alpha \geq 0$ and every $s \in S$.

Then T is a contraction with modulus β on $(\mathcal{B}(S,\mathbb{R}), \|\cdot\|_\infty)$, and therefore has a unique fixed point, reached from any starting point by iteration.

Proof. Let $f, g \in \mathcal{B}(S,\mathbb{R})$. For every s, $g(s) - f(s) \leq |g(s) - f(s)| \leq \|f - g\|_\infty$, so

$$g \leq f + \|f - g\|_\infty .$$

Applying monotonicity and then discounting with $\alpha = \|f - g\|_\infty \geq 0$,

$$Tg \;\leq\; T\big(f + \|f - g\|_\infty\big) \;\leq\; Tf + \beta\,\|f - g\|_\infty .$$

Interchanging the roles of f and g gives $Tf \leq Tg + \beta\,\|f - g\|_\infty$. The two inequalities together say that $|(Tf)(s) - (Tg)(s)| \leq \beta\,\|f - g\|_\infty$ for every $s \in S$, and taking the supremum over s on the left yields $\|Tf - Tg\|_\infty \leq \beta\,\|f - g\|_\infty$. Since $\mathcal{B}(S,\mathbb{R})$ is complete by Theorem 6.3.2, Theorem 6.4.1 applies. ■

This is Proposition 8.20 of the slides and Proposition 8.21 of Fukuda (2026d, p. 289); the proof is the one sketched on (Fukuda 2026g, slide 76).

Remark 6.5.1 (The conditions are sufficient, not necessary). A contraction on $\mathcal{B}(S,\mathbb{R})$ need not be monotone: on $\mathcal{B}(S,\mathbb{R})$ with S a single point, $Tf = -\frac{1}{2}f$ is a contraction with modulus $\frac{1}{2}$ and reverses order. What Blackwell's conditions provide is a criterion adapted to operators that arise from maximisation, for which both properties are transparent. Monotonicity of a Bellman operator says that a pointwise higher continuation value cannot lower today's optimised value, which holds because the same feasible choices remain available. Discounting says that adding a constant α to every continuation value raises today's value by at most $\beta\alpha$, which holds because the constant enters only through the discounted continuation term. Neither statement requires knowing the maximiser.

Example 6.5.1 (Blackwell in $\mathbb{R}^n$). Equip $\mathbb{R}^n$ with $\|x\|_\infty = \max_i |x_i|$ and let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ satisfy monotonicity ($x \leq y$ componentwise implies $Tx \leq Ty$) and discounting ($T(x + \alpha\mathbf{1}) \leq Tx + \beta\alpha\mathbf{1}$ for $\alpha \geq 0$, where $\mathbf{1} = (1, \dots, 1)^\top$). The proof of Theorem 6.5.1 applies verbatim with $S = \{1, \dots, n\}$, since $\mathcal{B}(\{1, \dots, n\}, \mathbb{R}) = \mathbb{R}^n$ with the supremum norm. This is Exercise 8.26 of Fukuda (2026d, p. 290), and it is the finite-state case: a Markov decision problem with n states, where the Bellman operator is a map of $\mathbb{R}^n$ into itself and value iteration is a matrix–vector recursion.

6.6 Economic Applications

6.6.1 The Bellman operator

Environment. A planner observes a state $s \in S$, chooses an action a from a feasible set $\Gamma(s) \subseteq A$, receives a period return $r(s,a)$, and moves to the state $g(s,a)$ next period. Future returns are discounted at $\beta \in (0,1)$. The problem is

$$\sup_{(a_t)_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t r(s_t, a_t) \qquad \text{subject to} \quad a_t \in \Gamma(s_t), \quad s_{t+1} = g(s_t, a_t), \quad s_0 \text{ given.} \tag{6.2}$$

The operator. Define T on $\mathcal{B}(S,\mathbb{R})$ by

$$(Tv)(s) := \sup_{a \in \Gamma(s)} \Big\{ r(s,a) + \beta\, v\big(g(s,a)\big) \Big\}. \tag{6.3}$$

Proposition 6.6.1 (The Bellman operator is a contraction). Suppose r is bounded, $\Gamma(s) \neq \emptyset$ for every s, and $\beta \in [0,1)$. Then T maps $\mathcal{B}(S,\mathbb{R})$ into itself and is a contraction with modulus β; consequently (6.3) has a unique bounded solution v^*, and $T^n v_0 \to v^*$ uniformly from any bounded v_0.

Proof. Self-map. If $|r| \leq M_r$ and $\|v\|_\infty \leq M_v$, then the expression in braces lies in $[-M_r - \beta M_v,\, M_r + \beta M_v]$ for every admissible a, so its supremum over the non-empty set $\Gamma(s)$ is finite and bounded by $M_r + \beta M_v$ uniformly in s. Hence $Tv \in \mathcal{B}(S,\mathbb{R})$.

Monotonicity. If $v \le w$ then for every s and every $a \in \Gamma(s)$, $r(s,a)+\beta v(g(s,a)) \le r(s,a)+\beta w(g(s,a))$; taking suprema over the same set $\Gamma(s)$ preserves the inequality, so $Tv \le Tw$.

Discounting. For $\alpha \ge 0$,

$$\big(T(v+\alpha)\big)(s) = \sup_{a\in\Gamma(s)} \big\{r(s,a)+\beta v(g(s,a))+\beta\alpha\big\} = (Tv)(s)+\beta\alpha,$$

since an additive constant passes through a supremum. The required inequality holds with equality.

Apply Theorem 6.5.1. ∎

Scope of the fixed-point argument. The theorem establishes that the functional equation has exactly one bounded solution and that value iteration converges to it geometrically, with the error after n steps bounded by (6.1). It does not by itself establish that v^* is the value of the sequence problem (6.2) — that equivalence requires a separate argument, given in Chapter 15 — nor that an optimal policy exists, which requires the supremum in (6.3) to be attained and hence the Weierstrass theorem (Theorem 5.4.1) applied to $\Gamma(s)$. Boundedness of r is restrictive: it excludes logarithmic utility on an unbounded state space, for which a weighted supremum norm is used instead (Stokey et al. 1989, Sec. 4.3).

6.6.2 Asset pricing

In the Lucas tree economy the price of a claim to a dividend stream satisfies

$$p(s) = \beta\,\mathbb{E}\big[\,m(s,s')\{d(s')+p(s')\} \mid s\,\big], \tag{6.4}$$

where s is the current state, d the dividend and m a positive stochastic discount factor. Define $(Tp)(s)$ as the right-hand side. If $\beta\,\sup_s \mathbb{E}[m(s,s') \mid s] := \bar\beta < 1$ and d is bounded, then for bounded p,q,

$$|(Tp)(s)-(Tq)(s)| = \beta\,\big|\mathbb{E}[m(s,s')\{p(s')-q(s')\} \mid s]\big| \le \bar\beta\,\|p-q\|_\infty\,,$$

using positivity of m and the conditional Jensen inequality of Proposition 14.4.2. So T is a contraction on $\mathcal{B}(S,\mathbb{R})$ and (6.4) has a unique bounded price function. Uniqueness here is an economic statement: it rules out rational bubbles among bounded price functions. Bubbles reappear precisely when the boundedness restriction is dropped, since the homogeneous equation $b(s) = \beta\mathbb{E}[mb(s') \mid s]$ then admits non-zero solutions (Lucas 1978).

6.6.3 The steady-state Kalman filter

Environment. A hidden state follows $\theta_{t+1} = \rho\,\theta_t + \eta_{t+1}$ with $\rho \in (-1,1)$ and $\eta_{t+1} \sim N(0,\tau_2)$, and is observed each period through a signal $y_t = \theta_t + \varepsilon_t$ with $\varepsilon_t \sim N(0,\tau_1^{-1})$, all shocks independent. Following the source exercise, τ_2 is the *variance* of the state innovation and τ_1 the *precision* of the signal. An observer forms Bayesian beliefs about θ_t from the signal history; let x_t denote the variance of that belief *before* seeing y_t.

Recursion. Normal–normal updating (Theorem 14.6.1) adds precisions: after observing y_t the posterior variance is $\big(x_t^{-1}+\tau_1\big)^{-1}$. The transition then scales the variance by ρ^2 and adds the innovation variance τ_2, so $x_{t+1} = T(x_t)$ with

$$T(x) = \rho^2\big(x^{-1}+\tau_1\big)^{-1} + \tau_2 = \frac{\rho^2 x}{1+\tau_1 x} + \tau_2, \qquad x \in \mathbb{R}_{++}. \tag{6.5}$$

This is the scalar Riccati equation of the Kalman filter.

Proposition 6.6.2 (The Riccati operator is a contraction). Let $\rho \in (-1,1)$ and $\tau_1,\tau_2 > 0$. Then T in (6.5) maps $\mathbb{R}_{++}$ into itself and is a contraction with modulus $\rho^2 \in [0,1)$. Consequently the conditional variance converges, from any initial belief, to the unique fixed point

$$x^* = \frac{(\rho^2+\tau_1\tau_2-1)+\sqrt{(1-\rho^2-\tau_1\tau_2)^2+4\tau_1\tau_2}}{2\tau_1}\,, \tag{6.6}$$

> the unique positive root of $\tau_1 x^2 + (1-\rho^2-\tau_1\tau_2)x - \tau_2 = 0$.

Proof. *Self-map.* For $x > 0$ both terms of $\rho^2 x/(1+\tau_1 x) + \tau_2$ are non-negative and the second is strictly positive, so $T(x) \geq \tau_2 > 0$.

Contraction. For $x, y > 0$,

$$T(x) - T(y) = \rho^2 \left(\frac{x}{1+\tau_1 x} - \frac{y}{1+\tau_1 y} \right) = \frac{\rho^2(x-y)}{(1+\tau_1 x)(1+\tau_1 y)},$$

after clearing the common denominator and cancelling the terms in $\tau_1 xy$. Since $\tau_1 > 0$ and $x, y > 0$, the denominator exceeds 1, so $|T(x) - T(y)| \leq \rho^2 |x-y|$ with $\rho^2 < 1$. The set $[\tau_2, \rho^2\tau_1^{-1} + \tau_2]$ is closed, non-empty, and invariant under T, hence complete by Proposition 6.3.2; Theorem 6.4.1 applies on it, and Corollary 6.4.2 places the fixed point there.

Fixed point. Setting $x = \rho^2 x/(1+\tau_1 x) + \tau_2$ and multiplying by $1 + \tau_1 x > 0$ gives $x + \tau_1 x^2 = \rho^2 x + \tau_2 + \tau_1\tau_2 x$, that is $\tau_1 x^2 + (1-\rho^2-\tau_1\tau_2)x - \tau_2 = 0$. The product of the two roots is $-\tau_2/\tau_1 < 0$, so exactly one root is positive, and it is (6.6). As a check, $\rho = 0$ gives $\tau_1 x^2 + (1-\tau_1\tau_2)x - \tau_2 = (\tau_1 x + 1)(x - \tau_2) = 0$, hence $x^* = \tau_2$: with no persistence, yesterday's signal is worthless and the conditional variance is the innovation variance. ∎

Economic content. First, uncertainty neither vanishes nor explodes. However long the observer has been learning, the conditional variance settles at $x^* > 0$: new innovations arrive at exactly the rate at which signals resolve old ones. Permanent learning does not produce perfect information, which is why models of learning about a persistent fundamental sustain non-degenerate belief dispersion indefinitely.

Second, the rate of convergence is ρ^2, the square of the persistence. Beliefs about a near-random-walk state ($\rho \to 1$) converge to their steady state arbitrarily slowly, so the transitional dynamics of learning are economically relevant over long horizons precisely when the fundamental is persistent.

Third, comparative statics are read off (6.6) through the quadratic. Write $F(x) := \tau_1 x^2 + (1-\rho^2-\tau_1\tau_2)x - \tau_2$, so that $F(x^*) = 0$ and $F'(x^*) > 0$ at the larger root. Since $\partial F/\partial \rho^2 = -x^* < 0$, the implicit function theorem (Theorem 8.6.1) gives $\partial x^*/\partial \rho^2 = x^*/F'(x^*) > 0$: a more persistent state is harder to track, not easier. Likewise $\partial F/\partial \tau_1 = x^*(x^* - \tau_2)$, which is strictly positive because $x^* = \rho^2 x^*/(1+\tau_1 x^*) + \tau_2 > \tau_2$; hence $\partial x^*/\partial \tau_1 < 0$ and a more precise signal lowers steady-state uncertainty, without qualification. This is Exercise 8.27 of Fukuda (2026d, p. 290), with the interpretation supplied.

6.6.4 Input–output systems revisited

Example 2.8.1 solved $x = Ax + d$ by a Neumann series under $\rho(A) < 1$. Theorem 6.4.1 gives the same conclusion with a different argument and a computable error bound: if $\|A\| < 1$ in some operator norm, then $Tx = Ax + d$ satisfies $\|Tx - Ty\| = \|A(x-y)\| \leq \|A\| \, \|x-y\|$, so T is a contraction on the complete space $\mathbb{R}^n$ and the iteration $x_{k+1} = Ax_k + d$ converges to the unique solution from any starting point, with $\|x_k - x^*\| \leq \|A\|^k (1 - \|A\|)^{-1} \|x_1 - x_0\|$. The two hypotheses are not identical: $\rho(A) \leq \|A\|$ always, so the spectral condition is weaker, and by Corollary 6.4.1 the gap is exactly accounted for by passing to a power of T, since $\rho(A) < 1$ implies $\|A^N\| < 1$ for some N.

6.7 Chapter Summary

A Cauchy sequence is one whose terms cluster without reference to any limit; every convergent sequence is Cauchy and every Cauchy sequence is bounded, but the converse of the first implication holds only in complete spaces. $\mathbb{R}^n$ is complete, as a consequence of the least upper bound axiom via Bolzano–Weierstrass; a subset of a complete space is complete exactly when it is closed (Proposition 6.3.2); the bounded functions on any set form a Banach space under the supremum norm (Theorem 6.3.2), and the bounded continuous functions form a closed subspace of it.

A contraction on a non-empty complete metric space has exactly one fixed point, reached by iteration from any starting point at geometric rate β with the computable error bound (6.1). Completeness delivers existence and $\beta < 1$ delivers uniqueness; each is indispensable. Three refinements extend the reach of the theorem: a map with a contracting power has a unique fixed point; a closed invariant set contains the fixed point, which is how qualitative properties of value functions are established; and fixed points of uniformly close contractions are within $a/(1-\beta)$ of each other, which bounds the error of numerical approximation. Blackwell's monotonicity and discounting conditions are sufficient for the contraction property and are verifiable directly on operators defined by maximisation, which is what makes them the standard tool for Bellman operators, asset-pricing operators and Bayesian updating recursions.

> **Exercise 6.7.1.** Let X be a complete normed space and $T\colon X \to X$ a contraction with modulus β. Show that for every $y \in X$ there is a unique $x \in X$ with $Tx - x = y$, and that the solution map $y \mapsto x$ is Lipschitz with constant $(1-\beta)^{-1}$. Interpret the second statement as a stability property of the linear system $x = Ax + d$ with respect to perturbations of d (Fukuda 2026d, Ex. 8.31).

> **Exercise 6.7.2.** Show that $Tx = x + 1/x$ on $[1, \infty)$ satisfies $|Tx - Ty| < |x - y|$ for all $x \neq y$ and has no fixed point, and that $Tx = \sqrt{1 + x^2}$ on $\mathbb{R}$ does the same. Identify the supremum of $|Tx - Ty| \,/\, |x - y|$ in each case and explain why Theorem 6.4.1 does not apply.

> **Exercise 6.7.3.** Let $S = \{1, \ldots, n\}$ and let Q be an $n \times n$ row-stochastic matrix. Show that $Tv = r + \beta Q v$ is a contraction on $(\mathbb{R}^n, \|\cdot\|_\infty)$ with modulus β, that its fixed point is $v^* = (I - \beta Q)^{-1} r$, and that $I - \beta Q$ is invertible for every $\beta \in [0, 1)$. Verify Blackwell's two conditions for T directly, and explain which property of Q delivers each.

Chapter 7

Differential Calculus in One Variable

7.1 Motivation: The Two Firm Problems

Two problems organise this chapter (Fukuda 2026a, slide 3). A price-taking firm solves

$$\max_{q\in[0,\infty)} \; pq - c(q), \tag{7.1}$$

and a price-setting firm facing inverse demand P solves

$$\max_{q\in[0,\infty)} \; P(q)q - c(q). \tag{7.2}$$

Assumption 7.1.1 (Standing assumptions on the firm)**.** The cost function $c\colon [0,\infty) \to \mathbb{R}$ is differentiable, strictly increasing and strictly convex, with $c(0) = c'(0) = 0$ and $\lim_{q\to\infty} c(q) = \lim_{q\to\infty} c'(q) = \infty$. The revenue function $q \mapsto P(q)q$ is concave and differentiable; the leading case is $P(q) = a - bq$ with $a, b > 0$.

Each assumption is used at an identifiable point. Differentiability makes the first-order condition available; strict convexity of c and concavity of revenue make the first-order condition sufficient (Theorem 7.2.3) and the solution unique; $c'(0) = 0$ with $c'(q) \to \infty$ places the optimum in the interior of $[0,\infty)$, where Theorem 7.2.1 applies; and $c(0) = 0$ makes the value of the problem the profit rather than profit net of a fixed cost.

7.2 The Derivative

Definition 7.2.1 (Derivative)**.** Let $f\colon (a,b) \to \mathbb{R}$ with $-\infty \le a < b \le \infty$ and let $x_0 \in (a,b)$. The function f is *differentiable at* x_0 if

$$f'(x_0) := \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

exists and is finite; the limit is the *derivative* of f at x_0, also written $Df(x_0)$ or $\frac{\mathrm{d}f}{\mathrm{d}x}(x_0)$. It is the slope of the tangent line to the graph of f at $\big(x_0, f(x_0)\big)$.

This is Definition 3.1 of Fukuda (2026d) (Fukuda 2026a, slide 4). That x_0 is required to be *interior* to the domain is part of the definition: the two-sided limit must exist, so the difference quotient must be defined for h of both signs.

Proposition 7.2.1 (Equivalent formulation and continuity)**.** Let $f\colon (a,b) \to \mathbb{R}$ and $x_0 \in (a,b)$. Then f is differentiable at x_0 with derivative L if and only if

$$f(x_0+h) = f(x_0) + Lh + r(h), \qquad \text{where} \quad \lim_{h\to 0} \frac{r(h)}{h} = 0. \tag{7.3}$$

In particular, differentiability at x_0 implies continuity at x_0.

Proof. Define $r(h) := f(x_0 + h) - f(x_0) - Lh$. Then $r(h)/h$ equals the difference quotient minus L, so $r(h)/h \to 0$ if and only if the difference quotient tends to L. For continuity, (7.3) gives $f(x_0+h)-f(x_0) = Lh + r(h) \to 0$ as $h \to 0$, since $r(h) = h \cdot (r(h)/h) \to 0 \cdot 0$. ■

Formulation (7.3) is the one that generalises: it says that f is well approximated near x_0 by an affine function, with an error small relative to h. In one variable the approximating linear map is multiplication by a number; in $\mathbb{R}^n$ it is a linear map, and Definition 8.3.1 takes (7.3) verbatim as the definition. The converse of the last sentence fails: $f(x) = |x|$ is continuous at 0 and not differentiable there, since the difference quotient equals $\operatorname{sgn}(h)$.

7.2.1 The first-order condition

Theorem 7.2.1 (Fermat)**.** Let $A \subseteq \mathbb{R}$, let x_0 be an interior point of A at which $f\colon A \to \mathbb{R}$ is differentiable, and suppose x_0 is a local maximiser or a local minimiser of f on A. Then $f'(x_0) = 0$.

Proof. Suppose x_0 is a local maximiser, so there is $\delta > 0$ with $B_\delta(x_0) \subseteq A$ and $f(x) \leq f(x_0)$ for $x \in B_\delta(x_0)$. For $0 < h < \delta$,

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0,$$

because the numerator is non-positive and the denominator positive; letting $h \downarrow 0$ gives $f'(x_0) \leq 0$. For $-\delta < h < 0$ the denominator is negative and the quotient is non-negative; letting $h \uparrow 0$ gives $f'(x_0) \geq 0$. Since f is differentiable at x_0 the two one-sided limits coincide with $f'(x_0)$, so $f'(x_0) = 0$. The minimiser case follows by applying this to $-f$. ■

A point with $f'(x_0) = 0$ is a *stationary point*, or critical point, of f (Fukuda 2026a, slide 5).

Remark 7.2.1 (Three ways the conclusion fails)**.** Each hypothesis of Theorem 7.2.1 is indispensable, and each failure has an economic counterpart.

(i) *Not interior.* $f(x) = x$ on $[0, 1]$ has its maximum at $x = 1$ with $f'(1) = 1 \neq 0$. The proof used h of both signs, and at a boundary point only one sign is admissible; the conclusion weakens to the inequality $f'(1) \geq 0$. In economics this is the corner solution, and the corresponding first-order conditions are the Kuhn–Tucker inequalities of Chapter 12.

(ii) *Not differentiable.* $f(x) = -|x|$ has a global maximum at $x = 0$ where no derivative exists. Non-differentiable kinks arise in economics from piecewise-linear tax schedules and from Leontief technology.

(iii) *The condition is necessary, not sufficient.* $f(x) = x^3$ has $f'(0) = 0$ and no local extremum at 0. Sufficiency requires either a second-order condition or a global structural hypothesis such as concavity.

7.2.2 Concavity and sufficiency

Definition 7.2.2 (Concave and convex functions of one variable)**.** A function $f\colon I \to \mathbb{R}$ on an interval $I \subseteq \mathbb{R}$ is *concave* if

$$f\big(\lambda x + (1-\lambda)y\big) \geq \lambda f(x) + (1-\lambda)f(y) \qquad \text{for all } x, y \in I,\ \lambda \in [0,1],$$

strictly concave if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$, and *convex* (respectively strictly convex) if $-f$ is concave (respectively strictly concave). The general theory, in $\mathbb{R}^n$, is Chapter 9.

Theorem 7.2.2 (Characterisation of differentiable concavity)**.** Let $f\colon (a, b) \to \mathbb{R}$ be differentiable. The following are equivalent.

(i) f is concave on (a, b).

> (ii) $f(x) \le f(x_0) + f'(x_0)(x - x_0)$ for all $x, x_0 \in (a, b)$.
>
> (iii) f' is non-increasing on (a, b).
>
> If moreover f is twice differentiable, each is equivalent to $f'' \le 0$ on (a, b). The strict versions correspond to strict inequality in (ii) for $x \ne x_0$ and to strictly decreasing f'; $f'' < 0$ is sufficient but not necessary for strict concavity, as $f(x) = -x^4$ shows at $x = 0$.

Proof. (i)⇒(ii). For $\lambda \in (0, 1]$, concavity gives $f\big(x_0 + \lambda(x - x_0)\big) \ge \lambda f(x) + (1 - \lambda) f(x_0)$, hence

$$\frac{f\big(x_0 + \lambda(x - x_0)\big) - f(x_0)}{\lambda} \ge f(x) - f(x_0).$$

Letting $\lambda \downarrow 0$, the left side tends to $f'(x_0)(x - x_0)$ by Definition 7.2.1 applied with $h = \lambda(x - x_0)$.

(ii)⇒(iii). Apply (ii) twice, at x_0 evaluated at x and at x evaluated at x_0, and add: $0 \le \big(f'(x_0) - f'(x)\big)(x - x_0)$. For $x > x_0$ this forces $f'(x) \le f'(x_0)$.

(iii)⇒(i). Let $x < y$ and $z = \lambda x + (1 - \lambda) y$ with $\lambda \in (0, 1)$. By the mean value theorem (Theorem 7.2.5) there are $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with $f(z) - f(x) = f'(\xi_1)(z - x)$ and $f(y) - f(z) = f'(\xi_2)(y - z)$. Since $\xi_1 < \xi_2$, (iii) gives $f'(\xi_1) \ge f'(\xi_2)$, so

$$\frac{f(z) - f(x)}{z - x} \ge \frac{f(y) - f(z)}{y - z}.$$

Substituting $z - x = (1 - \lambda)(y - x)$ and $y - z = \lambda(y - x)$ and rearranging gives $f(z) \ge \lambda f(x) + (1 - \lambda) f(y)$.

The twice-differentiable statement is (iii) together with the fact that a differentiable function is non-increasing exactly when its derivative is non-positive, itself a consequence of Theorem 7.2.5. ∎

> **Theorem 7.2.3** (The first-order condition is sufficient under concavity)**.** Let $f\colon (a, b) \to \mathbb{R}$ be differentiable and concave. Then $x_0 \in (a, b)$ is a global maximiser of f on (a, b) if and only if $f'(x_0) = 0$. If f is convex, the same statement holds with "maximiser" replaced by "minimiser". If f is strictly concave, the maximiser is unique when it exists.

Proof. Necessity is Theorem 7.2.1. For sufficiency, suppose $f'(x_0) = 0$. By Theorem 7.2.2(ii), $f(x) \le f(x_0) + f'(x_0)(x - x_0) = f(x_0)$ for every $x \in (a, b)$. Uniqueness under strict concavity: if $x_0 \ne x_1$ were both maximisers with common value M, then $f\big(\tfrac{1}{2}x_0 + \tfrac{1}{2}x_1\big) > \tfrac{1}{2}M + \tfrac{1}{2}M = M$, contradicting maximality. ∎

This is the content of (Fukuda 2026a, slides 7–8). The geometric statement — the tangent line at any point lies above the graph, so a horizontal tangent lies above the whole graph — is the proof, once Theorem 7.2.2 makes "lies above" precise.

7.2.3 Mean value theorems

> **Theorem 7.2.4** (Rolle)**.** Let $f\colon [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b), with $f(a) = f(b)$. Then $f'(\xi) = 0$ for some $\xi \in (a, b)$.

Proof. $[a, b]$ is compact and f continuous, so by Corollary 5.4.1 f attains a maximum at some $x_{\max}$ and a minimum at some $x_{\min}$ in $[a, b]$. If both are endpoints then f is constant, since $f(a) = f(b)$ forces $\max f = \min f$, and any $\xi \in (a, b)$ works. Otherwise one of them is interior, and Theorem 7.2.1 applies there. ∎

> **Theorem 7.2.5** (Mean value theorem)**.** Let $f\colon [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there is $\xi \in (a, b)$ with
>
> $$f(b) - f(a) = f'(\xi)(b - a).$$

Proof. Apply Theorem 7.2.4 to $g(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$, which is continuous on $[a, b]$, differentiable on (a, b), and satisfies $g(a) = g(b) = 0$. Then $0 = g'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{b - a}$. ∎

The mean value theorem is the device by which local information about f' is converted into global information about f: it is what makes "$f' \ge 0$ on an interval" imply "f is non-decreasing there", a step used without comment whenever a comparative-static sign is read off a derivative.

7.3 Rules of Differentiation

Proposition 7.3.1 (Linearity). If $f, g\colon (a,b) \to \mathbb{R}$ are differentiable at x and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g$ is differentiable at x and $(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x)$.

Proof. The difference quotient of $\alpha f + \beta g$ equals α times that of f plus β times that of g, and limits are linear by Proposition 5.2.2(i). ∎

Proposition 7.3.2 (Product rule). If f, g are differentiable at x, so is fg, and $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$.

Proof. Add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \frac{f(x+h) - f(x)}{h}\, g(x) + f(x+h)\, \frac{g(x+h) - g(x)}{h}.$$

As $h \to 0$ the first quotient tends to $f'(x)$ and the second to $g'(x)$, while $f(x+h) \to f(x)$ by Proposition 7.2.1. ∎

The choice of which factor to hold fixed is the substance of the proof: the decomposition corresponds to splitting the area increment of a rectangle of sides f and g into two strips, and the omitted corner term $(f(x+h) - f(x))(g(x+h) - g(x))$ is of order h^2 (Fukuda 2026a, slide 11).

Proposition 7.3.3 (Quotient rule). If f, g are differentiable at x and $g(x) \neq 0$, then f/g is differentiable at x and

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Proof. Since g is continuous at x and $g(x) \neq 0$, g is non-zero on a neighbourhood of x, so f/g is defined there. Write $h := f/g$, so $f = gh$. By Proposition 7.3.2, $f' = g'h + gh'$, whence

$$h' = \frac{f' - g'h}{g} = \frac{f' - g'f/g}{g} = \frac{f'g - fg'}{g^2}.$$

The step presumes h differentiable; to avoid that, compute directly

$$\frac{1}{h}\left(\frac{1}{g(x+h)} - \frac{1}{g(x)}\right) = -\frac{1}{g(x+h)g(x)} \cdot \frac{g(x+h) - g(x)}{h} \longrightarrow -\frac{g'(x)}{g(x)^2},$$

which establishes $(1/g)' = -g'/g^2$, and then apply Proposition 7.3.2 to $f \cdot (1/g)$. ∎

Both derivations are those of (Fukuda 2026a, slides 13–14, 16); the second is logically self-contained, whereas the first assumes what it computes and is a mnemonic rather than a proof. The distinction is recorded because the first is frequently presented as though it were the second.

Proposition 7.3.4 (Chain rule). Let $f\colon (a,b) \to \mathbb{R}$ and $g\colon (c,d) \to \mathbb{R}$ with $f\big((a,b)\big) \subseteq (c,d)$. If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 and

$$(g \circ f)'(x_0) = g'\big(f(x_0)\big) f'(x_0).$$

Proof. Write $y_0 = f(x_0)$ and define

$$\varphi(y) := \begin{cases} \dfrac{g(y) - g(y_0)}{y - y_0}, & y \neq y_0, \\ g'(y_0), & y = y_0. \end{cases}$$

Then φ is continuous at y_0 by differentiability of g, and $g(y) - g(y_0) = \varphi(y)(y - y_0)$ for *all* y, including $y = y_0$. Hence for $h \neq 0$,

$$\frac{g(f(x_0+h)) - g(f(x_0))}{h} = \varphi\big(f(x_0+h)\big) \cdot \frac{f(x_0+h) - f(x_0)}{h}.$$

As $h \to 0$, $f(x_0+h) \to y_0$ by continuity of f, so the first factor tends to $\varphi(y_0) = g'(y_0)$ by Proposition 5.2.3, and the second to $f'(x_0)$. ∎

The auxiliary function φ exists to handle the case $f(x_0+h) = f(x_0)$, at which the naive multiply-and-divide argument would divide by zero. The naive argument is the one behind the suggestive notation $\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x}$ of (Fukuda 2026a, slide 18); it gives the right answer and is not a proof.

Proposition 7.3.5 (Derivative of an inverse function)**.** Let $f\colon (a,b) \to \mathbb{R}$ be continuous and strictly monotone, hence injective with continuous inverse f^{-1} on the interval $f\big((a,b)\big)$. If f is differentiable at x_0 with $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and

$$\big(f^{-1}\big)'(y_0) = \frac{1}{f'\big(f^{-1}(y_0)\big)} = \frac{1}{f'(x_0)}.$$

Proof. Let $y_k \to y_0$ with $y_k \neq y_0$ and set $x_k := f^{-1}(y_k)$, so $x_k \neq x_0$ and $x_k \to x_0$ by continuity of f^{-1}. Then

$$\frac{f^{-1}(y_k) - f^{-1}(y_0)}{y_k - y_0} = \frac{x_k - x_0}{f(x_k) - f(x_0)} = \left(\frac{f(x_k) - f(x_0)}{x_k - x_0}\right)^{-1} \longrightarrow \frac{1}{f'(x_0)},$$

the last step by $f'(x_0) \neq 0$ and continuity of $t \mapsto 1/t$ away from 0. Proposition 5.2.3 converts the sequential statement into the limit. ■

The hypothesis $f'(x_0) \neq 0$ cannot be dropped: $f(x) = x^3$ is a strictly increasing bijection of $\mathbb{R}$ with $f'(0) = 0$, and $f^{-1}(y) = y^{1/3}$ has a vertical tangent at 0. This is the one-dimensional case of the inverse function theorem, and the condition $f'(x_0) \neq 0$ becomes invertibility of the Jacobian in Corollary 8.6.1. Economically it is the statement that a demand function D and an inverse demand function P have reciprocal slopes, $P'(q) = 1/D'(p)$ at $q = D(p)$ (Fukuda 2026a, slides 19–20).

7.4 Taylor Expansion

The problem is to approximate f near x_0 by a polynomial in h,

$$f(x_0 + h) \approx \alpha_0 + \alpha_1 h + \alpha_2 h^2 + \cdots + \alpha_n h^n. \tag{7.4}$$

Setting $h = 0$ gives $\alpha_0 = f(x_0)$. Differentiating (7.4) k times with respect to h and again setting $h = 0$ kills every term of degree below k and every term of degree above k, leaving $f^{(k)}(x_0) = k!\,\alpha_k$, so

$$\alpha_k = \frac{f^{(k)}(x_0)}{k!}, \qquad k = 0, 1, \ldots, n. \tag{7.5}$$

This is the derivation of (Fukuda 2026a, slides 28–31). It identifies the coefficients on the assumption that an expansion of the stated form exists; it does not establish that one does, nor how large the error is. Both are the content of the following theorem.

Theorem 7.4.1 (Taylor's theorem with Lagrange remainder)**.** Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be an open interval containing x_0, and let $f\colon I \to \mathbb{R}$ be $n+1$ times differentiable on I. Define the *Taylor polynomial* of order n,

$$T_n f(x_0; h) := \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!} h^k.$$

Then for every h with $x_0 + h \in I$ there exists ξ strictly between x_0 and $x_0 + h$ (or $\xi = x_0$ if $h = 0$) such that

$$f(x_0 + h) = T_n f(x_0; h) + \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}. \tag{7.6}$$

Proof. Fix $h \neq 0$ and let J be the closed interval with endpoints x_0 and $x_0 + h$. Define M by $f(x_0+h) = T_n f(x_0; h) + M h^{n+1}$, which determines M uniquely since $h \neq 0$, and set

$$g(t) := f(t) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(t - x_0)^k - M(t - x_0)^{n+1}, \qquad t \in J.$$

By construction $g^{(k)}(x_0) = 0$ for $k = 0, 1, \ldots, n$, and $g(x_0 + h) = 0$ by the choice of M. Since $g(x_0) = g(x_0 + h) = 0$, Theorem 7.2.4 gives ξ_1 strictly between them with $g'(\xi_1) = 0$. Since also $g'(x_0) = 0$, Rolle applied to g' on the interval with endpoints x_0 and ξ_1 gives ξ_2 with $g''(\xi_2) = 0$. Iterating $n+1$ times produces $\xi := \xi_{n+1}$ strictly between x_0 and $x_0 + h$ with $g^{(n+1)}(\xi) = 0$. But $g^{(n+1)}(t) = f^{(n+1)}(t) - (n+1)!\,M$, so $M = f^{(n+1)}(\xi)/(n+1)!$, which is (7.6). ■

Corollary 7.4.1 (Peano remainder)**.** If f is n times differentiable at x_0, then $f(x_0+h) = T_n f(x_0; h) + o(h^n)$ as $h \to 0$; that is, the error divided by h^n tends to 0.

The two remainder forms answer different questions. The Lagrange form (7.6) is an exact identity with an unknown evaluation point, and it yields *numerical* error bounds once $f^{(n+1)}$ is bounded on the relevant interval. The Peano form is an asymptotic statement with no evaluation point, and it is the one used when the argument is a limit as a parameter goes to zero — which is the situation in Section 7.6 below.

Example 7.4.1 (Exponential and logarithm)**.** For $f(x) = e^x$ every derivative is e^x and $f^{(k)}(0) = 1$, so

$$e^x = 1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3 + \cdots + \tfrac{1}{n!}x^n + \frac{e^\xi}{(n+1)!}x^{n+1}.$$

For $f(x) = \log(1+x)$ on $(-1, \infty)$, $f'(x) = (1+x)^{-1}$, $f''(x) = -(1+x)^{-2}$, $f'''(x) = 2(1+x)^{-3}$, so $f^{(k)}(0) = (-1)^{k-1}(k-1)!$ and

$$\log(1+x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \cdots + \frac{(-1)^{n-1}}{n}x^n + \frac{(-1)^n}{n+1}\frac{x^{n+1}}{(1+\xi)^{n+1}}.$$

The first-order truncations $e^x \approx 1+x$ and $\log(1+x) \approx x$ carry an error of order x^2, with sign determined by the second derivative: the exponential approximation understates and the logarithmic one overstates (Fukuda 2026a, slides 32–36).

7.4.1 Four uses of the logarithmic approximation

Example 7.4.2 (Growth accounting)**.** Since $(\log f)' = f'/f$ by Proposition 7.3.4, taking logarithms of the Cobb–Douglas aggregate $Y(t) = A(t)K(t)^\alpha L(t)^{1-\alpha}$ and differentiating in t gives the exact identity

$$\frac{Y'(t)}{Y(t)} = \frac{A'(t)}{A(t)} + \alpha\frac{K'(t)}{K(t)} + (1-\alpha)\frac{L'(t)}{L(t)}.$$

No approximation is involved: the growth rate of a Cobb–Douglas aggregate is exactly the factor-share-weighted sum of the growth rates of its arguments, and the residual A'/A is what growth accounting measures (Fukuda 2026a, slide 37).

Example 7.4.3 (Log differences as growth rates)**.** For a positive sequence (Y_t) with growth rate $r_t = (Y_t - Y_{t-1})/Y_{t-1}$,

$$\Delta \log Y_t = \log Y_t - \log Y_{t-1} = \log\left(\frac{Y_t}{Y_{t-1}}\right) = \log(1 + r_t) \approx r_t.$$

By Example 7.4.1 the error is $-r_t^2/2 + O(r_t^3)$, so log differences *understate* growth, by about half a percentage point at $r_t = 10\%$ and by five at $r_t = 30\%$. The approximation is a good one at quarterly frequencies and a poor one for hyperinflations (Fukuda 2026a, slide 38).

Example 7.4.4 (Fisher equation)**.** The exact relation between the nominal rate i, the inflation rate π and the real rate r is $(1+i) = (1+\pi)(1+r)$. Taking logarithms turns the product into a sum exactly, $\log(1+i) = \log(1+\pi) + \log(1+r)$, and applying $\log(1+x) \approx x$ three times gives the Fisher equation $i \approx \pi + r$. The neglected term is πr, which is second order in the two rates (Fukuda 2026a, slide 39).

Example 7.4.5 (Log-linearisation)**.** First-order Taylor expansion gives $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$. Writing the absolute change as x_0 times the proportional change and using $\log x - \log x_0 \approx (x - x_0)/x_0$,

$$f(x) \approx f(x_0) + f'(x_0)x_0 \cdot \big(\log x - \log x_0\big).$$

The coefficient $f'(x_0)x_0$ is the semi-elasticity of f at x_0. This is the step by which a non-linear dynamic model is converted into a linear system in log deviations from steady state, whose stability is then read off the eigenvalues as in Proposition 2.8.1 (Fukuda 2026a, slide 40).

7.5 Economic Application: The Firm's Problem

7.5.1 The price-taking firm

Proposition 7.5.1. Under Assumption 7.1.1 and for every $p > 0$, problem (7.1) has a unique solution $q^*(p) > 0$, characterised by

$$c'(q^*) = p, \qquad \text{equivalently} \qquad q^*(p) = (c')^{-1}(p). \tag{7.7}$$

The supply function q^* is strictly increasing and differentiable with $\big(q^*\big)'(p) = 1/c''\big(q^*(p)\big) > 0$ whenever c is twice differentiable with $c'' > 0$.

Proof. The objective $\pi(q) = pq - c(q)$ is strictly concave, being linear minus strictly convex, and differentiable, with $\pi'(q) = p - c'(q)$. Since $c'(0) = 0 < p$ and $c'(q) \to \infty$, the continuous strictly increasing function c' takes the value p at exactly one point $q^* > 0$, by the intermediate value theorem (Corollary 5.2.1) and injectivity. At that point $\pi' = 0$, and Theorem 7.2.3 makes it the unique global maximiser on $(0, \infty)$; comparison with $\pi(0) = 0 < \pi(q^*)$ rules out the boundary. The supply function is the inverse of c', and its derivative is Proposition 7.3.5. ∎

The economic content of (7.7) is that the firm equates marginal cost to price. The comparative static $(q^*)' = 1/c'' > 0$ makes precise the sense in which the slope of the supply curve is the reciprocal of the curvature of the cost function: a firm with nearly linear costs responds strongly to price, one with sharply increasing marginal cost hardly at all.

7.5.2 The price-setting firm and the markup

Proposition 7.5.2. Under Assumption 7.1.1, an interior solution $q^* > 0$ of (7.2) satisfies

$$P(q^*)\left(1 + \frac{1}{\eta}\right) = c'(q^*), \qquad \text{equivalently} \qquad P(q^*) = \underbrace{\frac{1}{1+\eta^{-1}}}_{\text{markup}} c'(q^*), \tag{7.8}$$

where

$$\eta := \frac{\mathrm{d}q}{\mathrm{d}p}\frac{p}{q}\bigg|_{q=q^*} < 0$$

is the price elasticity of demand. If $\eta < -1$ the markup exceeds 1 and $P(q^*) > c'(q^*)$.

Proof. The objective is $\pi(q) = P(q)q - c(q)$, concave by Assumption 7.1.1. Its derivative, by Proposition 7.3.2, is $\pi'(q) = P'(q)q + P(q) - c'(q)$, so Theorem 7.2.1 gives

$$\underbrace{P'(q^*)q^* + P(q^*)}_{\text{marginal revenue}} = \underbrace{c'(q^*)}_{\text{marginal cost}} . \tag{7.9}$$

Factoring $P(q^*) > 0$ out of the left side,

$$P(q^*)\left(1 + \frac{P'(q^*)q^*}{P(q^*)}\right) = c'(q^*).$$

By Proposition 7.3.5 applied to the demand–inverse-demand pair, $P'(q^*) = 1/D'(p^*)$ at $p^* = P(q^*)$, so

$$\frac{P'(q^*)q^*}{P(q^*)} = \frac{\mathrm{d}p}{\mathrm{d}q}\frac{q}{p} = \left(\frac{\mathrm{d}q}{\mathrm{d}p}\frac{p}{q}\right)^{-1} = \frac{1}{\eta},$$

which is (7.8). Solving for $P(q^*)$ gives the markup form, and $\eta < -1$ makes $1 + \eta^{-1} \in (0, 1)$, hence the markup exceeds 1. ∎

First, (7.8) is the Lerner condition, usually written $\big(P - c'\big)/P = -1/\eta$; the two forms are algebraically identical. Second, the derivation uses the inverse function rule twice over — once to convert P' into $1/D'$ and once implicitly in reading the elasticity in either direction — so Proposition 7.3.5 carries the derivation. Third, profit maximisation requires $\eta < -1$: if demand were inelastic, marginal revenue $P(1 + \eta^{-1})$ would be negative while marginal cost is positive, and (7.9) would have no solution. A monopolist never operates on the inelastic portion of its demand curve, and this is a theorem about (7.9) rather than an assumption.

Example 7.5.1 (The dual formulation). Choosing the price rather than the quantity, the firm solves $\max_{p\geq 0}\ pD(p) - c\big(D(p)\big)$. The first-order condition is $D(p^*) + p^* D'(p^*) - c'\big(D(p^*)\big)D'(p^*) = 0$; dividing by $D'(p^*) < 0$ and rearranging gives

$$p^* = \left(\frac{1}{1+\eta^{-1}}\right) c'\big(D(p^*)\big), \qquad \eta = \left.\frac{D'(p)p}{D(p)}\right|_{p=p^*},$$

the same condition as (7.8). That the two formulations agree is a consequence of the chain rule and the strict monotonicity of D, which makes the change of variable $q = D(p)$ a bijection between the two feasible sets. This is Exercise 11.14 of Fukuda (2026d) (Fukuda 2026a, slide 25).

7.6 Economic Application: The Risk Premium

Environment. A decision maker with wealth w and utility $u\colon \mathbb{R} \to \mathbb{R}$, strictly increasing and twice continuously differentiable, faces a gamble $\tilde{\varepsilon}$ with $\mathbb{E}[\tilde{\varepsilon}] = 0$ and $\operatorname{Var}[\tilde{\varepsilon}] = \sigma^2$. The *risk premium* π is the sure amount whose surrender leaves the decision maker indifferent to bearing the gamble:

$$u(w-\pi) = \mathbb{E}\big[u(w+\tilde{\varepsilon})\big]. \tag{7.10}$$

The slides derive the approximation by expanding each side and matching (Fukuda 2026a, slides 26–27, 41). That derivation is correct in its conclusion and incomplete as an argument: it expands the left side to first order and the right to second order without saying why the orders may differ, and it does not say in what limit the approximation is valid. Both gaps are closed by scaling the gamble.

Theorem 7.6.1 (Arrow–Pratt). Let u be three times continuously differentiable with $u' > 0$ on a neighbourhood of w, and let $\tilde{z}$ be a bounded random variable with $\mathbb{E}[\tilde{z}] = 0$ and $\operatorname{Var}[\tilde{z}] = 1$. For $t > 0$ let $\pi(t)$ solve (7.10) with $\tilde{\varepsilon} = t\tilde{z}$. Then π is twice differentiable at $t = 0$ with $\pi(0) = \pi'(0) = 0$ and

$$\pi(t) = \frac{1}{2} A(w)\, t^2 + o(t^2) \qquad \text{as } t \downarrow 0, \qquad \text{where} \quad A(w) := -\frac{u''(w)}{u'(w)}$$

is the *coefficient of absolute risk aversion*. Writing $\sigma^2 = t^2$ for the variance of the gamble, this is $\pi \approx \frac{1}{2}A(w)\sigma^2$.

Proof. Define $\Phi(\pi, t) := u(w-\pi) - \mathbb{E}[u(w+t\tilde{z})]$. Boundedness of $\tilde{z}$ and continuity of u and its derivatives permit differentiation under the expectation on a neighbourhood of $(0,0)$, by Lemma 10.5.1. We have $\Phi(0,0) = 0$ and $\partial\Phi/\partial\pi = -u'(w) \neq 0$ at $(0,0)$, so by the implicit function theorem (Theorem 8.6.1) there is a twice continuously differentiable $\pi(t)$ near $t = 0$ with $\pi(0) = 0$ and $\Phi(\pi(t), t) = 0$.

The order of π. Differentiating the identity $\Phi(\pi(t), t) = 0$ and using (8.9),

$$\pi'(t) = -\frac{\partial\Phi/\partial t}{\partial\Phi/\partial\pi} = -\frac{\mathbb{E}\big[u'(w+t\tilde{z})\,\tilde{z}\big]}{u'\big(w-\pi(t)\big)}.$$

At $t = 0$ the numerator is $u'(w)\mathbb{E}[\tilde{z}] = 0$, so $\pi'(0) = 0$. Since π is twice continuously differentiable with $\pi(0) = \pi'(0) = 0$, Theorem 7.4.1 gives $\pi(t) = O(t^2)$ as $t \downarrow 0$. This is what licenses the truncation below; it is a conclusion, not an assumption.

Expand the right-hand side of (7.10) with the Lagrange remainder of Theorem 7.4.1 at order 2: for each realisation z,

$$u(w+tz) = u(w) + u'(w)tz + \tfrac{1}{2}u''(w)t^2z^2 + \tfrac{1}{6}u'''(\xi_z)t^3z^3.$$

Taking expectations and using $\mathbb{E}[\tilde{z}] = 0$, $\mathbb{E}[\tilde{z}^2] = 1$ and boundedness of $\tilde{z}$ and of u''' near w,

$$\mathbb{E}\big[u(w+t\tilde{z})\big] = u(w) + \tfrac{1}{2}u''(w)t^2 + O(t^3).$$

Expanding the left-hand side at order 2 in π, $u(w-\pi) = u(w) - u'(w)\pi + O(\pi^2)$. Equating and using the bound $\pi = O(t^2)$ established above, which makes $O(\pi^2) = O(t^4)$,

$$-u'(w)\pi(t) + O(t^4) = \tfrac{1}{2}u''(w)t^2 + O(t^3),$$

so $\pi(t) = -\frac{1}{2}\big(u''(w)/u'(w)\big)t^2 + O(t^3)$, which is the claim. ■

Remark 7.6.1 (Why the two sides are expanded to different orders)**.** The theorem explains the asymmetry that the slide derivation leaves implicit. Because π is itself of order t^2, the term $-u'(w)\pi$ on the left is already of order t^2, and the next term $\frac{1}{2}u''(w)\pi^2$ is of order t^4, hence negligible at the order retained. On the right the linear term vanishes because $\mathbb{E}[\tilde{\varepsilon}] = 0$, so the leading non-trivial term is the quadratic one. Expanding the left side to first order and the right to second is therefore expanding *both* to order t^2: the orders in π and in t are different bookkeeping for the same expansion.

Remark 7.6.2 (The variance identity)**.** The step $\mathbb{E}[\tilde{\varepsilon}^2] = \sigma^2$ uses $\operatorname{Var}[X] = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2$, which follows by expanding $\mathbb{E}\big[(X - \mathbb{E}X)^2\big] = \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}X)^2$ and is Proposition 14.2.2. It is exactly here that the mean-zero normalisation of the gamble is used; for a gamble with non-zero mean the first-order term does not vanish and the premium contains a term of order t that reflects the change in expected wealth rather than risk (Fukuda 2026a, slide 42).

Economic reading. Three consequences. The premium is proportional to the variance, so risk aversion of this local kind prices only second moments; a gamble's skewness enters at order t^3 and is invisible in the approximation. The factor of proportionality is $A(w)/2$, a property of preferences alone, which is why A is comparable across decision makers and across wealth levels while u'' alone is not — u'' changes under the affine transformations of u that leave preferences unchanged, and the ratio $-u''/u'$ does not. Finally, the premium is second order in the size of the gamble: for small gambles a risk averse agent is approximately risk neutral, which is the formal statement behind the practice of ignoring risk in first-order comparative statics and the reason that observed rejections of small-stakes gambles are hard to reconcile with expected utility.

7.7 Chapter Summary

The derivative is the coefficient of the best affine approximation (7.3), and differentiability implies continuity but not conversely. At an interior maximiser or minimiser of a differentiable function the derivative vanishes (Theorem 7.2.1); the conclusion fails at boundaries, at kinks, and as a sufficient condition. Under concavity the first-order condition becomes sufficient and identifies the global maximiser (Theorem 7.2.3), because a differentiable concave function lies below each of its tangents. Rolle's theorem and the mean value theorem convert information about the derivative into information about the function, and are the engine behind both the monotonicity criterion and Taylor's theorem. The differentiation rules — linearity, product, quotient, chain, inverse — are proved here in the forms actually used later, with the inverse rule carrying the economic content that demand and inverse demand have reciprocal slopes.

Taylor's theorem states that the polynomial with coefficients (7.5) approximates f with an error given exactly by (7.6) or asymptotically by Corollary 7.4.1. Applied to $\log(1 + x)$ it justifies log differences as growth rates, the Fisher equation, and log-linearisation, in each case with a quantified second-order error. Applied to the certainty-equivalence condition it produces the Arrow–Pratt formula $\pi \approx \frac{1}{2}A(w)\sigma^2$, with the orders of expansion on the two sides reconciled by the observation that the premium is itself second order in the scale of the gamble. For the firm, the first-order condition gives price equal to marginal cost under price taking and the Lerner markup condition under price setting, the latter implying that a monopolist operates only where demand is elastic.

Exercise 7.7.1. Let $c(q) = \frac{1}{2}\gamma q^2$ with $\gamma > 0$ and $P(q) = a - bq$ with $a, b > 0$. Compute q^* and $P(q^*)$ for both firm problems, verify (7.8) directly, and show that the monopoly quantity is exactly half the competitive quantity when $\gamma = 0$. Identify where Assumption 7.1.1 fails when $\gamma = 0$ and why the conclusion survives anyway.

Exercise 7.7.2. Show that a function may be differentiable everywhere with discontinuous derivative, using $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Compute $f'(0)$ from the definition, compute $f'(x)$ for $x \neq 0$, and show that $\lim_{x \to 0} f'(x)$ does not exist. Explain why this does not contradict Theorem 7.2.5.

Exercise 7.7.3. For CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma > 0$, $\gamma \neq 1$, compute $A(w)$ and the coefficient of relative risk aversion $wA(w)$. Show that the risk premium for a multiplicative gamble $w(1 + t\tilde{z})$ is approximately $\frac{1}{2}\gamma w t^2$, and contrast with Theorem 7.6.1. Verify that $\lim_{\gamma \to 1} u(c)$, after subtracting the constant $1/(1-\gamma)$, equals $\log c$.

Chapter 8

Differential Calculus in Several Variables

8.1 Motivation: Two Multivariate Problems

Two problems drive the chapter (Fukuda 2026a, slide 43). The representative firm chooses capital and labour to solve

$$\max_{K,L\geq 0} \Pi(K,L), \qquad \Pi(K,L) := AK^{\alpha}L^{1-\alpha} - (rK + wL), \tag{8.1}$$

and the econometrician fits a line by

$$\min_{(\beta_1,\beta_2)\in\mathbb{R}^2} \sum_{i=1}^{n} \bigl(y_i - \beta_1 - \beta_2 x_i\bigr)^2. \tag{8.2}$$

Neither is harder than the one-variable problems of Chapter 7 once the right notion of derivative is in place, and finding that notion is the substance of the chapter. Two distinct ideas are involved and are frequently conflated. The first is to vary one coordinate at a time, which produces partial derivatives. The second is to approximate f near a point by an affine function of the whole vector, which produces the total derivative. In one variable the two coincide. In several variables the second is strictly stronger, and it is the second that supports the chain rule, the first-order condition, and the implicit function theorem.

8.2 Partial Derivatives and the Gradient

Definition 8.2.1 (Partial derivative and gradient)**.** Let $f\colon A \to \mathbb{R}$ with $A \subseteq \mathbb{R}^n$ and let x be interior to A. For $i \in \{1,\dots,n\}$ the *ith partial derivative* of f at x is

$$\frac{\partial f}{\partial x_i}(x) := \lim_{h\to 0} \frac{f(x_1,\dots,x_i+h,\dots,x_n) - f(x_1,\dots,x_i,\dots,x_n)}{h},$$

provided the limit exists and is finite; equivalently, it is the ordinary derivative at x_i of the section $t \mapsto f(x_1,\dots,t,\dots,x_n)$. When all n partial derivatives exist, the *gradient* of f at x is the vector

$$\nabla f(x) := \left(\frac{\partial f}{\partial x_1}(x),\dots,\frac{\partial f}{\partial x_n}(x)\right)^{\mathsf{T}} \in \mathbb{R}^n.$$

This is Definition 3.5 of Fukuda (2026d) (Fukuda 2026a, slides 46, 48), where the alternatives $\partial f(x)/\partial x_i$, $f_{x_i}(x)$ and $f'(x)$ for the gradient are also recorded. This volume writes the gradient as a column vector, consistently with the front-matter chapter on notation and conventions; the row vector $\nabla f(x)^{\mathsf{T}}$ is the Jacobian of f viewed as a map into $\mathbb{R}^1$, and conflating the two is the most frequent source of transpose errors in comparative statics.

Example 8.2.1 (Cobb–Douglas). For $f(x,y) = x^\alpha y^{1-\alpha}$ with $\alpha \in (0,1)$ and $x, y > 0$, fixing y makes f a power function of x, so

$$\frac{\partial f}{\partial x} = \alpha x^{\alpha-1} y^{1-\alpha} = \alpha \left(\frac{y}{x}\right)^{1-\alpha}, \qquad \frac{\partial f}{\partial y} = (1-\alpha) x^\alpha y^{-\alpha} = (1-\alpha)\left(\frac{x}{y}\right)^{\alpha}.$$

Both are positive and each is decreasing in its own argument: marginal products are positive and diminishing. Both are homogeneous of degree zero, so they depend on the two inputs only through the ratio x/y — the property that makes the factor-price equations of Section 8.7 determine the capital–labour ratio and nothing else (Fukuda 2026a, slide 47).

Example 8.2.2 (Gradient of a quadratic form). Let $A = (a_{ij}) \in M_{n\times n}(\mathbb{R})$ and $f(x) = x^\mathsf{T} A x = \sum_{i,j} a_{ij} x_i x_j$. Differentiating with respect to x_k, the terms containing x_k are $a_{kk}x_k^2$ and, for $j \neq k$, $a_{kj}x_k x_j$ and $a_{jk}x_j x_k$, so

$$\frac{\partial f}{\partial x_k}(x) = 2a_{kk}x_k + \sum_{j\neq k}(a_{kj} + a_{jk})x_j = \sum_{j=1}^{n}(a_{kj} + a_{jk})x_j,$$

that is,

$$\nabla\big(x^\mathsf{T} A x\big) = \big(A + A^\mathsf{T}\big)x. \tag{8.3}$$

For $n = 2$ this is the matrix $\left(\begin{smallmatrix} 2a_{11} & a_{12}+a_{21} \\ a_{12}+a_{21} & 2a_{22}\end{smallmatrix}\right)$ applied to x, as in Exercise 3.3 and Example 11.2 of Fukuda (2026d) (Fukuda 2026a, slide 49). When A is symmetric (8.3) reduces to the familiar $2Ax$; when it is not, only the symmetric part $\frac{1}{2}(A + A^\mathsf{T})$ of A affects the quadratic form, which is Remark 2.5.1. The identities that recur in least squares, in the second-order conditions of Chapter 12 and in the Kalman recursion of subsection 14.6.2 are collected in Appendix B.

8.3 Differentiability

Definition 8.3.1 (Differentiability and the total derivative). Let $A \subseteq \mathbb{R}^n$ be open and $f\colon A \to \mathbb{R}$. The function f is *differentiable at* $x_0 \in A$ if there is a vector $\nabla f(x_0) \in \mathbb{R}^n$ such that

$$\beta_{x_0}(u) := \frac{f(x_0+u) - f(x_0) - \langle \nabla f(x_0), u\rangle}{\|u\|} \longrightarrow 0 \qquad \text{as } u \to 0,\ u \neq 0. \tag{8.4}$$

The associated linear functional $Df(x_0)\colon \mathbb{R}^n \to \mathbb{R}$, $Df(x_0)u := \langle \nabla f(x_0), u\rangle$, is the *total derivative* of f at x_0. The function is differentiable on A if it is differentiable at every point of A.

This is Definition 11.1 of Fukuda (2026d, p. 369). Condition (8.4) says that $f(x_0) + \langle \nabla f(x_0), u\rangle$ approximates $f(x_0+u)$ with an error small relative to $\|u\|$; geometrically, the graph has a tangent hyperplane at x_0 (Fukuda 2026a, slide 50). The existence of the vector $\nabla f(x_0)$ is part of the requirement, and by Theorem 3.2.1 any linear functional on $\mathbb{R}^n$ has such a representation, so nothing is lost by stating the definition with an inner product rather than an abstract linear map.

Proposition 8.3.1 (Consistency with the one-variable definition). For $n = 1$, Definition 8.3.1 holds at x_0 with $\nabla f(x_0) = L$ if and only if Definition 7.2.1 holds with $f'(x_0) = L$.

Proof. For $n = 1$, (8.4) reads $\big(f(x_0+u) - f(x_0) - Lu\big)/|u| \to 0$, which since $\|u\| = |u|$ is equivalent to $\big(f(x_0+u) - f(x_0)\big)/u \to L$. This is Proposition 7.2.1, and hence Definition 7.2.1. ∎

Proposition 8.3.2 (Differentiability implies continuity and the existence of partials). If f is differentiable at x_0, then f is continuous at x_0, every partial derivative of f exists at x_0, and the vector $\nabla f(x_0)$ in (8.4) is the gradient of Definition 8.2.1. Moreover the directional derivative in any direction $v \in \mathbb{R}^n$ exists and equals $\langle \nabla f(x_0), v\rangle$.

Proof. Continuity: from (8.4), $f(x_0+u) - f(x_0) = \langle \nabla f(x_0), u\rangle + \|u\|\,\beta_{x_0}(u)$, and both terms tend to 0 as $u \to 0$, the first by Cauchy–Schwarz (Lemma 3.2.1) and the second because $\beta_{x_0}(u) \to 0$.

Directional derivatives: put $u = tv$ with $t \neq 0$ and $v \neq 0$. Then

$$\frac{f(x_0+tv) - f(x_0)}{t} = \langle \nabla f(x_0), v\rangle + \frac{|t|\,\|v\|}{t}\,\beta_{x_0}(tv) \longrightarrow \langle \nabla f(x_0), v\rangle,$$

because $||t|/t| = 1$ and $\beta_{x_0}(tv) \to 0$. Taking $v = e_i$ gives the ith partial derivative and identifies the ith coordinate of $\nabla f(x_0)$. ■

Each implication in Proposition 8.3.2 is strictly one-way. The next three examples exhibit, in turn, partial derivatives without continuity, continuity and partial derivatives without directional derivatives in the remaining directions, and all directional derivatives without differentiability.

Example 8.3.1 (Partials without continuity)**.** Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be $f(x_1, x_2) = 0$ if $x_1 x_2 = 0$ and $f(x_1, x_2) = 1$ otherwise. Along each axis f is identically zero, so $\partial f/\partial x_1(0,0) = \partial f/\partial x_2(0,0) = 0$. Yet $f(\varepsilon, \varepsilon) = 1$ for every $\varepsilon \neq 0$, so f is not continuous at the origin. In one variable this cannot happen (Proposition 7.2.1); in several, the partial derivatives see only n lines through the point and are blind to everything else (Fukuda 2026d, Rem. 11.4, p. 375).

Example 8.3.2 (Continuity and partials without differentiability)**.** Let

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Since $|xy| \leq \frac{1}{2}(x^2+y^2)$, we have $|f(x,y)| \leq \frac{1}{2}\sqrt{x^2+y^2} \to 0$, so f is continuous at the origin. Both partial derivatives at the origin vanish, since f is identically zero on each axis. If f were differentiable at the origin, then by Proposition 8.3.2 the gradient would be 0 and (8.4) would require

$$\beta_0(x,y) = \frac{f(x,y)}{\sqrt{x^2+y^2}} = \frac{xy}{x^2+y^2} \to 0.$$

Along $(x,y) = (h,h)$ this ratio equals $\frac{1}{2}$ for every $h \neq 0$, so the limit is not 0 and f is not differentiable at the origin (Fukuda 2026d, Rem. 11.3, p. 374).

Here the failure is already visible in the directional derivatives. For a direction $v = (a,b)$,

$$\frac{f(tv) - f(0)}{t} = \frac{t^2 ab}{|t|\sqrt{a^2+b^2}} \cdot \frac{1}{t} = \operatorname{sgn}(t)\,\frac{ab}{\sqrt{a^2+b^2}},$$

so the two-sided limit as $t \to 0$ exists only when $ab = 0$: apart from the two coordinate axes, this f has no directional derivative at all.

Example 8.3.3 (All directional derivatives without differentiability)**.** The stronger failure — every directional derivative exists, yet f is not differentiable — needs a different function. Let

$$f(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Since $|f(x,y)| \leq |x|$, the function is continuous at the origin. For $v = (a,b) \neq 0$ and $t \neq 0$,

$$\frac{f(tv) - f(0)}{t} = \frac{1}{t} \cdot \frac{t^3 a^3}{t^2(a^2+b^2)} = \frac{a^3}{a^2+b^2},$$

which does not depend on t; the directional derivative $D_v f(0,0) = a^3/(a^2+b^2)$ therefore exists for every v. In particular $\partial f/\partial x(0,0) = 1$ and $\partial f/\partial y(0,0) = 0$, so the only candidate for a gradient is $(1,0)$. But $v \mapsto D_v f(0,0)$ is not linear: with $v = (1,0)$ and $w = (0,1)$ one has $D_{v+w} f(0,0) = \frac{1}{2}$ while $D_v f(0,0) + D_w f(0,0) = 1$. By Proposition 8.3.2 the map $v \mapsto D_v f(0,0)$ would have to equal $\langle \nabla f(0,0), \cdot \rangle$ and hence be linear, so f is not differentiable at the origin.

The two examples separate two distinct obstructions. In Example 8.3.2 the one-dimensional restrictions of f are themselves badly behaved; here each restriction to a line is differentiable and the failure is only that the derivatives along different lines do not assemble into a linear functional. Both are ruled out by Proposition 8.3.3, and the economic reading is the same: a comparative-statics exercise conducted one variable at a time can be internally consistent along every ray and still fail to describe the response to a joint perturbation.

Proposition 8.3.3 (Continuous partials suffice)**.** Let $A \subseteq \mathbb{R}^n$ be open and let $f : A \to \mathbb{R}$ have partial derivatives with respect to every variable at every point of A, all of them continuous at $x_0 \in A$. Then f is differentiable at x_0.

Proof. Write $u = (u_1, \dots, u_n)$ and interpolate one coordinate at a time. Set $v^0 := x_0$ and $v^k := x_0 + \sum_{i \le k} u_i e_i$, so $v^n = x_0 + u$. Then

$$f(x_0 + u) - f(x_0) = \sum_{k=1}^{n} \big(f(v^k) - f(v^{k-1})\big),$$

and each summand differs only in the kth coordinate. Applying the one-variable mean value theorem (Theorem 7.2.5) to that section gives ξ^k on the segment from v^{k-1} to v^k with $f(v^k) - f(v^{k-1}) = \frac{\partial f}{\partial x_k}(\xi^k)\, u_k$. Hence

$$f(x_0 + u) - f(x_0) - \langle \nabla f(x_0), u \rangle = \sum_{k=1}^{n} \left(\frac{\partial f}{\partial x_k}(\xi^k) - \frac{\partial f}{\partial x_k}(x_0) \right) u_k.$$

Since $|u_k| \le \|u\|$ and $\xi^k \to x_0$ as $u \to 0$, continuity of the partials makes each bracket tend to 0, so the whole expression divided by $\|u\|$ tends to 0. ∎

This is Proposition 11.4 of Fukuda (2026d, p. 375). It supplies the practical criterion: a function given by an explicit formula built from elementary operations has continuous partials wherever those operations are defined, hence is differentiable there, and the pathologies of Example 8.3.1 and Example 8.3.2 arise only at points where the formula changes.

Definition 8.3.2 (Continuous differentiability)**.** f is *continuously differentiable* on an open A, written $f \in C^1(A)$, if all partial derivatives exist and are continuous on A. Inductively, $f \in C^k(A)$ if all partial derivatives of order k exist and are continuous.

The implications established so far, none of which reverses, are

$$f \in C^1 \implies f \text{ differentiable} \implies \begin{cases} f \text{ continuous,} \\ \text{all partials exist} \end{cases}$$

and neither of the two consequences on the right, alone or together, implies differentiability.

Remark 8.3.1 (The total differential)**.** Writing $\mathrm{d}y$ for the increment of f and $\mathrm{d}x_i$ for the increments of the arguments, (8.4) is the statement

$$\mathrm{d}y = \frac{\partial f}{\partial x_1}\, \mathrm{d}x_1 + \cdots + \frac{\partial f}{\partial x_n}\, \mathrm{d}x_n$$

up to an error small relative to $\|\mathrm{d}x\|$ (Fukuda 2026a, slide 51). The classical notation is convenient and hides the content: the identity is not an algebraic manipulation of infinitesimals but the assertion that the error term in (8.4) vanishes faster than $\|u\|$, which by Example 8.3.2 is a genuine restriction on f and not a consequence of the existence of the partial derivatives appearing in it.

8.3.1 Vector-valued maps and the Jacobian

Definition 8.3.3 (Jacobian)**.** Let $A \subseteq \mathbb{R}^n$ be open and $f = (f_1, \dots, f_m) : A \to \mathbb{R}^m$. Then f is differentiable at x_0 if there is a linear map $Df(x_0) : \mathbb{R}^n \to \mathbb{R}^m$ with $\|f(x_0 + u) - f(x_0) - Df(x_0)u\| \,/\, \|u\| \to 0$ as $u \to 0$. Equivalently, each component f_k is differentiable at x_0; and the matrix of $Df(x_0)$ in the canonical bases is the *Jacobian matrix*

$$Df(x_0) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1}(x_0) & \cdots & \dfrac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & & \vdots \\ \dfrac{\partial f_m}{\partial x_1}(x_0) & \cdots & \dfrac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix} \in M_{m\times n}(\mathbb{R}).$$

The row-wise reading of a matrix flagged in Theorem 2.3.1 appears here: row k of $Df(x_0)$ is $\nabla f_k(x_0)^{\mathsf{T}}$. For $m = 1$ the Jacobian is the *row* vector $\nabla f(x_0)^{\mathsf{T}}$, which is why gradients and Jacobians differ by a transpose and why the chain rule below is stated for matrices rather than vectors.

Theorem 8.3.1 (Chain rule)**.** Let $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^m$ be open, $f\colon A \to \mathbb{R}^m$ with $f(A) \subseteq B$, and $\varphi\colon B \to \mathbb{R}^p$. If f is differentiable at x_0 and φ is differentiable at $f(x_0)$, then $\varphi \circ f$ is differentiable at x_0 and

$$D(\varphi \circ f)(x_0) = D\varphi\big(f(x_0)\big)\, Df(x_0), \tag{8.5}$$

a product of a $p \times m$ by an $m \times n$ matrix. Componentwise, for $p = 1$,

$$\frac{\partial(\varphi \circ f)}{\partial x_i}(x_0) = \sum_{k=1}^{m} \frac{\partial \varphi}{\partial y_k}\big(f(x_0)\big)\, \frac{\partial f_k}{\partial x_i}(x_0). \tag{8.6}$$

Proof. Write $y_0 = f(x_0)$, $S := Df(x_0)$ and $T := D\varphi(y_0)$. By differentiability, $f(x_0 + u) = y_0 + Su + \|u\|\,\varepsilon_1(u)$ with $\varepsilon_1(u) \to 0$, and $\varphi(y_0 + v) = \varphi(y_0) + Tv + \|v\|\,\varepsilon_2(v)$ with $\varepsilon_2(v) \to 0$. Put $v(u) := Su + \|u\|\,\varepsilon_1(u)$; then

$$\varphi\big(f(x_0 + u)\big) - \varphi(y_0) = T\big(Su\big) + \|u\|\; T\varepsilon_1(u) + \|v(u)\|\; \varepsilon_2\big(v(u)\big).$$

Divide by $\|u\|$. The second term tends to 0 because T is linear on a finite-dimensional space, hence continuous by Proposition 5.2.1(ii), and $\varepsilon_1(u) \to 0$. For the third, $\|v(u)\| \le \big(\|S\| + \|\varepsilon_1(u)\|\big)\,\|u\|$ stays bounded relative to $\|u\|$, while $v(u) \to 0$ forces $\varepsilon_2(v(u)) \to 0$. Hence the quotient tends to 0 and $D(\varphi \circ f)(x_0) = TS$. ∎

This is Theorem 11.1 of Fukuda (2026d, p. 380). Identity (8.6) is the rule "sum over all channels through which x_i affects the value", and it is the formal content of every total-derivative computation in comparative statics: when a policy parameter enters an equilibrium condition both directly and through an endogenous variable, (8.6) is what enumerates the channels.

Proposition 8.3.4 (Gradients are orthogonal to level sets)**.** Let $f\colon A \to \mathbb{R}$ be differentiable at x_0 and let $\gamma\colon (-\delta, \delta) \to A$ be a differentiable curve with $\gamma(0) = x_0$ lying in the level set $\{x : f(x) = f(x_0)\}$. Then $\langle \nabla f(x_0), \gamma'(0)\rangle = 0$. Moreover, among unit vectors v, the directional derivative $\langle \nabla f(x_0), v\rangle$ is maximised at $v = \nabla f(x_0)/\,\|\nabla f(x_0)\|$ when $\nabla f(x_0) \neq 0$.

Proof. The function $t \mapsto f(\gamma(t))$ is constant, so by Theorem 8.3.1 its derivative $\langle \nabla f(x_0), \gamma'(0)\rangle$ vanishes. The second claim is the equality case of Cauchy–Schwarz (Lemma 3.2.1): $\langle \nabla f(x_0), v\rangle \le \|\nabla f(x_0)\|$ with equality exactly when v is the stated unit vector. ∎

The economic reading is that the gradient of a utility function is orthogonal to the indifference curve and points in the direction of steepest preference increase, and that the marginal rate of substitution between goods i and j is the ratio $\big(\partial u/\partial x_i\big)/\big(\partial u/\partial x_j\big)$ because the level set is orthogonal to the gradient. This is Proposition 11.6 of Fukuda (2026d, p. 385).

8.3.2 Homogeneous functions

Definition 8.3.4 (Homogeneity)**.** Let $X \subseteq \mathbb{R}^n$ be a cone, that is $\lambda x \in X$ whenever $x \in X$ and $\lambda > 0$. A function $f\colon X \to \mathbb{R}$ is *homogeneous of degree k* if $f(\lambda x) = \lambda^k f(x)$ for all $x \in X$ and $\lambda > 0$.

Theorem 8.3.2 (Euler)**.** Let $X \subseteq \mathbb{R}^n$ be an open cone and $f\colon X \to \mathbb{R}$ differentiable. Then f is homogeneous of degree k if and only if

$$\langle \nabla f(x), x\rangle = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x)\, x_i = k\, f(x) \qquad \text{for all } x \in X. \tag{8.7}$$

Moreover, if f is homogeneous of degree k then each partial derivative $\partial f/\partial x_i$ is homogeneous of degree $k-1$.

Proof. ($\Rightarrow$) Differentiate $f(\lambda x) = \lambda^k f(x)$ with respect to λ using Theorem 8.3.1: $\langle \nabla f(\lambda x), x\rangle = k\lambda^{k-1} f(x)$. Setting $\lambda = 1$ gives (8.7). Differentiating instead with respect to x_i gives $\lambda\, \partial_i f(\lambda x) = \lambda^k \partial_i f(x)$, which is homogeneity of degree $k-1$ of $\partial_i f$.

($\Leftarrow$) Fix x and set $g(\lambda) := \lambda^{-k} f(\lambda x)$ for $\lambda > 0$. By Theorem 8.3.1 and (8.7) applied at λx,

$$g'(\lambda) = -k\lambda^{-k-1} f(\lambda x) + \lambda^{-k}\,\langle \nabla f(\lambda x), x\rangle = -k\lambda^{-k-1} f(\lambda x) + \lambda^{-k-1}\, k f(\lambda x) = 0,$$

using $\langle \nabla f(\lambda x), \lambda x\rangle = k f(\lambda x)$. So g is constant on $(0, \infty)$ by Theorem 7.2.5, and $g(1) = f(x)$ gives $f(\lambda x) = \lambda^k f(x)$. ■

Homogeneity of degree 1 of a production function is constant returns to scale, and (8.7) then reads $Y = F_K K + F_L L$: output is exhausted by payments to factors at their marginal products. This is Definition 11.3 of Fukuda (2026d, p. 385), and it is the identity behind the factor shares computed in Section 8.7.

8.4 Optimisation Without Constraints

Theorem 8.4.1 (Fermat, several variables)**.** Let $A \subseteq \mathbb{R}^n$, let x^* be an interior point of A at which $f\colon A \to \mathbb{R}$ is differentiable, and suppose x^* is a local maximiser or local minimiser of f on A. Then $\nabla f(x^*) = 0$.

Proof. Fix i and consider the section $g_i(t) := f(x^* + te_i)$, defined for $|t|$ small because x^* is interior. Then g_i is differentiable at 0 with $g_i'(0) = \partial f/\partial x_i(x^*)$ by Proposition 8.3.2, and 0 is a local extremum of g_i. By Theorem 7.2.1, $g_i'(0) = 0$. As i was arbitrary, $\nabla f(x^*) = 0$. ■

Theorem 8.4.2 (Sufficiency under concavity)**.** Let $C \subseteq \mathbb{R}^n$ be convex, $f\colon C \to \mathbb{R}$ continuous on C, differentiable on $\operatorname{Int} C$, and concave. Then $x^* \in \operatorname{Int} C$ is a global maximiser of f on C if and only if $\nabla f(x^*) = 0$. The convex case with minimisers is obtained by replacing f with $-f$.

Proof. Necessity is Theorem 8.4.1. For sufficiency, the multivariate analogue of Theorem 7.2.2(ii), proved as Theorem 9.3.1, gives $f(x) \le f(x^*) + \langle \nabla f(x^*), x - x^*\rangle = f(x^*)$ for all $x \in C$, first for $x \in \operatorname{Int} C$ and then for $x \in C$ by continuity. ■

This is the content of (Fukuda 2026a, slide 52).

Example 8.4.1 (The four canonical surfaces)**.** At the origin, all four functions below have vanishing gradient (Fukuda 2026a, slide 53).

(i) $f(x, y) = x^2 + y^2$: a strict global minimum. $\mathrm{D}^2 f = 2I \succ 0$.

(ii) $f(x, y) = -x^2 - y^2$: a strict global maximum. $\mathrm{D}^2 f = -2I \prec 0$.

(iii) $f(x, y) = x^2 - y^2$: a saddle. $\mathrm{D}^2 f = \operatorname{diag}(2, -2)$ is indefinite; the origin is a minimum along the x-axis and a maximum along the y-axis, so it is neither.

(iv) $f(x, y) = -x^3$: neither, and the Hessian at the origin is the zero matrix, which is both positive and negative semidefinite. The second-order test is silent, and the third derivative decides.

Cases (iii) and (iv) are the two ways in which a stationary point can fail to be an extremum, and they are distinguished by whether the Hessian is indefinite (the test rules out an extremum) or merely singular (the test is inconclusive).

8.5 Second-Order Theory

Definition 8.5.1 (Hessian). Let $A \subseteq \mathbb{R}^n$ be open and $f\colon A \to \mathbb{R}$ twice differentiable at $x \in A$. The *Hessian* of f at x is the matrix of second partial derivatives

$$\mathrm{D}^2 f(x) := \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x)\right)_{i,j=1}^{n} \in M_{n\times n}(\mathbb{R}),$$

also written $Hf(x)$ or $\nabla^2 f(x)$; it is the Jacobian of the map $x \mapsto \nabla f(x)$.

Theorem 8.5.1 (Schwarz). If $f \in C^2(A)$ then $\mathrm{D}^2 f(x)$ is symmetric for every $x \in A$: $\partial^2 f/\partial x_i \partial x_j = \partial^2 f/\partial x_j \partial x_i$.

Symmetry is not automatic without the continuity hypothesis; the standard counterexample is $f(x,y) = xy(x^2-y^2)/(x^2+y^2)$ with $f(0,0)=0$, for which the two mixed partials at the origin are -1 and $+1$. See Rudin (1976, Thm. 9.41). Symmetry of the Hessian is what produces the Slutsky symmetry conditions and the reciprocity relations that make the Hessian of an expenditure function testable.

Theorem 8.5.2 (Second-order Taylor expansion). Let $A \subseteq \mathbb{R}^n$, let $\bar{x} \in \operatorname{Int} A$, and let $f\colon A \to \mathbb{R}$ be twice differentiable at $\bar{x}$. Then

$$f(x) = f(\bar{x}) + \langle \nabla f(\bar{x}), x-\bar{x}\rangle + \tfrac{1}{2}(x-\bar{x})^{\mathsf{T}} \mathrm{D}^2 f(\bar{x})(x-\bar{x}) + \|x-\bar{x}\|^2 \alpha_{\bar{x}}(x-\bar{x}), \tag{8.8}$$

where $\alpha_{\bar{x}}(x-\bar{x}) \to 0$ as $x \to \bar{x}$. If $f \in C^2$ on a neighbourhood of the segment $[\bar{x}, x]$, there is $\theta \in (0,1)$ with the exact expression

$$f(x) = f(\bar{x}) + \langle \nabla f(\bar{x}), x-\bar{x}\rangle + \tfrac{1}{2}(x-\bar{x})^{\mathsf{T}} \mathrm{D}^2 f\big(\bar{x}+\theta(x-\bar{x})\big)(x-\bar{x}).$$

Proof. For the exact form, fix x and set $\varphi(t) := f\big(\bar{x}+t(x-\bar{x})\big)$ for $t \in [0,1]$. By Theorem 8.3.1, $\varphi'(t) = \langle \nabla f(\bar{x}+t(x-\bar{x})), x-\bar{x}\rangle$ and, applying the chain rule again to the gradient map, $\varphi''(t) = (x-\bar{x})^{\mathsf{T}} \mathrm{D}^2 f(\bar{x}+t(x-\bar{x}))(x-\bar{x})$. Applying the one-variable Taylor theorem (Theorem 7.4.1 with $n=1$) to φ on $[0,1]$ gives $\varphi(1) = \varphi(0) + \varphi'(0) + \frac{1}{2}\varphi''(\theta)$ for some $\theta \in (0,1)$, which is the claim. The Peano form (8.8) follows from twice differentiability at the single point $\bar{x}$ by the same argument applied with Corollary 7.4.1. ■

This is Theorem 11.5 of Fukuda (2026d, p. 413).

Theorem 8.5.3 (Second-order conditions). Let $A \subseteq \mathbb{R}^n$ be open, $f \in C^2(A)$, and $x^* \in A$ with $\nabla f(x^*) = 0$.

(i) If $\mathrm{D}^2 f(x^*) \prec 0$, then x^* is a strict local maximiser.

(ii) If $\mathrm{D}^2 f(x^*) \succ 0$, then x^* is a strict local minimiser.

(iii) If $\mathrm{D}^2 f(x^*)$ is indefinite, then x^* is neither.

(iv) If x^* is a local maximiser then $\mathrm{D}^2 f(x^*) \preceq 0$; if a local minimiser then $\mathrm{D}^2 f(x^*) \succeq 0$.

Proof. (i) Let $H := \mathrm{D}^2 f(x^*) \prec 0$. The function $v \mapsto v^{\mathsf{T}} H v$ is continuous on the compact unit sphere, so by Corollary 5.4.1 it attains a maximum -2μ there, and $\mu > 0$ by negative definiteness. By homogeneity, $u^{\mathsf{T}} H u \le -2\mu \|u\|^2$ for all u. From (8.8) with $u = x - x^*$ and $\nabla f(x^*) = 0$,

$$f(x) - f(x^*) \le -\mu \|u\|^2 + \|u\|^2 \alpha(u) = \|u\|^2 \big(\alpha(u) - \mu\big),$$

which is negative for $u \neq 0$ small enough that $|\alpha(u)| < \mu$. Part (ii) applies (i) to $-f$.

(iii) Indefiniteness supplies v_+ with $v_+^{\mathsf{T}} H v_+ > 0$ and v_- with $v_-^{\mathsf{T}} H v_- < 0$. Restricting f to the line $t \mapsto x^* + tv_+$ and applying (8.8) shows $f(x^* + tv_+) > f(x^*)$ for small $t \neq 0$, and symmetrically along v_-; so x^* is neither a local maximum nor a local minimum.

(iv) If x^* is a local maximiser and some v had $v^\mathsf{T} H v > 0$, the argument of (iii) along v would contradict maximality. ■

The gap between the sufficient conditions (i)–(ii) and the necessary condition (iv) is exactly the semidefinite case, and it cannot be closed: $f(x) = x^3$ and $f(x) = x^4$ both have $f''(0) = 0$ and differ in their behaviour. Definiteness is checked by the eigenvalue criterion (Proposition 2.5.1) or by Sylvester's criterion (Proposition 2.5.2), with the warning of Remark 2.5.2 about the semidefinite case.

Theorem 8.5.4 (Hessian characterisation of convexity)**.** Let $A \subseteq \mathbb{R}^n$ be open and convex and $f \in C^2(A)$. Then f is convex on A if and only if $\mathrm{D}^2 f(x) \succeq 0$ for every $x \in A$, and concave if and only if $\mathrm{D}^2 f(x) \preceq 0$ for every $x \in A$. If $\mathrm{D}^2 f(x) \succ 0$ everywhere then f is strictly convex; the converse fails, as $f(x) = x^4$ shows.

Proof. Convexity of f on the convex set A is equivalent to convexity of $\varphi_{x,y}(t) := f\big(x + t(y-x)\big)$ on $[0,1]$ for every $x, y \in A$, since the two statements are the same inequality. By Theorem 7.2.2 applied to $-\varphi_{x,y}$, that holds if and only if $\varphi''_{x,y} \geq 0$, and $\varphi''_{x,y}(t) = (y-x)^\mathsf{T}\, \mathrm{D}^2 f(x + t(y-x))(y-x)$ as computed in the proof of Theorem 8.5.2. Ranging over $x, y \in A$ makes $y - x$ an arbitrary direction and $x + t(y-x)$ an arbitrary point of A, which is exactly $\mathrm{D}^2 f \succeq 0$ on A. ■

This is Theorem 11.6 of Fukuda (2026d, p. 415).

8.6 The Implicit and Inverse Function Theorems

Theorem 8.6.1 (Implicit function theorem)**.** Let $\Lambda \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open and $f\colon \Lambda \to \mathbb{R}^m$ continuously differentiable. Let $(\bar{x}, \bar{y}) \in \Lambda$ satisfy $f(\bar{x}, \bar{y}) = 0$ and suppose the $m \times m$ matrix $\nabla_y f(\bar{x}, \bar{y})$ is non-singular. Then there are an open neighbourhood $\Gamma \subseteq \mathbb{R}^n$ of $\bar{x}$ and a continuously differentiable map $e\colon \Gamma \to \mathbb{R}^m$ such that

(i) $(x, e(x)) \in \Lambda$ and $f\big(x, e(x)\big) = 0$ for all $x \in \Gamma$;

(ii) $e(\bar{x}) = \bar{y}$;

(iii) $\nabla_y f\big(x, e(x)\big)$ is non-singular for all $x \in \Gamma$; and

(iv) for all $x \in \Gamma$,
$$De(x) = -\big[\nabla_y f\big(x, e(x)\big)\big]^{-1} \nabla_x f\big(x, e(x)\big). \tag{8.9}$$

Moreover e is the unique such function on a possibly smaller neighbourhood: the solutions of $f(x, y) = 0$ near $(\bar{x}, \bar{y})$ are exactly the points $(x, e(x))$.

This is Theorem 11.7 of Fukuda (2026d, p. 419). The proof constructs e as the fixed point of a contraction: for each x near $\bar{x}$ the map $y \mapsto y - [\nabla_y f(\bar{x}, \bar{y})]^{-1} f(x, y)$ is a contraction on a small closed ball, by continuity of Df, and Theorem 6.4.1 applies on that complete set, with Corollary 6.4.2 keeping the fixed point inside the ball; the regularity of e then follows from the regularity of f. It is carried out in full in Appendix A, where it is placed because it is long and purely technical, and because it is the use of the contraction mapping theorem advertised on (Fukuda 2026g, slide 65); the classical treatment is Rudin (1976, Thm. 9.28).

Formula (8.9) is not an extra: it is obtained by differentiating the identity $f(x, e(x)) = 0$ with respect to x using Theorem 8.3.1,
$$\nabla_x f\big(x, e(x)\big) + \nabla_y f\big(x, e(x)\big) De(x) = 0,$$
and solving, which is legitimate because (iii) guarantees the inverse exists. The differentiation is valid only once e is known to be differentiable, which is why (iv) is a conclusion of the theorem and not a derivation of it.

Corollary 8.6.1 (Inverse function theorem)**.** Let $A \subseteq \mathbb{R}^n$ be open, $g\colon A \to \mathbb{R}^n$ continuously differentiable, and $\bar{x} \in A$ with $Dg(\bar{x})$ non-singular. Then there are open neighbourhoods $U \ni \bar{x}$ and $V \ni g(\bar{x})$ such that $g|_U : U \to V$ is a bijection with continuously differentiable inverse, and
$$D\big(g^{-1}\big)(y) = \big[Dg\big(g^{-1}(y)\big)\big]^{-1}, \qquad y \in V.$$

Proof. Apply Theorem 8.6.1 to $f(y,x) := g(x) - y$ on $\mathbb{R}^n \times A$, at the point $(g(\bar{x}), \bar{x})$. Here the roles are exchanged: the "parameter" is y and the "unknown" is x, and $\nabla_x f = Dg(\bar{x})$ is non-singular by hypothesis. The implicit function e is the local inverse, and (8.9) gives $De(y) = -[Dg]^{-1}(-I) = [Dg]^{-1}$. ■

For $n = 1$ this is Proposition 7.3.5, with $Dg(\bar{x}) \neq 0$ in place of $f'(x_0) \neq 0$. The theorem is local and cannot be made global: the map $g(x_1, x_2) = (e^{x_1}\cos x_2, e^{x_1}\sin x_2)$ has everywhere non-singular Jacobian and is not injective, being 2π-periodic in x_2.

Remark 8.6.1 (Comparative statics is the implicit function theorem)**.** The standard comparative-statics exercise has this form. An endogenous vector $y \in \mathbb{R}^m$ and an exogenous vector $x \in \mathbb{R}^n$ satisfy m equilibrium conditions $f(x,y) = 0$ — first-order conditions, market-clearing conditions, or both. At a reference equilibrium $(\bar{x}, \bar{y})$ one asks how y responds to x. The theorem answers three separate questions.

Does an equilibrium exist nearby? Yes, and it varies continuously: this is conclusion (i)–(ii).

Is it locally unique? Yes: this is the uniqueness clause, and it is what makes "the" comparative static well posed. Uniqueness is local only; the theorem says nothing about other equilibria far away.

What is the response? It is (8.9). The economic content is in the two factors. The matrix $\nabla_x f$ collects the direct effects of the parameters on the equilibrium conditions; the matrix $[\nabla_y f]^{-1}$ propagates them through the simultaneous system. When $f = \nabla_y \Pi$ for an objective Π, the propagating matrix is the inverse Hessian, and Theorem 8.5.3 makes it negative definite at a maximum — which is the source of the standard sign restrictions on comparative statics.

The condition that can fail is non-singularity of $\nabla_y f$. What follows from its failure is only that the theorem is silent: the hypotheses no longer deliver a locally unique solution branch, and (8.9) has no meaning because the matrix to be inverted is singular. It does *not* follow that the equilibrium is non-unique, or that comparative statics fail to exist. Take $n = m = 1$ and $f(x,y) = y^3 - x$ at $(\bar{x}, \bar{y}) = (0,0)$: here $\partial f/\partial y = 3y^2$ vanishes at the origin, yet $f(x,y) = 0$ has the unique global solution $y = x^{1/3}$, which is continuous everywhere and differentiable away from 0. The correct reading is that a singular $\nabla_y f$ marks the points at which the implicit function theorem stops certifying anything, so that uniqueness, smoothness and the sign of the response must be established by other means — and it is among such points, not at every such point, that multiplicity and bifurcation can occur.

8.7 Economic Application: The Representative Firm

Environment. A firm rents capital $K \geq 0$ at rate $r > 0$ and hires labour $L \geq 0$ at wage $w > 0$, producing $Y = F(K,L) = AK^\alpha L^{1-\alpha}$ with $A > 0$ and $\alpha \in (0,1)$. Output is the numeraire. The firm solves (8.1).

Proposition 8.7.1 (First-order conditions and factor shares)**.** Let (K^*, L^*) be an interior solution of (8.1). Then

$$r = \frac{\partial Y}{\partial K} = \alpha A \left(\frac{K^*}{L^*}\right)^{\alpha-1}, \qquad w = \frac{\partial Y}{\partial L} = (1-\alpha)A\left(\frac{K^*}{L^*}\right)^{\alpha}, \tag{8.10}$$

and consequently

$$\alpha = \frac{rK^*}{Y^*}, \qquad 1-\alpha = \frac{wL^*}{Y^*}, \qquad \text{and} \qquad \Pi(K^*, L^*) = 0. \tag{8.11}$$

Proof. Π is differentiable on $\mathbb{R}^2_{++}$ and (K^*, L^*) is interior, so Theorem 8.4.1 gives $\nabla\Pi = 0$, which is (8.10) after computing the partials as in Example 8.2.1. Multiplying the first equation by K^* gives $rK^* = \alpha A(K^*)^\alpha (L^*)^{1-\alpha} = \alpha Y^*$, and similarly for labour. Adding, $rK^* + wL^* = Y^*$, so profit is zero. ■

Reading the result. Three points, each with a mathematical source.

The zero-profit conclusion is Theorem 8.3.2 with $k = 1$: F is homogeneous of degree 1, so $\langle \nabla F, (K,L)\rangle = F(K,L)$, and payment of each factor at its marginal product exhausts output exactly. Nothing about Cobb–Douglas is used beyond constant returns.

The factor shares in (8.11) are constants, independent of r, w and A. This is special to Cobb–Douglas, and it is the reason the function is used: it is the production function whose elasticity of substitution between capital and labour equals 1, and constancy of shares is equivalent to that.

The scale of the firm is indeterminate. Equations (8.10) involve K^* and L^* only through their ratio, because the marginal products are homogeneous of degree 0 by the second part of Theorem 8.3.2. Given (r, w), either no ratio satisfies both equations, or one does and then every scalar multiple of (K^*, L^*) is also optimal. Formally, the Hessian

$$\mathrm{D}^2\,\Pi = \mathrm{D}^2\,F = \alpha(1-\alpha)\frac{Y}{KL}\begin{pmatrix} -L/K & 1 \\ 1 & -K/L \end{pmatrix}$$

has determinant 0 and trace $-\alpha(1-\alpha)\frac{Y}{KL}(L/K + K/L) < 0$, so it is negative *semi*definite with a one-dimensional null space spanned by (K, L) — the direction of proportional scaling. The maximum is therefore not strict, and Theorem 8.5.3(i) does not apply. This is not a defect of the mathematics but the correct statement that a constant-returns firm in a competitive market has an indeterminate size: only the aggregate is pinned down, by demand. This is the optional exercise of (Fukuda 2026a, slides 43, 54).

8.8 Economic Application: Least Squares by Calculus

Problem. Given data $(x_i, y_i)_{i=1}^n$ with at least two distinct x_i, minimise the sum of squared residuals (8.2).

Proposition 8.8.1. Let $S(\beta_1, \beta_2) := \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$. If the x_i are not all equal, then S has a unique global minimiser, given by

$$\hat{\beta}_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}, \tag{8.12}$$

where $\bar{x} = n^{-1}\sum_i x_i$ and $\bar{y} = n^{-1}\sum_i y_i$.

Proof. S is a polynomial, hence C^∞. Its partial derivatives are

$$\frac{\partial S}{\partial \beta_1} = -2\sum_i (y_i - \beta_1 - \beta_2 x_i), \qquad \frac{\partial S}{\partial \beta_2} = -2\sum_i x_i(y_i - \beta_1 - \beta_2 x_i),$$

so $\nabla S = 0$ is the pair of *normal equations*

$$n\beta_1 + \Big(\sum_i x_i\Big)\beta_2 = \sum_i y_i, \qquad \Big(\sum_i x_i\Big)\beta_1 + \Big(\sum_i x_i^2\Big)\beta_2 = \sum_i x_i y_i. \tag{8.13}$$

The Hessian is

$$\mathrm{D}^2\,S = 2\begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix},$$

constant in β, with $\det \mathrm{D}^2\,S = 4\big(n\sum_i x_i^2 - (\sum_i x_i)^2\big) = 4n\sum_i (x_i - \bar{x})^2 > 0$ when the x_i are not all equal, and leading entry $2n > 0$. By Proposition 2.5.2, $\mathrm{D}^2\,S \succ 0$, so S is strictly convex by Theorem 8.5.4 and the unique stationary point is the global minimiser by Theorem 8.4.2. Solving (8.13) by Cramer's rule (Example 2.4.2) yields (8.12). ■

Scope of the calculus route. The calculus route delivers existence, uniqueness and a formula, and it makes the role of the data condition explicit: $\sum_i (x_i - \bar{x})^2 > 0$ is simultaneously the non-degeneracy condition on the regressors, the invertibility condition for the normal equations, and the positive definiteness condition for the Hessian. It is the two-variable case of the no-perfect-multicollinearity condition of Remark 2.3.1.

What the calculus route does not deliver is the interpretation. Formula (8.12) says that $\hat{\beta}_2$ is the ratio of a sample covariance to a sample variance, and Example 3.2.2 identified that ratio as a cosine rescaled by norms. The geometry behind it — that the fitted values are the orthogonal projection of y onto the span of the regressors, that the residual is orthogonal to that span, and that the normal equations (8.13) *are* the orthogonality conditions — is developed in Chapter 11, where the same estimator is obtained without differentiating anything and in a form that survives to infinite dimensions (Fukuda 2026a, slides 55–64).

8.9 Chapter Summary

Partial derivatives differentiate along the coordinate axes; the total derivative requires a single linear map to approximate the increment in every direction at once. In one variable the two notions coincide; in several they do not, and the gap is genuine: a function can have all partial derivatives at a point and be discontinuous there (Example 8.3.1), be continuous with all partials and still lack directional derivatives off the axes (Example 8.3.2), or possess every directional derivative and still fail to be differentiable because those derivatives are not linear in the direction (Example 8.3.3). Continuity of the partials is sufficient (Proposition 8.3.3), which is why C^1 is the standard working hypothesis. Differentiability delivers continuity, all directional derivatives, the chain rule in matrix form (8.5), orthogonality of the gradient to level sets, and Euler's identity for homogeneous functions.

At an interior local extremum of a differentiable function the gradient vanishes; under concavity the condition is also sufficient and global. The second-order theory rests on the Hessian: symmetric under C^2 by Theorem 8.5.1, appearing as the quadratic term of the Taylor expansion (8.8), and determining local behaviour at a stationary point through definiteness, with the semidefinite case inconclusive. Global convexity or concavity is exactly semidefiniteness of the Hessian everywhere.

The implicit function theorem converts a system of equilibrium conditions into a locally unique, continuously differentiable solution mapping with derivative (8.9); it is the theorem underlying differentiable comparative statics, and where its hypothesis — non-singularity of the Jacobian in the endogenous variables — fails, the theorem certifies nothing, though a well-behaved solution branch may still exist (Remark 8.6.1). Applied to the constant-returns firm, the first-order conditions determine the capital–labour ratio, exhaust output through Euler's identity, and leave scale indeterminate, the last fact readable from the singular Hessian.

Exercise 8.9.1. Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ take the value 1 on the region $\{(x, y) : 0 < y < x^2\}$ and the value 0 elsewhere. Show that for every $v \in \mathbb{R}^2$ the directional derivative $D_v f(0, 0)$ exists and equals 0, so that $v \mapsto D_v f(0, 0)$ is linear, and yet f is not continuous at the origin. (Consider the curve $t \mapsto (t, t^2/2)$.) Conclude that neither the existence of all directional derivatives nor their linearity in v implies differentiability, and locate the step of Proposition 8.3.2 that this example does not contradict.

Exercise 8.9.2. Let F be C^2, homogeneous of degree 1, and strictly quasiconcave on $\mathbb{R}^2_{++}$. Show that $\mathrm{D}^2 F(K, L)$ is singular with (K, L) in its kernel, and deduce that no interior strict local maximum of $F(K, L) - rK - wL$ exists. Then show that the cost-minimisation problem $\min\{rK + wL : F(K, L) \geq \bar{Y}\}$ does have a unique solution, and explain the difference.

Exercise 8.9.3. A monopolist chooses q facing inverse demand $P(q, \theta)$ with a demand shifter θ, and cost $c(q)$. Write the first-order condition as $f(\theta, q) = 0$ and use (8.9) to compute $dq^*/d\theta$. State the exact sign restriction on $\partial^2\Pi/\partial q\, \partial\theta$ that makes the response positive, and verify that the second-order condition supplies the sign of the denominator. Relate the result to Remark 8.6.1.

Chapter 9

Convexity and Separation

9.1 Motivation: The Uses of Convexity

Three consequences of convexity are used throughout the volume, and nothing else delivers them.

It converts local information into global information. A stationary point of a concave function is a global maximiser (Theorem 8.4.2), so the first-order conditions that Chapter 8 derived as necessary become sufficient, and the search for an optimum reduces to solving equations.

It delivers uniqueness. Strict concavity of the objective on a convex feasible set makes the solution a point rather than a set (Proposition 9.3.5), which is what turns a demand correspondence into a demand function and makes comparative statics well posed.

It produces prices. The separation theorems say that two disjoint convex sets can be told apart by a linear functional — a vector of coefficients, one per coordinate. When the coordinates are commodities, that vector is a price system, and the existence of supporting prices in the second welfare theorem, of state prices under no arbitrage, and of Lagrange multipliers in Chapter 12 are three instances of one theorem.

9.2 Convex Sets

Definition 9.2.1 (Convex set). A subset C of a real vector space X is *convex* if

$$\lambda x + (1-\lambda)y \in C \qquad \text{for all } x, y \in C \text{ and } \lambda \in [0,1].$$

Equivalently, C contains the segment joining any two of its points. The empty set and every singleton are convex, vacuously and trivially.

Example 9.2.1 (Convex sets in economics). (i) The budget set $\mathcal{B}(p,m) = \{x \in \mathbb{R}^n_+ : \langle p, x\rangle \le m\}$ is convex, being the intersection of the half-space $\{\langle p, x\rangle \le m\}$ with the cone $\mathbb{R}^n_+$.

(ii) The unit simplex $\Delta_n = \{p \in \mathbb{R}^n : \sum_i p_i = 1,\ p_i \ge 0\}$, the set of probability distributions on n states, is convex and compact.

(iii) An upper contour set $\{x : u(x) \ge u(x_0)\}$ is convex precisely when u is quasiconcave (Definition 9.3.2); this is the formal content of "convex preferences".

(iv) A production set $Y \subseteq \mathbb{R}^n$ is convex when the technology admits time-sharing between any two feasible plans, and convexity of Y is what excludes increasing returns.

Proposition 9.2.1 (Operations preserving convexity). Let $C, D \subseteq \mathbb{R}^n$ be convex, $\lambda, \mu \in \mathbb{R}$, and $T\colon \mathbb{R}^n \to \mathbb{R}^m$ linear. Then

(i) $\bigcap_{i\in I} C_i$ is convex for any family of convex sets;

(ii) $\lambda C + \mu D := \{\lambda x + \mu y : x \in C,\ y \in D\}$ is convex;

(iii) $T(C) \subseteq \mathbb{R}^m$ and $T^{-1}(E) \subseteq \mathbb{R}^n$ are convex, the latter for any convex $E \subseteq \mathbb{R}^m$;

(iv) $\operatorname{Int} C$ and $\overline{C}$ are convex.

Proof. (i) If $x, y \in \bigcap_i C_i$ then the segment lies in each C_i.

(ii) Let $z_k = \lambda x_k + \mu y_k$ for $k = 1, 2$ and $\theta \in [0, 1]$. Then $\theta z_1 + (1-\theta) z_2 = \lambda\big(\theta x_1 + (1-\theta)x_2\big) + \mu\big(\theta y_1 + (1-\theta) y_2\big)$, and both brackets lie in the respective sets.

(iii) Immediate from linearity of T, which commutes with convex combinations.

(iv) For the closure: let $x, y \in \overline{C}$ and take $(x_k), (y_k) \subseteq C$ with $x_k \to x$, $y_k \to y$ (Theorem 4.3.1). Then $\lambda x_k + (1-\lambda) y_k \in C$ converges to $\lambda x + (1-\lambda) y$, which is therefore in $\overline{C}$. For the interior: let $x, y \in \operatorname{Int} C$ with $B_\varepsilon(x), B_\varepsilon(y) \subseteq C$ and set $z = \lambda x + (1-\lambda) y$. For $\|u\| < \varepsilon$, $z + u = \lambda(x+u) + (1-\lambda)(y+u) \in C$, so $B_\varepsilon(z) \subseteq C$. ■

This is Proposition 6.3 and the results of Section 7.6 of Fukuda (2026d). Part (i) is the one used most often: a feasible set described by a list of convex constraints is convex, however long the list, and part (ii) is what makes the aggregate of convex production sets convex.

Definition 9.2.2 (Convex combination and convex hull)**.** A point $a \in \mathbb{R}^n$ is a *convex combination* of $x_1, \ldots, x_k$ if $a = \sum_{i=1}^k \lambda_i x_i$ with $\lambda_i \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$. The *convex hull* $\operatorname{co} A$ of $A \subseteq \mathbb{R}^n$ is the set of all convex combinations of finitely many points of A; equivalently, by Proposition 9.2.1(i), the intersection of all convex sets containing A, hence the smallest such set.

Theorem 9.2.1 (Carathéodory)**.** Let $A \subseteq \mathbb{R}^n$. Every point of $\operatorname{co} A$ is a convex combination of at most $n+1$ points of A.

Proof. Let $x = \sum_{i=1}^k \lambda_i x_i$ with $x_i \in A$, $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and suppose $k \geq n+2$. The $k-1 \geq n+1$ vectors $x_2 - x_1, \ldots, x_k - x_1$ lie in $\mathbb{R}^n$ and are therefore linearly dependent: there are $\mu_2, \ldots, \mu_k$, not all zero, with $\sum_{i=2}^k \mu_i (x_i - x_1) = 0$. Setting $\mu_1 := -\sum_{i \geq 2} \mu_i$ gives $\sum_{i=1}^k \mu_i x_i = 0$ and $\sum_{i=1}^k \mu_i = 0$, with some $\mu_i > 0$ because they sum to zero and are not all zero.

For $t \geq 0$ the coefficients $\lambda_i - t\mu_i$ still sum to 1 and still represent x. Let $t^* := \min\{\lambda_i/\mu_i : \mu_i > 0\}$, attained at some index j. Then $\lambda_i - t^*\mu_i \geq 0$ for every i and equals 0 at $i = j$, so x is a convex combination of the remaining $k-1$ points. Iterate until $k \leq n+1$. ■

This is Theorem 6.1 of Fukuda (2026d, p. 190). The bound $n+1$ is sharp — a point in the interior of a simplex in $\mathbb{R}^n$ needs all $n+1$ vertices — and the theorem is the reason that finitely many agents suffice to generate any aggregate consumption bundle in an n-commodity economy.

Lemma 9.2.1 (Nearest point in a closed convex set)**.** Let $\emptyset \neq D \subseteq \mathbb{R}^n$ be closed and convex and let $x \in \mathbb{R}^n$. Then there is exactly one $P_D x \in D$ with

$$\|x - P_D x\| = \operatorname{dist}(x, D) := \inf_{y \in D} \|x - y\|,$$

and $P_D x$ is characterised by the variational inequality

$$\langle x - P_D x, y - P_D x \rangle \leq 0 \qquad \text{for all } y \in D. \tag{9.1}$$

Proof. Existence. Fix $y_0 \in D$ and set $R := \|x - y_0\|$. The set $D \cap \overline{B_R(x)}$ is non-empty, closed and bounded, hence compact by Theorem 4.4.2, and the continuous function $y \mapsto \|x - y\|$ attains its minimum there by Corollary 5.4.1. That minimum is the infimum over all of D, since points outside the ball are further away than y_0.

Uniqueness. Let $d := \operatorname{dist}(x, D)$ and suppose $\|x - y_1\| = \|x - y_2\| = d$ with $y_1 \neq y_2$. The midpoint $\bar{y} = \frac{1}{2}(y_1 + y_2)$ lies in D by convexity, and the parallelogram law (Remark 3.3.1) gives

$$\|x - \bar{y}\|^2 = \tfrac{1}{2}\|x - y_1\|^2 + \tfrac{1}{2}\|x - y_2\|^2 - \tfrac{1}{4}\|y_1 - y_2\|^2 = d^2 - \tfrac{1}{4}\|y_1 - y_2\|^2 < d^2,$$

contradicting minimality of d.

Characterisation. Let $z := P_D x$, $y \in D$ and $t \in (0, 1]$. Then $z + t(y - z) \in D$, so

$$\|x - z\|^2 \leq \|x - z - t(y - z)\|^2 = \|x - z\|^2 - 2t\,\langle x - z, y - z \rangle + t^2\,\|y - z\|^2.$$

Cancelling, dividing by $t > 0$ and letting $t \downarrow 0$ gives (9.1). Conversely, if $z \in D$ satisfies (9.1), then for any $y \in D$,

$$\|x-y\|^2 = \|x-z\|^2 - 2\langle x-z, y-z\rangle + \|y-z\|^2 \geq \|x-z\|^2,$$

so z is the minimiser. ■

This is the finite-dimensional case of Theorem 9.1 of Fukuda (2026d, p. 295). The Hilbert-space version, in which compactness is unavailable and completeness does the work instead, is Theorem 11.3.1; the uniqueness and variational arguments above carry over verbatim, and only the existence argument changes.

9.3 Convex and Concave Functions

Definition 9.3.1 (Convexity and concavity). Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : C \to \mathbb{R}$ is *convex* if

$$f\big(\lambda x + (1-\lambda)y\big) \leq \lambda f(x) + (1-\lambda) f(y) \qquad \text{for all } x, y \in C,\ \lambda \in [0,1],$$

strictly convex if the inequality is strict for $x \neq y$ and $\lambda \in (0,1)$, and *concave* (respectively strictly concave) if $-f$ is convex (respectively strictly convex).

Proposition 9.3.1 (Epigraph and Jensen). Let $C \subseteq \mathbb{R}^n$ be convex and $f : C \to \mathbb{R}$.

(i) f is convex if and only if $\operatorname{epi} f = \{(x,z) \in C \times \mathbb{R} : f(x) \leq z\}$ is a convex subset of $\mathbb{R}^{n+1}$.

(ii) If f is convex, then for all $x_1, \ldots, x_k \in C$ and weights $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$,

$$f\Big(\sum_{i=1}^k \lambda_i x_i\Big) \leq \sum_{i=1}^k \lambda_i f(x_i). \tag{9.2}$$

Proof. (i) Suppose f convex and $(x_1,z_1),(x_2,z_2) \in \operatorname{epi} f$. For $\lambda \in [0,1]$, $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2) \leq \lambda z_1 + (1-\lambda) z_2$, so the convex combination is in $\operatorname{epi} f$. Conversely, apply convexity of $\operatorname{epi} f$ to the points $(x_i, f(x_i))$.

(ii) Induction on k. The case $k = 2$ is the definition. For the step, if $\lambda_k < 1$ write $\sum_{i \leq k} \lambda_i x_i = (1-\lambda_k) \sum_{i<k} \frac{\lambda_i}{1-\lambda_k} x_i + \lambda_k x_k$, apply the definition, and then the inductive hypothesis to the inner combination, whose weights sum to 1. ■

Combining Proposition 9.3.1(i) with Proposition 5.3.2 gives the identification announced in Chapter 5: f is convex and lower semicontinuous exactly when $\operatorname{epi} f$ is a closed convex set. Inequality (9.2) is Jensen's inequality for a finitely supported distribution; the general version, for an arbitrary distribution and for conditional expectations, is Proposition 14.4.2.

Proposition 9.3.2 (Operations). Let $C \subseteq \mathbb{R}^n$ be convex.

(i) If f, g are convex and $\alpha, \beta \geq 0$, then $\alpha f + \beta g$ is convex.

(ii) If $(f_i)_{i \in I}$ are convex and $f = \sup_{i \in I} f_i$ is finite on C, then f is convex. The corresponding statement for infima of concave functions holds; suprema of concave functions need not be concave.

(iii) If f is convex and $\varphi : \mathbb{R} \to \mathbb{R}$ is convex and non-decreasing, then $\varphi \circ f$ is convex.

(iv) If T is affine, $f \circ T$ is convex whenever f is.

Proof. (i) and (iv) are immediate from the definition. (ii) is $\operatorname{epi}\big(\sup_i f_i\big) = \bigcap_i \operatorname{epi} f_i$ together with Proposition 9.2.1(i) and Proposition 9.3.1(i). (iii): $\varphi(f(\lambda x + (1-\lambda)y)) \leq \varphi(\lambda f(x) + (1-\lambda) f(y)) \leq \lambda \varphi(f(x)) + (1-\lambda)\varphi(f(y))$, the first step by monotonicity of φ and convexity of f, the second by convexity of φ. Monotonicity cannot be dropped: $\varphi(t) = t^2$ is convex, $f(x) = -x$ is convex, and $\varphi \circ f = x^2$ happens to be convex, but with $f(x) = |x| - 1$ and $\varphi(t) = -t$ the composition is concave. ■

Part (ii) is the source of the profit function's convexity in prices, as Exercise 5.6.3 noted: $\pi(p) = \sup_{y \in Y} \langle p, y\rangle$ is a supremum of linear, hence convex, functions of p, and the conclusion holds for any production set whatsoever, convex or not.

Proposition 9.3.3 (Continuity). A convex function on an open convex set $C \subseteq \mathbb{R}^n$ is continuous on C, and Lipschitz on every compact subset of C.

The proof is a standard argument bounding a convex function on a simplex by its values at the vertices; see Rockafellar (1970, Thm. 10.4). The openness hypothesis is essential: $f(x) = 0$ on $[0, 1)$ and $f(1) = 1$ is convex on $[0, 1]$ and discontinuous at the endpoint. In economics this is why boundary behaviour of convex cost or value functions requires separate attention even when the interior behaviour is smooth.

Theorem 9.3.1 (Differentiable characterisation). Let $C \subseteq \mathbb{R}^n$ be open and convex and $f : C \to \mathbb{R}$ differentiable. The following are equivalent.

(i) f is concave on C.

(ii) $f(x) \leq f(x_0) + \langle \nabla f(x_0), x - x_0 \rangle$ for all $x, x_0 \in C$.

(iii) $\langle \nabla f(x) - \nabla f(x_0), x - x_0 \rangle \leq 0$ for all $x, x_0 \in C$, that is, ∇f is monotone decreasing.

If f is twice differentiable, each is equivalent to $\mathrm{D}^2 f \preceq 0$ on C (Theorem 8.5.4). The convex case is obtained by reversing every inequality.

Proof. Fix $x, x_0 \in C$ and set $\varphi(t) := f\big(x_0 + t(x - x_0)\big)$ on an open interval containing $[0, 1]$. By Theorem 8.3.1, $\varphi'(t) = \langle \nabla f(x_0 + t(x - x_0)), x - x_0 \rangle$.

f is concave on C if and only if every such φ is concave, since the defining inequality for f at the pair (x, x_0) is the inequality for φ at the pair $(1, 0)$ and conversely. By Theorem 7.2.2, φ concave is equivalent to $\varphi(1) \leq \varphi(0) + \varphi'(0)$, which is (ii), and to φ' non-increasing, which evaluated at $t = 0$ and $t = 1$ is (iii). ∎

This is Theorem 11.3 of Fukuda (2026d, p. 388). Statement (ii) is the tangent-plane picture of (Fukuda 2026a, slides 8, 52): a concave function lies below each of its tangent hyperplanes, which is why a horizontal tangent certifies a global maximum.

9.3.1 Quasiconcavity

Definition 9.3.2 (Quasiconcavity). Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : C \to \mathbb{R}$ is *quasiconcave* if every upper contour set $\{x \in C : f(x) \geq \alpha\}$ is convex, equivalently if

$$f\big(\lambda x + (1 - \lambda)y\big) \geq \min\{f(x), f(y)\} \qquad \text{for all } x, y \in C,\ \lambda \in [0, 1].$$

It is *strictly quasiconcave* if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$. Quasiconvexity is the same with lower contour sets and max.

Proposition 9.3.4. Every concave function is quasiconcave, and quasiconcavity is preserved by strictly increasing transformations of the values, whereas concavity is not.

Proof. If f is concave then $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda) f(y) \geq \min\{f(x), f(y)\}$. For the second claim, let ψ be strictly increasing. The upper contour sets of $\psi \circ f$ are upper contour sets of f, since $\psi(f(x)) \geq \alpha$ if and only if $f(x) \geq \psi^{-1}(\alpha)$ on the range; convexity of the latter is quasiconcavity of f. That concavity is not preserved is shown by $f(x) = x$ on $\mathbb{R}$, concave, and $\psi(t) = t^3$, giving the non-concave x^3. ∎

The asymmetry in Proposition 9.3.4 is the reason quasiconcavity, not concavity, is the right hypothesis on a utility function. Utility is an ordinal representation of a preference relation, unique only up to strictly increasing transformation, and convexity of preferences — convexity of the sets $\{y : y \succsim x\}$ — is exactly quasiconcavity of every representation. Concavity, by contrast, is a property of a particular representation and has no preference-level meaning.

Proposition 9.3.5 (Uniqueness of the maximiser). Let $C \subseteq \mathbb{R}^n$ be convex and $f : C \to \mathbb{R}$ strictly quasiconcave. Then f has at most one maximiser on C. If in addition C is compact and non-empty and f is upper semicontinuous, the maximiser exists and is unique.

Proof. Suppose $x \neq y$ both maximise f on C, with common value M. Then $\frac{1}{2}x + \frac{1}{2}y \in C$ and strict quasiconcavity gives $f\left(\frac{1}{2}x + \frac{1}{2}y\right) > \min\{f(x), f(y)\} = M$, contradicting maximality. Existence is Theorem 5.4.1(ii). ∎

Proposition 9.3.6 (Differentiable quasiconcavity)**.** Let $C \subseteq \mathbb{R}^n$ be open and convex and $f\colon C \to \mathbb{R}$ differentiable. Then f is quasiconcave if and only if

$$f(x) \geq f(x_0) \quad \Longrightarrow \quad \langle \nabla f(x_0), x - x_0 \rangle \geq 0 \qquad \text{for all } x, x_0 \in C.$$

This is Proposition 11.7 of Fukuda (2026d, p. 396). The condition says that moving from x_0 towards any weakly preferred point does not initially decrease f; geometrically the gradient at x_0 supports the upper contour set at x_0. It is weaker than Theorem 9.3.1(ii), which bounds f globally, and correspondingly it does *not* make a stationary point a global maximiser: $f(x) = x^3$ is quasiconcave on $\mathbb{R}$ with $f'(0) = 0$ and 0 is not a maximiser. Under quasiconcavity the first-order condition remains only necessary, which is why Chapter 12 states its sufficiency results under concavity and treats quasiconcavity separately.

9.4 Hyperplanes and Separation

Definition 9.4.1 (Hyperplane and half-spaces)**.** A *hyperplane* in $\mathbb{R}^n$ is a set of the form

$$H_{a,\alpha} := \{x \in \mathbb{R}^n : \langle a, x \rangle = \alpha\}, \qquad a \in \mathbb{R}^n \setminus \{0\},\ \alpha \in \mathbb{R},$$

equivalently an affine subspace of dimension $n-1$. It determines the two closed *half-spaces* $\{\langle a, x \rangle \geq \alpha\}$ and $\{\langle a, x \rangle \leq \alpha\}$. The hyperplane *separates* A and B if

$$\langle a, x \rangle \geq \alpha \geq \langle a, y \rangle \qquad \text{for all } x \in A,\ y \in B,$$

and *strictly separates* them if both inequalities can be taken strict with a gap, that is if $\sup_{y \in B} \langle a, y \rangle < \alpha < \inf_{x \in A} \langle a, x \rangle$.

That every hyperplane has the stated form is Proposition 10.1 of Fukuda (2026d, p. 334), and it is Theorem 3.2.1 in disguise: an affine subspace of codimension 1 is a level set of a non-zero linear functional, and every such functional is an inner product with a fixed vector.

Lemma 9.4.1 (Separating a point from a closed convex set)**.** Let $D \subseteq \mathbb{R}^n$ be non-empty, closed and convex and let $x \notin D$. Then there are $a \in \mathbb{R}^n \setminus \{0\}$ and $\alpha \in \mathbb{R}$ with

$$\langle a, x \rangle > \alpha > \langle a, y \rangle \qquad \text{for all } y \in D.$$

Proof. Let $z := P_D x$ from Lemma 9.2.1 and put $a := x - z \neq 0$, since $x \notin D$ and $z \in D$. By (9.1), $\langle a, y - z \rangle \leq 0$, that is $\langle a, y \rangle \leq \langle a, z \rangle$ for every $y \in D$. Also

$$\langle a, x \rangle - \langle a, z \rangle = \langle a, x - z \rangle = \|a\|^2 > 0.$$

Any α strictly between $\langle a, z \rangle$ and $\langle a, x \rangle$ — for instance $\alpha = \langle a, z \rangle + \frac{1}{2}\|a\|^2$ — works. ∎

Theorem 9.4.1 (Strict separation)**.** Let $C, D \subseteq \mathbb{R}^n$ be non-empty, convex and disjoint, with C compact and D closed. Then there is a hyperplane strictly separating them.

Proof. The set $E := C + (-D) = \{x - y : x \in C,\ y \in D\}$ is convex by Proposition 9.2.1(ii), and $0 \notin E$ because $C \cap D = \emptyset$. It is closed: if $x_k - y_k \to w$ with $x_k \in C$ and $y_k \in D$, compactness of C gives a subsequence $x_{k_j} \to x \in C$, whence $y_{k_j} = x_{k_j} - (x_{k_j} - y_{k_j}) \to x - w$, which lies in D because D is closed; so $w = x - (x - w) \in E$.

Apply Lemma 9.4.1 to the closed convex set E and the point $0 \notin E$: there are $b \neq 0$ and α_0 with $0 = \langle b, 0 \rangle > \alpha_0 > \langle b, w \rangle$ for every $w \in E$, that is

$$\langle b, x \rangle - \langle b, y \rangle < \alpha_0 < 0 \qquad \text{for all } x \in C,\ y \in D.$$

Taking the supremum over $x \in C$ and the infimum over $y \in D$,

$$\sup_{x\in C} \langle b, x\rangle \;\le\; \inf_{y\in D} \langle b, y\rangle + \alpha_0 \;<\; \inf_{y\in D} \langle b, y\rangle ,$$

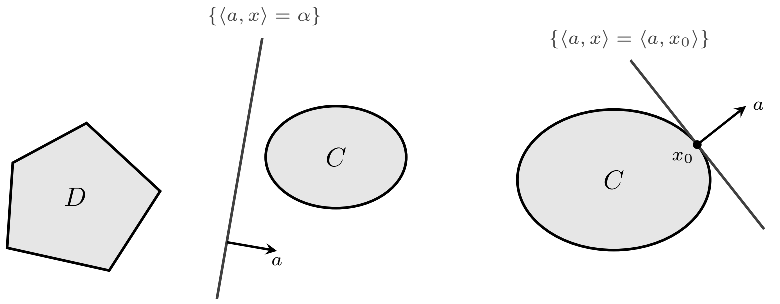
so the two quantities are separated by the strictly positive gap $|\alpha_0|$. Setting $a := -b$ and taking α to be -1 times the midpoint of the two bounds gives $\inf_{x\in C} \langle a, x\rangle > \alpha > \sup_{y\in D} \langle a, y\rangle$, which is strict separation in the orientation of Definition 9.4.1. ■

Theorem 9.4.2 (Supporting hyperplane)**.** Let $C \subseteq \mathbb{R}^n$ be non-empty and convex, not necessarily closed, and let $x_0 \in \partial C$. Then there is $a \in \mathbb{R}^n \setminus \{0\}$ with

$$\langle a, x\rangle \le \langle a, x_0\rangle \qquad \text{for all } x \in C.$$

Proof. Since $x_0 \in \partial C = \overline{C} \setminus \operatorname{Int} C$, there is a sequence $(x_k) \subseteq \left(\overline{C}\right)^{\mathsf{c}}$ with $x_k \to x_0$. For each k, Lemma 9.4.1 applied to the closed convex set $\overline{C}$ and the point x_k supplies $a_k \neq 0$ with $\langle a_k, x\rangle < \langle a_k, x_k\rangle$ for all $x \in \overline{C}$; normalising, $\|a_k\| = 1$. The unit sphere in $\mathbb{R}^n$ is compact by Theorem 4.4.2, so a subsequence converges, $a_{k_j} \to a$ with $\|a\| = 1$, hence $a \neq 0$. For fixed $x \in C$, $\langle a_{k_j}, x\rangle < \langle a_{k_j}, x_{k_j}\rangle$ for every j; letting $j \to \infty$ and using continuity of the inner product gives $\langle a, x\rangle \le \langle a, x_0\rangle$. ■

This is Theorem 10.3 of Fukuda (2026d, p. 343). The proof is an instance of a recurring pattern: obtain the result at nearby points where a stronger theorem applies, normalise, and extract a convergent subsequence by compactness of the sphere — a step available only in finite dimensions, which is why the infinite-dimensional supporting hyperplane theorem requires non-empty interior or a different argument (Aliprantis and Border 2006, Ch. 7). Figure 9.1 contrasts the two conclusions: strict separation puts a gap on each side of the hyperplane, support allows the hyperplane to touch.



(a) strict separation (b) supporting hyperplane

Figure 9.1: The two geometries. In (a) the sets are disjoint, one of them compact, and the hyperplane passes strictly between them with a gap on each side, so $\sup_{y\in D} \langle a, y\rangle < \alpha < \inf_{x\in C} \langle a, x\rangle$ (Theorem 9.4.1). In (b) the hyperplane touches C at a boundary point and leaves the whole set in one closed half-space, so $\langle a, x\rangle \le \langle a, x_0\rangle$ for every $x \in C$ (Theorem 9.4.2). In both panels a points into the half-space on which $\langle a, \cdot\rangle$ is larger, which is toward C in (a) and away from C in (b): the two statements orient the normal oppositely, and replacing a by $-a$ converts either into the other.

Theorem 9.4.3 (Separation)**.** Let $C, D \subseteq \mathbb{R}^n$ be non-empty, convex and disjoint. Then there is a hyperplane separating them; moreover the separating inequality is strict on $\operatorname{Int} C$ whenever $\operatorname{Int} C \neq \emptyset$.

Proof. As above, $E = C + (-D)$ is convex with $0 \notin E$. If $0 \notin \overline{E}$, apply Lemma 9.4.1 to $\overline{E}$, which is convex by Proposition 9.2.1(iv). If $0 \in \overline{E}$, then $0 \in \partial E$, since $0 \notin E$ excludes $0 \in \operatorname{Int} E$, and Theorem 9.4.2 applied to E at its boundary point 0 supplies $a \neq 0$ with $\langle a, w\rangle \le \langle a, 0\rangle = 0$ for all $w \in E$. In both cases $\langle a, x\rangle \le \langle a, y\rangle$ for all $x \in C$ and $y \in D$, so with $\alpha := \sup_{x\in C} \langle a, x\rangle$, which is finite and at most $\inf_{y\in D} \langle a, y\rangle$,

$$\langle a, x\rangle \le \alpha \le \langle a, y\rangle \qquad \text{for all } x \in C,\ y \in D,$$

which is separation in the sense of Definition 9.4.1, with the roles of the two sets interchanged.

Strictness on $\operatorname{Int} C$: suppose $\langle a, x_0\rangle = \alpha$ for some $x_0 \in \operatorname{Int} C$. Then $x_0 + \varepsilon a \in C$ for small $\varepsilon > 0$, and $\langle a, x_0 + \varepsilon a\rangle = \alpha + \varepsilon \|a\|^2 > \alpha$, contradicting the inequality just established. ■

These are Theorems 10.1 and 10.2 of Fukuda (2026d, pp. 337, 339), and Boyd and Vandenberghe (2004, Sec. 2.5, p. 46) states the same pair in the form used by the optimisation literature; the compactness hypothesis in Theorem 9.4.1 is what upgrades separation to strict separation and cannot be dropped. The standard witness is $C = \{(x_1, x_2) : x_2 \geq e^{x_1}\}$ and $D = \{(x_1, 0) : x_1 \in \mathbb{R}\}$: both are closed and convex and disjoint, the separating hyperplane $x_2 = 0$ is unique, and no strict separation exists because $\operatorname{dist}(C, D) = 0$.

9.5 The Farkas–Minkowski Theorem

Theorem 9.5.1 (Farkas–Minkowski). Let $f, g_1, \ldots, g_m : \mathbb{R}^n \to \mathbb{R}$ be concave, and let $q \in \{0, 1, \ldots, m\}$ be such that g_i is affine for $i \in \{q+1, \ldots, m\}$. Assume the system

$$g_i(x) > 0 \ \ (i \leq q), \qquad g_i(x) \geq 0 \ \ (i > q) \tag{9.3}$$

has a solution. Then the system

$$f(x) > 0 \quad \text{and} \quad g_i(x) \geq 0 \ \ (i = 1, \ldots, m) \tag{9.4}$$

has no solution in $\mathbb{R}^n$ if and only if there exists $y = (y_1, \ldots, y_m) \in \mathbb{R}^m_+$ such that

$$\sup_{x \in \mathbb{R}^n} \left\{ f(x) + \sum_{i=1}^{m} y_i g_i(x) \right\} \leq 0.$$

This is Theorem 10.9 of Fukuda (2026d, p. 353), proved there from Theorem 9.4.3 applied to the convex set $\left\{(f(x), g_1(x), \ldots, g_m(x)) + \mathbb{R}^{m+1}_{--} : x \in \mathbb{R}^n\right\}$ and the non-negative orthant. The proof is not reproduced here; it is a page of careful bookkeeping with no additional economic content, and the statement is used only through Chapter 12.

The theorem is the mechanism by which multipliers appear. Read (9.4) as "no feasible point strictly improves on the candidate": f is the improvement in the objective and the g_i are the constraints. The conclusion produces non-negative numbers y_i — the multipliers — such that the aggregate $f + \sum_i y_i g_i$ is nowhere positive, which is the Lagrangian inequality of the Kuhn–Tucker theorem. Condition (9.3) is Slater's constraint qualification, and Example 9.5.1 shows what goes wrong without it.

Example 9.5.1 (Failure of Slater's condition). Take $n = 1$, $m = 1$, $g_1(x) = -x^2$, and $f(x) = x$. The constraint set is $\{x : -x^2 \geq 0\} = \{0\}$, and (9.3) fails: no x has $-x^2 > 0$. The system (9.4) indeed has no solution, since it requires $x > 0$ and $x = 0$. But no $y \geq 0$ makes $\sup_x\{x - yx^2\} \leq 0$: for $y > 0$ the supremum is $1/(4y) > 0$, and for $y = 0$ it is $+\infty$. The conclusion of Theorem 9.5.1 fails, and with it the existence of a multiplier at the constrained optimum $x = 0$.

9.6 Subgradients and Conjugate Duality

Theorem 9.3.1 characterises concavity by the gradient inequality and assumes differentiability. Value functions, profit functions and indirect utility functions are convex or concave by construction and differentiable only under further hypotheses, as Section 13.5 shows. The subgradient replaces the gradient by the whole family of supporting affine functions, and it exists wherever the function is convex, differentiable or not.

Definition 9.6.1 (Subgradient and supergradient). Let $C \subseteq \mathbb{R}^n$ be convex and $f : C \to \mathbb{R}$ convex. A vector $v \in \mathbb{R}^n$ is a *subgradient* of f at $x \in C$ if

$$f(y) \geq f(x) + \langle v, y - x \rangle \qquad \text{for every } y \in C. \tag{9.5}$$

The set of subgradients is the *subdifferential* $\partial f(x)$. For concave g, a *supergradient* at x is a v with $g(y) \leq g(x) + \langle v, y - x \rangle$ for every $y \in C$, and the *superdifferential* is $\bar{\partial} g(x) = -\partial(-g)(x)$.

Inequality (9.5) makes sense for an arbitrary $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ at a point $x \in \operatorname{dom} f$, with y ranging over all of $\mathbb{R}^n$, and $\partial f(x)$ is defined by it in that generality throughout subsection 9.6.1. The requirement is global: the affine function must lie below f everywhere, not merely near x. Without convexity the set is frequently empty — $f(x) = -x^2$ has $\partial f(x) = \emptyset$ at every x — and the results below that assume $\partial f(x) \neq \emptyset$ are assuming something.

Geometrically, $v \in \partial f(x)$ says that the affine function $y \mapsto f(x) + \langle v, y - x\rangle$ lies below f and touches it at x; equivalently, $(v, -1)$ is normal to a hyperplane supporting $\operatorname{epi} f$ at $(x, f(x))$.

Theorem 9.6.1 (Existence of subgradients)**.** Let $C \subseteq \mathbb{R}^n$ be convex, $f : C \to \mathbb{R}$ convex, and $x \in \operatorname{Int} C$. Then $\partial f(x)$ is non-empty, convex and compact.

Proof. Non-emptiness. The set $\operatorname{epi} f \subseteq \mathbb{R}^{n+1}$ is convex by Proposition 9.3.1(i), and $(x, f(x)) \in \partial(\operatorname{epi} f)$, since $(x, f(x) - \varepsilon) \notin \operatorname{epi} f$ for every $\varepsilon > 0$. By Theorem 9.4.2 there is $(a, \beta) \in \mathbb{R}^n \times \mathbb{R}$, not both zero, with

$$\langle a, y\rangle + \beta z \leq \langle a, x\rangle + \beta f(x) \qquad \text{for all } (y, z) \in \operatorname{epi} f. \tag{9.6}$$

Taking $y = x$ and $z \to +\infty$, which is admissible because $(x, z) \in \operatorname{epi} f$ for every $z \geq f(x)$, forces $\beta \leq 0$. If $\beta = 0$ then $\langle a, y - x\rangle \leq 0$ for every $y \in C$; since $x \in \operatorname{Int} C$, the point $y = x + \varepsilon a$ lies in C for small $\varepsilon > 0$ and gives $\varepsilon \|a\|^2 \leq 0$, hence $a = 0$, contradicting $(a, \beta) \neq 0$. So $\beta < 0$, and dividing (9.6) by $-\beta$ and setting $v := a/(-\beta)$ gives, with $z = f(y)$, $f(y) \geq f(x) + \langle v, y - x\rangle$ for every $y \in C$.

Convexity and closedness. For each fixed y, (9.5) is a closed half-space condition on v, and $\partial f(x)$ is the intersection over $y \in C$ of these half-spaces, hence closed and convex by Proposition 9.2.1.

Boundedness. Choose $r > 0$ with $\overline{B_r(x)} \subseteq C$. By Proposition 9.3.3, f is continuous on C, so $M := \max_{\overline{B_r(x)}} f$ is finite by Corollary 5.4.1. For $v \in \partial f(x)$ with $v \neq 0$, take $y = x + rv/\|v\|$ in (9.5): $M \geq f(y) \geq f(x) + r\|v\|$, so $\|v\| \leq (M - f(x))/r$. ∎

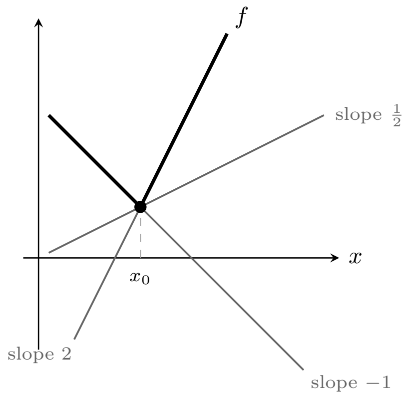


Figure 9.2: The convex function $f(x) = \max\{1.5 - x,\ 2x - 1.5\}$ and three of the affine functions supporting it at the kink $x_0 = 1$. Every slope between the one-sided derivatives -1 and 2 gives a supporting line, so $\partial f(x_0) = [-1, 2]$; at every other point the supporting line is unique and equals the tangent. By Proposition 9.6.2 and its converse, non-differentiability and multiplicity of supporting slopes are the same thing.

Proposition 9.6.1 (Subgradients and optimality)**.** Let C be convex and $f : C \to \mathbb{R}$ convex. Then $x^* \in C$ minimises f over C if and only if $0 \in \partial f(x^*)$.

Proof. Both statements read $f(y) \geq f(x^*)$ for every $y \in C$, the second by putting $v = 0$ in (9.5). ∎

The proposition replaces "the derivative vanishes" by "zero is one of the slopes of the supporting affine functions", and it holds with no differentiability and no interiority. It is the form in which first-order conditions survive at kinks — a piecewise-linear cost function, a portfolio problem with proportional transaction costs, a utility function with a subsistence kink.

Proposition 9.6.2 (Subdifferential and gradient)**.** Let $C \subseteq \mathbb{R}^n$ be open and convex and $f : C \to \mathbb{R}$ convex. If f is differentiable at $x \in C$ then $\partial f(x) = \{\nabla f(x)\}$.

Proof. By Theorem 9.3.1 applied to the convex function f, the gradient inequality $f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle$ holds, so $\nabla f(x) \in \partial f(x)$. Conversely let $v \in \partial f(x)$ and fix $d \in \mathbb{R}^n$. For small $t > 0$, (9.5) with $y = x + td$ gives $\big(f(x+td) - f(x)\big)/t \geq \langle v, d\rangle$; letting $t \downarrow 0$ and using differentiability, $\langle \nabla f(x), d\rangle \geq \langle v, d\rangle$. Applying this to d and to $-d$ gives $\langle \nabla f(x) - v, d\rangle = 0$ for every d, so $v = \nabla f(x)$. ∎

The converse also holds: a convex function whose subdifferential at an interior point is a singleton is differentiable there. The proof requires the theory of directional derivatives of convex functions and is not reproduced here; it is Rockafellar (1970, Thm. 25.1), in the section on differentiability of convex functions. Both directions together say that non-differentiability of a convex function is precisely multiplicity of supporting slopes, which is how Section 13.5 reads the kink in a value function at a point where the binding constraint changes.

9.6.1 The conjugate function

Extended-real-valued convex analysis uses three adjectives that are easy to confuse. Call $f\colon \mathbb{R}^n \to \mathbb{R} \cup \{\pm\infty\}$ *proper* if it never takes $-\infty$ and $\operatorname{dom} f := \{x : f(x) < +\infty\}$ is non-empty; *closed* if $\operatorname{epi} f$ is a closed subset of $\mathbb{R}^{n+1}$, equivalently if f is lower semicontinuous; and convex if $\operatorname{epi} f$ is convex, which for a proper f is Definition 9.3.1 on $\operatorname{dom} f$. A proper convex f is automatically continuous on $\operatorname{Int}(\operatorname{dom} f)$ by Proposition 9.3.3, so closedness is a condition on the boundary of the domain alone.

Definition 9.6.2 (Fenchel conjugate)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ with $\operatorname{dom} f := \{x : f(x) < \infty\}$ non-empty. Its *conjugate* is

$$f^*(y) := \sup_{x \in \operatorname{dom} f} \big\{\langle y, x\rangle - f(x)\big\}, \qquad y \in \mathbb{R}^n. \tag{9.7}$$

Proposition 9.6.3 (Elementary properties)**.** Let f be as in Definition 9.6.2, so that f is proper: it never takes the value $-\infty$ and $\operatorname{dom} f \neq \emptyset$. Convexity of f is *not* assumed, and ∂f is the global subdifferential defined after Definition 9.6.1.

(i) f^* is convex and lower semicontinuous as an extended-real function on $\mathbb{R}^n$, with values in $\mathbb{R} \cup \{+\infty\}$.

(ii) *(Fenchel–Young)* $f(x) + f^*(y) \geq \langle x, y\rangle$ for all $x \in \operatorname{dom} f$ and $y \in \mathbb{R}^n$.

(iii) Equality holds in (ii) if and only if $y \in \partial f(x)$.

(iv) $f^{**} \leq f$ everywhere, with equality at every x at which $\partial f(x) \neq \emptyset$. The converse fails: on $f(x) = -\sqrt{x}$ for $x \geq 0$ and $f(x) = +\infty$ for $x < 0$, which is convex and lower semicontinuous, so that $f^{**} = f$ everywhere, one has $\partial f(0) = \emptyset$, because $-\sqrt{y} \geq vy$ for all $y > 0$ would force $v \leq -1/\sqrt{y} \to -\infty$.

Proof. (i) For each fixed x, $y \mapsto \langle y, x\rangle - f(x)$ is affine, hence convex and continuous; a pointwise supremum of convex functions is convex by Proposition 9.3.2 and lower semicontinuous by Proposition 5.3.3.

(ii) is the definition of the supremum: $f^*(y) \geq \langle y, x\rangle - f(x)$.

(iii) Equality says that the supremum in (9.7) is attained at x, that is $\langle y, z\rangle - f(z) \leq \langle y, x\rangle - f(x)$ for every $z \in \operatorname{dom} f$, which rearranges to $f(z) \geq f(x) + \langle y, z - x\rangle$ for every z, that is $y \in \partial f(x)$.

(iv) By (ii), $f(x) \geq \langle x, y\rangle - f^*(y)$ for every y, so $f(x) \geq \sup_y\{\langle x, y\rangle - f^*(y)\} = f^{**}(x)$. If $y_0 \in \partial f(x)$ then (iii) gives $f(x) = \langle x, y_0\rangle - f^*(y_0) \leq f^{**}(x)$, and the two inequalities combine to equality. ∎

By Theorem 9.6.1, a finite convex f has $\partial f(x) \neq \emptyset$ at every interior point of its domain, so $f^{**} = f$ on $\operatorname{Int}(\operatorname{dom} f)$. In general f^{**} is the largest convex lower semicontinuous function below f, the closed convex hull of f, and the conjugate cannot distinguish f from f^{**}: the map $f \mapsto f^*$ factors through f^{**}, since $f^* = f^{***}$. The general statement, including the boundary behaviour that the example above illustrates, is Boyd and Vandenberghe (2004, Sec. 3.3, p. 90). The clean form of that statement is the following.

Theorem 9.6.2 (Fenchel–Moreau)**.** Let $f\colon \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper. Then $f^{**} = f$ if and only if f is convex and closed.

Sufficiency of convexity and closedness is what Proposition 9.6.3(iv) fails to reach: the proof identifies f^{**} with the supremum of all affine functions below f and then uses the separating hyperplane theorem in $\mathbb{R}^{n+1}$ to show that a closed convex epigraph is the intersection of the closed half-spaces containing it, including at boundary points where no subgradient exists. That last step is the part not reproduced here; it is Rockafellar (1970, Thm. 12.2). Necessity is immediate, since f^{**} is convex and closed by Proposition 9.6.3(i) applied to f^*.

9.6.2 Economic application: the profit function

Let a competitive firm have cost function $c\colon \mathbb{R}^n_+ \to \mathbb{R}$, where $c(q)$ is the minimum cost of producing the output vector q, and let $p \in \mathbb{R}^n_{++}$ be the vector of output prices. Its profit function is

$$\pi(p) := \sup_{q \geq 0} \big\{ \langle p, q \rangle - c(q) \big\} = c^*(p), \tag{9.8}$$

the conjugate of the cost function, where c is extended by $+\infty$ off $\mathbb{R}^n_+$. Four conclusions follow from Proposition 9.6.3 alone.

Convexity of profit in prices. By (i), π is convex and lower semicontinuous in p whether or not c is convex, that is, whether or not the technology exhibits non-increasing returns. Convexity of the profit function is a theorem about maximisation, not a hypothesis about the firm; the economic reading is that a firm facing a mean-preserving spread in prices is better off in expectation, because it can adjust q after observing p.

Price equals marginal cost. By (iii), q^* attains the supremum in (9.8) if and only if $p \in \partial c(q^*)$. Where c is differentiable this is $p = \nabla c(q^*)$; where c has a kink, as with a capacity constraint, it is the statement that p lies in the interval between the left and right marginal costs, and the firm's supply is flat over that interval of prices.

Hotelling's lemma. If c is proper, convex and lower semicontinuous, then $c^{**} = c$ and $q \in \partial\pi(p)$ if and only if $p \in \partial c(q)$, so $\partial\pi(p)$ is the set of profit-maximising output vectors. Suppose in addition that $p \in \operatorname{Int}(\operatorname{dom}\pi)$, so that profit is finite in a neighbourhood of p and Theorem 9.6.1 applies to π there. Combining with Proposition 9.6.2 and its converse, π is differentiable at p if and only if the profit-maximising output is unique, and then

$$\nabla\pi(p) = q^*(p), \tag{9.9}$$

the supply function. Differentiating once more, $Dq^*(p) = \mathrm{D}^2\pi(p)$ is symmetric and positive semidefinite by Theorem 8.5.4, which is the statement that own-price supply responses are non-negative and cross-price responses are symmetric. These are the restrictions that producer theory tests, and they are consequences of maximisation with no assumption on the technology beyond what makes π twice differentiable.

What price data cannot reveal. Since $c^* = c^{***}$, the cost function c and its closed convex hull c^{**} generate the same profit function, so no observation of supply behaviour at any price vector distinguishes them. What that leaves unobserved is precisely the set where $c > c^{**}$, which by (iv) contains no point with a supporting affine function. If the technology has a region of increasing returns, an output level q in that region is never a global profit maximiser at any price: were it one, $p \in \partial c(q)$ would hold for some p and (iv) would give $c(q) = c^{**}(q)$. Output levels at which $c = c^{**}$ can lie in a locally non-convex region — the endpoints of the interval spanned by the convex hull are of this kind — so the statement is about supporting hyperplanes, not about local curvature. The same algebra explains why the expenditure function determines only the convex hull of the upper contour sets of a utility function, which is the duality statement of consumer theory.

9.7 Economic Applications

9.7.1 Supporting prices

Environment. An economy has n commodities, an endowment $\omega \in \mathbb{R}^n_+$, a single consumer with preferences represented by $u\colon \mathbb{R}^n_+ \to \mathbb{R}$, and an aggregate production set $Y \subseteq \mathbb{R}^n$ with $0 \in Y$. An *allocation* is a pair (x, y) with $x \in \mathbb{R}^n_+$ and $y \in Y$; it is *attainable* if $x \leq \omega + y$. An attainable allocation (x^*, y^*) is *efficient* if no attainable (x, y) satisfies $u(x) > u(x^*)$. Utility is *locally non-satiated* if every ball around every $x \in \mathbb{R}^n_+$ contains a point $x' \in \mathbb{R}^n_+$ with $u(x') > u(x)$.

Write $C := (\omega + Y) - \mathbb{R}^n_+$ for the set of consumption vectors that the technology and the endowment make available, disposal permitted. Since $0 \in Y$, C contains $\omega - \mathbb{R}^n_+$, so $\operatorname{Int} C \neq \emptyset$.

Theorem 9.7.1 (Supporting prices for an efficient allocation)**.** Let u be quasiconcave and locally non-satiated and let Y be convex, and let (x^*, y^*) be an efficient allocation. Then there is a price vector $p \in \mathbb{R}^n_+ \setminus \{0\}$ such that

(i) $u(x) > u(x^*)$ implies $\langle p, x\rangle \geq \langle p, x^*\rangle$ for $x \in \mathbb{R}^n_+$;

(ii) $\langle p, y\rangle \leq \langle p, y^*\rangle$ for every $y \in Y$; and

(iii) $\langle p, \omega + y^* - x^*\rangle = 0$, so unused resources are free.

Proof. Put $B := \{x \in \mathbb{R}^n_+ : u(x) > u(x^*)\}$, which is convex by quasiconcavity of u and Definition 9.3.2 and non-empty by local non-satiation, and recall $C = (\omega + Y) - \mathbb{R}^n_+$, convex by Proposition 9.2.1(ii) and non-empty.

Step 1: $B \cap C = \emptyset$. A point $x \in B \cap C$ satisfies $x \geq 0$ and $x \leq \omega + y$ for some $y \in Y$, so (x, y) is attainable, and $u(x) > u(x^*)$ then contradicts efficiency.

Step 2: separation. By Theorem 9.4.3 applied to the disjoint non-empty convex sets C and B there are $p \neq 0$ and $\alpha \in \mathbb{R}$ with

$$\langle p, z\rangle \leq \alpha \leq \langle p, x\rangle \qquad \text{for all } z \in C,\ x \in B. \tag{9.10}$$

Step 3: $p \geq 0$. Fix i and note $\omega \in C$ (take $y = 0$), hence $\omega - te_i \in C$ for every $t \geq 0$ because $C - \mathbb{R}^n_+ = C$. Then $\langle p, \omega\rangle - tp_i \leq \alpha$ for all $t \geq 0$, which is possible only if $p_i \geq 0$.

Step 4: $\alpha = \langle p, x^*\rangle$. Local non-satiation produces $x_k \to x^*$ with $u(x_k) > u(x^*)$, that is $x_k \in B$; the right-hand inequality of (9.10) and continuity of $\langle p, \cdot\rangle$ give $\langle p, x^*\rangle \geq \alpha$. Attainability gives $x^* \leq \omega + y^*$ with $y^* \in Y$, so $x^* \in C$ and the left-hand inequality gives $\langle p, x^*\rangle \leq \alpha$.

Step 5: the three conclusions. Claim (i) is the right-hand inequality of (9.10) together with Step 4. For (ii) and (iii), let $y \in Y$. Then $\omega + y \in C$, so $\langle p, \omega + y\rangle \leq \alpha = \langle p, x^*\rangle$. Taking $y = y^*$ and using $x^* \leq \omega + y^*$ with $p \geq 0$,

$$\langle p, \omega + y^*\rangle \leq \langle p, x^*\rangle \leq \langle p, \omega + y^*\rangle\,,$$

so both inequalities are equalities, which is (iii); and then $\langle p, \omega + y\rangle \leq \langle p, x^*\rangle = \langle p, \omega + y^*\rangle$ for every $y \in Y$ reduces to (ii). ∎

Reading the hypotheses. Convexity of Y and quasiconcavity of u are what make Theorem 9.4.3 applicable, and neither can be weakened to a topological condition, because separation is a theorem about convex sets and has no counterpart for arbitrary disjoint sets. Under increasing returns Y is not convex, and then an efficient allocation need not be supportable by any price vector at all; the resulting failure of decentralisation is one of the standard arguments for regulating a natural monopoly, though the argument identifies a market failure rather than prescribing a remedy, and the regulatory literature turns on informational constraints that this model does not contain. Local non-satiation enters only at Step 4, where it places x^* in the closure of B; without it the separating value α can strictly exceed $\langle p, x^*\rangle$ and conclusion (i) is lost. Continuity of u is used nowhere in the proof.

The conclusion is expenditure minimisation over the strictly preferred set, not utility maximisation: (i) says that nothing strictly better than x^* is strictly cheaper. The upgrade to "x^* maximises u on $\{x \geq 0 : \langle p, x\rangle \leq \langle p, x^*\rangle\}$" needs continuity of u and a cheaper point, that is some $x' \in \mathbb{R}^n_+$ with $\langle p, x'\rangle < \langle p, x^*\rangle$. Given both, suppose $u(x) > u(x^*)$ with $\langle p, x\rangle \leq \langle p, x^*\rangle$; the points $x_\lambda = \lambda x' + (1-\lambda)x$ lie in $\mathbb{R}^n_+$ and satisfy $\langle p, x_\lambda\rangle < \langle p, x^*\rangle$ for $\lambda \in (0, 1]$, and by continuity of u at x the inequality $u(x_\lambda) > u(x^*)$ persists for λ small, contradicting (i). The cheaper-point condition fails when $\langle p, x^*\rangle = 0$, which is why the result is usually stated with $p \gg 0$ and $x^* \neq 0$, or with the strengthening supplied by monotone preferences.

9.7.2 No arbitrage and state prices

Environment. There are S states tomorrow and J assets. Asset j pays R_{sj} in state s; write $R \in M_{S\times J}(\mathbb{R})$. A portfolio is $\theta \in \mathbb{R}^J$, costing $\langle q, \theta\rangle$ today at prices $q \in \mathbb{R}^J$ and paying $R\theta \in \mathbb{R}^S$ tomorrow.

Definition 9.7.1 (Arbitrage)**.** An *arbitrage* is a portfolio θ with $\langle q, \theta\rangle \leq 0$ and $R\theta \geq 0$, at least one of the $S + 1$ inequalities being strict.

Theorem 9.7.2 (Fundamental theorem of asset pricing)**.** There is no arbitrage if and only if there exists a strictly positive state price vector $\psi \in \mathbb{R}^S_{++}$ with

$$q = R^\mathsf{T}\psi, \qquad \text{that is} \qquad q_j = \sum_{s=1}^{S} \psi_s R_{sj} \quad \text{for every } j. \tag{9.11}$$

Proof. ($\Leftarrow$) Suppose (9.11) holds and θ satisfies $R\theta \geq 0$. Then $\langle q, \theta\rangle = \left\langle R^\mathsf{T}\psi, \theta\right\rangle = \langle \psi, R\theta\rangle \geq 0$, with strict inequality if $R\theta \neq 0$ because $\psi \gg 0$. So no arbitrage exists.

($\Rightarrow$) Consider in $\mathbb{R}^{1+S}$ the subspace $M := \left\{(-\langle q,\theta\rangle, R\theta) : \theta \in \mathbb{R}^J\right\}$, which is a linear subspace hence convex and closed (Corollary 6.3.1), and the set $K := \{v \in \mathbb{R}^{1+S}_+ : \sum_i v_i = 1\}$, which is convex and compact. No arbitrage says $M \cap K = \emptyset$: a point of the intersection would be a portfolio with $-\langle q, \theta\rangle \geq 0$, $R\theta \geq 0$ and not all components zero. By Theorem 9.4.1 there are $\phi = (\phi_0, \psi) \in \mathbb{R}^{1+S} \setminus \{0\}$ and α with

$$\langle \phi, v\rangle > \alpha > \langle \phi, w\rangle \qquad \text{for all } v \in K,\ w \in M.$$

M is a subspace, so $\langle \phi, w\rangle$ is bounded above on it only if it vanishes identically: if $\langle \phi, w_0\rangle \neq 0$ for some $w_0 \in M$, then $\lambda w_0 \in M$ for every $\lambda \in \mathbb{R}$ and $\langle \phi, \lambda w_0\rangle$ is unbounded above. Hence $\langle \phi, w\rangle = 0$ for all $w \in M$, and since $0 \in M$ the displayed inequality forces $\alpha > 0$ and therefore $\langle \phi, v\rangle > \alpha > 0$ for all $v \in K$. Taking v to be each standard basis vector of $\mathbb{R}^{1+S}$, which lies in K, gives $\phi_0 > 0$ and $\psi \gg 0$. Applying $\langle \phi, w\rangle = 0$ to w generated by $\theta = e_j$ gives $-\phi_0 q_j + \langle \psi, R_{\cdot j}\rangle = 0$, that is $q = R^\mathsf{T}(\psi/\phi_0)$. Replacing ψ by $\psi/\phi_0 > 0$ completes the proof. ∎

Economic content. The state prices ψ_s are the prices today of one unit of consumption in state s tomorrow, and (9.11) says every asset is priced by its payoff-weighted sum of state prices. Strict positivity of ψ is the whole content: it is equivalent to the absence of arbitrage, and it is what permits the normalisation $\psi_s = \beta\pi_s m_s$ into a discount factor, a probability and a stochastic discount factor, giving the Euler equation form used in subsection 6.6.2. If the market is complete — rank $R = S$ — then ψ is unique, since R^T is then injective on $\mathbb{R}^S$ by Theorem 2.3.2; if incomplete, the set of consistent state price vectors is the non-empty relatively open convex set $\{\psi \gg 0 : R^\mathsf{T}\psi = q\}$, and asset prices outside the span of R are not determined by no arbitrage alone. This is the connection to Example 3.2.4, where the same vector appeared as the Riesz representation of the pricing functional.

9.8 Chapter Summary

Convex sets are closed under intersection, sums, scalar multiples, linear images and preimages, and interiors and closures; convex hulls in $\mathbb{R}^n$ need at most $n+1$ points per element (Theorem 9.2.1). A non-empty closed convex set contains a unique nearest point to any given point, characterised by the variational inequality (9.1), and that lemma is the engine of every separation result: a point outside a closed convex set is strictly separated from it, disjoint convex sets with one compact and one closed are strictly separated, disjoint convex sets are separated, and a boundary point of a convex set carries a supporting hyperplane.

Convex functions are those with convex epigraph, are stable under non-negative combinations, arbitrary suprema, and composition with increasing convex maps, and are continuous on the interior of their domain. For differentiable functions, concavity is the tangent-plane inequality, is monotonicity of the gradient, and is negative semidefiniteness of the Hessian. Quasiconcavity — convexity of upper contour sets — is the ordinal weakening appropriate to utility, is preserved by increasing transformations, and yields uniqueness of the maximiser in its strict form but not sufficiency of the first-order condition.

The Farkas–Minkowski theorem converts the non-existence of an improving feasible direction into the existence of non-negative multipliers, under a Slater condition that cannot be dropped. Separation delivers supporting prices for a Pareto efficient allocation when both preferences and technology are convex, and delivers strictly positive state prices exactly when no arbitrage exists.

Exercise 9.8.1. Show that the set of positive semidefinite symmetric $n \times n$ matrices is a closed convex cone, and that $A \mapsto \lambda_{\min}(A)$ is concave on it. Deduce that $\{A : A \succeq \varepsilon I\}$ is convex for each $\varepsilon \geq 0$, and relate this to Proposition 2.5.1.

Exercise 9.8.2. Let $u\colon \mathbb{R}^n_+ \to \mathbb{R}$ be continuous and quasiconcave, and let $v(p,m) = \max_{x \in \mathcal{B}(p,m)} u(x)$. Show that v is quasiconvex in (p,m): that is, $\{(p,m) : v(p,m) \leq \alpha\}$ is convex. (Hint: show that $\mathcal{B}(\lambda p + (1-\lambda)p', \lambda m + (1-\lambda)m') \subseteq \mathcal{B}(p,m) \cup \mathcal{B}(p',m')$.) Explain why the conclusion is quasiconvexity rather than convexity, and contrast with the profit function of Exercise 5.6.3.

Exercise 9.8.3. In the setting of Theorem 9.7.2, suppose $S = 3$, $J = 2$, and $R = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{smallmatrix}\right)$ with $q = (1,2)^\mathsf{T}$. Determine the set of state price vectors consistent with no arbitrage, verify that it is a non-empty relatively open convex set, and compute the interval of prices at which a claim paying $(0,1,0)^\mathsf{T}$ can trade without creating an arbitrage. Identify where completeness fails.

Chapter 10

Integration

10.1 Motivation: Moments of a Distribution

Expectations are integrals. The question that organises this chapter is the one posed on (Fukuda 2026a, slide 65): if X is normally distributed with density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{10.1}$$

confirm that $\mathbb{E}[X] = \mu$ and $\mathrm{Var}[X] = \sigma^2$. Doing so requires three tools — integration by parts, integration by substitution, and a change of variables in the plane — and, before any of them, a definition of the integral precise enough that the manipulations are licensed rather than merely plausible.

10.2 The Riemann Integral

Definition 10.2.1 (Partitions and Darboux sums)**.** Let $f\colon [a,b] \to \mathbb{R}$ be bounded. A *partition* of $[a,b]$ is a finite set $P = \{a = t_0 < t_1 < \cdots < t_k = b\}$. Writing $m_i := \inf_{t\in[t_{i-1},t_i]} f(t)$ and $M_i := \sup_{t\in[t_{i-1},t_i]} f(t)$, the *lower* and *upper Darboux sums* are

$$L(f,P) := \sum_{i=1}^{k} m_i(t_i - t_{i-1}), \qquad U(f,P) := \sum_{i=1}^{k} M_i(t_i - t_{i-1}).$$

The function is *Riemann integrable* on $[a,b]$ if $\sup_P L(f,P) = \inf_P U(f,P)$, and the common value is $\int_a^b f(x)\,\mathrm{d}x$.

Boundedness is part of the definition: an unbounded f has $M_i = +\infty$ on some subinterval of every partition. Integrals over unbounded domains, or of unbounded functions, are defined as limits in Definition 10.3.2.

Proposition 10.2.1 (Riemann's criterion)**.** A bounded $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is a partition P with $U(f,P) - L(f,P) < \varepsilon$.

Proof. Refining a partition cannot increase U or decrease L, and any two partitions have a common refinement, so $L(f,P) \le U(f,Q)$ for all P, Q and hence $\sup_P L \le \inf_Q U$. If the criterion holds, the two are within ε of each other for every ε, hence equal. Conversely, if they are equal to I, choose P_1 with $L(f,P_1) > I - \varepsilon/2$ and P_2 with $U(f,P_2) < I + \varepsilon/2$ and take their common refinement. ■

Proposition 10.2.2 (Sufficient conditions for integrability)**.** Let $f\colon [a,b] \to \mathbb{R}$.

(i) If f is continuous, it is Riemann integrable.

(ii) If f is bounded with finitely many discontinuities, it is Riemann integrable.

(iii) If f is monotone, it is Riemann integrable.

Proof. (i) $[a, b]$ is compact, so f is bounded by Corollary 5.4.1 and uniformly continuous by Theorem 5.2.2. Given $\varepsilon > 0$ choose $\delta > 0$ with $|f(s) - f(t)| < \varepsilon/(b-a)$ whenever $|s - t| < \delta$, and take any partition of mesh less than δ. On each subinterval the supremum and infimum are attained (Corollary 5.4.1) at points less than δ apart, so $M_i - m_i \le \varepsilon/(b-a)$ and

$$U(f, P) - L(f, P) = \sum_i (M_i - m_i)(t_i - t_{i-1}) \le \frac{\varepsilon}{b-a} \sum_i (t_i - t_{i-1}) = \varepsilon.$$

Apply Proposition 10.2.1.

(ii) Let D be the finite set of discontinuities and $|f| \le M$. Cover D by finitely many intervals of total length less than $\varepsilon/(4M)$; on those intervals the contribution to $U - L$ is at most $2M \cdot \varepsilon/(4M) = \varepsilon/2$. On the complement, a finite union of closed intervals on which f is continuous, apply (i) to get a further $\varepsilon/2$.

(iii) Let f be non-decreasing and take the uniform partition of mesh $(b-a)/k$. Then $M_i = f(t_i)$ and $m_i = f(t_{i-1})$, so the sum telescopes:

$$U(f, P) - L(f, P) = \frac{b-a}{k} \sum_{i=1}^{k} \big(f(t_i) - f(t_{i-1})\big) = \frac{b-a}{k} \big(f(b) - f(a)\big) \longrightarrow 0.$$

■

This is Proposition 3.11 of Fukuda (2026d) (Fukuda 2026a, slide 66), with the proofs supplied. A monotone function may have countably many discontinuities and is still integrable, so (ii) and (iii) are genuinely different criteria. The exact characterisation — integrable if and only if the set of discontinuities has Lebesgue measure zero — is Lebesgue's criterion, for which see Rudin (1976, Thm. 11.33); it is the point at which the Riemann theory is superseded by the Lebesgue theory of Chapter 14.

Proposition 10.2.3 (Basic properties)**.** Let f, g be Riemann integrable on $[a, b]$ and $\alpha, \beta \in \mathbb{R}$. Then

(i) $\alpha f + \beta g$ is integrable with $\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$;

(ii) for $c \in (a, b)$, $\int_a^b f = \int_a^c f + \int_c^b f$;

(iii) $f \le g$ implies $\int_a^b f \le \int_a^b g$; in particular $\left|\int_a^b f\right| \le \int_a^b |f|$;

(iv) if f is continuous there is $\xi \in [a, b]$ with $\int_a^b f = f(\xi)(b-a)$.

Proof. (i)–(iii) follow from the corresponding statements for Darboux sums and Proposition 10.2.1; see Rudin (1976, Thm. 6.12). For (iv), let $m = \min_{[a,b]} f$ and $M = \max_{[a,b]} f$, both attained by Corollary 5.4.1. By (iii), $m(b-a) \le \int_a^b f \le M(b-a)$, so $\frac{1}{b-a} \int_a^b f \in [m, M]$, and the intermediate value theorem (Corollary 5.2.1) supplies ξ with $f(\xi)$ equal to it. ■

10.3 The Fundamental Theorem of Calculus

Definition 10.3.1 (Indefinite integral and antiderivative)**.** Let $f\colon [a, b] \to \mathbb{R}$ be integrable. The *indefinite integral* of f is $F(x) := \int_a^x f(t)\,\mathrm{d}t$ for $x \in [a, b]$. A function $G\colon [a, b] \to \mathbb{R}$ is an *antiderivative* (a primitive) of f if G is continuous on $[a, b]$ and $G'(x) = f(x)$ for every $x \in (a, b)$. Any two antiderivatives differ by a constant, by Theorem 7.2.5, which is why the generic antiderivative is written $\int f(x)\,\mathrm{d}x = G(x) + c$.

Theorem 10.3.1 (Fundamental theorem of calculus)**.** Let $f\colon [a, b] \to \mathbb{R}$ be Riemann integrable — hence bounded, by Definition 10.2.1 — and let $F(x) = \int_a^x f(t)\,\mathrm{d}t$.

(i) F is Lipschitz, hence continuous, on $[a, b]$; and F is differentiable with $F'(x_0) = f(x_0)$ at every point x_0 at which f is continuous, one-sidedly at $x_0 \in \{a, b\}$. In particular, if f is continuous on $[a, b]$ then $F' = f$ throughout.

(ii) Let $G\colon [a,b] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b) with $G'(x) = f(x)$ for every $x \in (a,b)$. Then

$$\int_a^b f(t)\,\mathrm{d}t = G(b) - G(a) := \big[G(t)\big]_a^b.$$

The regularity demanded of G in (ii) is what the mean value theorem consumes and no more: continuity up to the endpoints, differentiability strictly inside. Nothing is assumed about G' at a or b, and the values $f(a)$ and $f(b)$ are irrelevant to both sides — altering an integrable function at finitely many points changes neither its integral nor the hypotheses on G.

Proof. (i) With $|f| \le M$, Proposition 10.2.3(ii)–(iii) gives $|F(y) - F(x)| = \left|\int_x^y f\right| \le M\,|y-x|$, which is the Lipschitz bound. Let f be continuous at x_0 and $\varepsilon > 0$; choose $\delta > 0$ with $|f(t) - f(x_0)| < \varepsilon$ for $|t - x_0| < \delta$. For $0 < |h| < \delta$,

$$\left|\frac{F(x_0+h) - F(x_0)}{h} - f(x_0)\right| = \left|\frac{1}{h}\int_{x_0}^{x_0+h} \big(f(t) - f(x_0)\big)\,\mathrm{d}t\right| \le \varepsilon,$$

using Proposition 10.2.3(iii) and $\frac{1}{h}\int_{x_0}^{x_0+h} \mathrm{d}t = 1$.

(ii) Let $P = \{a = t_0 < \cdots < t_k = b\}$ be any partition. By Theorem 7.2.5 applied to G on each $[t_{i-1}, t_i]$ there is $\xi_i \in (t_{i-1}, t_i)$ with $G(t_i) - G(t_{i-1}) = G'(\xi_i)(t_i - t_{i-1}) = f(\xi_i)(t_i - t_{i-1})$. Summing, the left side telescopes to $G(b) - G(a)$, and since $m_i \le f(\xi_i) \le M_i$,

$$L(f,P) \le G(b) - G(a) \le U(f,P)$$

for every P. Integrability makes the two outer quantities have the common supremum and infimum $\int_a^b f$, so $G(b) - G(a) = \int_a^b f$. ∎

The two halves say different things, and the slides' formulation $\frac{\mathrm{d}}{\mathrm{d}x}\int f(x)\,\mathrm{d}x = f$ and $\int f'(x)\,\mathrm{d}x = f + c$ (Fukuda 2026a, slide 68) compresses them. Part (i) constructs an antiderivative and requires continuity of f at the point in question; part (ii) evaluates an integral given an antiderivative and requires only integrability of f together with the existence of G. Neither implies the other: there are differentiable G whose derivative is not Riemann integrable, for which (ii) does not apply, and integrable f with no antiderivative, for which (i) fails at the discontinuities.

10.3.1 Integration by parts

Theorem 10.3.2 (Integration by parts). Let $u, v\colon [a,b] \to \mathbb{R}$ be differentiable with u' and v' Riemann integrable. Then

$$\int_a^b u'(x)v(x)\,\mathrm{d}x = \big[u(x)v(x)\big]_a^b - \int_a^b u(x)v'(x)\,\mathrm{d}x. \tag{10.2}$$

Proof. By the product rule (Proposition 7.3.2), uv is differentiable with $(uv)' = u'v + uv'$, and the right side is integrable as a sum of products of integrable functions. Apply Theorem 10.3.1(ii) to uv and then Proposition 10.2.3(i). ∎

Identity (10.2) is usually applied in the form of (Fukuda 2026a, slide 71): given a product fg to integrate, take an antiderivative F of f and differentiate g,

$$\int_a^b f(x)g(x)\,\mathrm{d}x = \big[F(x)g(x)\big]_a^b - \int_a^b F(x)g'(x)\,\mathrm{d}x,$$

choosing the split so that Fg' is simpler than fg. Two standard choices: $g = \log x$ with $f = 1$ reduces $\int \log x\,\mathrm{d}x$ to $\int 1\,\mathrm{d}x$, giving $x(\log x - 1) + c$ (Fukuda 2026a, slide 78); and g a power with f an exponential reduces the power by one at each application, which is the recursion used in subsection 10.7.1.

10.3.2 Substitution

Theorem 10.3.3 (Change of variable in one dimension). Let $\varphi\colon [a,b] \to \mathbb{R}$ be continuously differentiable and let f be continuous on an interval containing $\varphi([a,b])$. Then

$$\int_{\varphi(a)}^{\varphi(b)} f(x)\,\mathrm{d}x = \int_a^b f\big(\varphi(t)\big)\varphi'(t)\,\mathrm{d}t. \tag{10.3}$$

Proof. Let F be an antiderivative of f, which exists by Theorem 10.3.1(i) since f is continuous. By the chain rule (Proposition 7.3.4), $(F\circ\varphi)'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t)$, which is continuous hence integrable. Applying Theorem 10.3.1(ii) twice,

$$\int_a^b f\big(\varphi(t)\big)\varphi'(t)\,\mathrm{d}t = \big[F(\varphi(t))\big]_a^b = F(\varphi(b)) - F(\varphi(a)) = \int_{\varphi(a)}^{\varphi(b)} f(x)\,\mathrm{d}x.$$

■

This is Proposition 3.16 of Fukuda (2026d) (Fukuda 2026a, slide 82). Note what the theorem does *not* require: φ need not be injective, and the limits of integration are $\varphi(a)$ and $\varphi(b)$ in that order, which may be reversed relative to $a<b$. The mnemonic $x=\varphi(t)$, $\mathrm{d}x = \varphi'(t)\,\mathrm{d}t$ (Fukuda 2026a, slide 81) is the identity (10.3) written in differentials; the substance is the chain rule, and the differentials are notation for it.

Example 10.3.1 (Two substitutions)**.** With $\varphi(t) = 2t+3$, $\varphi' = 2$, and $f(x) = x^4$,

$$\int_1^2 (2t+3)^4\,\mathrm{d}t = \frac{1}{2}\int_5^7 x^4\,\mathrm{d}x = \frac{1}{10}\big[x^5\big]_5^7 = \frac{7^5-5^5}{10} = \frac{6841}{10}.$$

With $\varphi(x) = -x^2$, $\varphi'(x) = -2x$, and $f(y) = e^y$,

$$\int xe^{-x^2}\,\mathrm{d}x = -\frac{1}{2}\int e^y\,\mathrm{d}y = -\frac{1}{2}e^{-x^2} + c.$$

These are the computations of (Fukuda 2026a, slides 83–84). The second is the pattern used for the normal density in subsection 10.7.3: an integrand of the form $g'(x)h(g(x))$ integrates to $H(g(x))$ with $H' = h$.

10.3.3 Improper integrals

Definition 10.3.2 (Improper integral)**.** Let f be Riemann integrable on $[a,c]$ for every $c>a$. Then

$$\int_a^\infty f(x)\,\mathrm{d}x := \lim_{c\to\infty}\int_a^c f(x)\,\mathrm{d}x$$

when the limit exists in $\mathbb{R}$, and analogously at $-\infty$. The integral over $\mathbb{R}$ is defined as $\int_{-\infty}^0 f + \int_0^\infty f$, requiring *both* limits to exist separately. The integral *converges absolutely* if $\int |f| < \infty$, and absolute convergence implies convergence.

The two halves must converge separately. For $f(x) = x$ the symmetric limit $\lim_{c\to\infty}\int_{-c}^c x\,\mathrm{d}x = 0$ exists while $\int_{-\infty}^\infty x\,\mathrm{d}x$ does not, and a distribution with such a density tail has no mean — the Cauchy distribution being the standard example. The distinction is the one between the principal value and the integral, and it is what makes "the expectation does not exist" a meaningful statement.

10.4 Multivariate Integration

Two results are needed for the Gaussian computation. Both are stated without proof and used only where cited; complete treatments are in Rudin (1976, Ch. 10) and, in the measure-theoretic form, Le Gall (2022, Ch. 5).

Theorem 10.4.1 (Fubini–Tonelli)**.** Let $f\colon \mathbb{R}^m\times\mathbb{R}^n\to\mathbb{R}$ be measurable. Suppose either that $f\geq 0$, or that the iterated integral of $|f|$ is finite. Then

$$\int_{\mathbb{R}^{m+n}} f = \int_{\mathbb{R}^m}\left(\int_{\mathbb{R}^n} f(x,y)\,\mathrm{d}y\right)\mathrm{d}x = \int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^m} f(x,y)\,\mathrm{d}x\right)\mathrm{d}y,$$

all three integrals existing and being equal.

The non-negativity or absolute integrability hypothesis is essential; without it the two iterated integrals can exist and differ. In economics the theorem is what licenses interchanging an expectation over states with an integral over time, or summing a double sum in either order.

Theorem 10.4.2 (Change of variables). Let $U, V \subseteq \mathbb{R}^n$ be open and $\Phi\colon U \to V$ a continuously differentiable bijection with $\det D\Phi(u) \neq 0$ for every $u \in U$. Then for every integrable $f\colon V \to \mathbb{R}$,

$$\int_V f(x)\,\mathrm{d}x = \int_U f\big(\Phi(u)\big)\,\big|\det D\Phi(u)\big|\,\mathrm{d}u. \tag{10.4}$$

The factor $|\det D\Phi|$ is the local volume magnification of Φ, which is the interpretation of the determinant established in Example 2.4.1: Φ maps an infinitesimal box of volume $\mathrm{d}u$ to a parallelepiped of volume $|\det D\Phi(u)|\,\mathrm{d}u$ (Fukuda 2026a, slide 93). For $n = 1$, $|\det D\Phi| = |\varphi'|$ and (10.4) is (10.3) with the absolute value accounting for orientation, which the one-dimensional statement instead encodes in the order of the limits.

10.5 Differentiating Under the Integral Sign

Lemma 10.5.1 (Differentiation under the integral). Let $f\colon [a,b] \times [\underline{r}, \bar{r}] \to \mathbb{R}$ be such that $f(\cdot, r)$ is continuous on $[a,b]$ for each r, and $\partial f/\partial r$ exists and is continuous on the whole rectangle. Then $r \mapsto \int_a^b f(x,r)\,\mathrm{d}x$ is differentiable on $(\underline{r}, \bar{r})$ with

$$\frac{\mathrm{d}}{\mathrm{d}r}\int_a^b f(x,r)\,\mathrm{d}x = \int_a^b \frac{\partial f}{\partial r}(x,r)\,\mathrm{d}x.$$

Proof. Fix r and let $h \neq 0$ be small enough that $r + h$ stays in the interval. By Theorem 7.2.5 applied to $s \mapsto f(x,s)$, for each x there is $\theta_x \in (0,1)$ with $\big(f(x, r+h) - f(x,r)\big)/h = \frac{\partial f}{\partial r}(x, r + \theta_x h)$. Hence

$$\left|\frac{1}{h}\int_a^b \big(f(x,r+h) - f(x,r)\big)\,\mathrm{d}x - \int_a^b \frac{\partial f}{\partial r}(x,r)\,\mathrm{d}x\right| \le (b-a)\sup_{x\in[a,b]}\left|\frac{\partial f}{\partial r}(x, r+\theta_x h) - \frac{\partial f}{\partial r}(x,r)\right|.$$

The rectangle is compact, so $\partial f/\partial r$ is uniformly continuous on it by Theorem 5.2.2. The two arguments differ by at most $|h|$, uniformly in x, so the supremum tends to 0 as $h \to 0$. ∎

Theorem 10.5.1 (Leibniz formula). Under the hypotheses of Lemma 10.5.1, let $A, B\colon [\underline{r}, \bar{r}] \to [a,b]$ be continuously differentiable and set $V(r) := \int_{A(r)}^{B(r)} f(x,r)\,\mathrm{d}x$. Then V is differentiable with

$$V'(r) = f\big(B(r), r\big)B'(r) - f\big(A(r), r\big)A'(r) + \int_{A(r)}^{B(r)} \frac{\partial f}{\partial r}(x,r)\,\mathrm{d}x. \tag{10.5}$$

Proof. Let $g(x,r) := \int_a^x f(u,r)\,\mathrm{d}u$, so that

$$\frac{\partial g}{\partial x} = f(x,r) \quad \text{by Theorem 10.3.1(i)}, \qquad \frac{\partial g}{\partial r} = \int_a^x \frac{\partial f}{\partial r}(u,r)\,\mathrm{d}u \quad \text{by Lemma 10.5.1};$$

both are continuous, so g is differentiable in the sense of Definition 8.3.1 by Proposition 8.3.3. By Proposition 10.2.3(ii), $V(r) = g(B(r), r) - g(A(r), r)$, and the chain rule (Theorem 8.3.1) gives

$$V'(r) = \frac{\partial g}{\partial x}(B(r), r)B'(r) + \frac{\partial g}{\partial r}(B(r), r) - \frac{\partial g}{\partial x}(A(r), r)A'(r) - \frac{\partial g}{\partial r}(A(r), r).$$

Substituting the two partial derivatives of g and combining the two $\partial g/\partial r$ terms into a single integral from $A(r)$ to $B(r)$ yields (10.5). ∎

This is Theorem 3.3 of Fukuda (2026d), and the proof is the route mapped out in Exercise 3.6 there (Fukuda 2026a, slides 95–96), carried out. The intuitive derivation on (Fukuda 2026a, slides 97–98) identifies the three terms correctly — the value gained at the moving upper endpoint, the value lost at the moving lower endpoint, and the change in the integrand over the fixed region — and the proof above supplies the hypotheses under which the approximations become identities.

10.6 Economic Application: Continuation Values

Proposition 10.6.1 (Differential form of a continuation value)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be bounded and continuous and $\rho > 0$, and define the continuation value

$$V(t) := \int_t^{\infty} e^{-\rho(s-t)} f(s)\,\mathrm{d}s.$$

Then the integral converges absolutely for every t, V is differentiable, and

$$V'(t) = \rho V(t) - f(t), \qquad \text{equivalently} \qquad \rho V(t) = f(t) + V'(t). \tag{10.6}$$

Proof. Let $|f| \le M$. Then $\int_t^{\infty} e^{-\rho(s-t)}\,|f(s)|\,\mathrm{d}s \le M\int_t^{\infty} e^{-\rho(s-t)}\,\mathrm{d}s = M/\rho < \infty$, so the integral converges absolutely.

Write $W(t) := \int_t^{\infty} e^{-\rho s} f(s)\,\mathrm{d}s$, which converges absolutely by the same bound, so that $V(t) = e^{\rho t}W(t)$. Fixing any t_0, $W(t) = W(t_0) - \int_{t_0}^{t} e^{-\rho s} f(s)\,\mathrm{d}s$, and the integrand is continuous, so Theorem 10.3.1(i) gives $W'(t) = -e^{-\rho t}f(t)$. By the product rule (Proposition 7.3.2),

$$V'(t) = \rho e^{\rho t}W(t) + e^{\rho t}W'(t) = \rho V(t) - f(t).$$

■

Reading (10.6)**.** In the form $\rho V = f + V'$ the identity is an asset-pricing equation: the required return ρV on holding the "asset" whose value is V equals the flow payoff f plus the capital gain V'. It is the deterministic, uncontrolled special case of the Hamilton–Jacobi–Bellman equation of Chapter 16, and the same three terms reappear there with a maximisation over controls inserted in front of f. This is Example 3.22 of Fukuda (2026d) (Fukuda 2026a, slide 94).

The proof deliberately avoids differentiating under an improper integral. The Leibniz formula (10.5) applies on compact rectangles, and $[t, \infty)$ is not one; a direct application would require a domination argument. Rewriting $V = e^{\rho t}W$ moves the t-dependence out of the integrand entirely, leaving only a moving endpoint, and Theorem 10.3.1(i) handles that with no additional hypotheses.

10.7 Economic Application: Moments of Two Distributions

10.7.1 The exponential distribution

Proposition 10.7.1. Let X have density $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$ and $f(x) = 0$ for $x < 0$, with $\lambda > 0$. Then its distribution function is $F(x) = 1 - e^{-\lambda x}$ for $x > 0$, and

$$\mathbb{E}[X] = \frac{1}{\lambda}, \qquad \mathbb{E}[X^2] = \frac{2}{\lambda^2}, \qquad \operatorname{Var}[X] = \frac{1}{\lambda^2}.$$

Proof. For $x > 0$, $F(x) = \int_0^x \lambda e^{-\lambda t}\,\mathrm{d}t = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x}$ by Theorem 10.3.1(ii).

For the first moment, write $\lambda e^{-\lambda x} = \left(-e^{-\lambda x}\right)'$ and apply Theorem 10.3.2 on $[0, c]$:

$$\int_0^c x\lambda e^{-\lambda x}\,\mathrm{d}x = \left[-xe^{-\lambda x}\right]_0^c + \int_0^c e^{-\lambda x}\,\mathrm{d}x = -ce^{-\lambda c} + \frac{1 - e^{-\lambda c}}{\lambda}.$$

Letting $c \to \infty$ and using $ce^{-\lambda c} \to 0$ gives $\mathbb{E}[X] = 1/\lambda$.

For the second moment, the same split with x^2 in place of x gives

$$\int_0^c x^2\lambda e^{-\lambda x}\,\mathrm{d}x = \left[-x^2 e^{-\lambda x}\right]_0^c + 2\int_0^c xe^{-\lambda x}\,\mathrm{d}x = -c^2e^{-\lambda c} + \frac{2}{\lambda}\int_0^c x\lambda e^{-\lambda x}\,\mathrm{d}x,$$

so letting $c \to \infty$ and substituting the first moment yields $\mathbb{E}[X^2] = 2/\lambda^2$. Finally $\operatorname{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$ by Proposition 14.2.2. ■

These are the computations of (Fukuda 2026a, slides 73–77), with the improper integrals handled as limits of integrals over $[0, c]$ in accordance with Definition 10.3.2 rather than by evaluating boundary terms at ∞ directly. The exponential distribution is the continuous-time counterpart of a constant hazard rate: $\mathbb{P}(X > s + t \mid X > s) = e^{-\lambda t}$ does not depend on s, which is why it is the standard model of the arrival of a price-adjustment opportunity in Calvo pricing and of job separation in search models.

10.7.2 The Gaussian integral

Proposition 10.7.2 (Gaussian integral).

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}, \qquad \text{and more generally} \qquad \int_{-\infty}^{\infty} e^{-a(x+b)^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}$$

for every $a > 0$ and $b \in \mathbb{R}$. Consequently the function (10.1) integrates to 1 and is a probability density.

Proof. Let $I := \int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x$, which converges because $e^{-x^2} \le e^{-|x|}$ for $|x| \ge 1$. The integrand of the product I^2 is non-negative, so Theorem 10.4.1 applies and

$$I^2 = \left(\int_{\mathbb{R}} e^{-x^2}\,\mathrm{d}x\right)\left(\int_{\mathbb{R}} e^{-y^2}\,\mathrm{d}y\right) = \int_{\mathbb{R}}\int_{\mathbb{R}} e^{-(x^2+y^2)}\,\mathrm{d}y\,\mathrm{d}x.$$

Apply Theorem 10.4.2 with polar coordinates $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$, a continuously differentiable bijection from $U = (0,\infty)\times(0,2\pi)$ onto $\mathbb{R}^2$ minus a half-line, a set of measure zero which does not affect the integral. Its Jacobian is

$$D\Phi(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \qquad \det D\Phi = r\big(\cos^2\theta + \sin^2\theta\big) = r > 0.$$

Since $x^2 + y^2 = r^2$,

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\, r\,\mathrm{d}r\,\mathrm{d}\theta = 2\pi\left[-\tfrac{1}{2}e^{-r^2}\right]_0^{\infty} = 2\pi\cdot\tfrac{1}{2} = \pi.$$

As $I > 0$, $I = \sqrt{\pi}$. For the general case substitute $x = y/\sqrt{a} - b$ in Theorem 10.3.3, so that $\mathrm{d}x = \mathrm{d}y/\sqrt{a}$ and $a(x+b)^2 = y^2$, giving $\int e^{-a(x+b)^2}\,\mathrm{d}x = I/\sqrt{a} = \sqrt{\pi/a}$. Taking $a = 1/(2\sigma^2)$ and $b = -\mu$ gives $\int_{\mathbb{R}} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)\,\mathrm{d}x = \sqrt{2\pi\sigma^2}$, so (10.1) integrates to 1. ■

This is the derivation of (Fukuda 2026a, slides 89–93), with the interchange of the two integrals justified by Theorem 10.4.1 and the polar substitution by Theorem 10.4.2 rather than by the differential mnemonic $\mathrm{d}y\,\mathrm{d}x = |\det J|\,\mathrm{d}r\,\mathrm{d}\theta$. The trick is that e^{-x^2} has no elementary antiderivative, so Theorem 10.3.1(ii) is unavailable; squaring converts the problem to two dimensions, where the radial symmetry of $e^{-(x^2+y^2)}$ makes the extra factor r from the Jacobian exactly the derivative needed for a substitution.

10.7.3 Mean and variance of the normal

Proposition 10.7.3. Let X have the density (10.1) with $\sigma > 0$. Then $\mathbb{E}[X] = \mu$ and $\mathrm{Var}[X] = \sigma^2$.

Proof. Mean. Since the density integrates to 1 (Proposition 10.7.2),

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x)\,\mathrm{d}x = \int_{\mathbb{R}} (x-\mu) f(x)\,\mathrm{d}x + \mu.$$

The remaining integral vanishes. Substituting $t = (x-\mu)/\sigma$, so $\mathrm{d}x = \sigma\,\mathrm{d}t$ by Theorem 10.3.3, turns it into $\sigma\int_{\mathbb{R}} t\,\phi(t)\,\mathrm{d}t$ with $\phi(t) = (2\pi)^{-1/2}e^{-t^2/2}$; the integrand is odd and absolutely integrable, so each half contributes the negative of the other and the integral is 0. Equivalently, and without appealing to symmetry, note that

$$(x-\mu)f(x) = -\sigma^2\frac{\mathrm{d}}{\mathrm{d}x}f(x), \qquad \text{so} \qquad \int_{-c}^{c}(x-\mu)f(x)\,\mathrm{d}x = -\sigma^2\big[f(x)\big]_{-c}^{c} \longrightarrow 0$$

as $c \to \infty$, since $f(x) \to 0$ at both ends.

Variance. Using the same identity $f'(x) = -(x-\mu)\sigma^{-2}f(x)$ and integrating by parts (Theorem 10.3.2) on $[-c,c]$,

$$\int_{-c}^{c}(x-\mu)^2 f(x)\,\mathrm{d}x = \int_{-c}^{c}(x-\mu)\big(-\sigma^2 f'(x)\big)\,\mathrm{d}x = \big[-\sigma^2(x-\mu)f(x)\big]_{-c}^{c} + \sigma^2\int_{-c}^{c} f(x)\,\mathrm{d}x.$$

The boundary term tends to 0 because f decays faster than any polynomial grows, and the remaining integral tends to 1. Hence $\mathrm{Var}[X] = \mathbb{E}[(X-\mu)^2] = \sigma^2$. ■

These are the computations of (Fukuda 2026a, slides 85–88). The identity $f' = -(x-\mu)\sigma^{-2}f$ is what makes the normal tractable: the density is its own antiderivative up to the linear factor, so every moment computation is an integration by parts that reduces the power by one. Both boundary terms are evaluated as limits over $[-c, c]$; writing $\big[\,\cdot\,\big]_{-\infty}^{\infty}$ directly, as the slides do, presupposes the convergence that Definition 10.3.2 requires be checked, and here it holds because of the Gaussian decay.

10.8 Chapter Summary

The Riemann integral is defined by squeezing a bounded function between step functions; continuity, boundedness with finitely many discontinuities, and monotonicity each suffice for integrability, by arguments that use uniform continuity, a covering of the bad set, and telescoping respectively. The fundamental theorem has two halves with different hypotheses: the indefinite integral of an integrable function is Lipschitz and differentiates back to the integrand wherever the integrand is continuous; and any antiderivative evaluates the definite integral. Integration by parts is the product rule integrated, and substitution is the chain rule integrated; in several variables the substitution rule acquires the factor $|\det D\Phi|$, the local volume magnification established in Chapter 2. Improper integrals are limits and require both tails to converge separately, which is what makes non-existence of a mean a meaningful statement.

Differentiation under the integral sign holds when the parameter derivative is continuous on a compact rectangle, by the mean value theorem and uniform continuity; with moving endpoints it becomes the Leibniz formula (10.5), whose three terms are the flows at the two endpoints and the change in the integrand. Applied to a discounted flow it gives $\rho V = f + V'$, the deterministic Hamilton–Jacobi–Bellman equation.

The exponential distribution's moments follow from repeated integration by parts; the normal density integrates to 1 by the polar-coordinate evaluation of the Gaussian integral, which needs both Fubini's theorem and the multivariate change of variables; and its mean and variance follow from the identity $f' = -(x-\mu)\sigma^{-2}f$ together with one integration by parts each.

> **Exercise 10.8.1.** Show that $f(x) = 1$ for $x \in \mathbb{Q} \cap [0, 1]$ and $f(x) = 0$ otherwise is bounded and not Riemann integrable, by computing $L(f, P)$ and $U(f, P)$ for an arbitrary partition. Then show that it is Lebesgue integrable with integral 0, and identify which hypothesis of Proposition 10.2.2 fails.

> **Exercise 10.8.2.** Compute the moment generating function $M(s) = \mathbb{E}[e^{sX}]$ for X normal with mean μ and variance σ^2, by completing the square in the exponent and applying Proposition 10.7.2. Verify $M'(0) = \mu$ and $M''(0) = \mu^2 + \sigma^2$, stating the hypothesis under which the two differentiations under the integral sign are legitimate.

> **Exercise 10.8.3.** A firm's value is $V(t) = \int_t^{T(t)} e^{-\rho(s-t)}\pi(s, t)\,\mathrm{d}s$, where the exit date T and the flow profit π both depend on t. Apply (10.5) to compute $V'(t)$, identify the three terms economically, and show that when T is chosen optimally the term involving $T'(t)$ vanishes. Relate the last observation to the envelope theorem of Chapter 12.

Chapter 11

Projection, Orthogonality, and Least Squares

11.1 Motivation: Least Squares Is Not About Calculus

Proposition 8.8.1 solved the least-squares problem by setting a gradient to zero. The solution is correct, and it conceals the structure that generalises. Writing $\mathbf{1} = (1, \ldots, 1)^\top$, $x = (x_1, \ldots, x_n)^\top$ and $y = (y_1, \ldots, y_n)^\top$ in $\mathbb{R}^n$, the objective is

$$S(\beta_1, \beta_2) = \sum_{i=1}^{n} \bigl(y_i - \beta_1 - \beta_2 x_i\bigr)^2 = \|y - \beta_1 \mathbf{1} - \beta_2 x\|^2, \tag{11.1}$$

so the problem is to find the point of the plane $\operatorname{span}\{\mathbf{1}, x\}$ closest to y (Fukuda 2026a, slide 61). That is a question about distance in an inner product space, and its answer — drop a perpendicular — does not involve differentiation, holds in infinite dimensions where the calculus argument would not, and explains why the first-order conditions look the way they do: the normal equations *are* the statement that the residual is orthogonal to the regressors.

This chapter develops that geometry. Its three payoffs are the orthogonal decomposition theorem, the projection formula $P = X(X^\top X)^{-1}X^\top$, and the identification of conditional expectation as a projection, which Chapter 14 uses to characterise the best predictor.

11.2 Orthogonality

Throughout, H is a real inner product space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$ (Proposition 3.3.1), and it is a Hilbert space when complete (Definition 3.5.1). The leading cases are $H = \mathbb{R}^n$ and $H = L^2$.

Definition 11.2.1 (Orthogonal complement). For $\emptyset \neq A \subseteq H$, the *orthogonal complement* is

$$A^\perp := \{x \in H : \langle x, a \rangle = 0 \text{ for all } a \in A\}.$$

A set A is *orthogonal* if $\langle x, y \rangle = 0$ for all distinct $x, y \in A$, and *orthonormal* if in addition $\|x\| = 1$ for every $x \in A$.

Proposition 11.2.1 (Elementary properties). Let $A, B \subseteq H$ be non-empty.

(i) $A^\perp$ is a closed vector subspace of H, whether or not A is.

(ii) $A \subseteq B$ implies $B^\perp \subseteq A^\perp$, and $A \subseteq A^{\perp\perp}$, and $A^\perp = A^{\perp\perp\perp}$.

(iii) $x \in \bigl(\operatorname{span}\{x_1, \ldots, x_m\}\bigr)^\perp$ if and only if $\langle x, x_i \rangle = 0$ for every i.

(iv) An orthogonal set not containing 0 is linearly independent.

Proof. (i) Linearity of the inner product in its first argument makes $A^\perp$ a subspace. For closedness, let $x_k \to x$ with $x_k \in A^\perp$; for each $a \in A$, $\langle x, a\rangle = \lim_k \langle x_k, a\rangle = 0$ by continuity of the inner product, which follows from Cauchy–Schwarz (Lemma 3.2.1).

(ii) The first claim is immediate. For $A \subseteq A^{\perp\perp}$: if $a \in A$ then $\langle a, z\rangle = 0$ for every $z \in A^\perp$ by definition of $A^\perp$, so $a \in A^{\perp\perp}$. Applying the first claim to $A \subseteq A^{\perp\perp}$ gives $A^{\perp\perp\perp} \subseteq A^\perp$, and applying $A \subseteq A^{\perp\perp}$ with $A^\perp$ in place of A gives the reverse inclusion.

(iii) If $\langle x, x_i\rangle = 0$ for every i, then $\langle x, \sum_i \lambda_i x_i\rangle = 0$ by linearity. The converse is the special case of singleton combinations.

(iv) Let $x_1, \dots, x_m$ be orthogonal and non-zero with $\sum_i \lambda_i x_i = 0$. Taking the inner product with x_j kills every term but the jth, giving $\lambda_j \|x_j\|^2 = 0$, hence $\lambda_j = 0$. ■

These are Propositions 9.4 and 9.6 of Fukuda (2026d, pp. 299–301). Part (i) contains a fact used repeatedly: orthogonal complements are automatically closed, which is why the decomposition theorem below can be stated for an arbitrary closed subspace M without any hypothesis on $M^\perp$.

Proposition 11.2.2 (Pythagoras). If $x \perp y$ then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$. More generally, for a finite orthogonal set $\{x_1, \dots, x_m\}$, $\|\sum_i x_i\|^2 = \sum_i \|x_i\|^2$.

Proof. Expand $\|x+y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$ and use $\langle x, y\rangle = 0$; iterate for the general case. ■

11.3 The Projection Theorem

Theorem 11.3.1 (Projection onto a closed convex set). Let H be a Hilbert space, $\emptyset \neq M \subseteq H$ closed and convex, and $x \in H$. Then there is a unique $P_M x \in M$ with

$$\|x - P_M x\| = \inf_{y \in M} \|x - y\|,$$

and $y_0 = P_M x$ is characterised by

$$\langle x - y_0, z - y_0\rangle \le 0 \qquad \text{for all } z \in M. \tag{11.2}$$

Proof. Existence. Let $d := \inf_{y\in M} \|x-y\|$ and take $(y_k) \subseteq M$ with $\|x - y_k\| \to d$. The parallelogram law (Remark 3.3.1) applied to $x - y_j$ and $x - y_k$ gives

$$\|y_j - y_k\|^2 = 2\|x-y_j\|^2 + 2\|x-y_k\|^2 - 4\left\|x - \frac{y_j+y_k}{2}\right\|^2 \le 2\|x-y_j\|^2 + 2\|x-y_k\|^2 - 4d^2,$$

where the inequality uses $\frac{1}{2}(y_j + y_k) \in M$ by convexity. As $j, k \to \infty$ the right side tends to $2d^2 + 2d^2 - 4d^2 = 0$, so (y_k) is Cauchy. Completeness of H and closedness of M (Proposition 6.3.2) give a limit $y_0 \in M$, and $\|x - y_0\| = \lim_k \|x - y_k\| = d$ by continuity of the norm.

Uniqueness and characterisation. Identical to the corresponding parts of the proof of Lemma 9.2.1, which used only the parallelogram law and convexity. ■

This is Theorem 9.1 with Proposition 9.1 of Fukuda (2026d, pp. 295–298). The comparison with Lemma 9.2.1 is instructive. There, existence came from compactness: a closed bounded subset of $\mathbb{R}^n$ is compact and a continuous function attains its minimum. Here compactness is unavailable — the closed unit ball of an infinite-dimensional space is not compact (Remark 4.4.1) — and the parallelogram law does the work instead, converting a minimising sequence into a Cauchy sequence, which completeness then converges. Convexity is used exactly once, to place the midpoint $\frac{1}{2}(y_j + y_k)$ in M, and the theorem is false without it: in $\mathbb{R}^2$ the point 0 has no unique nearest point in the closed non-convex set $\{y : \|y\| = 1\}$.

Proposition 11.3.1 (Characterisation for subspaces). Let $M \subseteq H$ be a closed subspace and $x \in H$. Then $y_0 \in M$ minimises $\|x - y\|$ over M if and only if

$$\langle x - y_0, z\rangle = 0 \qquad \text{for all } z \in M, \tag{11.3}$$

that is, if and only if $x - y_0 \in M^\perp$.

Proof. (⇒) Let $u \in M$ and $t \in \mathbb{R}$. Then $y_0 + tu \in M$, so

$$\|x - y_0\|^2 \le \|x - y_0 - tu\|^2 = \|x - y_0\|^2 - 2t\langle x - y_0, u\rangle + t^2\|u\|^2 .$$

Hence $t^2\|u\|^2 - 2t\langle x - y_0, u\rangle \ge 0$ for *every* real t, including t of both signs. A quadratic in t with a non-negative value everywhere and a zero at $t = 0$ must have vanishing linear coefficient, so $\langle x - y_0, u\rangle = 0$.

(⇐) If $x - y_0 \perp M$ then for any $y \in M$, $y_0 - y \in M$ and Pythagoras (Proposition 11.2.2) gives

$$\|x - y\|^2 = \|(x - y_0) + (y_0 - y)\|^2 = \|x - y_0\|^2 + \|y_0 - y\|^2 \ge \|x - y_0\|^2 .$$

■

This is Proposition 9.8 of Fukuda (2026d, p. 301). Comparing (11.3) with (11.2): on a subspace the variational inequality holds for z and for $-z$ simultaneously, so it collapses to an equality. The last display also proves more than was asked: it quantifies the loss from using any other $y \in M$, and that identity is the decomposition of total variation used in Equation 11.7 below.

Theorem 11.3.2 (Orthogonal decomposition). Let M be a closed subspace of a Hilbert space H. Then every $x \in H$ has a unique decomposition

$$x = y + z, \qquad y = P_M x \in M, \quad z \in M^\perp,$$

so that $H = M \oplus M^\perp$, and $M^{\perp\perp} = M$.

Proof. *Existence.* If $x \in M$ take $(y, z) = (x, 0)$. Otherwise let $y = P_M x$ from Theorem 11.3.1, which applies since a closed subspace is closed and convex, and let $z := x - y$; then $z \in M^\perp$ by Proposition 11.3.1.

Uniqueness. If $y_1 + z_1 = y_2 + z_2$ with $y_i \in M$, $z_i \in M^\perp$, then $y_1 - y_2 = z_2 - z_1 \in M \cap M^\perp$. Any w in that intersection satisfies $\langle w, w\rangle = 0$, hence $w = 0$.

$M^{\perp\perp} = M$. The inclusion $M \subseteq M^{\perp\perp}$ is Proposition 11.2.1(ii). Conversely, let $x \in M^{\perp\perp}$ and decompose $x = y + z$ as above. Then $z = x - y \in M^\perp$, and also $z \in M^{\perp\perp}$ since both x and $y \in M \subseteq M^{\perp\perp}$ lie there. So $\langle z, z\rangle = 0$ and $x = y \in M$. ■

This is Theorem 9.2 of Fukuda (2026d, p. 302). The hypothesis that M be *closed* cannot be dropped, and in infinite dimensions it is not automatic: by Corollary 6.3.1 every subspace of $\mathbb{R}^n$ is closed, but the subspace of finitely supported sequences in ℓ^2 is dense and proper, so its orthogonal complement is $\{0\}$ and no decomposition of the stated form exists.

Proposition 11.3.2 (The projection operator). Let $M \subseteq H$ be a closed subspace. The map $P_M : H \to M$ is

(i) linear and surjective onto M;

(ii) idempotent, $P_M^2 = P_M$, with $\operatorname{Ker} P_M = M^\perp$ and $\operatorname{Im} P_M = M$;

(iii) self-adjoint, $\langle P_M x, w\rangle = \langle x, P_M w\rangle$ for all $x, w \in H$;

(iv) norm-reducing, $\|P_M x\| \le \|x\|$, with equality if and only if $x \in M$.

Proof. (i) Let $x = y + z$ and $x' = y' + z'$ be the decompositions of Theorem 11.3.2 and $\alpha, \beta \in \mathbb{R}$. Then $\alpha x + \beta x' = (\alpha y + \beta y') + (\alpha z + \beta z')$ with the first bracket in M and the second in $M^\perp$, both being subspaces; by uniqueness this is *the* decomposition, so $P_M(\alpha x + \beta x') = \alpha P_M x + \beta P_M x'$. Surjectivity: $P_M y = y$ for $y \in M$.

(ii) Immediate from $P_M y = y$ on M and $P_M z = 0$ on $M^\perp$.

(iii) Write $x = y + z$, $w = y' + z'$. Then $\langle P_M x, w\rangle = \langle y, y' + z'\rangle = \langle y, y'\rangle$ since $y \in M$ and $z' \in M^\perp$, and symmetrically $\langle x, P_M w\rangle = \langle y + z, y'\rangle = \langle y, y'\rangle$.

(iv) $\|x\|^2 = \|y\|^2 + \|z\|^2 \ge \|y\|^2$ by Proposition 11.2.2, with equality exactly when $z = 0$. ■

This is Proposition 9.9 of Fukuda (2026d, p. 302), with the self-adjointness added because it is what identifies the projection matrix in Proposition 11.4.1 and what makes the residual maker symmetric.

Theorem 11.3.3 (Riesz representation). Let H be a Hilbert space and $f : H \to \mathbb{R}$ a continuous linear functional. Then there is a unique $a \in H$ with $f(x) = \langle a, x\rangle$ for all $x \in H$, and $\|a\|$ equals the operator norm of f.

Proof. If $f \equiv 0$ take $a = 0$. Otherwise $M := \operatorname{Ker} f$ is a closed proper subspace — closed because f is continuous and $M = f^{-1}(\{0\})$ (Theorem 5.2.1) — so by Theorem 11.3.2 there is $u \in M^\perp$ with $\|u\| = 1$. For any $x \in H$ the vector $f(x)u - f(u)x$ lies in M, since f of it is $f(x)f(u) - f(u)f(x) = 0$; taking the inner product with $u \in M^\perp$ gives $f(x) - f(u)\langle u, x\rangle = 0$, that is $f(x) = \langle f(u)u, x\rangle$. So $a = f(u)u$ works. Uniqueness: if $\langle a, x\rangle = \langle a', x\rangle$ for all x, take $x = a - a'$.

The norm. Write $\|f\| := \sup\{|f(x)| : \|x\| \le 1\}$ for the operator norm. Cauchy–Schwarz (Lemma 3.2.1) gives $|f(x)| = |\langle a, x\rangle| \le \|a\|\,\|x\|$, so $\|f\| \le \|a\|$. For the reverse inequality: if $a = 0$ then $f \equiv 0$ and both norms vanish; otherwise $x := a/\|a\|$ has $\|x\| = 1$ and $f(x) = \langle a, a\rangle / \|a\| = \|a\|$, so $\|f\| \ge \|a\|$. Hence $\|f\| = \|a\|$, and the supremum defining $\|f\|$ is attained. ■

This is Theorem 9.3 of Fukuda (2026d, p. 304) and the promised generalisation of Theorem 3.2.1; in $\mathbb{R}^n$ continuity was automatic by Proposition 5.2.1(ii), and in infinite dimensions it must be assumed, since discontinuous linear functionals exist there.

11.4 Projection in $\mathbb{R}^n$ and the Projection Matrix

Proposition 11.4.1 (Projection matrix)**.** Let $X \in M_{n\times k}(\mathbb{R})$ have linearly independent columns, and let $M := \operatorname{Im} X = \operatorname{span}\{X_{\cdot 1}, \dots, X_{\cdot k}\} \subseteq \mathbb{R}^n$. Then $X^\top X$ is invertible, and for every $y \in \mathbb{R}^n$,

$$P_M y = X\bigl(X^\top X\bigr)^{-1} X^\top y, \qquad P := X\bigl(X^\top X\bigr)^{-1} X^\top. \tag{11.4}$$

The matrix P is symmetric and idempotent with $\operatorname{rank} P = k$, and $Q := I_n - P$ is the projection onto $M^\perp$, symmetric and idempotent with $\operatorname{rank} Q = n - k$.

Proof. Invertibility. Suppose $X^\top X v = 0$ for some $v \in \mathbb{R}^k$. Then $0 = v^\top X^\top X v = \|Xv\|^2$, so $Xv = 0$, and linear independence of the columns forces $v = 0$. Hence $\operatorname{Ker}(X^\top X) = \{0\}$ and Theorem 2.3.2 makes the square matrix $X^\top X$ invertible.

The formula. Set $\hat y := X(X^\top X)^{-1}X^\top y$, which lies in M because it is X times a vector. By Proposition 11.3.1 it suffices to check $y - \hat y \perp M$, and by Proposition 11.2.1(iii) it suffices to check orthogonality to each column of X, that is $X^\top(y - \hat y) = 0$:

$$X^\top y - X^\top X\bigl(X^\top X\bigr)^{-1} X^\top y = X^\top y - X^\top y = 0.$$

Properties. Symmetry: $P^\top = X\bigl((X^\top X)^{-1}\bigr)^\top X^\top = P$, since $X^\top X$ is symmetric and so is its inverse. Idempotence: $P^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = P$. Rank: $\operatorname{Im} P = M$ has dimension k. The statements for Q follow from Theorem 11.3.2, $Q = I - P$ being the projection onto $M^\perp$, whose dimension is $n - k$ by Theorem 2.3.2. ■

This is Theorem 9.4 of Fukuda (2026d, p. 305), and it is the formula displayed on (Fukuda 2026a, slide 62). Both hypotheses on X carry economic content: linear independence of the columns is the no-perfect-multicollinearity condition of Remark 2.3.1, and $k \le n$ is the requirement of at least as many observations as parameters, which is forced because k independent vectors cannot live in $\mathbb{R}^n$ with $k > n$ by Theorem 2.2.1.

11.4.1 Gram–Schmidt and QR

Proposition 11.4.2 (Gram–Schmidt)**.** Let $v_1, \dots, v_k \in H$ be linearly independent. Define recursively

$$u_1 := v_1, \qquad u_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, u_i\rangle}{\|u_i\|^2}\, u_i \quad (j = 2, \dots, k), \qquad e_j := \frac{u_j}{\|u_j\|}.$$

Then $\{e_1, \dots, e_k\}$ is orthonormal and $\operatorname{span}\{e_1, \dots, e_j\} = \operatorname{span}\{v_1, \dots, v_j\}$ for every j.

Proof. Induction on j. Each u_j is v_j minus its projection onto $\operatorname{span}\{u_1, \dots, u_{j-1}\}$, expressed through the orthogonal basis by Proposition 11.2.2; by Proposition 11.3.1 the difference is orthogonal to that span. It is non-zero because $v_j \notin \operatorname{span}\{v_1, \dots, v_{j-1}\} = \operatorname{span}\{u_1, \dots, u_{j-1}\}$ by independence, so the normalisation is legitimate, and the span statement follows since the transformation is triangular with unit diagonal. ■

Corollary 11.4.1 (QR decomposition). Every $X \in M_{n\times k}(\mathbb{R})$ with independent columns factors as $X = QR$ with $Q \in M_{n\times k}(\mathbb{R})$ having orthonormal columns and $R \in M_{k\times k}(\mathbb{R})$ upper triangular with positive diagonal. Then $P = QQ^\top$ and the least-squares coefficient solves $R\hat{\beta} = Q^\top y$.

Proof. Apply Proposition 11.4.2 to the columns of X and collect the coefficients of v_j in terms of $e_1, \dots, e_j$ into the jth column of R, which is upper triangular by construction. Then $X^\top X = R^\top Q^\top QR = R^\top R$ and (11.4) gives $P = QR(R^\top R)^{-1}R^\top Q^\top = QQ^\top$; substituting into $X^\top X\hat{\beta} = X^\top y$ gives $R^\top R\hat{\beta} = R^\top Q^\top y$, and $R^\top$ is invertible. ■

This is Section 9.5 of Fukuda (2026d, pp. 320–323). The practical point is numerical: forming $X^\top X$ squares the condition number of X, so the explicit inverse in (11.4) is a formula for analysis and a poor algorithm, whereas solving the triangular system $R\hat{\beta} = Q^\top y$ is stable. Statistical software computes the QR factorisation, not the inverse.

11.5 Econometric Application: Least Squares

Setting. Observations $(x_i, y_i)_{i=1}^n$ are stacked as $y \in \mathbb{R}^n$ and $X = [\mathbf{1}\ \ x] \in M_{n\times 2}(\mathbb{R})$. The least-squares problem is (11.1).

Proposition 11.5.1 (Least squares as projection). Suppose the columns of X are linearly independent, equivalently that the x_i are not all equal. Then (11.1) has the unique solution

$$\hat{\beta} = \left(X^\top X\right)^{-1} X^\top y, \tag{11.5}$$

the fitted vector is $\hat{y} = X\hat{\beta} = Py$, the residual is $\hat{e} = y - \hat{y} = Qy$, and

$$X^\top \hat{e} = 0, \qquad \text{that is} \qquad \sum_{i=1}^n \hat{e}_i = 0 \quad \text{and} \quad \sum_{i=1}^n x_i\hat{e}_i = 0. \tag{11.6}$$

Moreover

$$\|y - \bar{y}\mathbf{1}\|^2 = \|\hat{y} - \bar{y}\mathbf{1}\|^2 + \|\hat{e}\|^2. \tag{11.7}$$

Proof. Minimising (11.1) is minimising $\|y - v\|$ over $v \in M = \operatorname{Im} X$, so the solution is $\hat{y} = P_M y$, unique by Theorem 11.3.1, and given by (11.4). Since X has independent columns, the representation $\hat{y} = X\hat{\beta}$ has a unique coefficient vector, namely (11.5). Orthogonality (11.6) is Proposition 11.3.1 combined with Proposition 11.2.1(iii), the two scalar equations being orthogonality to $\mathbf{1}$ and to x.

For (11.7): the first equation of (11.6) gives $\bar{\hat{e}} = 0$, hence $\bar{\hat{y}} = \bar{y}$, so $\hat{y} - \bar{y}\mathbf{1} \in M$ and $\hat{e} \in M^\perp$ are orthogonal; $y - \bar{y}\mathbf{1} = (\hat{y} - \bar{y}\mathbf{1}) + \hat{e}$ and Proposition 11.2.2 apply. ■

Consequences of the projection reading. Four things that the calculus derivation leaves opaque.

First, the normal equations (11.6) are not computational artefacts. They say the residual is orthogonal to every regressor, which is the defining property of the projection, and they are the sample analogue of the population moment conditions $\mathbb{E}[\hat{e}] = 0$, $\mathbb{E}[x\hat{e}] = 0$ that identify the estimand. The first equation holds *because* $\mathbf{1}$ is a regressor: a regression without an intercept has residuals that need not sum to zero, and then $\bar{\hat{y}} \neq \bar{y}$ and (11.7) fails.

Second, (11.7) is the analysis-of-variance decomposition, total sum of squares equals explained plus residual, and it is Pythagoras' theorem. The coefficient of determination $R^2 = \|\hat{y} - \bar{y}\mathbf{1}\|^2 / \|y - \bar{y}\mathbf{1}\|^2$ is therefore $\cos^2$ of the angle between the demeaned data vector and the demeaned fit, which by Example 3.2.2 is the squared sample correlation between y and $\hat{y}$. That $R^2 \in [0, 1]$ is Cauchy–Schwarz, not a convention.

Third, adding a regressor cannot raise the residual sum of squares, because it enlarges M and Theorem 11.3.1 minimises over a larger set. This is a geometric fact about nested subspaces, which is why R^2 is non-decreasing in the number of regressors and why model selection needs a penalty rather than a fit criterion.

Fourth, the estimator is linear in y: $\hat\beta = (X^\top X)^{-1}X^\top y$ applies a fixed matrix to the data. Linearity is Proposition 11.3.2(i), and it is the property that makes the sampling distribution of $\hat\beta$ computable from that of y — normal if y is normal, by the linear-image property of the Gaussian family.

Example 11.5.1 (The simulation)**.** Example 8.1 of Fukuda (2026d) (Fukuda 2026a, slides 63–64) draws $n = 500$ independent standard normal regressors x_i and errors ε_i, sets $y_i = 1 + 3x_i + \varepsilon_i$, and computes $\hat\beta$ from (11.5). The accompanying programs — `regression.m` for MATLAB, `regression_Octave.m`, and `regression_Python.py` — build $X = [\mathbf{1}\ \ x]$ and evaluate `inv(X'*X)*X'*y`, comparing the result with the built-in `regress` in MATLAB and with `numpy.linalg.lstsq` in Python. All three fix a seed — `rng(0)` in the MATLAB and Octave versions, `np.random.seed(1)` in the Python one — but the three generators are different implementations, so the draws are not comparable across languages and the three realisations of $\hat\beta$ are not meant to agree. What agrees within each program is the hand-coded normal-equation solution and the library routine. Executed on MATLAB R2025b, `regression.m` returns $\hat\beta = (0.955343, 2.940044)^\top$ from both routes, agreeing to 1.6×10^{-15}; `regression_Python.py` returns $\hat\beta = (1.023019, 3.021874)^\top$ from both. The Octave variant, translated line by line into MATLAB syntax and run there, reproduces the MATLAB figures exactly, as its seed and construction are identical. Its second `fprintf` is labelled "`regress`" but prints `beta_hat` rather than `beta_hat_2`, so the comparison it advertises is not the one it displays; the two vectors coincide to 1.6×10^{-15}, so the printed numbers are right for the wrong reason.

The realisation differs from the true $(1,3)^\top$ by an amount that is itself the projection of the error vector ε,

$$\hat\beta - \beta = \left(X^\top X\right)^{-1} X^\top \varepsilon.$$

Every result in the asymptotic theory of least squares is a statement about that expression.

11.5.1 The best linear predictor

Proposition 11.5.2 (Population least squares)**.** Let (X, Y) be square-integrable random variables with $\mathrm{Var}[X] > 0$. The minimiser of $\mathbb{E}\left[(Y - a - bX)^2\right]$ over $(a, b) \in \mathbb{R}^2$ is

$$b^* = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}[X]}, \qquad a^* = \mathbb{E}[Y] - b^*\mathbb{E}[X],$$

and the prediction error $Y - a^* - b^*X$ has mean zero and is uncorrelated with X.

Proof. Work in the Hilbert space L^2 with $\langle U, V\rangle = \mathbb{E}[UV]$ (Example 3.2.1) and let $M := \mathrm{span}\{1, X\}$, a two-dimensional, hence closed, subspace by Corollary 6.3.1 applied inside the finite-dimensional span. Minimising $\mathbb{E}[(Y - a - bX)^2] = \|Y - a - bX\|^2$ is projecting Y onto M, so by Proposition 11.3.1 the solution is characterised by $\mathbb{E}[(Y - a - bX)\cdot 1] = 0$ and $\mathbb{E}[(Y - a - bX)X] = 0$. The first gives $a = \mathbb{E}[Y] - b\mathbb{E}[X]$; substituting into the second gives $\mathbb{E}[(Y - \mathbb{E}Y)X] - b\,\mathbb{E}[(X - \mathbb{E}X)X] = 0$, that is $\mathrm{Cov}(X,Y) = b\,\mathrm{Var}[X]$. ∎

The formulas are the population counterparts of (8.12), and the argument is the same projection applied in L^2 rather than in $\mathbb{R}^n$: the sample version replaces $\mathbb{E}$ by the sample average, that is, replaces the L^2 inner product by $\langle u, v\rangle = n^{-1}\sum_i u_i v_i$. Enlarging M from $\mathrm{span}\{1, X\}$ to the set of all square-integrable functions of X turns the best *linear* predictor into the best predictor, and Theorem 14.5.1 identifies it as $\mathbb{E}[Y \mid X]$.

11.6 Chapter Summary

Orthogonal complements are always closed subspaces; orthogonal sets of non-zero vectors are independent; and orthogonality makes norms add, which is Pythagoras. In a Hilbert space, every non-empty closed convex set contains a unique nearest point to any given point, with existence obtained from the parallelogram law and completeness rather than from compactness; on a subspace the characterising variational inequality becomes the orthogonality condition (11.3). Hence $H = M \oplus M^\perp$ for every closed subspace M, the projection operator is linear, idempotent, self-adjoint and norm-reducing, and every continuous linear functional on a Hilbert space is an inner product with a fixed vector.

In $\mathbb{R}^n$ the projection onto the column space of a full-column-rank X is $X(X^\top X)^{-1}X^\top$, and invertibility of $X^\top X$ is exactly independence of the columns. Least squares is that projection: the normal equations are orthogonality of the residual to the regressors, the analysis-of-variance identity is Pythagoras, R^2 is a squared cosine, adding regressors cannot worsen the fit because it enlarges the subspace, and the estimator is linear in the data. The same projection performed in L^2 gives the best linear predictor, with covariance over variance replacing the sample formula.

Exercise 11.6.1. Let $M_1 \subseteq M_2$ be closed subspaces of a Hilbert space. Show that $P_{M_1}P_{M_2} = P_{M_2}P_{M_1} = P_{M_1}$, and deduce that $\|y - P_{M_2}y\| \leq \|y - P_{M_1}y\|$. Give the econometric reading, and explain why the inequality is generally strict even when the added regressor is independent of y in the population.

Exercise 11.6.2. (Frisch–Waugh–Lovell.) Partition $X = [X_1 \;\; X_2]$ and let $Q_1 = I - X_1(X_1^\mathsf{T} X_1)^{-1} X_1^\mathsf{T}$. Show that the sub-vector $\hat{\beta}_2$ of (11.5) equals the least-squares coefficient from regressing $Q_1 y$ on $Q_1 X_2$. Identify at which step Proposition 11.3.2(ii) and (iii) are used, and interpret the result as "controlling for X_1".

Exercise 11.6.3. Let $H = L^2([0,1])$ and M the subspace of polynomials of degree at most 2. Compute $P_M f$ for $f(t) = e^t$ by applying Proposition 11.4.2 to $\{1, t, t^2\}$, and compare the resulting quadratic with the second-order Taylor polynomial of e^t at $t = 0$. Explain in terms of the two different optimality criteria why they differ.

Chapter 12

Constrained Optimisation: Lagrange and Kuhn–Tucker

12.1 The Programming Problem

Definition 12.1.1 (The programming problem)**.** Let $f, g_1, \ldots, g_n \colon \mathbb{R}^m \to \mathbb{R}$. The *programming problem*, henceforth *Problem 1*, is

$$\max_{x \in \mathbb{R}^m} f(x) \qquad \text{subject to} \quad g_j(x) \geq 0 \ \text{ for each } j \in \{1, \ldots, n\}. \tag{12.1}$$

The inequalities are the *constraints*, f is the *objective function*, and

$$D := \{x \in \mathbb{R}^m : g_j(x) \geq 0 \text{ for each } j\}$$

is the *domain* of the program. The program is *bounded* if f is bounded above on D.

The sign convention $g_j \geq 0$ is not restrictive. A constraint $h(x) \leq 0$ is $-h(x) \geq 0$, and an equality constraint $h(x) = 0$ is the pair $h(x) \geq 0$ and $-h(x) \geq 0$; nothing else is needed (Fukuda 2026c, slide 4). The convention is chosen so that the multipliers below are non-negative, and it must be applied consistently: writing a budget constraint as $m - \langle p, x\rangle \geq 0$ rather than $\langle p, x\rangle - m \geq 0$ is what makes the multiplier on wealth positive rather than negative.

Definition 12.1.2 (Binding and slack)**.** The constraint $g_j(x) \geq 0$ is *binding* at $x^* \in D$ if $g_j(x^*) = 0$, and *slack* if $g_j(x^*) > 0$.

Remark 12.1.1 (Identify the binding constraints first)**.** Before differentiating anything, determine which constraints bind at the optimum. The recommendation of (Fukuda 2026c, slides 6–9) is not a matter of convenience; it is where the economics of a problem is located, and three cases recur.

Non-negativity. With $\lim_{c \downarrow 0} u'(c) = \infty$ — an Inada condition — the constraint $c \geq 0$ cannot bind at an optimum, since moving to $c = \varepsilon$ raises utility without bound relative to the cost. The first-order conditions are then equalities and the analysis is standard. When instead a non-negativity-type constraint does bind — a borrowing constraint, a short-sale constraint, limited liability, the zero lower bound on nominal rates — the character of the solution changes, and solving the unconstrained first-order condition gives the wrong answer rather than an approximation to the right one.

Incentive constraints. A mechanism-design or agency problem may carry dozens of incentive-compatibility constraints. Establishing which of them bind — typically the local downward constraints under a single-crossing condition — reduces the problem to a tractable one and is the substantive step.

Direction of an equality. If a constraint is known to hold with equality at the optimum, it still matters which inequality is imposed. Writing the budget set as $\langle p, x\rangle \leq m$ rather than $\langle p, x\rangle \geq m$ yields a multiplier with the sign that makes it the marginal utility of wealth.

12.2 Five Formulations

The Kuhn–Tucker theorem is the statement that five apparently different problems have the same solutions once the objective and constraints are concave and a constraint qualification holds. Stating them separately makes visible exactly which hypothesis is responsible for which implication (Fukuda 2026c, slides 10–20).

Definition 12.2.1 (Problem 2: local maximum)**.** Find $x \in D$ such that for every $u \in \mathbb{R}^m$ there is $\lambda_0 > 0$ with

$$0 \le \lambda \le \lambda_0 \text{ and } x + \lambda u \in D \quad \Longrightarrow \quad f(x + \lambda u) \le f(x).$$

The quantifier structure matters: λ_0 may depend on the direction u. This is weaker than requiring a single ball around x on which $f(x)$ dominates, and it is the formulation that matches the way local optimality is verified in economics — by considering one-dimensional deviations. The one-shot deviation principle in repeated games and the perturbation of a consumption path in a growth model are both instances of checking Definition 12.2.1 along a particular family of directions u.

Definition 12.2.2 (Problem 3: no improving feasible direction)**.** Suppose $f, g_1, \dots, g_n$ are differentiable. Find $x \in D$ such that

$$\langle \nabla f(x), u \rangle \le 0 \quad \text{for every } u \in \mathbb{R}^m \text{ satisfying} \quad \big[\, g_j(x) = 0 \implies \langle \nabla g_j(x), u \rangle \ge 0 \,\big].$$

In words: every direction that does not immediately violate a binding constraint fails to increase the objective to first order. The geometry of (Fukuda 2026c, slides 12–13) is that $\nabla f(x)$ makes an angle of at least 90° with every such u, while each binding $\nabla g_j(x)$ makes an angle of at most 90°. With a single binding constraint this forces $\nabla f(x)$ and $\nabla g_1(x)$ to be antiparallel, that is $\nabla f(x) + y_1 \nabla g_1(x) = 0$ for some $y_1 \ge 0$, which is the Lagrange condition.

Definition 12.2.3 (Problem 4: Lagrange's problem)**.** Suppose $f, g_1, \dots, g_n$ are differentiable. Find $x \in D$ and $y = (y_1, \dots, y_n) \in \mathbb{R}^n_+$ such that

$$\frac{\partial f}{\partial x_i}(x) + \sum_{j=1}^{n} y_j \frac{\partial g_j}{\partial x_i}(x) = 0 \qquad \text{for each } i \in \{1, \dots, m\}, \tag{12.2}$$

$$\sum_{j=1}^{n} y_j g_j(x) = 0. \tag{12.3}$$

Definition 12.2.4 (Problem 5: saddle point)**.** Define the *Lagrangian* $F\colon \mathbb{R}^m \times \mathbb{R}^n_+ \to \mathbb{R}$ by

$$F(x, y) := f(x) + \sum_{j=1}^{n} y_j g_j(x). \tag{12.4}$$

Find $(\bar{x}, \bar{y}) \in \mathbb{R}^m \times \mathbb{R}^n_+$ that is a *saddle point* of F:

$$F(x, \bar{y}) \le F(\bar{x}, \bar{y}) \le F(\bar{x}, y) \qquad \text{for all } x \in \mathbb{R}^m \text{ and } y \in \mathbb{R}^n_+. \tag{12.5}$$

Condition (12.5) has two halves: $\bar{x}$ maximises $F(\cdot, \bar{y})$ over the *whole* of $\mathbb{R}^m$, with no constraint, and $\bar{y}$ minimises $F(\bar{x}, \cdot)$ over $\mathbb{R}^n_+$. This is the saddle-point interpretation of Boyd and Vandenberghe (2004, Sec. 5.4, p. 237), whose Section 5.5 (p. 241) states the optimality conditions (12.2)–(12.3) in the same form and whose Section 5.6 (p. 249) develops the sensitivity reading of the multiplier recovered below in Theorem 12.3.2. The first is what makes the Lagrangian method a method: an unconstrained problem replaces a constrained one. The second encodes feasibility, as the next proposition shows.

Proposition 12.2.1 (Complementary slackness at a saddle point)**.** Let $(\bar{x}, \bar{y})$ solve Problem 5. Then $\bar{x} \in D$ and $\bar{y}_j g_j(\bar{x}) = 0$ for every j; in particular (12.3) holds.

Proof. Feasibility. Suppose $g_{j_0}(\bar{x}) < 0$ for some j_0. Taking $y = \bar{y} + te_{j_0}$ with $t > 0$ gives $F(\bar{x}, y) = F(\bar{x}, \bar{y}) + tg_{j_0}(\bar{x}) \to -\infty$ as $t \to \infty$, contradicting the right-hand inequality of (12.5). So $\bar{x} \in D$.

Complementary slackness. Taking $y = 0 \in \mathbb{R}^n_+$ in (12.5) gives $F(\bar{x}, \bar{y}) \le F(\bar{x}, 0) = f(\bar{x})$, that is $\sum_j \bar{y}_j g_j(\bar{x}) \le 0$. Every summand is a product of the non-negative numbers $\bar{y}_j$ and $g_j(\bar{x})$, so every summand is non-negative; a non-negative sum that is at most zero has all terms zero. ∎

This is Exercise 12.3 of Fukuda (2026d) (Fukuda 2026c, slide 19). The economic reading of $\bar{y}_j g_j(\bar{x}) = 0$ is exactly the dichotomy on (Fukuda 2026c, slide 16): either the constraint binds, or its marginal valuation is zero. A resource in excess supply has price zero, and a resource with a positive price is exhausted.

Proposition 12.2.2 (The unconditional implications)**.** (i) If $(\bar{x}, \bar{y})$ solves Problem 5, then $\bar{x}$ solves Problem 1.

(ii) If $\bar{x}$ solves Problem 1, then $\bar{x}$ solves Problem 2.

Neither implication requires any hypothesis on f or the g_j.

Proof. (i) By Proposition 12.2.1, $\bar{x} \in D$ and $F(\bar{x}, \bar{y}) = f(\bar{x})$. For any $x \in D$, the left inequality of (12.5) gives

$$f(x) \le f(x) + \sum_j \bar{y}_j g_j(x) = F(x, \bar{y}) \le F(\bar{x}, \bar{y}) = f(\bar{x}),$$

the first step because $\bar{y}_j \ge 0$ and $g_j(x) \ge 0$ for feasible x.

(ii) A global maximiser over D satisfies the local condition of Definition 12.2.1 with λ_0 arbitrary. ∎

This is Proposition 12.1 of Fukuda (2026d) (Fukuda 2026c, slide 19). Part (i) is the part used in practice: exhibiting a saddle point *proves* global optimality with no concavity, no differentiability and no constraint qualification. The converse implications are where hypotheses are needed.

12.3 The Kuhn–Tucker Theorem

Assumption 12.3.1 (Slater's condition)**.** There exists $\hat{x} \in \mathbb{R}^m$ such that $g_j(\hat{x}) \ge 0$ for every j, and $g_j(\hat{x}) > 0$ for every j for which g_j is not affine.

Theorem 12.3.1 (Kuhn–Tucker)**.** Suppose $f, g_1, \ldots, g_n$ are concave and Assumption 12.3.1 holds.

(i) Problems 1, 2 and 5 are equivalent: $\bar{x}$ solves one if and only if it solves the others, where for Problem 5 this means that some $\bar{y} \in \mathbb{R}^n_+$ makes $(\bar{x}, \bar{y})$ a saddle point.

(ii) If in addition $f, g_1, \ldots, g_n$ are differentiable, all five problems are equivalent.

Proof. By Proposition 12.2.2 it remains to prove that a solution of Problem 2 is a solution of Problem 5, and, under differentiability, to relate Problems 3 and 4 to the rest.

Problem 2 ⇒ Problem 1 under concavity. Let $\bar{x}$ solve Problem 2 and let $x \in D$. Concavity of each g_j makes D convex (it is an intersection of upper contour sets of concave, hence quasiconcave, functions), so $\bar{x} + \lambda(x - \bar{x}) \in D$ for $\lambda \in [0, 1]$. Applying Definition 12.2.1 with $u = x - \bar{x}$ gives, for small $\lambda > 0$,

$$f(\bar{x}) \ge f\big(\bar{x} + \lambda(x - \bar{x})\big) \ge (1 - \lambda)f(\bar{x}) + \lambda f(x),$$

the second inequality by concavity of f. Rearranging, $\lambda f(\bar{x}) \ge \lambda f(x)$, so $f(\bar{x}) \ge f(x)$.

Problem 1 ⇒ Problem 5. Let $\bar{x}$ solve Problem 1 and set $\tilde{f}(x) := f(x) - f(\bar{x})$, which is concave. Optimality of $\bar{x}$ says the system

$$\tilde{f}(x) > 0 \quad \text{and} \quad g_j(x) \ge 0 \ (j = 1, \ldots, n)$$

has no solution. Assumption 12.3.1 is the solvability hypothesis (9.3) of the Farkas–Minkowski theorem, so Theorem 9.5.1 supplies $\bar{y} \in \mathbb{R}^n_+$ with

$$\sup_{x \in \mathbb{R}^m} \Big\{ \tilde{f}(x) + \sum_j \bar{y}_j g_j(x) \Big\} \le 0, \qquad \text{that is} \qquad F(x, \bar{y}) \le f(\bar{x}) \ \text{ for all } x \in \mathbb{R}^m.$$

Evaluating at $x = \bar{x}$ gives $\sum_j \bar{y}_j g_j(\bar{x}) \le 0$; every term is non-negative, so all vanish and $F(\bar{x}, \bar{y}) = f(\bar{x})$. Hence $F(x, \bar{y}) \le F(\bar{x}, \bar{y})$ for all x, the left half of (12.5). For the right half, $y \in \mathbb{R}^n_+$ gives $F(\bar{x}, y) = f(\bar{x}) + \sum_j y_j g_j(\bar{x}) \ge f(\bar{x}) = F(\bar{x}, \bar{y})$, since each $g_j(\bar{x}) \ge 0$.

Problems 3, 4 and 5 under differentiability. If $(\bar{x}, \bar{y})$ is a saddle point, then $\bar{x}$ maximises the differentiable function $F(\cdot, \bar{y})$ over the open set $\mathbb{R}^m$, so Theorem 8.4.1 gives (12.2), and (12.3) is Proposition 12.2.1: Problem 5 implies Problem 4. Conversely, if (12.2)–(12.3) hold, then $F(\cdot, \bar{y})$ is concave — a non-negative combination of concave functions, by Proposition 9.3.2(i) applied to $-F$ — with vanishing gradient at $\bar{x}$, so Theorem 8.4.2 makes $\bar{x}$ a global maximiser of $F(\cdot, \bar{y})$, and the right half of (12.5) follows as above: Problem 4 implies Problem 5. That Problem 4 implies Problem 3 is immediate on taking the inner product of (12.2) with an admissible u; the converse uses Theorem 9.5.1 applied to the linearised system at $\bar{x}$. ■

This is Theorem 12.1 of Fukuda (2026d, p. 436) (Fukuda 2026c, slide 20). The proof locates each hypothesis. Concavity of the g_j makes D convex, which is what turns local into global. Concavity of f is used in the same step and again in Theorem 8.4.2. Slater's condition is used once, to invoke Theorem 9.5.1, and it is what guarantees a multiplier *exists*; Example 9.5.1 exhibits a program in which it does not.

> **Remark 12.3.1** (Constraint qualifications). Assumption 12.3.1 is one of several conditions under which multipliers exist; they are collectively the *constraint qualifications*. Three points about them.
>
> First, they are conditions on the *constraint functions*, not on the feasible set. The set $\{x \in \mathbb{R} : -x^2 \ge 0\} = \{0\}$ and the set $\{x \in \mathbb{R} : x \ge 0,\ -x \ge 0\} = \{0\}$ are identical, but the second description satisfies Assumption 12.3.1 — both constraints are affine — while the first does not, as Example 9.5.1 showed. A change of description changes whether multipliers exist.
>
> Second, the affine exemption in Assumption 12.3.1 is not cosmetic. Linear programs and problems with only budget and non-negativity constraints satisfy the qualification automatically, which is why the constraint qualification is almost never mentioned in consumer theory and is essential in problems with non-linear constraints such as incentive compatibility.
>
> Third, when the objective and constraints are not concave the relevant qualification is local and differential rather than global. The standard condition is that the gradients of the binding constraints at $\bar{x}$ be linearly independent, and under it the conclusion is weaker: the first-order conditions (12.2)–(12.3) are necessary at a local maximum, but not sufficient, and Problem 5 is not equivalent to Problem 1. See Mangasarian (1994, Ch. 7).

Example 12.3.1 (Failure of the differential constraint qualification). Maximise $f(x_1, x_2) = -x_1$ subject to

$$g_1(x) = -x_2 \ge 0, \qquad g_2(x) = x_2 - (1 - x_1)^3 \ge 0.$$

The two constraints require $(1 - x_1)^3 \le x_2 \le 0$, and the left inequality forces $x_1 \ge 1$; conversely every $x_1 \ge 1$ admits $x_2 \in [(1 - x_1)^3, 0]$. The feasible set is therefore

$$D = \{(x_1, x_2) : x_1 \ge 1,\ (1 - x_1)^3 \le x_2 \le 0\},$$

on which $f = -x_1 \le -1$ with equality only at $x_1 = 1$, where the interval $[(1 - x_1)^3, 0]$ collapses to $\{0\}$. So $\bar{x} = (1, 0)$ is the unique *global* maximiser.

Both constraints bind at $\bar{x}$, and

$$\nabla g_1(\bar{x}) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad \nabla g_2(\bar{x}) = \begin{pmatrix} 3(1 - x_1)^2 \\ 1 \end{pmatrix}\Bigg|_{x_1 = 1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

which are linearly dependent, so the linear-independence qualification of Remark 12.3.1 fails. It fails consequentially: since $\nabla f(\bar{x}) = (-1, 0)^\mathsf{T}$ has non-zero first component while both constraint gradients have first component 0, no $y \in \mathbb{R}^2_+$ — indeed no $y \in \mathbb{R}^2$ — satisfies $\nabla f(\bar{x}) + y_1 \nabla g_1(\bar{x}) + y_2 \nabla g_2(\bar{x}) = 0$. The first-order conditions (12.2) have no solution at a point that is globally optimal, and a search over Kuhn–Tucker points would miss it.

Two features of this example deserve separating from each other. Assumption 12.3.1 is *not* what fails: $x = (2, -\frac{1}{2})$ gives $g_1 = \frac{1}{2} > 0$ and $g_2 = \frac{1}{2} > 0$, so the program is strictly feasible. What fails is concavity, since g_2 has second derivative $-6(1 - x_1) > 0$ in x_1 for $x_1 > 1$, and with it the hypotheses of Theorem 12.3.1. Outside the concave class the relevant qualification is the local one, and here it is violated: the feasible set has an outward cusp at $\bar{x}$ whose linearisation, the set $\{u : \langle \nabla g_1, u \rangle \ge 0,\ \langle \nabla g_2, u \rangle \ge 0\} = \{u : u_2 = 0\}$, is strictly larger than the cone of directions along which D can actually be entered from $\bar{x}$, namely $\{u : u_1 \geq 0,\ u_2 = 0\}$. It is that discrepancy, not the geometry of D on its own, that destroys the multiplier rule.

12.3.1 Interpretation of the multipliers

Theorem 12.3.2 (Envelope theorem). Consider the parametrised program

$$V(b,\alpha) := \max_{x\in\mathbb{R}^m} f(x,\alpha) \qquad \text{subject to} \quad g(x,\alpha) = b,$$

with f, g continuously differentiable, and suppose that on a neighbourhood of $(\bar{b}, \bar{\alpha})$ the solution $x^*(b,\alpha)$ and the multiplier $\lambda^*(b,\alpha)$ are unique and continuously differentiable. Then

$$\frac{\partial V}{\partial \alpha} = \frac{\partial f}{\partial \alpha}(x^*,\alpha) - \lambda^* \frac{\partial g}{\partial \alpha}(x^*,\alpha), \qquad \frac{\partial V}{\partial b} = \lambda^*. \tag{12.6}$$

Proof. Write $V(b,\alpha) = f(x^*,\alpha) + \lambda^*\big(b - g(x^*,\alpha)\big)$, which is legitimate because the bracket vanishes at the optimum. Differentiate with respect to α using Theorem 8.3.1:

$$\frac{\partial V}{\partial \alpha} = \underbrace{\big(f_x - \lambda^* g_x\big)}_{=0 \text{ by } (12.2)} \cdot \frac{\partial x^*}{\partial \alpha} + \underbrace{\big(b - g(x^*,\alpha)\big)}_{=0 \text{ by feasibility}} \cdot \frac{\partial \lambda^*}{\partial \alpha} + f_\alpha - \lambda^* g_\alpha.$$

The first bracket vanishes by the first-order condition and the second by the constraint, leaving the last two terms. Differentiating instead with respect to b leaves only the term $\lambda^* \cdot \partial b/\partial b = \lambda^*$. ■

This is Proposition 13.9 and the constrained envelope theorem of Fukuda (2026d, pp. 479–481), and the computation of (Fukuda 2026c, slide 28). The content is that the indirect effects, operating through the response $\partial x^*/\partial\alpha$ of the optimal choice, cancel: at an optimum the marginal value of moving x is zero, so a marginal change in the parameter has only its direct effect. This is what makes λ^* the *shadow price* of the constraint: it is the rate at which the maximised value rises when the constraint is relaxed, and it is computable without differentiating the solution.

Remark 12.3.2 (Differentiability of V is an assumption, not a conclusion). Theorem 12.3.2 presumes that x^* and λ^* are differentiable in the parameters, which is itself a conclusion of the implicit function theorem (Theorem 8.6.1) applied to the first-order conditions, and requires the bordered Hessian to be non-singular at the optimum. When the solution set is not a singleton or when the active set changes, V can have a kink and (12.6) fails as stated. The general form due to Milgrom and Segal (2002) replaces the derivative by one-sided derivatives and requires only that $f_\alpha(x^*,\alpha)$ exist at a single maximiser: $V'(\alpha^-) \leq f_\alpha(x^*,\alpha) \leq V'(\alpha^+)$ whenever the one-sided derivatives exist, with equality when V is differentiable. That version is what applies to mechanism design, where X may be a discrete set with no differentiable structure at all.

12.4 Economic Application: The Consumer's Problem

Problem. With $f(x_1,x_2) = x_1^\alpha x_2^{1-\alpha}$, $\alpha \in (0,1)$, and constraints

$$g_1(x) = m - p_1x_1 - p_2x_2 \geq 0, \qquad g_2(x) = x_1 \geq 0, \qquad g_3(x) = x_2 \geq 0,$$

with $p_1, p_2, m > 0$, solve (12.1).

Proposition 12.4.1. The problem has the unique solution

$$x_1^* = \frac{\alpha m}{p_1}, \qquad x_2^* = \frac{(1-\alpha)m}{p_2}, \tag{12.7}$$

with multipliers $y_2^* = y_3^* = 0$ and

$$y_1^* = \frac{(x_1^*)^\alpha (x_2^*)^{1-\alpha}}{m} = \frac{v(p,m)}{m}, \tag{12.8}$$

> where $v(p,m)$ is the indirect utility. Expenditure shares are constant: $p_1 x_1^*/m = \alpha$ and $p_2 x_2^*/m = 1-\alpha$.

Proof. Existence and uniqueness. The budget set is compact (Example 4.4.1) and non-empty, and f is continuous, so a solution exists by Corollary 5.4.1. On $\mathbb{R}^2_{++}$ the function f is strictly quasiconcave, so the solution is unique by Proposition 9.3.5 (Fukuda 2026c, slide 21).

Which constraints bind. If $x_1^* = 0$ then $f = 0$, while the feasible point $(m/(2p_1), m/(2p_2))$ gives $f > 0$; so g_2 is slack, and symmetrically g_3. The budget constraint binds: f is strictly increasing in each argument on $\mathbb{R}^2_{++}$, so an unspent unit of wealth could be spent to raise utility.

First-order conditions. With $y_2 = y_3 = 0$ by Proposition 12.2.1, (12.2) reads

$$\alpha(x_1^*)^{\alpha-1}(x_2^*)^{1-\alpha} = y_1 p_1, \qquad (1-\alpha)(x_1^*)^{\alpha}(x_2^*)^{-\alpha} = y_1 p_2.$$

Dividing eliminates y_1 and gives $\dfrac{x_2^*}{x_1^*} = \dfrac{1-\alpha}{\alpha}\cdot\dfrac{p_1}{p_2}$, that is $p_2 x_2^* = \frac{1-\alpha}{\alpha} p_1 x_1^*$. Substituting into the binding budget constraint $p_1 x_1^* + p_2 x_2^* = m$ yields $p_1 x_1^*\big(1 + \frac{1-\alpha}{\alpha}\big) = m$, hence (12.7).

Multiplier. From the first condition,

$$y_1^* = \frac{\alpha(x_1^*)^{\alpha-1}(x_2^*)^{1-\alpha}}{p_1} = \frac{\alpha(x_1^*)^{\alpha}(x_2^*)^{1-\alpha}}{p_1 x_1^*} = \frac{(x_1^*)^{\alpha}(x_2^*)^{1-\alpha}}{m},$$

using $p_1 x_1^* = \alpha m$. The second condition gives the same value, using $p_2 x_2^* = (1-\alpha)m$.

Verification via Problem 5. f is concave on $\mathbb{R}^2_+$ and the g_j are affine, so Assumption 12.3.1 holds automatically and Theorem 12.3.1 applies; alternatively, the candidate with its multipliers satisfies (12.2)–(12.3), which by the last part of that proof makes it a saddle point, and Proposition 12.2.2(i) then certifies global optimality without any further argument. ∎

Reading the multiplier. Formula (12.8) says $y_1^* = v(p,m)/m$, and since Cobb–Douglas indirect utility is $v(p,m) = m\cdot(\alpha/p_1)^{\alpha}((1-\alpha)/p_2)^{1-\alpha}$, the multiplier is independent of m: the marginal utility of wealth is constant, because the utility function is homogeneous of degree one in the bundle and hence in wealth. Theorem 12.3.2 confirms this directly, $\partial v/\partial m = y_1^*$, which is the computation of (Fukuda 2026c, slides 28–29) in which every term involving $\mathrm{d}x_i^*/\mathrm{d}m$ cancels against a first-order condition.

The constant expenditure shares are the economic signature of Cobb–Douglas preferences and are the reason the functional form is used: demand for each good is independent of the price of the other, income elasticities are unity, and the own-price elasticity is exactly -1. Each of these is a strong restriction, and all three follow from the single fact that the elasticity of substitution is 1.

Numerical counterpart. Three of the accompanying programs solve the same problem by sequential quadratic programming (Fukuda 2026c, slides 33–36): `consumer_choice_fmincon.m`, `consumer_choice_sqp_Octave.m` and `consumer_choice_python.py`. All three fix $\alpha = 0.7$, $p_1 = p_2 = 1$ and $m = 1$. Run as supplied, `consumer_choice_fmincon.m` returns $x^* = (0.700, 0.300)$ with objective value 0.54288 and constraint multiplier 0.54288; `consumer_choice_python.py` returns the same allocation with objective 0.54285 and multiplier 0.54295, the discrepancy in the fifth decimal being the solver tolerance. The closed form (12.7) gives $x^* = (\alpha m/p_1, (1-\alpha)m/p_2) = (0.7, 0.3)$ and

$$v(p,m) = \frac{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}{p_1^{\alpha}p_2^{1-\alpha}}\, m = 0.7^{0.7}\, 0.3^{0.3} = 0.542884,$$

which is linear in m, so $\partial v/\partial m = v(p,m)/m = 0.542884$, matching (12.8): the reported multiplier is the shadow price of income to five decimals, as Theorem 12.3.2 requires.

The Octave variant, translated into MATLAB syntax and run there, returns $x^* = (0.700000, 0.300000)$ with utility and multiplier both 0.542881, and zero multipliers on the two non-negativity constraints.

The three programs do not share a sign convention, and neither matches Definition 12.1.1. All three minimise, so each maximisation is passed as $-u$. `fmincon` takes inequality constraints in the form $c(x) \le 0$ and returns non-negative multipliers for them; Octave's `sqp` takes them in the form $h(x) \ge 0$, so its h is the negative of `fmincon`'s c and its multipliers are again non-negative; SciPy's `SLSQP` follows the Octave convention. The three multipliers therefore agree in sign and magnitude here only because each program supplies the constraint in the form its own solver expects. Scaling matters as well: replacing $p_1x_1 + p_2x_2 \le m$ by $2p_1x_1 + 2p_2x_2 \le 2m$ leaves the solution unchanged and halves the reported multiplier, so a multiplier is a shadow price only relative to the units in which its constraint is written.

12.5 Economic Application: Intertemporal Choice

Problem. A consumer with wealth $W > 0$ chooses consumption in two periods, facing gross interest rate $R > 0$ and discount factor $\beta \in (0, 1)$:

$$\max_{c_1, c_2 \geq 0} \; u(c_1) + \beta u(c_2) \qquad \text{subject to} \quad c_1 + \frac{c_2}{R} \leq W, \tag{12.9}$$

with u strictly increasing, strictly concave, differentiable and satisfying $\lim_{c \downarrow 0} u'(c) = \infty$.

Proposition 12.5.1 (Euler equation)**.** Problem (12.9) has a unique solution, it is interior, the budget constraint binds, and the solution satisfies

$$u'(c_1^*) = \beta R\, u'(c_2^*), \tag{12.10}$$

with multiplier $\lambda^* = u'(c_1^*)$. For $u(c) = c^{1-\gamma}/(1-\gamma)$ this gives $c_2^*/c_1^* = (\beta R)^{1/\gamma}$ and

$$c_1^* = \frac{W}{1 + \beta^{1/\gamma} R^{(1-\gamma)/\gamma}}.$$

Proof. The feasible set is compact and convex and the objective continuous and strictly concave, so a unique solution exists by Corollary 5.4.1 and Proposition 9.3.5. The Inada condition rules out $c_i^* = 0$: at such a point the derivative of the objective in c_i is $+\infty$ while the marginal cost is finite, so a small increase is feasible and improving. Monotonicity makes the budget constraint bind. The constraints are affine, so Assumption 12.3.1 holds and Theorem 12.3.1 applies. The Lagrangian $F = u(c_1) + \beta u(c_2) + \lambda(W - c_1 - c_2/R)$ gives $u'(c_1^*) = \lambda^*$ and $\beta u'(c_2^*) = \lambda^*/R$; eliminating λ^* yields (12.10). With CRRA utility, $u'(c) = c^{-\gamma}$ and (12.10) reads $(c_1^*)^{-\gamma} = \beta R (c_2^*)^{-\gamma}$, that is $c_2^* = (\beta R)^{1/\gamma} c_1^*$; substituting into the binding budget constraint gives the stated c_1^*. ■

Content of (12.10)**.** The marginal utility cost of saving one more unit today equals its discounted marginal utility benefit tomorrow, scaled by the gross return. The consumption growth rate $(\beta R)^{1/\gamma}$ is increasing in R with elasticity $1/\gamma$, the elasticity of intertemporal substitution. The effect of R on c_1^* is ambiguous in sign: differentiating the closed form of Proposition 12.5.1, $\mathrm{d}c_1^*/\mathrm{d}R > 0$ if and only if $\gamma > 1$, the income effect dominating the substitution effect. The multiplier $\lambda^* = u'(c_1^*)$ is the marginal utility of wealth, so by Theorem 12.3.2 $\partial v/\partial W = u'(c_1^*)$, where $v(W)$ denotes the maximum value of (12.9); the same identity in the infinite-horizon case is the Benveniste–Scheinkman formula of Chapter 15.

The corresponding programs are `two_periods_consumption_fmincon.m`, `two_periods_consumption_sqp_Octave.m` and `two_periods_consumption_python.py` (Fukuda 2026c, slides 37–40). They take $\beta = 0.9$, $R = 1.05$, $W = 1$ and logarithmic utility, the case $\gamma = 1$ in which the two offsetting effects cancel and $c_1^* = W/(1+\beta)$ is independent of R. Run as supplied, both the MATLAB and the Python version return $c^* = (0.526, 0.497)$ with objective -1.27044 and multiplier 1.90000. The closed form gives $c_1^* = 1/1.9 = 0.526316$ and $c_2^* = \beta R W/(1+\beta) = 0.497368$, lifetime utility $\log c_1^* + \beta \log c_2^* = -1.270437$, and $\lambda^* = u'(c_1^*) = 1/c_1^* = 1 + \beta = 1.9$ exactly. The Octave variant, translated into MATLAB syntax and run there, returns $c^* = (0.526316, 0.497368)$, lifetime utility -1.270436 and multiplier 1.900000.

12.6 Economic Application: Nash Bargaining

Problem. Two players divide a surplus. The set of feasible utility pairs is $S \subseteq \mathbb{R}^2$, compact and convex, and the disagreement point is $d \in S$. The *Nash bargaining solution* solves

$$\max_{(u_1, u_2) \in S} \; (u_1 - d_1)(u_2 - d_2) \qquad \text{subject to} \quad u_1 \geq d_1, \; u_2 \geq d_2. \tag{12.11}$$

Proposition 12.6.1. Suppose S is compact and convex and there is $u \in S$ with $u \gg d$. Then (12.11) has a unique solution u^*, it satisfies $u^* \gg d$, and if $S = \{u \in \mathbb{R}^2_+ : u_1/a_1 + u_2/a_2 \leq 1\}$ with $a_1, a_2 > 0$ and $d = 0$, then $u^* = (a_1/2, a_2/2)$: the surplus is split equally in normalised units.

Proof. Existence follows from compactness of the feasible set, which is the intersection of S with two closed half-spaces, and continuity of the objective (Corollary 5.4.1). The objective is not concave, but its logarithm $\log(u_1 - d_1) + \log(u_2 - d_2)$ is strictly concave on the region where both factors are positive, and by Proposition 9.3.4 the strictly increasing transformation preserves the maximiser; uniqueness is Proposition 9.3.5. The hypothesis $u \gg d$ makes the maximum positive, so $u^* \gg d$ and the transformation is legitimate at the optimum.

For the linear frontier with $d = 0$, maximise $\log u_1 + \log u_2$ subject to $u_1/a_1 + u_2/a_2 \leq 1$. This is Proposition 12.4.1 with $\alpha = \frac{1}{2}$, $p_i = 1/a_i$ and $m = 1$, giving $u_i^* = a_i/2$. ■

Maximising a product subject to a linear constraint is literally the Cobb–Douglas consumer problem, so the equal split of the Nash solution and the equal expenditure shares of symmetric Cobb–Douglas preferences are one computation seen twice.

> **Remark 12.6.1** (A curved frontier). Linearity of the frontier plays no role in the split. Take $S = \{u \in \mathbb{R}^2_+ : u_1^2/a_1^2 + u_2^2/a_2^2 \leq 1\}$ with $a_1, a_2 > 0$ and $d = 0$. The set is the intersection of an ellipse's interior with the positive orthant, hence compact and convex, and $(a_1, a_2)/\sqrt{2} \gg 0$ lies in it, so Proposition 12.6.1 gives a unique solution. Maximising $\log u_1 + \log u_2$ subject to the constraint, the first-order conditions read $1/u_i = 2\lambda u_i/a_i^2$, so $u_1^2/a_1^2 = u_2^2/a_2^2$; the constraint binds, each term equals $\frac{1}{2}$, and
>
> $$u^* = \left(\frac{a_1}{\sqrt{2}}, \frac{a_2}{\sqrt{2}}\right).$$
>
> The normalised shares u_i^*/a_i are again equal across players. What produces this in both cases is the invariance of the Nash product under the rescaling $u_i \mapsto u_i/a_i$, which turns either frontier into a symmetric one; it is the scale-invariance axiom of Nash (1950), not the shape of S.

The numerical counterparts are `Nash_bargaining_fmincon.m`, `Nash_bargaining_sqp_Octave.m` and `Nash_bargaining_python.py` (Fukuda 2026c, slides 41–44). They solve the curved case of Remark 12.6.1 with $a_1 = a_2 = 1$: the objective passed to the solver is $-u_1 u_2$ and the non-linear constraint is $u_1^2 + u_2^2 \leq 1$, with lower bounds $u \geq 0$ and starting point $(0.1, 0.1)$. Run as supplied, both the MATLAB and the Python version return $u^* = (0.7071, 0.7071)$, and the Python version reports multiplier 0.50000; the Octave variant, translated into MATLAB syntax and run there, returns $(0.707107, 0.707107)$ with product 0.500000 and multiplier 0.500000. All agree with $u^* = (1/\sqrt{2}, 1/\sqrt{2}) = (0.707107, 0.707107)$ and, from the first-order condition $u_2^* = 2\lambda^* u_1^*$ for the product objective, $\lambda^* = \frac{1}{2}$.

The starting point is not incidental. The gradient of $(u_1 - d_1)(u_2 - d_2)$ vanishes at $u = d$, so a solver launched from the disagreement point sees a stationary point and stops there; $(0.1, 0.1)$ is chosen to be interior to the region where both factors are positive. Solving (12.11) directly is also worse conditioned than solving the logarithmic version, the product objective being flat along the axes.

12.7 Chapter Summary

A programming problem in the standard form (12.1) admits five formulations: global maximum, local maximum, absence of an improving feasible direction, the Lagrange conditions with complementary slackness, and a saddle point of the Lagrangian. Two implications hold with no hypotheses at all: a saddle point yields a global maximum, and a global maximum is a local maximum. The converses require concavity of the objective and constraints and a constraint qualification, and under those hypotheses the Kuhn–Tucker theorem makes all five equivalent, with differentiability needed only for the two formulations that mention derivatives. The proof of the essential implication — from global optimality to the existence of multipliers — is an application of the Farkas–Minkowski theorem, itself a consequence of the separating hyperplane theorem, so the multipliers are the coefficients of a separating hyperplane.

Complementary slackness says a constraint either binds or has zero multiplier. The multiplier is the shadow price of the constraint, and the envelope theorem makes this precise: at an optimum, the derivative of the value function with respect to a parameter is the partial derivative of the Lagrangian, with all indirect effects through the optimal choice cancelling against the first-order conditions. Applied to the Cobb–Douglas consumer, the machinery yields constant expenditure shares and a multiplier equal to indirect utility divided by wealth; applied to two-period consumption, the Euler equation $u'(c_1) = \beta R u'(c_2)$ with the multiplier equal to the marginal utility of current consumption; and applied to bargaining, the Nash solution, which is formally the same optimisation as symmetric Cobb–Douglas demand.

Exercise 12.7.1. A firm maximises $pF(K,L) - rK - wL$ subject to $K \leq \bar{K}$, with F concave and increasing. Write the Kuhn–Tucker conditions, and show that the multiplier on the capacity constraint equals $pF_K - r$ when the constraint binds and 0 otherwise. Interpret it as the shadow rental rate, and use Theorem 12.3.2 to compute the value of an extra unit of capacity.

Exercise 12.7.2. Consider $\max x_1$ subject to $g_1(x) = x_2 - x_1^3 \geq 0$ and $g_2(x) = -x_2 \geq 0$. Show that $\bar{x} = (0,0)$ is the unique feasible point with $x_1 \geq 0$ maximal, that both constraints bind there, and that no multipliers satisfy (12.2). Identify which hypothesis of Theorem 12.3.1 fails, and verify that adding the affine constraint $x_1 \leq 0$ makes the program satisfy Assumption 12.3.1.

Exercise 12.7.3. Derive the expenditure-minimisation dual of Proposition 12.4.1: minimise $p_1x_1 + p_2x_2$ subject to $x_1^\alpha x_2^{1-\alpha} \geq \bar{u}$. Show that the multiplier on the utility constraint is the reciprocal of y_1^* from (12.8) evaluated at the corresponding wealth, and that the expenditure function $e(p,\bar{u})$ satisfies $\partial e/\partial p_i = x_i^h$, the Hicksian demand. State which part of Theorem 12.3.2 delivers the last identity.

Chapter 13

Correspondences, the Maximum Theorem and Fixed Points

13.1 Why Set-Valued Maps

The optimisation of Chapter 12 produced a maximiser and treated it as a point. Two features of economic models make that treatment inadequate.

The maximiser need not be unique. Consumer demand is single-valued only when preferences are strictly convex; with linear indifference curves the consumer is indifferent over a whole face of the budget set, and the demand *set* moves discontinuously as relative prices cross the ratio at which the linear indifference curves are parallel to the budget line. In game theory the best response to a mixed strategy is generically a vertex of the simplex but is a whole face whenever the player is indifferent, and indifference is what characterises equilibrium in mixed strategies. Nothing can be said about the existence of equilibrium if the objects being compared are required to be single-valued.

The feasible set itself moves with the parameters. In the consumer's problem the budget set depends on prices and income; in a dynamic programme the set of feasible continuations depends on the current state. Asking whether the value of the problem varies continuously with the parameters is therefore a question about the joint behaviour of a function and a moving domain, and it cannot be posed until the movement of the domain has been given a topology.

This chapter supplies both. It defines continuity for set-valued maps, proves Berge's theorem that the value function of a parametrised programme is continuous and the maximiser correspondence upper hemicontinuous, and states the two fixed-point theorems — Brouwer's for functions and Kakutani's for correspondences — on which the existence of Walrasian and Nash equilibrium rests. Theorem 6.4.1 gave a fixed-point theorem of a different kind, and Proposition 6.4.1 recorded the trade: contraction buys uniqueness and an algorithm at the price of a Lipschitz hypothesis, whereas Brouwer and Kakutani buy existence alone, from compactness and convexity, with no uniqueness and no constructive procedure.

13.2 Correspondences

Definition 13.2.1 (Correspondence). A *correspondence* $\varphi\colon X \rightrightarrows Y$ from a set X into a set Y is a function $\varphi\colon X \to \mathcal{P}(()Y)$ into the power set of Y; with each $x \in X$ it associates a subset $\varphi(x)$ of Y. Its *graph* is

$$\operatorname{gr}\varphi := \{(x,y) \in X \times Y : y \in \varphi(x)\} \subseteq X \times Y,$$

and φ is recovered from it by $\varphi(x) = \{y \in Y : (x,y) \in \operatorname{gr}\varphi\}$.

The definition permits $\varphi(x) = \emptyset$; φ is *non-empty-valued* if $\varphi(x) \neq \emptyset$ for every x. A function $f\colon X \to Y$ is the single-valued correspondence $x \mapsto \{f(x)\}$, so every statement below specialises to a statement about functions. Identifying φ with $\operatorname{gr}\varphi$ also identifies it with a binary relation from X to Y in the sense of Chapter 1, which is the reason the graph, rather than the map into $\mathcal{P}(()Y)$, is the object one topologises.

Definition 13.2.2 (Value restrictions). Let X, Y be topological spaces. A correspondence $\varphi\colon X \rightrightarrows Y$ is *open-valued*, *closed-valued*, *compact-valued* or *convex-valued* if $\varphi(x)$ is respectively open, closed, compact or convex in Y for every $x \in X$ (convexity requiring Y to be a vector space).

Example 13.2.1 (Budget and demand correspondences). Define $B\colon \mathbb{R}^2_{++} \times \mathbb{R}_+ \rightrightarrows \mathbb{R}^2_+$ by

$$B(p,m) := \{x \in \mathbb{R}^2_+ : \langle p, x\rangle \le m\}. \tag{13.1}$$

By Example 4.4.1 the budget set is compact when $p \gg 0$, and it is convex as the intersection of $\mathbb{R}^2_+$ with a half-space, so B is compact-valued, closed-valued and convex-valued. The associated *demand correspondence*

$$D(p,m) := \operatorname*{arg\,max}_{x \in B(p,m)} \; u(x) \tag{13.2}$$

is the leading example of an *argmax* correspondence, which associates with each parameter the set of maximisers. It is non-empty-valued whenever u is continuous, by Corollary 5.4.1, and it is single-valued whenever u is strictly quasiconcave, by Proposition 9.3.5; without strict quasiconcavity it need be neither.

Example 13.2.2 (A correspondence with a closed graph). $\varphi\colon \mathbb{R} \rightrightarrows \mathbb{R}$, $\varphi(x) = [x-1, x+1]$, has graph $\operatorname{gr}\varphi = \{(x,y) : |y - x| \le 1\}$, a closed band about the diagonal. It is compact-valued and convex-valued.

13.2.1 Two inverses

A function has one inverse image; a correspondence has two, according to whether the whole value or merely part of it is required to land in the target set. The distinction is the source of the two distinct notions of continuity.

Definition 13.2.3 (Upper and lower inverse). Let $\varphi\colon X \rightrightarrows Y$ and $A \subseteq Y$. The *upper inverse* (or strong inverse) and *lower inverse* (or weak inverse) of A are

$$\varphi^u(A) := \{x \in X : \varphi(x) \subseteq A\}, \qquad \varphi^\ell(A) := \{x \in X : \varphi(x) \cap A \ne \emptyset\}. \tag{13.3}$$

If φ is single-valued, $\varphi^u(A) = f^{-1}(A) = \varphi^\ell(A)$, so the two notions coincide precisely on functions.

Proposition 13.2.1 (Calculus of the two inverses). Let $\varphi\colon X \rightrightarrows Y$ and let $A, B \subseteq Y$ and $(A_i)_{i\in I}$ be subsets of Y.

(i) If φ is non-empty-valued then $\varphi^u(A) \subseteq \varphi^\ell(A)$.

(ii) $\varphi^u(A) = \big(\varphi^\ell(A^c)\big)^c$ and $\varphi^\ell(A) = \big(\varphi^u(A^c)\big)^c$.

(iii) $\varphi^u\big(\bigcap_i A_i\big) = \bigcap_i \varphi^u(A_i)$ and $\varphi^\ell\big(\bigcup_i A_i\big) = \bigcup_i \varphi^\ell(A_i)$.

(iv) $A \subseteq B$ implies $\varphi^u(A) \subseteq \varphi^u(B)$ and $\varphi^\ell(A) \subseteq \varphi^\ell(B)$; consequently $\bigcup_i \varphi^u(A_i) \subseteq \varphi^u\big(\bigcup_i A_i\big)$ and $\varphi^\ell\big(\bigcap_i A_i\big) \subseteq \bigcap_i \varphi^\ell(A_i)$, both inclusions being strict in general.

Proof. (i) If $\varphi(x) \subseteq A$ and $\varphi(x) \ne \emptyset$ then $\varphi(x) \cap A = \varphi(x) \ne \emptyset$.

(ii) $x \notin \varphi^\ell(A^c)$ says $\varphi(x) \cap A^c = \emptyset$, that is $\varphi(x) \subseteq A$, which is $x \in \varphi^u(A)$. The second identity is the first applied to A^c and complemented.

(iii) $\varphi(x) \subseteq \bigcap_i A_i$ if and only if $\varphi(x) \subseteq A_i$ for every i; and $\varphi(x) \cap \bigcup_i A_i \ne \emptyset$ if and only if $\varphi(x) \cap A_i \ne \emptyset$ for some i.

(iv) Monotonicity is immediate from (13.3), and the two inclusions follow. For strictness take $X = \{x\}$, $Y = \{1,2\}$, $\varphi(x) = \{1,2\}$, $A_1 = \{1\}$, $A_2 = \{2\}$: then $\varphi^u(A_1) \cup \varphi^u(A_2) = \emptyset$ while $\varphi^u(A_1 \cup A_2) = \{x\}$, and dually $\varphi^\ell(A_1 \cap A_2) = \emptyset$ while $\varphi^\ell(A_1) \cap \varphi^\ell(A_2) = \{x\}$. ∎

Part (iii) is the reason the two inverses divide the work of Proposition 1.2.2, where a single preimage commuted with both unions and intersections. Set-valuedness breaks that symmetry, and the break is what forces two continuity concepts rather than one.

13.3 Hemicontinuity

Definition 13.3.1 (Upper and lower hemicontinuity)**.** Let X, Y be topological spaces and $\varphi\colon X \rightrightarrows Y$.

(i) φ is *upper hemicontinuous* at $x \in X$ if for every open $U \supseteq \varphi(x)$ there is an open neighbourhood V of x with $\varphi(z) \subseteq U$ for every $z \in V$.

(ii) φ is *lower hemicontinuous* at $x \in X$ if for every open U with $\varphi(x) \cap U \neq \emptyset$ there is an open neighbourhood V of x with $\varphi(z) \cap U \neq \emptyset$ for every $z \in V$.

(iii) φ is *continuous* at x if it is both; and φ is upper hemicontinuous, lower hemicontinuous or continuous if it is so at every point of X.

Both conditions compare the values at points *near* x with the value *at* x, and neither restricts $\varphi(x)$ itself. Upper hemicontinuity says that the nearby values do not stick out: they are contained in any prescribed open set around $\varphi(x)$ once z is close enough, so no point of $\varphi(z)$ can stay at a fixed distance from $\varphi(x)$ as $z \to x$. Lower hemicontinuity says that the nearby values do not leave gaps: they meet every open set that $\varphi(x)$ meets, so every point of $\varphi(x)$ is approximated by points of $\varphi(z)$. In the sequential form of Theorem 13.3.2 and Theorem 13.3.3 the asymmetry is plainest: upper hemicontinuity constrains the limits of arbitrary selections $y_n \in \varphi(x_n)$ to lie in $\varphi(x)$, while lower hemicontinuity requires each $y \in \varphi(x)$ to be the limit of some such selection. A value set that is large at x and small nearby therefore satisfies the first and violates the second, and one that is small at x and large nearby does the reverse, as Example 13.3.1 shows. Neither implies the other.

Proposition 13.3.1 (Global characterisations)**.** Let $\varphi\colon X \rightrightarrows Y$ between topological spaces.

(i) φ is upper hemicontinuous if and only if $\varphi^u(V)$ is open in X for every open $V \subseteq Y$, if and only if $\varphi^\ell(F)$ is closed in X for every closed $F \subseteq Y$.

(ii) φ is lower hemicontinuous if and only if $\varphi^\ell(V)$ is open in X for every open $V \subseteq Y$, if and only if $\varphi^u(F)$ is closed in X for every closed $F \subseteq Y$.

Consequently a single-valued correspondence is upper hemicontinuous if and only if it is lower hemicontinuous, if and only if the underlying function is continuous.

Proof. (i) Suppose φ is upper hemicontinuous and $V \subseteq Y$ is open. If $x \in \varphi^u(V)$ then V is an open set containing $\varphi(x)$, so there is an open $W \ni x$ with $\varphi(z) \subseteq V$ for all $z \in W$, that is $W \subseteq \varphi^u(V)$; hence $\varphi^u(V)$ is open. Conversely, if $\varphi^u(V)$ is open for every open V and $U \supseteq \varphi(x)$ is open, then $\varphi^u(U)$ is an open set containing x and serves as the required neighbourhood. The equivalence with the closed-set statement is Proposition 13.2.1(ii): $\varphi^u(V)$ open for all open V means $\varphi^\ell(V^c) = \varphi^u(V)^c$ closed for all such V, and $V \mapsto V^c$ is a bijection between open and closed sets.

(ii) The same argument with the roles of the two inverses exchanged.

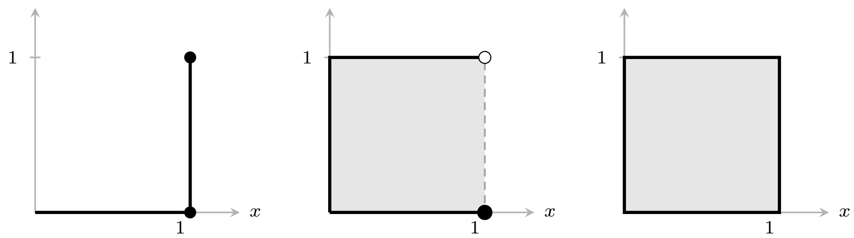
The final claim follows because for a single-valued φ the two inverses agree with f^{-1}, and Theorem 5.2.1 identifies continuity of f with openness of $f^{-1}(V)$ for open V. ∎

Example 13.3.1 (The two notions are independent)**.** Define $\varphi, \psi, \gamma\colon [0,1] \rightrightarrows [0,1]$ by

$$\varphi(x) = \begin{cases} \{0\}, & x \in [0,1), \\ [0,1], & x = 1, \end{cases} \qquad \psi(x) = \begin{cases} [0,1], & x \in [0,1), \\ \{0\}, & x = 1, \end{cases} \qquad \gamma(x) = [0,1].$$

Then φ is upper hemicontinuous everywhere but not lower hemicontinuous at $x = 1$: the open set $U = (\frac{1}{2}, 1]$ meets $\varphi(1)$ but misses $\varphi(z) = \{0\}$ for every $z < 1$. Symmetrically ψ is lower hemicontinuous everywhere but not upper hemicontinuous at $x = 1$: the open set $U = [0, \frac{1}{2})$ contains $\psi(1)$ but contains no $\psi(z)$ with $z < 1$. The constant correspondence γ is continuous.

The two failures are the two ways a jump at $x = 1$ can go. For φ the value at 1 is larger than every value nearby; nearby values therefore cannot stick out of a neighbourhood of $\varphi(1)$, and upper hemicontinuity survives, while the part of $\varphi(1)$ away from 0 is approximated by nothing, and lower hemicontinuity fails. For ψ the value at 1 is smaller than every value nearby; every point of $\psi(1)$ is trivially approximated, and lower hemicontinuity survives, while the nearby values stick out of any small neighbourhood of $\psi(1) = \{0\}$, and upper hemicontinuity fails. The three graphs are drawn in Figure 13.1.



(a) φ: upper but not lower hemicontinuous at $x = 1$

(b) ψ: lower but not upper hemicontinuous at $x = 1$

(c) γ: continuous

Figure 13.1: The graphs of the three correspondences of Example 13.3.1. Shading marks the values; a dashed edge is not part of the graph. In (a) the value at $x = 1$ is larger than the neighbouring values, which is compatible with upper hemicontinuity and destroys lower hemicontinuity; in (b) it is smaller, and the two verdicts are exchanged. Both properties compare $\varphi(z)$ for z near 1 with $\varphi(1)$, not the reverse.

Proposition 13.3.2 (Compactness is preserved). Let $\varphi\colon X \rightrightarrows Y$ be compact-valued and upper hemicontinuous, and let $A \subseteq X$ be compact. Then $\varphi(A) := \bigcup_{x\in A} \varphi(x)$ is compact in Y.

Proof. Let $\mathcal{V}$ be an open cover of $\varphi(A)$. Fix $x \in A$. The compact set $\varphi(x)$ is covered by finitely many members of $\mathcal{V}$; let U_x be their union, an open set containing $\varphi(x)$. Upper hemicontinuity supplies an open $W_x \ni x$ with $\varphi(z) \subseteq U_x$ for $z \in W_x$. The family $(W_x)_{x\in A}$ covers the compact set A, so finitely many $W_{x_1}, \dots, W_{x_k}$ suffice, and then $\varphi(A) \subseteq \bigcup_{i=1}^k U_{x_i}$, a union of finitely many finite subfamilies of $\mathcal{V}$. ∎

This generalises the statement that a continuous image of a compact set is compact (Proposition 5.2.4), and both hypotheses are needed: the upper hemicontinuous but non-compact-valued $\varphi(x) = \mathbb{R}$ has non-compact image, and the compact-valued but not upper hemicontinuous ψ of Example 13.3.1 would fail if $[0, 1]$ were replaced by an unbounded set.

13.3.1 Closed graphs

Definition 13.3.2 (Closed graph). A correspondence $\varphi\colon X \rightrightarrows Y$ between topological spaces *has a closed graph* if $\operatorname{gr}\varphi$ is closed in $X \times Y$ with the product topology.

Theorem 13.3.1 (Closed graph theorem). Let X be a metric space and Y a compact metric space. A correspondence $\varphi\colon X \rightrightarrows Y$ has a closed graph if and only if it is upper hemicontinuous and closed-valued.

Proof. ($\Leftarrow$) A closed subset of a compact metric space is compact, so a closed-valued φ into Y is compact-valued. Let $(x, y) \notin \operatorname{gr}\varphi$. Since $\varphi(x)$ is compact and $y \notin \varphi(x)$, the number $\varepsilon := \operatorname{dist}(y, \varphi(x))$ is strictly positive, and $U := \bigcup_{z\in\varphi(x)} B_{\varepsilon/2}(z)$ is an open set containing $\varphi(x)$ and disjoint from $B_{\varepsilon/2}(y)$. Upper hemicontinuity gives an open $W \ni x$ with $\varphi(w) \subseteq U$ for $w \in W$, so $\big(W \times B_{\varepsilon/2}(y)\big) \cap \operatorname{gr}\varphi = \emptyset$ and $(\operatorname{gr}\varphi)^c$ is open.

($\Rightarrow$) A closed graph gives closed values, since $\varphi(x) = \{y : (x, y) \in \operatorname{gr}\varphi\}$ is the slice of a closed set. Suppose φ fails to be upper hemicontinuous at some x: there is an open $V \supseteq \varphi(x)$ such that every neighbourhood of x contains a point whose value is not contained in V. Taking the neighbourhoods $B_{1/n}(x)$ produces $x_n \to x$ and $y_n \in \varphi(x_n) \setminus V$. Since Y is compact, Theorem 4.4.1 gives a subsequence $y_{n_k} \to y$, and $y \in V^c$ because V^c is closed, so $y \notin \varphi(x)$. Then $(x_{n_k}, y_{n_k}) \in \operatorname{gr}\varphi$ converges to $(x, y) \notin \operatorname{gr}\varphi$, and by Theorem 4.3.1 the graph is not closed. ∎

Remark 13.3.1 (Where compactness of Y is used). Without compactness of Y a closed graph does not give upper hemicontinuity. Take $X = Y = \mathbb{R}$ and $\varphi(x) = \{1/x\}$ for $x \neq 0$, $\varphi(0) = \{0\}$. The graph is the hyperbola together with the origin, which is closed in $\mathbb{R}^2$, but φ is not upper hemicontinuous at 0: the open set $(-1, 1) \supseteq \varphi(0)$ contains no $\varphi(z)$ with z small. Economic models in which quantities are unbounded require the closed-graph criterion to be supplemented by a compactness or boundedness argument before upper hemicontinuity may be inferred.

13.3.2 Sequential criteria

In finite-dimensional applications the definitions are almost always verified through sequences.

Theorem 13.3.2 (Sequential criterion for upper hemicontinuity)**.** Let $X \subseteq \mathbb{R}^k$ and $Y \subseteq \mathbb{R}^d$ and let $\varphi\colon X \rightrightarrows Y$ be compact-valued. Then φ is upper hemicontinuous at $x \in X$ if and only if for every sequence $(x_n, y_n) \in \operatorname{gr}\varphi$ with $x_n \to x$ the sequence (y_n) has a subsequence converging to a point of $\varphi(x)$.

Proof. ($\Rightarrow$) Let $x_n \to x$ with $y_n \in \varphi(x_n)$. Since $\varphi(x)$ is compact it is bounded, so it is contained in some open ball B; upper hemicontinuity gives a neighbourhood W of x with $\varphi(z) \subseteq B$ for $z \in W$, and $x_n \in W$ for $n \geq n_0$, whence $y_n \in B$ for such n and (y_n) is bounded. By Theorem 4.3.2 a subsequence $y_{n_k} \to y$ exists. Suppose $y \notin \varphi(x)$. As $\varphi(x)$ is closed and non-empty, $\varepsilon := \operatorname{dist}(y, \varphi(x)) > 0$, and the open set $U := \bigcup_{z\in\varphi(x)} B_{\varepsilon/2}(z)$ contains $\varphi(x)$ but not y; indeed $\operatorname{dist}(y, U) \geq \varepsilon/2$. Upper hemicontinuity gives n_1 with $y_n \in \varphi(x_n) \subseteq U$ for $n \geq n_1$, so $y = \lim_k y_{n_k}$ lies in $\overline{U}$, contradicting $\operatorname{dist}(y, U) \geq \varepsilon/2$. Hence $y \in \varphi(x)$.

($\Leftarrow$) Suppose φ is not upper hemicontinuous at x: there is an open $V \supseteq \varphi(x)$ such that for every n some $x_n \in B_{1/n}(x)$ has $\varphi(x_n) \nsubseteq V$. Pick $y_n \in \varphi(x_n) \cap V^c$. Then $x_n \to x$, and any convergent subsequence of (y_n) has its limit in the closed set V^c, hence outside $\varphi(x) \subseteq V$. So no subsequence converges into $\varphi(x)$. ■

Theorem 13.3.3 (Sequential criterion for lower hemicontinuity)**.** Let $X \subseteq \mathbb{R}^k$, $Y \subseteq \mathbb{R}^d$ and $\varphi\colon X \rightrightarrows Y$. Then φ is lower hemicontinuous at x if and only if for every sequence $x_n \to x$ in X and every $y \in \varphi(x)$ there exist $y_n \in \varphi(x_n)$ with $y_n \to y$.

Proof. ($\Rightarrow$) Let $x_n \to x$ and $y \in \varphi(x)$. For each $k \in \mathbb{N}$ the open ball $B_{1/k}(y)$ meets $\varphi(x)$, so lower hemicontinuity gives a neighbourhood U_k of x with $\varphi(z) \cap B_{1/k}(y) \neq \emptyset$ for $z \in U_k$. Choose $n_1 < n_2 < \cdots$ with $x_n \in U_k$ for all $n \geq n_k$. For $n_k \leq n < n_{k+1}$ pick $y_n \in \varphi(x_n) \cap B_{1/k}(y)$, and for $n < n_1$ pick $y_n \in \varphi(x_n)$ arbitrarily. Then $\|y_n - y\| < 1/k$ once $n \geq n_k$, so $y_n \to y$.

($\Leftarrow$) Suppose φ is not lower hemicontinuous at x: there is an open V with $\varphi(x) \cap V \neq \emptyset$ such that every neighbourhood of x contains a point z with $\varphi(z) \cap V = \emptyset$. Taking $B_{1/n}(x)$ produces $x_n \to x$ with $\varphi(x_n) \cap V = \emptyset$. Fix $y \in \varphi(x) \cap V$. Any $y_n \in \varphi(x_n)$ lies in the closed set V^c, so no such sequence can converge to $y \in V$. ■

Remark 13.3.2 (The asymmetry of the two criteria)**.** The two criteria are not mirror images, and the difference is exactly the difference between the two inverses. Upper hemicontinuity asks that limits of *arbitrary* selections stay inside the value at the limit; it is a statement about the graph and, with compact values, is equivalent to closedness of the graph by Theorem 13.3.1. Lower hemicontinuity asks that *every* point of the value at the limit be approachable by a selection; it is a statement about the existence of continuous selections and has no closed-graph reformulation. In economic terms: upper hemicontinuity rules out the sudden appearance of new optima, lower hemicontinuity rules out the sudden disappearance of existing ones.

Corollary 13.3.1 (Closed sub-correspondences)**.** Let $\varphi, \psi\colon X \rightrightarrows Y$ between subsets of finite-dimensional Euclidean spaces with $\psi(x) \subseteq \varphi(x)$ for all x. If φ is compact-valued and upper hemicontinuous and $\operatorname{gr}\psi$ is closed, then ψ is upper hemicontinuous and compact-valued.

Proof. $\psi(x)$ is a closed subset of the compact set $\varphi(x)$, hence compact. Let $(x_n, y_n) \in \operatorname{gr}\psi$ with $x_n \to x$. Then $(x_n, y_n) \in \operatorname{gr}\varphi$, so by Theorem 13.3.2 a subsequence $y_{n_k} \to y \in \varphi(x)$. Since $(x_{n_k}, y_{n_k}) \to (x, y)$ and $\operatorname{gr}\psi$ is closed, $(x, y) \in \operatorname{gr}\psi$, that is $y \in \psi(x)$. Apply Theorem 13.3.2 again. ■

13.3.3 Operations

Proposition 13.3.3 (Composition, union, intersection, product)**.** Let X, Y, Z be topological spaces.

(i) For $\varphi\colon X \rightrightarrows Y$ and $\psi\colon Y \rightrightarrows Z$ define $(\psi \circ \varphi)(x) := \bigcup_{y \in \varphi(x)} \psi(y)$. Then $(\psi \circ \varphi)^u(A) = \varphi^u(\psi^u(A))$ and $(\psi \circ \varphi)^\ell(A) = \varphi^\ell(\psi^\ell(A))$, and the composition inherits upper, respectively lower, hemicontinuity from both factors.

(ii) A union of lower hemicontinuous correspondences is lower hemicontinuous; a union of two upper hemicontinuous correspondences is upper hemicontinuous.

(iii) If φ has a closed graph and ψ is upper hemicontinuous and compact-valued, then $\varphi \cap \psi$ is upper hemicontinuous.

(iv) A product of upper hemicontinuous compact-valued correspondences is upper hemicontinuous and compact-valued; a product of lower hemicontinuous correspondences is lower hemicontinuous.

Proof. (i) For the upper inverse, $x \in (\psi \circ \varphi)^u(A)$ says $\psi(y) \subseteq A$ for every $y \in \varphi(x)$, that is $\varphi(x) \subseteq \psi^u(A)$; the lower inverse is analogous with "some" in place of "every". Hemicontinuity then follows from Proposition 13.3.1: openness of $\psi^u(V)$ and of φ^u of an open set compose.

(ii) For unions, $\left(\bigcup_i \varphi_i\right)^\ell(V) = \bigcup_i \varphi_i^\ell(V)$ is open when each summand is; for two upper hemicontinuous correspondences, $(\varphi \cup \psi)^u(V) = \varphi^u(V) \cap \psi^u(V)$ is a finite intersection of open sets.

(iii) Let $V \subseteq Y$ be open with $(\varphi \cap \psi)(x) \subseteq V$, and put $K := \psi(x) \setminus V$, a compact set. If $K = \emptyset$ take $N := \psi^u(V)$. Otherwise each $y \in K$ lies outside $\varphi(x)$, so $(x, y) \notin \operatorname{gr} \varphi$, and closedness of the graph supplies open $U_y \ni x$ and $W_y \ni y$ with $(U_y \times W_y) \cap \operatorname{gr} \varphi = \emptyset$. Finitely many $W_{y_1}, \dots, W_{y_r}$ cover K; set $W := \bigcup_i W_{y_i}$ and $N := \psi^u(V \cup W) \cap \bigcap_i U_{y_i}$. Since $\psi(x) \subseteq V \cup W$, upper hemicontinuity of ψ makes N an open neighbourhood of x. For $z \in N$ we have $\psi(z) \subseteq V \cup W$ and $\varphi(z) \cap W = \emptyset$, hence $(\varphi \cap \psi)(z) \subseteq V$.

(iv) See Aliprantis and Border (2006, Thm. 17.28); the compact-valued case in a finite product follows from Theorem 13.3.2 applied coordinatewise together with Proposition 4.3.4. ∎

13.4 Economic Application: Continuity of the Budget Correspondence

Proposition 13.4.1. Let $B\colon \mathbb{R}^n_{++} \times \mathbb{R}_{++} \rightrightarrows \mathbb{R}^n_+$ be the budget correspondence (13.1). Then B is non-empty-valued, compact-valued, convex-valued and continuous.

Proof. Non-emptiness, compactness and convexity are Example 13.2.1.

Upper hemicontinuity. Let $(p_k, m_k) \to (p, m)$ in $\mathbb{R}^n_{++} \times \mathbb{R}_{++}$ and $x_k \in B(p_k, m_k)$. Since $p \gg 0$, there are $\underline{p} > 0$ and $\bar{m} < \infty$ with $p_k \geq \underline{p}\mathbb{1}$ and $m_k \leq \bar{m}$ for large k, so $\|x_k\|_1 \leq \bar{m}/\underline{p}$ and (x_k) is bounded. By Theorem 4.3.2 a subsequence converges, say $x_{k_j} \to x \geq 0$, and passing to the limit in $\langle p_{k_j}, x_{k_j} \rangle \leq m_{k_j}$ gives $\langle p, x \rangle \leq m$, so $x \in B(p, m)$. Apply Theorem 13.3.2.

Lower hemicontinuity. Let $(p_k, m_k) \to (p, m)$ and $x \in B(p, m)$. If $\langle p, x \rangle < m$, then $\langle p_k, x \rangle < m_k$ for large k and the constant sequence $x_k := x$ works for those k; for the finitely many remaining indices take $x_k := 0$. If instead $\langle p, x \rangle = m$, set

$$\lambda_k := \min\left\{1, \frac{m_k}{\langle p_k, x \rangle}\right\}, \qquad x_k := \lambda_k x.$$

Then $x_k \in \mathbb{R}^n_+$ and $\langle p_k, x_k \rangle \leq m_k$, so $x_k \in B(p_k, m_k)$; and $\langle p_k, x \rangle \to \langle p, x \rangle = m > 0$, so $\lambda_k \to \min\{1, m/m\} = 1$ and $x_k \to x$. Apply Theorem 13.3.3. ∎

Example 13.4.1 (The two boundaries of $\mathbb{R}^n_{++} \times \mathbb{R}_{++}$)**.** Proposition 13.4.1 is stated on $\mathbb{R}^n_{++} \times \mathbb{R}_{++}$. The two boundaries behave differently.

Zero wealth is harmless. The conclusion extends to $m = 0$. Upper hemicontinuity: the boundedness argument of the proof needs only $p \gg 0$ and $m_k \leq \bar{m}$, and the limit x of a convergent subsequence satisfies $\langle p, x \rangle \leq 0$, hence $x = 0 \in B(p, 0)$. Lower hemicontinuity: $B(p, 0) = \{0\}$ and $0 \in B(p_k, m_k)$ for every k, so the constant sequence $x_k := 0$ suffices. So B is continuous on $\mathbb{R}^n_{++} \times \mathbb{R}_+$, and the restriction $m > 0$ in Proposition 13.4.1 is a convenience of the proof rather than a boundary of the result.

Zero prices are not. Take $n = 2$, $p^k = (1, 1/k) \to p = (1, 0)$ and $m = 1$. Then $B(p^k, 1)$ is the triangle with vertices $(0,0)$, $(1,0)$, $(0,k)$, while $B\big((1,0), 1\big) = \{x \in \mathbb{R}^2_+ : x_1 \leq 1\}$ is closed and unbounded, hence not compact. Compact-valuedness therefore fails, and with it both Theorem 13.3.2, whose statement presupposes compact values, and the hypotheses of Berge's theorem (Theorem 13.5.1). This is a loss of the tools, not by itself a proof that upper hemicontinuity fails; what does fail, and is the substantive point, is existence: a strictly increasing u has no maximiser on the unbounded set $B\big((1,0),1\big)$, so demand is undefined at a zero price. General-equilibrium existence proofs respond either by bounding consumption sets artificially or by working on the interior of the price simplex and handling the boundary separately.

Where lower hemicontinuity genuinely fails. The budget correspondence on $\mathbb{R}^n_+$ is lower hemicontinuous because 0 is always available and always cheapest. Replace the consumption set by $X := \{x \in \mathbb{R}^2_+ : x_1 \geq 1\}$, a subsistence requirement, and set $B_X(p,m) := \{x \in X : \langle p, x\rangle \leq m\}$. At $p = (1,1)$ and $m = 1$ one has $B_X(p,1) = \{(1,0)\}$, whereas $B_X(p, 1-1/k) = \emptyset$ for every k, since $x_1 \geq 1$ and $x_2 \geq 0$ force $\langle p, x\rangle \geq 1$. No selection converges to $(1,0)$, so B_X is not lower hemicontinuous at $(p,1)$. The failure is exactly the absence of a *cheaper point*: some $x \in X$ with $\langle p, x\rangle < m$. That condition, and not positivity of wealth, is what lower hemicontinuity of a budget correspondence rests on.

13.5 Berge's Theorem of the Maximum

Recall from Definition 5.3.1 that $m\colon X \to \mathbb{R}$ is lower semicontinuous if $\{x : m(x) > \alpha\}$ is open for every $\alpha \in \mathbb{R}$, and upper semicontinuous if $-m$ is lower semicontinuous. The maximum theorem is proved by establishing the two halves separately, each from one half of the continuity of the constraint correspondence.

Lemma 13.5.1 (Lower semicontinuity of the value)**.** Let $\varphi\colon X \rightrightarrows Y$ be lower hemicontinuous and let $f\colon \operatorname{gr}\varphi \to \mathbb{R}$ be lower semicontinuous. Then $m(x) := \sup_{y\in\varphi(x)} f(x,y)$, with $m(x) = -\infty$ when $\varphi(x) = \emptyset$, is lower semicontinuous.

Proof. Fix $\alpha \in \mathbb{R}$ and x_0 with $m(x_0) > \alpha$. Then $f(x_0, y_0) > \alpha$ for some $y_0 \in \varphi(x_0)$, so $\varphi(x_0) \neq \emptyset$. The set $W := \{(x,y) \in \operatorname{gr}\varphi : f(x,y) > \alpha\}$ is open in $\operatorname{gr}\varphi$ by lower semicontinuity of f and contains (x_0, y_0), so there are open $U \ni x_0$ in X and $V \ni y_0$ in Y with $(U \times V) \cap \operatorname{gr}\varphi \subseteq W$. Put $N := U \cap \varphi^\ell(V)$, which is open by Proposition 13.3.1(ii) and contains x_0 because $y_0 \in \varphi(x_0) \cap V$. For $x \in N$ pick $y \in \varphi(x) \cap V$; then $(x,y) \in (U\times V) \cap \operatorname{gr}\varphi \subseteq W$, so $f(x,y) > \alpha$ and $m(x) > \alpha$. Hence $\{m > \alpha\}$ is open. ∎

Lemma 13.5.2 (Upper semicontinuity of the value)**.** Let $\varphi\colon X \rightrightarrows Y$ be upper hemicontinuous, non-empty- and compact-valued, and let $f\colon \operatorname{gr}\varphi \to \mathbb{R}$ be upper semicontinuous. Then $m(x) := \max_{y\in\varphi(x)} f(x,y)$ is well defined and upper semicontinuous.

Proof. For fixed x the function $f(x,\cdot)$ is upper semicontinuous on the non-empty compact set $\varphi(x)$, so Theorem 5.4.1 gives the maximum. Fix $\alpha \in \mathbb{R}$ and x_0 with $m(x_0) < \alpha$, and put $W := \{(x,y) \in \operatorname{gr}\varphi : f(x,y) < \alpha\}$, open in $\operatorname{gr}\varphi$ and containing $\{x_0\} \times \varphi(x_0)$. For each $y \in \varphi(x_0)$ choose open $U_y \ni x_0$ and $V_y \ni y$ with $(U_y \times V_y) \cap \operatorname{gr}\varphi \subseteq W$. Compactness of $\varphi(x_0)$ gives $y_1, \dots, y_r$ with $\varphi(x_0) \subseteq V := \bigcup_i V_{y_i}$; set $U := \bigcap_i U_{y_i}$ and $N := U \cap \varphi^u(V)$, open by Proposition 13.3.1(i) and containing x_0. For $x \in N$ and $y \in \varphi(x)$ we have $(x,y) \in (U \times V) \cap \operatorname{gr}\varphi \subseteq W$, so $f(x,y) < \alpha$ and therefore $m(x) < \alpha$. ∎

Theorem 13.5.1 (Berge's theorem of the maximum)**.** Let X, Y be topological spaces, let $\varphi\colon X \rightrightarrows Y$ be continuous with non-empty compact values, and let $f\colon \operatorname{gr}\varphi \to \mathbb{R}$ be continuous. Define the *value function* and the *argmax correspondence*

$$m(x) := \max_{y\in\varphi(x)} f(x,y), \qquad \mu(x) := \{y \in \varphi(x) : f(x,y) = m(x)\}. \tag{13.4}$$

Then

(i) m is continuous on X;

(ii) μ is non-empty- and compact-valued; and

(iii) μ is upper hemicontinuous.

Proof. (i) f is both upper and lower semicontinuous and φ is both upper and lower hemicontinuous, so Lemma 13.5.1 and Lemma 13.5.2 make m both lower and upper semicontinuous, hence continuous by

Corollary 5.3.1.

(ii) Fix x. The map $y \mapsto (x, y)$ is continuous from $\varphi(x)$ into $\operatorname{gr}\varphi$, so $f(x, \cdot)$ is continuous on the non-empty compact set $\varphi(x)$ and attains its maximum there by Theorem 5.4.1; thus $\mu(x) \neq \emptyset$. Moreover $\mu(x) = \{y \in \varphi(x) : f(x, y) \geq m(x)\}$ is closed in $\varphi(x)$, hence compact.

(iii) Fix $x_0 \in X$ and an open $U \subseteq Y$ with $\mu(x_0) \subseteq U$. Put $K := \varphi(x_0) \setminus U$, a closed subset of the compact set $\varphi(x_0)$ and therefore compact. If $K = \emptyset$, then $\varphi(x_0) \subseteq U$ and upper hemicontinuity of φ supplies an open $N \ni x_0$ with $\mu(z) \subseteq \varphi(z) \subseteq U$ for $z \in N$.

Suppose $K \neq \emptyset$. Every $y \in K$ lies in $\varphi(x_0)$ but not in $\mu(x_0)$, so $f(x_0, y) < m(x_0)$; since $f(x_0, \cdot)$ is continuous on the compact set K it attains a maximum there, and hence there is $\varepsilon > 0$ with

$$f(x_0, y) < m(x_0) - 2\varepsilon \qquad \text{for all } y \in K. \tag{13.5}$$

The set $W := \{(x, y) \in \operatorname{gr}\varphi : f(x, y) < m(x) - \varepsilon\}$ is open in $\operatorname{gr}\varphi$, because f is continuous on $\operatorname{gr}\varphi$ and m is continuous on X by (i), and (13.5) puts $\{x_0\} \times K$ inside W. For each $y \in K$ choose open $U_y \ni x_0$ in X and $V_y \ni y$ in Y with $(U_y \times V_y) \cap \operatorname{gr}\varphi \subseteq W$. Compactness of K gives $y_1, \ldots, y_r$ with $K \subseteq V := \bigcup_{i \leq r} V_{y_i}$; set $N_1 := \bigcap_{i \leq r} U_{y_i}$, an open neighbourhood of x_0, and note that $(N_1 \times V) \cap \operatorname{gr}\varphi \subseteq W$.

Since $\varphi(x_0) \subseteq U \cup V$ and $U \cup V$ is open, upper hemicontinuity of φ supplies an open $N_2 \ni x_0$ with $\varphi(z) \subseteq U \cup V$ for $z \in N_2$. Let $z \in N_1 \cap N_2$ and $y \in \mu(z)$. Then $y \in \varphi(z) \subseteq U \cup V$. Were $y \in V$, then $(z, y) \in (N_1 \times V) \cap \operatorname{gr}\varphi \subseteq W$ would give $f(z, y) < m(z) - \varepsilon$, contradicting $f(z, y) = m(z)$. Hence $y \in U$, so $\mu(z) \subseteq U$ for every $z \in N_1 \cap N_2$. ■

The argument is purely topological: no metric, no sequences and no separation axiom on Y enters, so the theorem holds as stated for arbitrary topological spaces. In $\mathbb{R}^n$ the conclusion can also be reached through Corollary 13.3.1, since $\operatorname{gr}\mu$ is then closed, but that route needs the sequential criteria and is available only in the metric setting.

> **Remark 13.5.1** (μ is not lower hemicontinuous). Upper hemicontinuity is the strongest conclusion available. Let $X = [0, 1]$, $Y = [1, 2]$, $\varphi \equiv Y$ and $f(x, y) = y^x$. Then $\mu(x) = \{2\}$ for $x > 0$ and $\mu(0) = Y$, since $f(0, \cdot) \equiv 1$ is constant. The correspondence μ is upper hemicontinuous at 0, as the theorem asserts, and is not lower hemicontinuous there: every $y \in [1, 2)$ belongs to $\mu(0)$ and is approached by no selection from $\mu(x) = \{2\}$. Ties in the objective produce large optimal sets that collapse under an arbitrarily small perturbation, and no hypothesis of the theorem prevents this. In economics the phenomenon is the discontinuous collapse of demand when an indifference curve is tangent to a whole segment of the budget line.

Example 13.5.1 (Every hypothesis is used). Let $X = Y = [0, 1]$, $f(x, y) = y$, and

$$\varphi(x) = \begin{cases} [0, 1], & x = 0, \\ \{0\}, & x > 0. \end{cases}$$

Then φ is upper hemicontinuous and compact-valued but not lower hemicontinuous at 0. Consequently $m(0) = 1$ and $m(x) = 0$ for $x > 0$, so m is not continuous but merely upper semicontinuous, as Lemma 13.5.2 predicts; and $\mu(0) = \{1\}$, $\mu(x) = \{0\}$ for $x > 0$, so μ is not upper hemicontinuous. Losing lower hemicontinuity of the constraint correspondence loses both conclusions at once.

> **Corollary 13.5.1** (Continuity of Marshallian demand). Let $u \colon \mathbb{R}^n_+ \to \mathbb{R}$ be continuous. Then the indirect utility function $v(p, m) = \max_{x \in B(p,m)} u(x)$ is continuous on $\mathbb{R}^n_{++} \times \mathbb{R}_{++}$ and the demand correspondence D of (13.2) is non-empty- and compact-valued and upper hemicontinuous there. If in addition u is strictly quasiconcave, D is single-valued and the demand function is continuous.

Proof. Proposition 13.4.1 and Theorem 13.5.1 give the first two claims, with $f(x, y) := u(y)$ as the objective. If u is strictly quasiconcave, Proposition 9.3.5 makes D single-valued, and by Proposition 13.3.1 a single-valued upper hemicontinuous correspondence is a continuous function. ■

The corollary is the theoretical warrant for the comparative statics performed throughout demand theory. Without it, an expression such as $\partial x^*/\partial p$ could not be assumed even to exist, and the observation of Remark 8.6.1 — that the implicit function theorem delivers differentiability from a non-singular bordered Hessian — would have nothing to be applied to. Corollary 13.5.1 also shows what is available without any differentiability at all: continuity of v and upper hemicontinuity of D come from compactness and continuity alone.

Proposition 13.5.1 (Concavity of the value function)**.** Let $f\colon \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}$ be concave and suppose $f^*(\alpha) := \max_{x\in\mathbb{R}^n} f(x,\alpha)$ exists for every $\alpha \in \mathbb{R}^k$. Then f^* is concave.

Proof. Fix $\alpha, \alpha' \in \mathbb{R}^k$ and $\lambda \in [0,1]$, and choose maximisers $x^*(\alpha), x^*(\alpha')$. Then

$$\begin{aligned} f^*\big((1-\lambda)\alpha + \lambda\alpha'\big) &\geq f\big((1-\lambda)x^*(\alpha) + \lambda x^*(\alpha'),\, (1-\lambda)\alpha + \lambda\alpha'\big) \\ &\geq (1-\lambda)f\big(x^*(\alpha),\alpha\big) + \lambda f\big(x^*(\alpha'),\alpha'\big) = (1-\lambda)f^*(\alpha) + \lambda f^*(\alpha'), \end{aligned}$$

the first inequality because the displayed point is feasible and the second by concavity of f on $\mathbb{R}^n \times \mathbb{R}^k$. ■

Maximisation over an unconstrained variable therefore preserves concavity. This is what makes the value function of a concave dynamic programme concave, and hence — by Proposition 9.3.3 — continuous on the interior of its domain.

Differentiability is a separate matter, and it does not follow from uniqueness of the maximiser. Consider

$$V(\alpha) := \max\{x \in \mathbb{R} : x \leq \alpha,\ x \leq 1\} = \min\{\alpha, 1\}.$$

The objective is linear, the constraints are affine, the maximiser $x^*(\alpha) = \min\{\alpha,1\}$ is unique for every α, and V is concave; yet V is not differentiable at $\alpha = 1$, where the binding constraint changes. Concavity delivers a non-empty superdifferential at every interior point and differentiability exactly where that superdifferential is a singleton, which is a condition on the whole family of supporting affine functions rather than on the argmax. The theorems that do convert regularity of the objective into differentiability of the value — Danskin's theorem, the envelope theorem Theorem 12.3.2, and Benveniste–Scheinkman (Theorem 15.4.1) — all assume differentiability of the objective in the parameter and interiority of the maximiser, and the example above satisfies neither at $\alpha = 1$.

Theorem 13.5.2 makes the first of these precise and shows exactly how much regularity of the value function a smooth objective delivers when the maximiser is not unique.

Theorem 13.5.2 (Danskin)**.** Let $X \subseteq \mathbb{R}^n$ be open, U a non-empty compact metric space, and $\varphi\colon X \times U \to \mathbb{R}$ continuous with the partial gradient $\nabla_x\varphi$ existing and continuous on $X \times U$. Put

$$V(x) := \max_{u\in U} \varphi(x,u), \qquad M(x) := \arg\max_{u\in U} \varphi(x,u).$$

Then V is locally Lipschitz on X, $M(x)$ is non-empty and compact, and for every $x \in X$ and $d \in \mathbb{R}^n$ the one-sided directional derivative exists and

$$V'(x;d) := \lim_{t\downarrow 0} \frac{V(x+td) - V(x)}{t} = \max_{u\in M(x)} \left\langle \nabla_x\varphi(x,u), d\right\rangle. \tag{13.6}$$

If $M(x) = \{u^*\}$ is a singleton, then V is differentiable at x with $\nabla V(x) = \nabla_x\varphi(x,u^*)$.

Proof. Non-emptiness and compactness of $M(x)$, and continuity and upper hemicontinuity of V and M, are Theorem 13.5.1 applied to the constant correspondence $\Gamma \equiv U$.

Local Lipschitz property. Let $K \subseteq X$ be compact and convex with $x \in \operatorname{Int} K$, and put $L := \max_{K\times U} \|\nabla_x\varphi\|$, finite by Corollary 5.4.1 on the compact $K \times U$. For $y, z \in K$,

$$|V(y) - V(z)| \leq \max_{u\in U} |\varphi(y,u) - \varphi(z,u)| \leq L\,\|y - z\|\,,$$

the first inequality because $|\max_u a(u) - \max_u b(u)| \leq \max_u |a(u) - b(u)|$ and the second by the mean value theorem along the segment $[z,y] \subseteq K$.

Lower bound in (13.6). Fix $u \in M(x)$. For small $t > 0$,

$$V(x+td) - V(x) \geq \varphi(x+td,u) - \varphi(x,u) = t\left\langle \nabla_x\varphi(x,u), d\right\rangle + o(t),$$

so $\liminf_{t\downarrow 0}\big(V(x+td) - V(x)\big)/t \geq \langle \nabla_x\varphi(x,u), d\rangle$; take the maximum over $u \in M(x)$.

Upper bound. Let $t_k \downarrow 0$ realise the limit superior and choose $u_k \in M(x+t_kd)$. The mean value theorem applied to $s \mapsto \varphi(x+sd,u_k)$ gives $\theta_k \in (0,1)$ with

$$\frac{V(x+t_kd) - V(x)}{t_k} \leq \frac{\varphi(x+t_kd,u_k) - \varphi(x,u_k)}{t_k} = \left\langle \nabla_x\varphi(x+\theta_kt_kd,u_k), d\right\rangle.$$

By compactness of U, pass to a subsequence with $u_k \to \bar{u}$; upper hemicontinuity of M together with Theorem 13.3.1 gives $\bar{u} \in M(x)$, and continuity of $\nabla_x\varphi$ gives $\langle \nabla_x\varphi(x+\theta_k t_k d, u_k), d\rangle \to \langle \nabla_x\varphi(x,\bar{u}), d\rangle$. Hence the limit superior is at most $\max_{u\in M(x)} \langle \nabla_x\varphi(x,u), d\rangle$, and (13.6) follows.

Differentiability under uniqueness. Write $g := \nabla_x\varphi(x,u^*)$, so that (13.6) reads $V'(x;d) = \langle g, d\rangle$ for every d. Suppose V were not differentiable at x with derivative g. Then there are $\varepsilon > 0$ and $h_k \to 0$, $h_k \neq 0$, with

$$\left|V(x+h_k) - V(x) - \langle g, h_k\rangle\right| \geq \varepsilon \left\|h_k\right\|. \tag{13.7}$$

Write $h_k = t_k d_k$ with $t_k = \|h_k\| \downarrow 0$ and $\|d_k\| = 1$; by compactness of the unit sphere, pass to a subsequence with $d_k \to d$. The local Lipschitz bound gives $|V(x+t_kd_k) - V(x+t_kd)| \leq Lt_k \|d_k - d\| = o(t_k)$, while the directional derivative along the fixed direction d gives $V(x+t_kd) - V(x) = t_k \langle g, d\rangle + o(t_k)$. Therefore

$$V(x+h_k) - V(x) - \langle g, h_k\rangle = t_k \langle g, d - d_k\rangle + o(t_k) = o(t_k),$$

since $d_k \to d$, contradicting (13.7). ∎

> **Remark 13.5.2** (What Danskin adds, and what it does not). Formula (13.6) holds with no convexity and no uniqueness: the value function of a smooth family always has one-sided directional derivatives, and they are computed by freezing the maximiser and differentiating the objective. What multiplicity of maximisers costs is two-sidedness. Taking d and $-d$ in (13.6),
>
> $$V'(x;d) + V'(x;-d) = \max_{u\in M(x)} \langle \nabla_x\varphi(x,u), d\rangle - \min_{u\in M(x)} \langle \nabla_x\varphi(x,u), d\rangle \geq 0,$$
>
> with equality for every d if and only if $\nabla_x\varphi(x,\cdot)$ is constant on $M(x)$. A kink in the value function is therefore not caused by multiplicity of the maximiser as such, but by disagreement among the maximisers about the marginal effect of the parameter. In the example $V(\alpha) = \min\{\alpha, 1\}$ above the maximiser is unique at every α, and the kink at $\alpha = 1$ comes instead from the failure of the constraint set to be described by a fixed compact U independent of the parameter, which is the hypothesis Danskin's theorem imposes and Theorem 13.5.1 relaxes.
>
> When each $\varphi(\cdot, u)$ is convex, V is convex by Proposition 9.3.2, and every $\nabla_x\varphi(x,u)$ with $u \in M(x)$ is a subgradient of V at x: for any y,
>
> $$V(y) \geq \varphi(y,u) \geq \varphi(x,u) + \langle \nabla_x\varphi(x,u), y - x\rangle = V(x) + \langle \nabla_x\varphi(x,u), y - x\rangle,$$
>
> using Theorem 9.3.1 for the middle inequality. Since $\partial V(x)$ is closed and convex by Theorem 9.6.1, it contains $\operatorname{co}\{\nabla_x\varphi(x,u) : u \in M(x)\}$. Equality of the two sets is the identification of $V'(x;\cdot)$ with the support function of $\partial V(x)$ and is proved in Rockafellar (1970, Sec. 23); it is not used below.

For constrained problems, concavity of the value function in the parameter needs the constraint correspondence to have a convex *graph*, not merely convex values. The budget correspondence (13.1) has a convex graph in (x,m) for fixed p, so if u is concave then $v(p,\cdot)$ is concave in wealth. In prices the graph is not convex, and the conclusion available is different in kind: v is quasiconvex in (p,m) jointly (Exercise 9.8.2), homogeneous of degree zero in (p,m), non-increasing in each price and non-decreasing in wealth. It need not be convex in p. Take $n = 1$ and $u(x) = -x^{-2}$ on $(0,\infty)$, which is strictly increasing and strictly concave since $u'(x) = 2x^{-3} > 0$ and $u''(x) = -6x^{-4} < 0$. Demand exhausts the budget, so $x(p,m) = m/p$ and

$$v(p,m) = -\left(\frac{p}{m}\right)^2, \qquad \frac{\partial^2 v}{\partial p^2} = -\frac{2}{m^2} < 0,$$

strictly concave in p. It remains quasiconvex, being monotone in a single variable. Convexity in p does occur — $u(x) = \log x$ gives $v(p,m) = \log m - \log p$ with $\partial^2 v/\partial p^2 = p^{-2} > 0$ — but it is a feature of the particular utility function and not a general property.

13.6 Fixed-Point Theorems

> **Theorem 13.6.1** (Brouwer). Let $C \subseteq \mathbb{R}^n$ be non-empty, compact and convex. Every continuous $f\colon C \to C$ has a fixed point.

The proof is topological — it rests on the fact that the n-sphere is not a retract of the closed ball — and is not reproduced here; see Ok (2007, Ch. E) for a full treatment and Fukuda (2026d, p. 270) for the statement used in this course. What is reproduced is the exact role of each hypothesis, since the failures are what one must check in applications.

Example 13.6.1 (The three hypotheses of Brouwer). *Compactness.* $C = \mathbb{R}$ and $f(x) = x + 1$: continuous, C convex, no fixed point. Closedness alone is not enough, and neither is boundedness alone: on $C = (0, 1)$ the map $f(x) = x/2$ is continuous with no fixed point.

Convexity. $C = \{x \in \mathbb{R}^2 : \|x\| = 1\}$, the unit circle, is compact but not convex, and the rotation by π, $f(x) = -x$, is continuous with no fixed point.

Continuity. $C = [0, 1]$ and $f(x) = 1$ for $x < \frac{1}{2}$, $f(x) = 0$ for $x \geq \frac{1}{2}$: compact convex domain, no fixed point.

> **Corollary 13.6.1** (Brouwer on the simplex). Let $\Delta^{n-1} := \{p \in \mathbb{R}^n_+ : \sum_{i=1}^n p_i = 1\}$. Every continuous $f \colon \Delta^{n-1} \to \Delta^{n-1}$ has a fixed point.

Proof. Δ^{n-1} is convex as the intersection of $\mathbb{R}^n_+$ with a hyperplane, and compact by Theorem 4.4.2, being closed and contained in the unit ball of $\|\cdot\|_1$. ∎

> **Theorem 13.6.2** (Kakutani). Let $X \subseteq \mathbb{R}^n$ be non-empty, compact and convex, and let $\varphi \colon X \rightrightarrows X$ be non-empty-valued, convex-valued and have a closed graph. Then φ has a *fixed point*: some $x \in X$ with $x \in \varphi(x)$.

By Theorem 13.3.1 the closed-graph hypothesis is, on a compact X, the same as requiring φ upper hemicontinuous with closed values. Kakutani's theorem contains Brouwer's: a continuous function is a single-valued correspondence with convex values and, by Proposition 13.3.1 and Theorem 13.3.1, a closed graph. The statement is Fukuda (2026d, Thm. 13.2, p. 478); the proof, which approximates φ by continuous functions on successively finer simplicial subdivisions and applies Brouwer to each, is in Aliprantis and Border (2006, Ch. 17).

Example 13.6.2 (The four hypotheses of Kakutani). *Compactness.* $X = \mathbb{R}$, $\varphi(x) = \{x + 1\}$.

Convexity of X. X the unit circle, $\varphi(x) = \{-x\}$.

Convex values. $X = [0, 1]$ and $\varphi(x) = \{0, 1\} \setminus \{x\}$ for $x \in \{0, 1\}$, $\varphi(x) = \{0, 1\}$ otherwise. This has a closed graph and compact values, but $x \in \varphi(x)$ never holds. A cleaner instance is $X = [0, 1]$ with $\varphi(x) = \{1\}$ for $x < \frac{1}{2}$, $\varphi(1/2) = \{0, 1\}$ and $\varphi(x) = \{0\}$ for $x > \frac{1}{2}$: the graph is closed, the values are compact and non-empty, and only convexity at $x = \frac{1}{2}$ fails, which is where a fixed point would have to be.

Closed graph. $X = [0, 1]$ with $\varphi(x) = \{1\}$ for $x < \frac{1}{2}$ and $\varphi(x) = \{0\}$ for $x \geq \frac{1}{2}$: convex-valued, non-empty-valued, no fixed point, and the graph is not closed at $x = \frac{1}{2}$.

13.7 Economic Application: Existence of Nash Equilibrium

Environment. A finite game has players indexed by $i \in \{1, \dots, I\}$, finite action sets A_i, and payoffs $u_i \colon \prod_j A_j \to \mathbb{R}$. A mixed strategy for player i is an element σ_i of the simplex $\Delta(A_i)$ over A_i. Write $\Sigma := \prod_i \Delta(A_i)$ and extend u_i to Σ by expectation,

$$u_i(\sigma) := \sum_{a \in \prod_j A_j} \Big(\prod_{j=1}^{I} \sigma_j(a_j) \Big) u_i(a). \tag{13.8}$$

A *Nash equilibrium* is $\sigma^* \in \Sigma$ with $u_i(\sigma^*) \geq u_i(\sigma_i, \sigma^*_{-i})$ for every i and every $\sigma_i \in \Delta(A_i)$.

> **Theorem 13.7.1** (Nash). Every finite game has a Nash equilibrium in mixed strategies.

Proof. Define the best-response correspondence $\beta_i \colon \Sigma \rightrightarrows \Delta(A_i)$ by $\beta_i(\sigma) := \arg\max_{\sigma_i \in \Delta(A_i)} u_i(\sigma_i, \sigma_{-i})$ and set $\beta := \prod_i \beta_i \colon \Sigma \rightrightarrows \Sigma$. A fixed point of β is precisely a Nash equilibrium, by the definition of β_i. The hypotheses of Theorem 13.6.2 are verified in turn.

Σ is non-empty, compact and convex. Each $\Delta(A_i)$ is a simplex, compact and convex by Corollary 13.6.1's argument, and a finite product of compact convex sets is compact by Proposition 4.4.2 and convex by Proposition 9.2.1.

β is non-empty-valued. $u_i(\cdot, \sigma_{-i})$ is a polynomial, hence continuous, on the compact set $\Delta(A_i)$, so Corollary 5.4.1 applies.

β is convex-valued. By (13.8), u_i is *affine* in σ_i for fixed σ_{-i}. The set of maximisers of an affine function on a convex set is convex: if $u_i(\sigma_i, \sigma_{-i}) = u_i(\sigma_i', \sigma_{-i}) = \max$ then affinity gives the same value at every convex combination. A product of convex sets is convex.

β has a closed graph. The constraint correspondence $\sigma \mapsto \Delta(A_i)$ is constant, hence continuous with non-empty compact values, and u_i is continuous on $\Sigma \times \Delta(A_i)$; so Theorem 13.5.1(iii) makes β_i upper hemicontinuous with compact values, and Theorem 13.3.1 converts this into a closed graph on the compact space Σ. Closed graphs are preserved by finite products.

Theorem 13.6.2 now supplies $\sigma^* \in \beta(\sigma^*)$. ■

Two features of the argument deserve emphasis. Convex-valuedness rests entirely on the *linearity* of expected utility in own mixed strategy; it is what mixing is for, and it is why an equilibrium in pure strategies need not exist while one in mixed strategies always does. And upper hemicontinuity of the best response is an instance of Theorem 13.5.1 in the degenerate case of a constant constraint correspondence, so the maximum theorem is doing the work even where no constraint moves.

13.8 Economic Application: Existence of Walrasian Equilibrium

Environment. There are n goods and a price vector p normalised to lie in Δ^{n-1}. Let $Z\colon \Delta^{n-1} \to \mathbb{R}^n$ denote aggregate excess demand: $Z_i(p)$ is the quantity of good i demanded in aggregate, net of the aggregate endowment, at prices p.

Assumption 13.8.1 (Excess demand). Z is continuous on Δ^{n-1} and satisfies *Walras' law* $\langle p, Z(p)\rangle = 0$ for every $p \in \Delta^{n-1}$.

Walras' law is the aggregate budget identity: each consumer spends exactly their wealth, so the value of aggregate excess demand is zero at every price vector, whether or not markets clear. Continuity of Z is not primitive; it is the conclusion of Corollary 13.5.1 applied to each consumer, requiring strictly quasiconcave utility for single-valuedness and interior prices for compactness of the budget set (Example 13.4.1).

Theorem 13.8.1 (Existence of equilibrium prices). Under Assumption 13.8.1 there exists $p^* \in \Delta^{n-1}$ with $Z(p^*) \le 0$; and $Z_i(p^*) < 0$ only if $p_i^* = 0$.

Proof. Define $g\colon \Delta^{n-1} \to \Delta^{n-1}$ by

$$g_i(p) := \frac{p_i + \max\{0, Z_i(p)\}}{1 + \sum_{k=1}^{n} \max\{0, Z_k(p)\}}. \tag{13.9}$$

The denominator is at least 1, the numerators are non-negative and sum to the denominator, so g maps Δ^{n-1} into itself; g is continuous because Z is and $t \mapsto \max\{0, t\}$ is. By Corollary 13.6.1 there is p^* with $g(p^*) = p^*$. Writing $S := \sum_k \max\{0, Z_k(p^*)\}$, the fixed-point equation is $p_i^*(1+S) = p_i^* + \max\{0, Z_i(p^*)\}$, that is

$$p_i^* S = \max\{0, Z_i(p^*)\} \qquad \text{for each } i. \tag{13.10}$$

Multiply (13.10) by $Z_i(p^*)$ and sum over i:

$$S \underbrace{\langle p^*, Z(p^*)\rangle}_{=0 \text{ (Walras)}} = \sum_{i=1}^{n} Z_i(p^*) \max\{0, Z_i(p^*)\} = \sum_{i=1}^{n} \big(\max\{0, Z_i(p^*)\}\big)^2.$$

A sum of squares equal to zero forces $\max\{0, Z_i(p^*)\} = 0$ for every i, that is $Z(p^*) \le 0$. Finally, if $Z_i(p^*) < 0$ for some i with $p_i^* > 0$, then $\langle p^*, Z(p^*)\rangle < 0$ because every term is at most zero and this one is strictly negative, contradicting Walras' law. ■

The map (13.9) is a normalised *tâtonnement*: prices of goods in excess demand are raised in proportion to the excess, prices of the remaining goods fall through the normalisation. The theorem says the process has a rest point; it says nothing about whether the process converges to it, and in general it does not. The last sentence of the statement is free disposal in equilibrium: a good in strict excess supply must be free.

> **Remark 13.8.1** (What has and has not been proved)**.** Theorem 13.8.1 is a statement about a function Z that has been *assumed* continuous on the whole simplex, boundary included. Deriving that hypothesis from preferences and endowments is the substance of the existence problem, and it is where the difficulties of Example 13.4.1 bite: as some $p_i \to 0$ the budget set ceases to be compact and, under monotone preferences, demand for good i diverges, so Z is unbounded near the boundary. The standard resolutions — truncating consumption sets to a large compact box and showing the truncation does not bind, or replacing Z by a correspondence and invoking Theorem 13.6.2 in the form of the Gale–Nikaidô–Debreu lemma — are treated in Aliprantis and Border (2006, Ch. 17). What the argument above establishes cleanly is that once continuity and Walras' law are available, existence is a one-page consequence of Brouwer's theorem.

13.9 Chapter Summary

A correspondence is a map into subsets, identified with its graph. Because a correspondence has two inverse images — the upper inverse demanding containment of the whole value, the lower inverse demanding only non-empty intersection — there are two continuity notions, both comparing the values near a point with the value at it. Upper hemicontinuity, equivalent to openness of upper inverses of open sets and, on compact ranges, to closedness of the graph, keeps the nearby values inside any prescribed neighbourhood of the value at the point; lower hemicontinuity requires the nearby values to reach every part of it. In finite dimensions both are checked by sequences: upper hemicontinuity says every selection along a convergent sequence of parameters has a subsequence converging into the limit value, lower hemicontinuity says every point of the limit value is the limit of some selection.

The budget correspondence is continuous with compact convex values at strictly positive prices, wealth zero included, and loses compact-valuedness when a price is zero; lower hemicontinuity is what fails when the consumption set admits no point cheaper than the wealth level. Berge's theorem of the maximum then yields, for a continuous objective on the graph of a continuous compact-valued constraint correspondence, a continuous value function and an upper hemicontinuous compact-valued argmax; lower hemicontinuity of the argmax is false in general, and the counterexample is ties in the objective. Applied to the consumer, this gives continuity of indirect utility and upper hemicontinuity of Marshallian demand, single-valued and continuous under strict quasiconcavity. Maximising out a variable preserves concavity of the objective in the remaining parameters, but not differentiability, which requires the separate hypotheses of the envelope and Benveniste–Scheinkman theorems.

Existence of equilibrium comes from fixed points. Brouwer's theorem gives a fixed point of a continuous self-map of a non-empty compact convex set; Kakutani's extends this to correspondences that are non-empty- and convex-valued with a closed graph. Kakutani applied to the best-response correspondence gives Nash equilibrium in mixed strategies, convex-valuedness being the linearity of expected utility in own strategy. Brouwer applied to a normalised tâtonnement map gives Walrasian equilibrium prices from continuity of excess demand and Walras' law. Neither theorem gives uniqueness, and neither gives an algorithm — in contrast to Theorem 6.4.1, which gives both at the cost of a contraction hypothesis.

> **Exercise 13.9.1.** Prove Proposition 13.3.1 in the pointwise form: φ is upper hemicontinuous at x if and only if $\varphi^u(U)$ is a neighbourhood of x for every neighbourhood U of $\varphi(x)$; and dually for lower hemicontinuity. Then show directly from the definitions that a correspondence is upper hemicontinuous and lower hemicontinuous at x if and only if it is continuous there.

> **Exercise 13.9.2.** Let $\varphi\colon X \rightrightarrows Y$ be a correspondence and $A \subseteq X$. Prove that $A \subseteq \varphi^u(\varphi(A))$, and that if φ is non-empty-valued then $A \subseteq \varphi^\ell(\varphi(A))$. Give an example in which both inclusions are strict, and reconcile the result with Proposition 1.2.2.

Exercise 13.9.3. Let $\Gamma\colon S \rightrightarrows S$ be the feasible-continuation correspondence of a dynamic programme on a compact state space S, assumed continuous with non-empty compact values, and let $F\colon \operatorname{gr}\Gamma \to \mathbb{R}$ be continuous and $\beta \in (0,1)$. Using Theorem 13.5.1, show that the Bellman operator $(Tv)(s) = \max_{s' \in \Gamma(s)}\{F(s,s') + \beta v(s')\}$ maps $\mathcal{C}(S,\mathbb{R})$ into itself, and combine this with Proposition 6.3.3 and Theorem 6.5.1 to conclude that the value function is the unique continuous fixed point and that the policy correspondence is upper hemicontinuous. This is the argument completed in Chapter 15.

Chapter 14

Probability, Conditional Expectation and Bayesian Updating

14.1 Probability Spaces

Uncertainty enters an economic model as a set of states, a rule assigning likelihoods to statements about the state, and functions of the state that the agent cares about. The three objects are the sample space, the probability measure and the random variables, and the first difficulty is that the second cannot in general be defined on all statements.

Definition 14.1.1 (σ-algebra)**.** Let Ω be a set. A collection $\mathcal{F}$ of subsets of Ω is a *σ-algebra* on Ω if

(i) $\Omega \in \mathcal{F}$;

(ii) $E \in \mathcal{F}$ implies $E^c \in \mathcal{F}$;

(iii) $E_n \in \mathcal{F}$ for every $n \in \mathbb{N}$ implies $\bigcap_{n\in\mathbb{N}} E_n \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is then a *measurable space* and the members of $\mathcal{F}$ are *events*.

Proposition 14.1.1 (Closure properties)**.** Let $\mathcal{F}$ be a σ-algebra on Ω. Then $\emptyset \in \mathcal{F}$ and $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{F}$ whenever every $E_n \in \mathcal{F}$; consequently $\mathcal{F}$ is closed under finite unions, finite intersections and set differences.

Proof. $\emptyset = \Omega^c \in \mathcal{F}$ by (i) and (ii). For unions, De Morgan's law (Proposition 1.2.1) gives $\bigcup_n E_n = \left(\bigcap_n E_n^c\right)^c$, and each $E_n^c \in \mathcal{F}$ by (ii), so the intersection lies in $\mathcal{F}$ by (iii) and its complement by (ii). Finite operations follow by padding the sequence with Ω or $\emptyset$, and $E \setminus F = E \cap F^c$. ∎

An event is a statement about the experiment admitting a yes-or-no answer, and the three axioms are the closure of such statements under negation and countable conjunction. The restriction to *countable* conjunction is what makes the definition non-trivial: closure under arbitrary unions would force $\mathcal{F}$ to contain every subset of Ω, since every set is the union of its singletons.

That the whole power set is inadmissible is a statement about a *particular* measure, not about probability measures in general. The Dirac measure $\delta_{\omega_0}(E) = \mathbb{1}\{\omega_0 \in E\}$ is a countably additive probability measure on $\mathcal{P}(()\Omega)$ for any Ω whatsoever. What fails is the following: there is no countably additive probability measure on $\mathcal{P}()([0,1])$ that is invariant under translation modulo 1 and assigns to each interval its length. The standard witness is a *Vitali set* — a set of representatives of the cosets of $\mathbb{Q}$ in $\mathbb{R}$, whose existence uses the axiom of choice — for which countable additivity and translation invariance force the measure of $[0,1]$ to be either 0 or ∞. Lebesgue measure therefore has to be restricted to the Borel or Lebesgue σ-algebra, and every uniform-distribution argument in economics inherits that restriction. See Le Gall (2022, Ch. 1). Chapter 4 recorded the formal parallel with Definition 4.2.2: a topology is closed under arbitrary unions and finite intersections, a σ-algebra under countable operations of both kinds and under complementation, and neither contains the other.

Definition 14.1.2 (Probability measure and probability space)**.** Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mathbb{P}\colon \mathcal{F} \to [0,1]$ is a *probability measure* if

(i) $\mathbb{P}(\Omega) = 1$;

(ii) (σ-additivity) $\mathbb{P}\big(\bigcup_{n\in\mathbb{N}} E_n\big) = \sum_{n=1}^{\infty} \mathbb{P}(E_n)$ whenever $(E_n)_{n\in\mathbb{N}}$ is pairwise disjoint.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Proposition 14.1.2 (Elementary properties)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $E, F \in \mathcal{F}$.

(i) $\mathbb{P}(\emptyset) = 0$.

(ii) (Monotonicity) $E \subseteq F$ implies $\mathbb{P}(E) \le \mathbb{P}(F)$.

(iii) (Finite additivity) $E \cap F = \emptyset$ implies $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$; in particular $\mathbb{P}(E) + \mathbb{P}(E^c) = 1$.

(iv) (Inclusion–exclusion) $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$, hence $\mathbb{P}(E \cup F) \le \mathbb{P}(E) + \mathbb{P}(F)$.

(v) (Continuity from below) If $E_1 \subseteq E_2 \subseteq \cdots$ and $E = \bigcup_n E_n$, then $\mathbb{P}(E_n) \to \mathbb{P}(E)$.

(vi) (Continuity from above) If $F_1 \supseteq F_2 \supseteq \cdots$ and $F = \bigcap_n F_n$, then $\mathbb{P}(F_n) \to \mathbb{P}(F)$.

Proof. (i) Suppose $\mathbb{P}(\emptyset) = a > 0$ and take $E_n = \emptyset$ for every n. These are pairwise disjoint, so σ-additivity gives $a = \mathbb{P}(\emptyset) = \sum_{n=1}^{\infty} a = \infty$, contradicting $\mathbb{P} \le 1$.

(ii) Put $E_1 = E$, $E_2 = F \setminus E$ and $E_n = \emptyset$ for $n \ge 3$; these are disjoint with union F, so $\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F \setminus E) \ge \mathbb{P}(E)$ by (i) and non-negativity.

(iii) Put $E_1 = E$, $E_2 = F$, $E_n = \emptyset$ for $n \ge 3$. Taking $F = E^c$ gives $\mathbb{P}(E) + \mathbb{P}(E^c) = \mathbb{P}(\Omega) = 1$.

(iv) Decompose $E \cup F$ into the three disjoint events $E \setminus F$, $E \cap F$, $F \setminus E$ and apply (iii) three times: $\mathbb{P}(E \cup F) = \mathbb{P}(E \setminus F) + \mathbb{P}(E \cap F) + \mathbb{P}(F \setminus E)$, while $\mathbb{P}(E) = \mathbb{P}(E \setminus F) + \mathbb{P}(E \cap F)$ and similarly for F; adding the last two and subtracting $\mathbb{P}(E \cap F)$ gives the identity. The inequality follows since $\mathbb{P}(E \cap F) \ge 0$.

(v) Put $A_1 := E_1$ and $A_n := E_n \setminus E_{n-1}$ for $n \ge 2$. The A_n are pairwise disjoint, $\bigcup_{k\le n} A_k = E_n$ and $\bigcup_k A_k = E$, so σ-additivity gives $\mathbb{P}(E) = \sum_{k\ge 1} \mathbb{P}(A_k) = \lim_n \sum_{k\le n} \mathbb{P}(A_k) = \lim_n \mathbb{P}(E_n)$.

(vi) Apply (v) to $E_n := F_n^{\mathsf{c}}$, which increase to F^{c}, and use (iii). ■

Remark 14.1.1 (The countable case)**.** When Ω is at most countable one may take $\mathcal{F} = \mathcal{P}(()\Omega)$ and specify $\mathbb{P}$ by the numbers $\mathbb{P}(\{\omega\})$, $\omega \in \Omega$, which are non-negative and sum to 1; every event then has $\mathbb{P}(E) = \sum_{\omega\in E} \mathbb{P}(\{\omega\})$ and σ-additivity is automatic. The four properties listed on (Fukuda 2026f, slide 4) — $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, monotonicity and additivity on disjoint events — are exactly Definition 14.1.2 together with Proposition 14.1.2 specialised to this case. The material of Section 14.3 and Section 14.4 is developed in this countable setting, where every expectation is a sum and conditioning is carried out only on events of strictly positive probability. A countable space still admits null events — a point mass $\mathbb{P}(\{\omega\}) = 0$ is allowed by the definition, as is a countable union of such points — so the restriction is a choice made in those sections and not a property of countability. The passage to general probability spaces is not a matter of rewriting sums as integrals: conditioning on an event of probability zero has no meaning under Definition 14.3.1, and the object that replaces it is defined through a σ-algebra and is unique only up to a null set. subsection 14.4.1 carries this out, and it is what licenses the continuous-state calculations of Section 14.6 and of Chapter 15.

14.2 Random Variables and Expectation

Definition 14.2.1 (Random variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *random variable* is a function $X\colon \Omega \to \mathbb{R}$ that is *measurable*: $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$. When Ω is at most countable and $\mathcal{F} = \mathcal{P}(()\Omega)$ the measurability requirement is vacuous. Writing $X(\Omega) = \{x_i\}_i$

for the set of realisations, the *probability mass function* of X is

$$f(x) := \mathbb{P}\big(\{\omega \in \Omega : X(\omega) = x\}\big) = \mathbb{P}\big(X^{-1}(\{x\})\big), \qquad x \in X(\Omega), \tag{14.1}$$

and one writes $p_i := f(x_i)$ and $[X = x] := X^{-1}(\{x\})$.

Measurability is the requirement that every statement about the value of X be an event, so that its probability is defined. It is the probabilistic counterpart of continuity: Theorem 5.2.1 characterised continuity by f^{-1}(open) open, and Remark 5.2.1 recorded that measurability replaces "open" by "Borel" and "open" by "in $\mathcal{F}$". On a countable sample space the condition is empty, which is why the slides define a random variable simply as a function (Fukuda 2026f, slide 15).

Definition 14.2.2 (Expectation)**.** For a random variable X on an at most countable probability space with $X(\Omega) = \{x_i\}_{i=1}^n$, the *expectation* is

$$\mathbb{E}[X] := \sum_{i=1}^{n} p_i x_i = \sum_{\omega\in\Omega} \mathbb{P}(\{\omega\})X(\omega), \tag{14.2}$$

the two expressions being equal because the events $[X = x_i]$ partition Ω.

Proof of the second equality in (14.2). Group the sum over Ω by the value of X:

$$\sum_{\omega\in\Omega} \mathbb{P}(\{\omega\})X(\omega) = \sum_{i=1}^{n} \sum_{\omega\,:\,X(\omega)=x_i} \mathbb{P}(\{\omega\})X(\omega) = \sum_{i=1}^{n} x_i \sum_{\omega\,:\,X(\omega)=x_i} \mathbb{P}(\{\omega\}) = \sum_{i=1}^{n} x_i p_i,$$

using (14.1) in the last step. Rearrangement is legitimate because the family $\big([X = x_i]\big)_i$ partitions Ω and, for a countable Ω, absolute convergence of $\sum_\omega \mathbb{P}(\{\omega\})\,|X(\omega)|$ is exactly the integrability of X. ■

Proposition 14.2.1 (Properties of expectation)**.** Let X, Y be integrable random variables and $a, b \in \mathbb{R}$.

(i) (Linearity) $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

(ii) (Monotonicity) $X \le Y$ pointwise implies $\mathbb{E}[X] \le \mathbb{E}[Y]$; and $X \ge 0$ with $\mathbb{E}[X] = 0$ implies $\mathbb{P}(X = 0) = 1$.

(iii) (Change of variables) $\mathbb{E}[g(X)] = \sum_{x\in X(\Omega)} g(x)f(x)$ for any $g\colon \mathbb{R} \to \mathbb{R}$ for which the sum converges absolutely.

(iv) (Independence) If X and Y are independent, meaning $\mathbb{P}([X \in A]\cap[Y \in B]) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all Borel A, B, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. (i)–(iii) are rearrangements of absolutely convergent sums, using (14.2) in its second form for (i) and its first form for (iii). For the second half of (ii), if $\mathbb{P}(X > 0) > 0$ then $\mathbb{P}(X > 1/k) > 0$ for some k by countable additivity applied to $[X > 0] = \bigcup_k [X > 1/k]$, and then $\mathbb{E}[X] \ge (1/k)\mathbb{P}(X > 1/k) > 0$. For (iv), independence makes the joint mass function factor, $f_{X,Y}(x, y) = f_X(x)f_Y(y)$, and $\mathbb{E}[XY] = \sum_{x,y} xyf_X(x)f_Y(y) = \big(\sum_x xf_X(x)\big)\big(\sum_y yf_Y(y)\big)$. ■

Definition 14.2.3 (Variance and covariance)**.** For square-integrable X, Y,

$$\operatorname{Var}(X) := \mathbb{E}\big[(X - \mathbb{E}[X])^2\big], \qquad \operatorname{Cov}(X, Y) := \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big],$$

$$\operatorname{Corr}(X, Y) := \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}, \qquad \text{defined when } \operatorname{Var}(X)\operatorname{Var}(Y) > 0.$$

Proposition 14.2.2 (The variance identity). For square-integrable X and Y,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2, \qquad \mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{14.3}$$

More generally, for any $c \in \mathbb{R}$, $\mathbb{E}\big[(X-c)^2\big] = \mathrm{Var}(X) + \big(\mathbb{E}[X]-c\big)^2$, so $c \mapsto \mathbb{E}[(X-c)^2]$ is minimised uniquely at $c = \mathbb{E}[X]$.

Proof. Write $\mu := \mathbb{E}[X]$ and expand, using Proposition 14.2.1(i) and the fact that μ is a constant:

$$\mathbb{E}\big[(X-c)^2\big] = \mathbb{E}\big[(X-\mu)^2\big] + 2(\mu-c)\underbrace{\mathbb{E}[X-\mu]}_{=0} + (\mu-c)^2 = \mathrm{Var}(X) + (\mu-c)^2.$$

Taking $c = 0$ gives the first identity in (14.3); the covariance identity is the same expansion applied to the product. The minimisation claim is immediate since $(\mu-c)^2 \geq 0$ with equality only at $c = \mu$. ■

Proposition 14.2.3 (Variance of a sum). Let $X_1, \dots, X_n$ be square-integrable and pairwise independent. Then

$$\mathrm{Var}\Big(\sum_{i=1}^n X_i\Big) = \sum_{i=1}^n \mathrm{Var}(X_i), \qquad \text{and} \qquad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X) \quad \text{for } a,b \in \mathbb{R}.$$

In particular, if the X_i are identically distributed with variance σ^2, the sample mean $\bar{X}_n$ has $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$.

Proof. Expanding the square and using linearity, $\mathrm{Var}\big(\sum_i X_i\big) = \sum_i \mathrm{Var}(X_i) + \sum_{i\neq j}\mathrm{Cov}(X_i, X_j)$. Pairwise independence gives $\mathbb{E}[X_iX_j] = \mathbb{E}[X_i]\mathbb{E}[X_j]$ by Proposition 14.2.1(iv), so $\mathrm{Cov}(X_i, X_j) = 0$ for $i \neq j$ by (14.3). The scaling identity is $\mathbb{E}[(aX+b-a\mathbb{E}[X]-b)^2] = a^2\mathbb{E}[(X-\mathbb{E}[X])^2]$, and the last claim combines the two with $\bar{X}_n = n^{-1}\sum_i X_i$. ■

The last sentence is the reason the mean, and not some other centre, is the object that appears throughout asset pricing and econometrics: it is the constant that best predicts X in mean square. Theorem 14.5.1 extends the statement from constants to functions of a conditioning variable, and Proposition 11.5.2 to affine functions.

Theorem 14.2.1 (Jensen's inequality). Let $I \subseteq \mathbb{R}$ be an interval, $\varphi\colon I \to \mathbb{R}$ convex, and X an integrable random variable with $\mathbb{P}(X \in I) = 1$ and $\mathbb{E}[X]$ in the interior of I. Then $\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X])$, with equality if and only if φ is affine on the smallest interval carrying all the mass of X. The inequality reverses for concave φ.

Proof. Put $\mu := \mathbb{E}[X]$. By Proposition 9.3.1 the set $\operatorname{epi}\varphi$ is convex, and $(\mu, \varphi(\mu))$ lies on its boundary, so Theorem 9.4.2 supplies a supporting hyperplane: there is $(a,b) \neq 0$ with $ax + bt \geq a\mu + b\varphi(\mu)$ for every $(x,t) \in \operatorname{epi}\varphi$. Since t may be taken arbitrarily large, $b \geq 0$; and $b = 0$ would force $ax \geq a\mu$ for all $x \in I$ with $a \neq 0$, impossible because μ is interior to I. Dividing by $b > 0$ and setting $s := -a/b$ gives the supporting line

$$\varphi(x) \geq \varphi(\mu) + s(x-\mu) \qquad \text{for all } x \in I. \tag{14.4}$$

Substituting X and taking expectations, monotonicity and linearity (Proposition 14.2.1) give $\mathbb{E}[\varphi(X)] \geq \varphi(\mu) + s\mathbb{E}[X-\mu] = \varphi(\mu)$. Equality forces $\mathbb{E}\big[\varphi(X) - \varphi(\mu) - s(X-\mu)\big] = 0$ with a non-negative integrand, hence by Proposition 14.2.1(ii) the integrand vanishes almost surely: φ agrees with the affine function (14.4) on the support of X. ■

Example 14.2.1 (Risk aversion is Jensen). If u is concave, Theorem 14.2.1 gives $\mathbb{E}[u(W)] \leq u(\mathbb{E}[W])$: the agent prefers the expected value of the gamble to the gamble. The certainty equivalent c defined by $u(c) = \mathbb{E}[u(W)]$ therefore satisfies $c \leq \mathbb{E}[W]$, and the risk premium $\mathbb{E}[W] - c$ is non-negative. The second-order expansion of Theorem 7.6.1 quantified this premium as $\frac{1}{2}A(\mathbb{E}[W])\,\mathrm{Var}(W)$ for small risks; Jensen gives the sign for risks of any size, with no differentiability and no smallness. Equality holds only if u is affine on the support of W, which is the definition of risk neutrality there.

14.3 Conditional Probability and Bayes' Rule

Definition 14.3.1 (Conditional probability)**.** For events D, H with $\mathbb{P}(H) > 0$,

$$\mathbb{P}(D \mid H) := \frac{\mathbb{P}(D \cap H)}{\mathbb{P}(H)}. \tag{14.5}$$

Events D and H are *independent* if $\mathbb{P}(D \cap H) = \mathbb{P}(D)\mathbb{P}(H)$, equivalently $\mathbb{P}(D \mid H) = \mathbb{P}(D)$ when $\mathbb{P}(H) > 0$.

For fixed H with $\mathbb{P}(H) > 0$ the map $E \mapsto \mathbb{P}(E \mid H)$ is itself a probability measure on $(\Omega, \mathcal{F})$: it assigns 1 to Ω and is σ-additive, both immediately from (14.5) and σ-additivity of $\mathbb{P}$. Everything proved for probability measures therefore applies to conditional probabilities without further argument.

Theorem 14.3.1 (Bayes' rule)**.** Let D, H be events with $\mathbb{P}(D) > 0$ and $\mathbb{P}(H) > 0$. Then

$$\mathbb{P}(H \mid D) = \frac{\mathbb{P}(H)\,\mathbb{P}(D \mid H)}{\mathbb{P}(D)}. \tag{14.6}$$

If moreover $0 < \mathbb{P}(H) < 1$,

$$\mathbb{P}(H \mid D) = \frac{\mathbb{P}(H)\,\mathbb{P}(D \mid H)}{\mathbb{P}(H)\,\mathbb{P}(D \mid H) + \mathbb{P}(H^c)\,\mathbb{P}(D \mid H^c)}. \tag{14.7}$$

More generally, if $(H_j)_{j=1}^n$ partitions Ω with $\mathbb{P}(H_j) > 0$ for every j, then for each i

$$\mathbb{P}(H_i \mid D) = \frac{\mathbb{P}(D \mid H_i)\mathbb{P}(H_i)}{\sum_{j=1}^n \mathbb{P}(D \mid H_j)\mathbb{P}(H_j)}. \tag{14.8}$$

Proof. By (14.5) applied twice, $\mathbb{P}(H \mid D)\mathbb{P}(D) = \mathbb{P}(D \cap H) = \mathbb{P}(D \mid H)\mathbb{P}(H)$; dividing by $\mathbb{P}(D) > 0$ gives (14.6). For the denominators, the events $D \cap H_j$ are pairwise disjoint with union D because (H_j) partitions Ω, so finite additivity (Proposition 14.1.2(iii)) gives the *law of total probability*

$$\mathbb{P}(D) = \sum_{j=1}^n \mathbb{P}(D \cap H_j) = \sum_{j=1}^n \mathbb{P}(D \mid H_j)\mathbb{P}(H_j). \tag{14.9}$$

Substituting (14.9) into (14.6) gives (14.8), and (14.7) is the case $n = 2$ with $H_1 = H$, $H_2 = H^c$. ■

The content of (14.6) is a reversal of the direction of conditioning. The likelihood $\mathbb{P}(D \mid H)$ runs from hypothesis to data, from cause to effect; the posterior $\mathbb{P}(H \mid D)$ runs from data back to hypothesis, and the theorem says the two are related by the ratio of the prior to the marginal likelihood of the data (Fukuda 2026f, slide 6).

Example 14.3.1 (Base rates dominate)**.** A disease affects 1% of the population. The test is positive with probability 0.95 on the diseased and negative with probability 0.9 on the healthy. Let H be "diseased" and D "tests positive". Then

$$\mathbb{P}(H \mid D) = \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.10} = \frac{0.0095}{0.1085} = \frac{19}{217} \approx 0.0876.$$

A test with 95% sensitivity and 90% specificity leaves the posterior probability of disease below 9%. The reason is arithmetic rather than statistical: the 99% healthy generate 9.9 false positives per hundred while the 1% diseased generate 0.95 true positives, and no reading of the likelihoods alone can reveal this. Neglect of the denominator in (14.7) is the base-rate fallacy, and it is the mechanism behind overconfidence in models of belief updating.

Example 14.3.2 (The airport carousel)**.** Bags are correctly loaded with probability 0.9. Conditional on the bag having been loaded, it fails to appear after nine minutes with probability 0.1; if it was not loaded it certainly does not appear. With H "not loaded" and D "bag missing",

$$\mathbb{P}(H \mid D) = \frac{0.1 \times 1}{0.1 \times 1 + 0.9 \times 0.1} = \frac{0.1}{0.19} = \frac{10}{19} \approx 0.5263.$$

Waiting turns a prior of 0.1 into a posterior above one half. The magnitude of the revision is governed by the likelihood ratio $\mathbb{P}(D \mid H)/\mathbb{P}(D \mid H^c) = 10$, which is the sufficient statistic for the evidence: (14.7) in odds form reads

$$\frac{\mathbb{P}(H \mid D)}{\mathbb{P}(H^c \mid D)} = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)} \cdot \frac{\mathbb{P}(D \mid H)}{\mathbb{P}(D \mid H^c)}, \tag{14.10}$$

posterior odds equal prior odds times the likelihood ratio.

Example 14.3.3 (A poll). Party A has 30% support, party B has 40%, and 30% support neither. The cabinet is supported by 80% of A's supporters, 20% of B's and 50% of the rest. By (14.8) with the three-element partition, a randomly chosen cabinet supporter belongs to party A with probability

$$\frac{0.3 \times 0.8}{0.3 \times 0.8 + 0.4 \times 0.2 + 0.3 \times 0.5} = \frac{0.24}{0.47} = \frac{24}{47} \approx 0.5106.$$

14.3.1 Economic application: order flow and the market maker's beliefs

Environment. An asset has value $V \in \{H, L\}$ with $H > L$. A market maker holds the prior $p := \mathbb{P}(V = H) \in (0,1)$ and quotes prices at which they will trade one unit. A trader arrives; with probability $q \in (0,1)$ the trader is *informed* and knows V, and with probability $1 - q$ the trader is *uninformed*. The informed trader buys when $V = H$ and sells when $V = L$; the uninformed trader buys or sells with probability $\frac{1}{2}$ each, independently of V.

Proposition 14.3.1 (Posterior after a sale). Let S denote the event that the first trade is a sale. Then

$$\mathbb{P}(V = H \mid S) = \frac{p(1-q)}{p(1-q) + (1-p)(1+q)}, \tag{14.11}$$

which is strictly less than p for every $q \in (0,1)$, is strictly decreasing in q, equals p at $q = 0$ and tends to 0 as $q \to 1$.

Proof. Conditioning on the trader's type and using (14.9) within each state,

$$\mathbb{P}(S \mid V = H) = q \cdot 0 + (1-q) \cdot \tfrac{1}{2} = \tfrac{1-q}{2}, \qquad \mathbb{P}(S \mid V = L) = q \cdot 1 + (1-q) \cdot \tfrac{1}{2} = \tfrac{1+q}{2},$$

because an informed trader never sells when the value is high and always sells when it is low. Substituting into (14.7) and cancelling the factor $\frac{1}{2}$ gives (14.11). Writing the posterior as $\left(1 + \frac{1-p}{p} \cdot \frac{1+q}{1-q}\right)^{-1}$ exhibits it as strictly decreasing in q, since $q \mapsto (1+q)/(1-q)$ is strictly increasing on $(0,1)$; the boundary values follow by inspection. ■

The comparative static in q is the informational content of order flow. When $q = 0$ every trade is noise and prices do not move; as $q \to 1$ a single sale reveals the state. The market maker who sets the bid at $\mathbb{E}[V \mid S] = L + (H - L)\mathbb{P}(V = H \mid S)$ and the ask at $\mathbb{E}[V \mid B]$ makes zero expected profit on each trade and covers the losses to informed traders with the gains from uninformed ones; the difference between the two quotes is the bid–ask spread, and (14.11) shows it widens in q. This is the adverse-selection account of the spread, and the calculation above is its entire mechanism (Fukuda 2026f, slide 12); see O'Hara (1998, Sec. 3.A.1) for the sequential-trade model built on it and Kyle (1985) for the continuous version in which the informed trader chooses order size strategically.

14.4 Conditional Expectation

Definition 14.4.1 (Conditional expectation given an event). For an event H with $\mathbb{P}(H) > 0$ and a random variable X,

$$\mathbb{E}[X \mid H] := \sum_{x \in X(H)} x\,\mathbb{P}\big([X = x] \mid H\big) = \sum_{\omega \in H} X(\omega)\,\mathbb{P}\big(\{\omega\} \mid H\big) = \frac{\mathbb{E}[X \mathbb{1}_H]}{\mathbb{P}(H)}, \tag{14.12}$$

where $X(H) := \{X(\omega) : \omega \in H\}$ and $\mathbb{1}_H$ is the indicator of H.

Proof of the equalities in (14.12). The first equals the second by grouping $\omega \in H$ according to the value of $X(\omega)$, exactly as in (14.2). For the third, expand $\mathbb{P}(\{\omega\} \mid H) = \mathbb{P}(\{\omega\})/\mathbb{P}(H)$ for $\omega \in H$ and note that $X(\omega)\mathbb{1}_H(\omega) = 0$ for $\omega \notin H$, so $\sum_{\omega \in H} X(\omega)\mathbb{P}(\{\omega\}) = \sum_{\omega \in \Omega} X(\omega)\mathbb{1}_H(\omega)\mathbb{P}(\{\omega\}) = \mathbb{E}[X\mathbb{1}_H]$. ■

Definition 14.4.2 (Conditional expectation given a random variable). Let X, Y be random variables. The *conditional expectation* $\mathbb{E}[Y \mid X]$ is the random variable $\omega \mapsto \mathbb{E}\big[Y \mid [X = X(\omega)]\big]$, that is, the function of X whose value at $x \in X(\Omega)$ is

$$\mathbb{E}[Y \mid X = x] := \sum_{y \in Y(\Omega)} y\, f_{Y|X}(y \mid x), \qquad f_{Y|X}(y \mid x) := \mathbb{P}\big([Y = y] \mid [X = x]\big). \tag{14.13}$$

The distinction between Definition 14.4.1 and Definition 14.4.2 is that the first is a number and the second a random variable — a function of X, hence itself measurable with respect to the information generated by X. Everything that follows turns on this: $\mathbb{E}[Y \mid X]$ is the best summary of Y available to someone who observes X.

Proposition 14.4.1 (Taking out what is known). Let X, Y be random variables with Y integrable, and let $g\colon \mathbb{R} \to \mathbb{R}$ be such that $g(X)Y$ is integrable. Then

$$\mathbb{E}\big[g(X)Y \mid X\big] = g(X)\,\mathbb{E}[Y \mid X]. \tag{14.14}$$

In particular $\mathbb{E}[g(X) \mid X] = g(X)$ and $\mathbb{E}[a \mid X] = a$ for constant a.

Proof. Fix $x_0 \in X(\Omega)$ and put $W := g(X)Y$. If $g(x_0) = 0$ then W is identically zero on $[X = x_0]$ and both sides of (14.14) vanish there. If $g(x_0) \neq 0$ then on the event $[X = x_0]$ we have $W = w$ if and only if $Y = w/g(x_0)$, so

$$\mathbb{P}\big([W = w] \mid [X = x_0]\big) = \mathbb{P}\Big(\big[Y = \tfrac{w}{g(x_0)}\big] \mid [X = x_0]\Big),$$

and substituting $y = w/g(x_0)$ in (14.13),

$$\mathbb{E}[W \mid X = x_0] = \sum_w w\,\mathbb{P}\big([W = w] \mid [X = x_0]\big) = g(x_0)\sum_y y\,\mathbb{P}\big([Y = y] \mid [X = x_0]\big) = g(x_0)\mathbb{E}[Y \mid X = x_0].$$

Since x_0 was arbitrary, (14.14) holds as an identity between random variables. Taking $Y \equiv 1$ gives $\mathbb{E}[g(X) \mid X] = g(X)$, and g constant gives the last claim. ■

Both integrability hypotheses are needed, and neither implies the other. With $g \equiv 0$ the product $g(X)Y$ is integrable whatever Y is, so the left side of (14.14) is 0, while the right side contains the factor $\mathbb{E}[Y \mid X]$, which is undefined for non-integrable Y. Conversely, an integrable Y and an unbounded g can make $g(X)Y$ non-integrable.

Theorem 14.4.1 (Law of iterated expectations). (i) Let $(H_j)_{j=1}^m$ partition Ω with $\mathbb{P}(H_j) > 0$ for every j, and let X be integrable. Then

$$\mathbb{E}[X] = \sum_{j=1}^m \mathbb{E}[X \mid H_j]\,\mathbb{P}(H_j). \tag{14.15}$$

(ii) For integrable X and any random variable Y,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big]. \tag{14.16}$$

Proof. (i) By the third form of (14.12), $\mathbb{E}[X \mid H_j]\mathbb{P}(H_j) = \mathbb{E}[X\mathbb{1}_{H_j}]$. Since (H_j) partitions Ω, $\sum_j \mathbb{1}_{H_j} = 1$ pointwise, so linearity (Proposition 14.2.1(i)) gives $\sum_j \mathbb{E}[X\mathbb{1}_{H_j}] = \mathbb{E}\big[X \sum_j \mathbb{1}_{H_j}\big] = \mathbb{E}[X]$.

(ii) Apply (i) to the partition $H_j := [Y = y_j]$, $y_j \in Y(\Omega)$, and observe that $\sum_j \mathbb{E}[X \mid Y = y_j]\mathbb{P}(Y = y_j)$ is by (14.2) and (14.13) precisely the expectation of the random variable $\mathbb{E}[X \mid Y]$, whose value on $[Y = y_j]$ is $\mathbb{E}[X \mid Y = y_j]$ and which takes that value with probability $\mathbb{P}(Y = y_j)$. ■

These are Theorems 2.4 and 2.5 of Fukuda (2026d, pp. 74–75) (Fukuda 2026f, slides 18–19). The statement to hold on to is that conditioning on less information cannot change the average: an expectation computed in two stages, first within each cell of a partition and then across cells, equals the expectation computed at once. Almost every argument in dynamic economics that replaces a many-period expectation by a one-period expectation of a continuation value is an application of (14.16).

Proposition 14.4.2 (Conditional Jensen)**.** Let $\varphi\colon I \to \mathbb{R}$ be convex on an interval I, and let Y be integrable with $\mathbb{P}(Y \in I) = 1$ and $\mathbb{E}[Y \mid X]$ almost surely interior to I. Then

$$\mathbb{E}\big[\varphi(Y) \mid X\big] \geq \varphi\big(\mathbb{E}[Y \mid X]\big) \qquad \text{almost surely,} \tag{14.17}$$

and the inequality reverses for concave φ.

Proof. Fix $x \in X(\Omega)$. Conditional on $[X = x]$ the map $E \mapsto \mathbb{P}(E \mid [X = x])$ is a probability measure, as noted after Definition 14.3.1, and $\mathbb{E}[\cdot \mid X = x]$ is the expectation under it. Theorem 14.2.1 applied under that measure, with $\mu := \mathbb{E}[Y \mid X = x]$ interior to I, gives $\mathbb{E}[\varphi(Y) \mid X = x] \geq \varphi(\mu)$. As x was arbitrary, the inequality holds pointwise as an inequality between random variables. ∎

Taking expectations in (14.17) and applying (14.16) recovers Theorem 14.2.1, so the conditional version is strictly stronger. It is what makes the value function of a dynamic programme concave in the state when the return function is concave, and it is what forces the martingale $\big(\mathbb{E}[Y \mid X_t]\big)_t$ to have a convex-increasing transform.

14.4.1 Conditioning on a σ-algebra

Everything above conditions on an event of positive probability. In the models of Section 14.6 and Chapter 15 the conditioning variable is continuously distributed, so $\mathbb{P}([X = x]) = 0$ for every x and Definition 14.3.1 does not apply. The construction that replaces it takes the conditioning *information* rather than a conditioning event as primitive.

Definition 14.4.3 (Conditional expectation given a σ-algebra)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a σ-algebra, and Y an integrable random variable. A random variable Z is a *conditional expectation* of Y given $\mathcal{G}$, written $Z = \mathbb{E}[Y \mid \mathcal{G}]$, if

(i) Z is $\mathcal{G}$-measurable and integrable; and

(ii) $\mathbb{E}[Z\mathbb{1}_G] = \mathbb{E}[Y\mathbb{1}_G]$ for every $G \in \mathcal{G}$.

Condition (i) is the requirement that the forecast use only the information in $\mathcal{G}$; condition (ii) is the requirement that it be unbiased on every event that information can distinguish. Nothing here refers to the probability of a single value of a conditioning variable.

Theorem 14.4.2 (Existence and almost sure uniqueness)**.** Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra and let Y be integrable. Then $\mathbb{E}[Y \mid \mathcal{G}]$ exists, and any two versions agree almost surely. If in addition $Y \in L^2$, then $\mathbb{E}[Y \mid \mathcal{G}]$ is the orthogonal projection of Y onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P}) \subseteq L^2(\Omega, \mathcal{F}, \mathbb{P})$.

Proof. Uniqueness. Let Z, Z' both satisfy (i) and (ii). The event $G := [Z > Z']$ lies in $\mathcal{G}$ because both variables are $\mathcal{G}$-measurable, and (ii) applied to each gives $\mathbb{E}[(Z - Z')\mathbb{1}_G] = \mathbb{E}[Y\mathbb{1}_G] - \mathbb{E}[Y\mathbb{1}_G] = 0$. The integrand is strictly positive on G and zero off it, so Proposition 14.2.1(ii) gives $\mathbb{P}(G) = 0$. Exchanging the roles of Z and Z' gives $\mathbb{P}([Z' > Z]) = 0$, hence $Z = Z'$ almost surely.

Existence for $Y \in L^2$. The set $L^2(\Omega, \mathcal{G}, \mathbb{P})$ of square-integrable $\mathcal{G}$-measurable random variables is a subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$, and it is closed: an L^2-convergent sequence has an almost surely convergent subsequence, whose limit is again $\mathcal{G}$-measurable as a pointwise limit of $\mathcal{G}$-measurable functions. Let $Z := P_{L^2(\mathcal{G})}Y$, which exists and is unique by Theorem 11.3.2. For $G \in \mathcal{G}$ the variable $\mathbb{1}_G$ lies in $L^2(\Omega, \mathcal{G}, \mathbb{P})$, so Proposition 11.3.1 gives $\mathbb{E}[(Y - Z)\mathbb{1}_G] = \langle Y - Z, \mathbb{1}_G \rangle = 0$, which is (ii); and Z is $\mathcal{G}$-measurable by construction, which is (i).

Existence for integrable Y. The extension from L^2 to L^1 is a monotone-convergence argument that this volume does not develop; the direct construction applies the Radon–Nikodym theorem to the finite measure $G \mapsto \mathbb{E}[Y^+\mathbb{1}_G]$ on $(\Omega, \mathcal{G})$, which is absolutely continuous with respect to the restriction of $\mathbb{P}$, and subtracts the corresponding object for Y^-. The statement and this proof are Theorem 11.3 of Le Gall (2022, p. 230), resting on the Radon–Nikodym theorem (Le Gall 2022, Thm. 4.11, p. 75). ∎

Two further facts complete the framework, and both are used without comment in applied work.

Theorem 14.4.3 (Doob–Dynkin). Let X be a random variable and $\sigma(X) := \{X^{-1}(B) : B \text{ Borel}\}$ the σ-algebra it generates. A random variable Z is $\sigma(X)$-measurable if and only if $Z = \varphi(X)$ for some Borel $\varphi\colon \mathbb{R} \to \mathbb{R}$.

Consequently $\mathbb{E}[Y \mid X] := \mathbb{E}[Y \mid \sigma(X)]$ can be written as $\varphi(X)$, and the symbol $\mathbb{E}[Y \mid X = x]$ is *defined* to mean $\varphi(x)$. The function φ is determined only up to sets of $\mathbb{P}_X$-measure zero, so $\mathbb{E}[Y \mid X = x]$ has no meaning at an individual x in isolation; statements about it are statements holding for $\mathbb{P}_X$-almost every x. This is the precise sense in which conditioning on a null event is legitimate.

Proposition 14.4.3 (Evaluation with densities). Let (X, Y) have joint density p on $\mathbb{R}^2$ and let $q(x) = \int_{\mathbb{R}} p(x, y)\mathrm{d}y$ be the marginal density of X. If Y is integrable, then a version of $\mathbb{E}[Y \mid X]$ is $\varphi(X)$ with

$$\varphi(x) = \frac{1}{q(x)} \int_{\mathbb{R}} y\, p(x, y)\mathrm{d}y \qquad \text{for every } x \text{ with } q(x) > 0, \tag{14.18}$$

and $\varphi(x)$ arbitrary where $q(x) = 0$; the exceptional set has $\mathbb{P}_X$-measure zero.

This is Proposition 11.11 of Le Gall (2022, p. 244), stated there for non-negative $h(X)$ and general dimensions. Formula (14.18) is the one used in Section 14.6, and it is the exact analogue of (14.13) with the conditional mass function $f_{Y|X}$ replaced by the conditional density $p(x, \cdot)/q(x)$.

Proposition 14.4.4 (The discrete case is a special case). Let X take countably many values and let Y be integrable. Then $\mathbb{E}[Y \mid \sigma(X)] = \varphi(X)$ almost surely, where $\varphi(x) = \mathbb{E}[Y \mid [X = x]]$ in the sense of Definition 14.4.1 for every x with $\mathbb{P}([X = x]) > 0$.

Proof. The atoms of $\sigma(X)$ are the events $[X = x]$ with $\mathbb{P}([X = x]) > 0$, together with a null set, so a $\sigma(X)$-measurable variable is constant on each atom and $Z := \varphi(X)$ satisfies (i). For (ii) it suffices, by countable additivity and σ-additivity of the expectation, to check $\mathbb{E}[Z\mathbb{1}_G] = \mathbb{E}[Y\mathbb{1}_G]$ on the atoms $G = [X = x]$, where both sides equal $\mathbb{E}[Y\mathbb{1}_{[X=x]}]$ by the third form of (14.12). ■

Every property established above in the countable setting — Proposition 14.4.1, Theorem 14.4.1 and Proposition 14.4.2 — holds for $\mathbb{E}[\cdot \mid \mathcal{G}]$ with "almost surely" appended, and the tower property takes the sharper form $\mathbb{E}\big[\mathbb{E}[Y \mid \mathcal{F}_2] \mid \mathcal{F}_1\big] = \mathbb{E}[Y \mid \mathcal{F}_1]$ for $\mathcal{F}_1 \subseteq \mathcal{F}_2$: coarser information wins. The proofs are those of Le Gall (2022, pp. 234–235, 238–241). In the L^2 case they are also immediate from Theorem 14.4.2, since composing orthogonal projections onto nested closed subspaces gives the projection onto the smaller one — which is the content of Exercise 11.6.1 and the reason Section 14.5 can treat conditioning as geometry.

14.5 Conditional Expectation as a Projection

Let $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ be the space of square-integrable random variables with the inner product $\langle U, V\rangle := \mathbb{E}[UV]$; it is a Hilbert space, and the geometry of Chapter 11 applies verbatim with $\|U\|^2 = \mathbb{E}[U^2]$.

Theorem 14.5.1 (Conditional expectation is the best predictor). Let $X, Y \in L^2$ and let $M(X) := \{\phi(X) : \phi \text{ Borel},\ \phi(X) \in L^2\}$, a closed subspace of L^2. Then for every ϕ with $\phi(X) \in L^2$,

$$\mathbb{E}\big[(Y - \phi(X))^2\big] = \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2\big] + \mathbb{E}\big[(\mathbb{E}[Y \mid X] - \phi(X))^2\big] \geq \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2\big], \tag{14.19}$$

with equality if and only if $\phi(X) = \mathbb{E}[Y \mid X]$ almost surely. Equivalently, $\mathbb{E}[Y \mid X]$ is the orthogonal projection of Y onto $M(X)$: $P_{M(X)}Y = \mathbb{E}[Y \mid X]$.

Proof. Write $\hat{Y} := \mathbb{E}[Y \mid X]$ and decompose $Y - \phi(X) = (Y - \hat{Y}) + (\hat{Y} - \phi(X))$. Expanding the square,

$$\mathbb{E}\big[(Y - \phi(X))^2\big] = \mathbb{E}\big[(Y - \hat{Y})^2\big] + 2\,\mathbb{E}\big[(Y - \hat{Y})(\hat{Y} - \phi(X))\big] + \mathbb{E}\big[(\hat{Y} - \phi(X))^2\big].$$

The cross term vanishes. Indeed $\hat{Y} - \phi(X)$ is a function of X, so Proposition 14.4.1 and then (14.16) give

$$\mathbb{E}\big[(Y - \hat{Y})(\hat{Y} - \phi(X))\big] = \mathbb{E}\Big[\mathbb{E}\big[(Y - \hat{Y})(\hat{Y} - \phi(X)) \mid X\big]\Big] = \mathbb{E}\Big[(\hat{Y} - \phi(X))\,\underbrace{(\mathbb{E}[Y \mid X] - \hat{Y})}_{=0}\Big] = 0,$$

using $\mathbb{E}[\hat{Y} \mid X] = \hat{Y}$ from Proposition 14.4.1. This proves the identity in (14.19); the inequality and its equality case follow because the second term is non-negative and vanishes only when $\hat{Y} = \phi(X)$ almost surely, by Proposition 14.2.1(ii).

The vanishing of the cross term for *every* ϕ says $Y - \hat{Y} \perp M(X)$, and $\hat{Y} \in M(X)$; by Proposition 11.3.1 this characterises $\hat{Y}$ as the orthogonal projection of Y onto $M(X)$, whose existence and uniqueness are Theorem 11.3.2. ■

Equation (14.19) carries three readings. As a minimisation, it says the conditional expectation is the mean-square-optimal forecast among all functions of the conditioning variable, extending the constant-forecast statement of Proposition 14.2.2. As a geometric statement, it identifies conditioning with orthogonal projection, so that all the structure of Chapter 11 — idempotence, the Pythagorean decomposition, contraction in norm — transfers to conditioning; the tower property (14.16) becomes the composition of projections onto nested subspaces. As a variance decomposition, taking $\phi \equiv \mathbb{E}[Y]$ in (14.19) and using (14.16) yields

$$\operatorname{Var}(Y) = \mathbb{E}\big[\operatorname{Var}(Y \mid X)\big] + \operatorname{Var}\big(\mathbb{E}[Y \mid X]\big), \tag{14.20}$$

the law of total variance: unexplained variation plus explained variation. In a regression context the second term over the first is the population R^2.

Remark 14.5.1 (Linear versus general prediction). Proposition 11.5.2 solved the same problem over the strictly smaller subspace $\operatorname{span}\{1, X\} \subseteq M(X)$ and produced

$$\mathbb{E}^*[Y \mid X] = \mathbb{E}[Y] + \frac{\operatorname{Cov}(X,Y)}{\operatorname{Var}(X)}\big(X - \mathbb{E}[X]\big). \tag{14.21}$$

Since the subspace is smaller, the linear predictor is weakly worse, and strictly worse whenever $\mathbb{E}[Y \mid X]$ is not affine in X. The two coincide exactly when $\mathbb{E}[Y \mid X]$ happens to be affine, which is the content of the next proposition and the reason the bivariate normal is the workhorse of applied work: it is the case in which linear regression estimates the conditional expectation rather than merely its best affine approximation.

Proposition 14.5.1 (The Gaussian case). Let (X, Y) be jointly normal with $\operatorname{Var}(X) > 0$. Then

$$\mathbb{E}[Y \mid X] = \mathbb{E}[Y] + \frac{\operatorname{Cov}(X,Y)}{\operatorname{Var}(X)}\big(X - \mathbb{E}[X]\big), \qquad \operatorname{Var}(Y \mid X) = \big(1 - \operatorname{Corr}(X,Y)^2\big)\operatorname{Var}(Y), \tag{14.22}$$

the conditional variance being a constant, not a function of X.

The multivariate statement, in which X and Y are vectors and the ratio $\operatorname{Cov}(X,Y)/\operatorname{Var}(X)$ becomes $\Sigma_{12}\Sigma_{22}^{-1}$, is Theorem C.2.1 of Appendix C, together with the affine stability of the family that makes the argument work.

Proof. Let $g(X)$ denote the right-hand side of (14.21). By construction $Y - g(X)$ is orthogonal to 1 and to X, that is $\mathbb{E}[Y - g(X)] = 0$ and $\operatorname{Cov}\big(X, Y - g(X)\big) = 0$. Now $Y - g(X)$ and X are jointly normal, being affine images of the jointly normal pair (X, Y), and for jointly normal variables zero correlation implies independence, since the joint density then factors. Hence $\mathbb{E}[Y - g(X) \mid X] = \mathbb{E}[Y - g(X)] = 0$, which is the first identity in (14.22).

For the variance, $Y - \mathbb{E}[Y] = \frac{\operatorname{Cov}(X,Y)}{\operatorname{Var}(X)}(X - \mathbb{E}[X]) + (Y - g(X))$ is an orthogonal decomposition, so Proposition 11.2.2 gives

$$\operatorname{Var}(Y) = \frac{\operatorname{Cov}(X,Y)^2}{\operatorname{Var}(X)^2}\operatorname{Var}(X) + \operatorname{Var}\big(Y - g(X)\big) = \frac{\operatorname{Cov}(X,Y)^2}{\operatorname{Var}(X)} + \operatorname{Var}(Y \mid X),$$

and rearranging gives the stated expression with $\operatorname{Corr}(X,Y)^2 = \operatorname{Cov}(X,Y)^2/(\operatorname{Var}(X)\operatorname{Var}(Y))$. ■

14.6 Normal–Normal Updating

Theorem 14.6.1 (Gaussian updating, one observation). Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ with $\sigma_x^2 > 0$ and let $Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon^2 > 0$, independent of X. Then

$$X \mid Y = y \;\sim\; \mathcal{N}\left(\frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_\varepsilon^2},\; \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_x^2 + \sigma_\varepsilon^2}\right). \tag{14.23}$$

Writing $\tau_x := 1/\sigma_x^2$ and $\tau_\varepsilon := 1/\sigma_\varepsilon^2$ for the *precisions* and $\tau := \tau_x + \tau_\varepsilon$, this reads

$$X \mid Y = y \;\sim\; \mathcal{N}\left(\frac{\tau_x \mu_x + \tau_\varepsilon y}{\tau},\; \tau^{-1}\right). \tag{14.24}$$

Proof. The prior density and the likelihood are

$$\pi(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2}\right), \qquad L(y \mid x) = \frac{1}{\sqrt{2\pi}\sigma_\varepsilon} \exp\left(-\frac{(y-x)^2}{2\sigma_\varepsilon^2}\right),$$

the second because $Y - X = \varepsilon$ is independent of X. By Bayes' rule in density form, $f(x \mid Y = y) = \pi(x)L(y \mid x) / \int_{-\infty}^{\infty} \pi(\tilde{x}) L(y \mid \tilde{x})\, \mathrm{d}\tilde{x}$, so as a function of x,

$$f(x \mid Y = y) \propto \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_\varepsilon^2}\right).$$

Complete the square in the exponent. Multiplying out,

$$\begin{aligned}
-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_\varepsilon^2} &= -\frac{1}{2\sigma_x^2\sigma_\varepsilon^2}\left[\sigma_\varepsilon^2(x-\mu_x)^2 + \sigma_x^2(y-x)^2\right] \\
&= -\frac{1}{2\sigma_x^2\sigma_\varepsilon^2}\left[(\sigma_x^2+\sigma_\varepsilon^2)x^2 - 2(\sigma_\varepsilon^2\mu_x + \sigma_x^2 y)x + (\sigma_\varepsilon^2\mu_x^2 + \sigma_x^2 y^2)\right] \\
&= -\frac{\sigma_x^2+\sigma_\varepsilon^2}{2\sigma_x^2\sigma_\varepsilon^2}\left(x - \frac{\sigma_\varepsilon^2\mu_x + \sigma_x^2 y}{\sigma_x^2+\sigma_\varepsilon^2}\right)^2 + c(y),
\end{aligned}$$

where $c(y)$ collects all terms free of x. Exponentiating, the posterior density is proportional to a Gaussian kernel in x with mean $(\sigma_\varepsilon^2\mu_x + \sigma_x^2 y)/(\sigma_x^2 + \sigma_\varepsilon^2)$ and variance $\sigma_x^2\sigma_\varepsilon^2/(\sigma_x^2+\sigma_\varepsilon^2)$; since a density is determined by its kernel up to the normalising constant, and by Proposition 10.7.2 that constant is $\big(2\pi\sigma_x^2\sigma_\varepsilon^2/(\sigma_x^2 + \sigma_\varepsilon^2)\big)^{-1/2}$, (14.23) follows. The precision form is the substitution $\sigma_x^2 = 1/\tau_x$, $\sigma_\varepsilon^2 = 1/\tau_\varepsilon$, under which $\sigma_x^2 + \sigma_\varepsilon^2 = \tau/(\tau_x\tau_\varepsilon)$. ∎

This is the conjugate normal model of Gelman et al. (2013, Sec. 2.5, pp. 39–42), whose multivariate counterpart with known variance is Section 3.5 of the same work; the precision form (14.24) is the statement there that the posterior precision is the prior precision plus the data precision. Three of its properties are the reason the Gaussian model is used wherever information is aggregated.

Precisions add. The posterior precision $\tau = \tau_x + \tau_\varepsilon$ does not depend on the realisation y: the informativeness of an observation is known before it is made. Consequently the posterior variance $\tau^{-1} < \sigma_x^2$ falls with every observation, deterministically.

The posterior mean is a precision-weighted average. It is linear in y,

$$\mathbb{E}[X \mid Y = y] = \mu_x + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\varepsilon^2}(y - \mu_x),$$

so the response to the surprise $y - \mu_x$ is the ratio of prior variance to total variance, between 0 and 1. A precise prior or a noisy signal produces sluggish updating; the converse produces aggressive updating. Linearity in y is also the statement that the best predictor is the best *linear* predictor, which is Proposition 14.5.1 applied to the jointly normal pair (X, Y).

Conjugacy. The posterior is again normal, so the operation can be iterated without leaving the family — and this is what makes recursive filtering tractable.

Theorem 14.6.2 (Gaussian updating, many observations). Let $X \sim \mathcal{N}(\mu_x, \tau_x^{-1})$ and let $Y_j = X + \varepsilon_j$ for $j = 1, \dots, n$, where the $\varepsilon_j \sim \mathcal{N}(0, \tau_j^{-1})$ are mutually independent and independent of X. Then with $\tau := \tau_x + \sum_{j=1}^{n} \tau_j$,

$$X \mid (Y_j = y_j)_{j=1}^{n} \;\sim\; \mathcal{N}\left(\frac{\tau_x}{\tau}\mu_x + \sum_{j=1}^{n} \frac{\tau_j}{\tau} y_j,\; \tau^{-1}\right). \tag{14.25}$$

Proof. Independence of the errors makes the joint likelihood the product $\prod_j L(y_j \mid x)$, so

$$f\big(x \mid (Y_j = y_j)_j\big) \propto \exp\left(-\tfrac{1}{2}\Big[\tau_x(x-\mu_x)^2 + \sum_{j=1}^{n} \tau_j (x-y_j)^2\Big]\right).$$

The bracket is a quadratic in x with leading coefficient $\tau_x + \sum_j \tau_j = \tau$ and linear coefficient $-2\big(\tau_x\mu_x + \sum_j \tau_j y_j\big)$, so completing the square as in Theorem 14.6.1 gives $\tau\big(x - (\tau_x\mu_x + \sum_j \tau_j y_j)/\tau\big)^2$ plus terms free of x. Identification of the Gaussian kernel gives (14.25). ∎

The posterior mean is a weighted average of the prior mean and all the observations, with weights proportional to precisions summing to one, and the posterior precision is the sum of all precisions. Applying Theorem 14.6.1 sequentially, using the posterior after j observations as the prior for observation $j+1$, gives the same answer — the order of arrival is irrelevant, which is a substantive restriction and not an accounting identity.

14.6.1 Economic application: CARA–normal portfolio choice

Environment. An investor with initial wealth W_0 and constant absolute risk aversion $a > 0$, $u(W) = -e^{-aW}$, chooses a quantity $\theta \in \mathbb{R}$ of a risky asset with payoff $V \sim \mathcal{N}(\mu, \sigma^2)$ at price p; the riskless rate is normalised to zero, so terminal wealth is $W = W_0 + \theta(V - p)$.

Proposition 14.6.1 (Gaussian demand)**.** The investor's optimal holding is

$$\theta^* = \frac{\mu - p}{a\sigma^2}, \tag{14.26}$$

and the certainty equivalent of the optimal portfolio is $W_0 + (\mu-p)^2/(2a\sigma^2)$. If instead the investor observes a signal $Y = V + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \tau_\varepsilon^{-1})$ independent of V, then θ^* is (14.26) with μ and σ^2 replaced by the posterior mean and variance of Theorem 14.6.1.

Proof. Terminal wealth is normal with mean $W_0 + \theta(\mu - p)$ and variance $\theta^2\sigma^2$, so by Proposition 10.7.1

$$\mathbb{E}[u(W)] = -\exp\big(-a\big(W_0 + \theta(\mu-p)\big) + \tfrac{1}{2}a^2\theta^2\sigma^2\big).$$

Maximising this is minimising the exponent, that is maximising $\theta(\mu - p) - \frac{1}{2}a\theta^2\sigma^2$, a strictly concave quadratic in θ; Theorem 7.2.3 gives the unique maximiser (14.26), and substituting back gives an exponent of $-a\big(W_0 + (\mu-p)^2/(2a\sigma^2)\big)$, whose certainty equivalent is the stated value. Conditioning on $Y = y$ replaces the distribution of V by its posterior, which is again normal by Theorem 14.6.1, and the same computation applies. ∎

Demand is linear in the price with slope $-1/(a\sigma^2)$, and initial wealth does not appear — the wealth effect is exactly zero under CARA, which is what makes the model aggregate. Two consequences follow. Because demand is linear in the posterior mean and the posterior mean is linear in the signal by Theorem 14.6.1, individual demands are linear in private signals, so aggregate demand is linear in the average signal and the market-clearing price is a linear statistic of the population's information. This is the mechanism by which prices aggregate dispersed information; adding noise traders to prevent the price from being fully revealing gives the noisy rational-expectations equilibrium of Grossman and Stiglitz (1980), in which the price is informative but not sufficient, so that costly information acquisition remains worthwhile. Second, the precision arithmetic of (14.24) means an investor with a more precise signal takes a larger position on the same price, which is the source of the informed trader's profit in subsection 14.3.1.

14.6.2 Economic application: recursive filtering

Conjugacy makes updating recursive. Let the state follow $X_{t+1} = \rho X_t + \eta_{t+1}$ with $\eta_{t+1} \sim \mathcal{N}(0, \tau_2)$ and let the observation be $Y_t = X_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \tau_1^{-1})$, all errors independent. If the prior on X_t given the history is $\mathcal{N}(\hat{x}_t, P_t)$, then Theorem 14.6.1 gives the posterior precision $P_t^{-1} + \tau_1$ after observing Y_t, and the transition inflates the variance by τ_2 after scaling by ρ^2:

$$P_{t+1} = \rho^2 \big(P_t^{-1} + \tau_1\big)^{-1} + \tau_2. \tag{14.27}$$

The right-hand side is exactly the operator T of (6.5), shown in Proposition 6.6.2 to be a contraction of modulus ρ^2 on $\mathbb{R}_{++}$. Theorem 6.4.1 therefore gives a unique steady-state variance x^* satisfying (6.6), reached from every initial uncertainty at the geometric rate ρ^2. The economics of the recursion is that the filter never learns the state exactly, no matter how long it observes: each period's transition injects fresh variance τ_2, and the steady state balances that injection against the precision τ_1 acquired from the new observation. Setting $\rho = 0$ makes the state i.i.d. and gives $x^* = \tau_2$, the variance of a state about which yesterday's data say nothing.

Equation (14.27) is the scalar case of the Kalman recursions of Brockwell and Davis (2016, Sec. 9.4, p. 270), whose general form replaces the variances by covariance matrices and the scalar ρ by a transition matrix A:

$$P_{t+1} = A\big(P_t^{-1} + C^\mathsf{T} R^{-1} C\big)^{-1} A^\mathsf{T} + Q,$$

with C the observation matrix, R the observation-noise covariance and Q the state-noise covariance. The contraction argument does *not* carry over verbatim. The scalar proof uses that $x \mapsto \rho^2(x^{-1} + \tau_1)^{-1} + \tau_2$ is a contraction for the Euclidean metric on $\mathbb{R}_{++}$, and the matrix analogue is false for the operator norm on positive definite matrices. What is true is that the discrete algebraic Riccati equation has a unique positive semidefinite stabilising solution, to which P_t converges from every positive semidefinite initialisation, provided the pair $(A, Q^{1/2})$ is stabilisable and the pair (A, C) is detectable; the natural argument uses a Riemannian metric on the cone of positive definite matrices rather than the Euclidean one. Stabilisability and detectability are the conditions under which unobserved directions of the state decay on their own and observed directions are pinned down by the data; without them the filter can carry an undamped, unmeasured mode and the error covariance need not settle. See Brockwell and Davis (2016, Sec. 9.4, p. 270) for the recursion and Ljungqvist and Sargent (2018, Sec. 5.4, pp. 136–137) for stabilisability and its role in the dual linear-regulator problem, whose Riccati equation is the same object.

14.7 Probability Inequalities

Theorem 14.7.1 (Markov and Chebyshev). Let $X \geq 0$ be a random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \tag{14.28}$$

Consequently, for any square-integrable Z and $k > 0$,

$$\mathbb{P}\big(|Z - \mathbb{E}[Z]| \geq k\big) \leq \frac{\operatorname{Var}(Z)}{k^2}. \tag{14.29}$$

Proof. Since $X \geq 0$, the pointwise inequality $X \geq a\mathbb{1}_{[X \geq a]}$ holds: on $[X \geq a]$ the left side is at least a, and elsewhere the right side is zero. Taking expectations and using monotonicity and linearity (Proposition 14.2.1) gives $\mathbb{E}[X] \geq a\,\mathbb{E}[\mathbb{1}_{[X \geq a]}] = a\mathbb{P}(X \geq a)$, which is (14.28). For (14.29), apply (14.28) to the non-negative variable $X := (Z - \mathbb{E}[Z])^2$ with threshold k^2 and use $\{|Z - \mathbb{E}[Z]| \geq k\} = \{X \geq k^2\}$ together with $\mathbb{E}[X] = \operatorname{Var}(Z)$. ■

Remark 14.7.1 (Sharpness and use). Markov's inequality is attained: if $X = a$ with probability $\mathbb{E}[X]/a$ and $X = 0$ otherwise, (14.28) holds with equality, so no distribution-free improvement is possible. Its role is to convert a statement about a mean into a statement about a tail using no distributional assumption, and this is what drives the weak law of large numbers: for i.i.d. square-integrable (Z_i) with mean μ and variance σ^2, the sample mean $\bar{Z}_n$ has $\mathbb{E}[\bar{Z}_n] = \mu$ and, by independence, $\operatorname{Var}(\bar{Z}_n) = \sigma^2/n$, so (14.29) gives $\mathbb{P}\big(|\bar{Z}_n - \mu| \geq k\big) \leq \sigma^2/(nk^2) \to 0$ for every $k > 0$.

> Convergence in probability of sample averages is therefore a two-line consequence of the variance identity and Markov's inequality, requiring no central limit theorem and no normality.

14.8 Convergence of Random Variables

Everything asymptotic in econometrics — consistency, asymptotic normality, the behaviour of a sample average as the sample grows — is a statement that a sequence of random variables converges. Unlike convergence in $\mathbb{R}^n$, there are several inequivalent senses in which it can do so, and the theorems that license exchanging a limit with an expectation carry hypotheses that are routinely needed and routinely omitted.

Definition 14.8.1 (Modes of convergence)**.** Let X and $X_1, X_2, \ldots$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) $X_n \to X$ *almost surely*, written $X_n \xrightarrow{\text{a.s.}} X$, if $\mathbb{P}\big(\{\omega : X_n(\omega) \to X(\omega)\}\big) = 1$.

(ii) $X_n \to X$ *in probability*, written $X_n \xrightarrow{\mathbb{P}} X$, if $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$.

(iii) $X_n \to X$ *in* $\mathcal{L}^p$, $p \geq 1$, if $\mathbb{E}\big[|X_n - X|^p\big] \to 0$.

(iv) $X_n \to X$ *in distribution*, written $X_n \xrightarrow{d} X$, if $\mathbb{E}[g(X_n)] \to \mathbb{E}[g(X)]$ for every bounded continuous $g\colon \mathbb{R} \to \mathbb{R}$; equivalently, if $F_n(x) \to F(x)$ at every continuity point x of the distribution function F of X.

Only (iv) is a statement about distributions rather than about the random variables themselves; X_n and X need not even live on the same probability space for it to make sense. This is why it is the mode in which limit theorems are stated, and the mode that supports no arithmetic. Let $Z \sim N(0,1)$ and set $X_n := Z$, $Y_n := -Z$ for every n, and let X, Y be *independent* standard normals. Then $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$, since each side has the same marginal law at every n, while $X_n + Y_n = 0$ has the degenerate limit and $X + Y \sim N(0,2)$. The marginal laws leave the joint law undetermined, and sums are a function of the joint law.

Proposition 14.8.1 (The hierarchy)**.** Let $X, X_1, X_2, \ldots$ be random variables and $p \geq 1$.

(i) $\mathcal{L}^p$ convergence implies convergence in probability.

(ii) Almost sure convergence implies convergence in probability.

(iii) Convergence in probability implies convergence in distribution.

(iv) Convergence in distribution to a constant c implies convergence in probability to c.

None of the converses of (i)–(iii) holds, and (i) and (ii) imply neither the other.

Proof. $\mathcal{L}^p \Rightarrow \mathbb{P}$. By Theorem 14.7.1 applied to $|X_n - X|^p$, $\mathbb{P}(|X_n - X| > \varepsilon) \leq \varepsilon^{-p}\mathbb{E}[|X_n - X|^p] \to 0$.

a.s. $\Rightarrow \mathbb{P}$. Fix $\varepsilon > 0$ and set $A_n := \bigcup_{m \geq n}\{|X_m - X| > \varepsilon\}$. The sets A_n decrease, and on the event $\{X_n \to X\}$ no ω lies in $\bigcap_n A_n$; hence $\mathbb{P}(\bigcap_n A_n) = 0$ and, by continuity from above (Proposition 14.1.2(vi)), $\mathbb{P}(A_n) \to 0$. Since $\{|X_n - X| > \varepsilon\} \subseteq A_n$, the claim follows.

$\mathbb{P} \Rightarrow d$. Let g be bounded by M and continuous, and fix $\varepsilon > 0$. Choose K with $\mathbb{P}(|X| > K) < \varepsilon$; g is uniformly continuous on $[-K-1, K+1]$ by Theorem 5.2.2, so there is $\delta \in (0,1)$ with $|g(x) - g(y)| < \varepsilon$ whenever $|x - y| < \delta$ and $|x| \leq K$. Splitting according to $\{|X_n - X| < \delta\}$, $\{|X| > K\}$ and the remainder,

$$\big|\mathbb{E}[g(X_n)] - \mathbb{E}[g(X)]\big| \leq \varepsilon + 2M\mathbb{P}(|X| > K) + 2M\mathbb{P}(|X_n - X| \geq \delta) \leq 3\varepsilon + 2M\mathbb{P}(|X_n - X| \geq \delta),$$

and the last term vanishes in the limit.

Failures of the converses. On $([0,1], \mathcal{B}, \mathrm{Leb})$ let $X_n = \mathbb{1}_{[j2^{-k},(j+1)2^{-k}]}$ where $n = 2^k + j$, $0 \leq j < 2^k$: the intervals sweep $[0,1]$ repeatedly with lengths $2^{-k} \to 0$, so $X_n \xrightarrow{\mathbb{P}} 0$, while $\limsup_n X_n(\omega) = 1$ for every ω, so X_n converges almost surely to nothing. Taking instead $X_n = n\mathbb{1}_{[0,1/n]}$ gives $X_n \xrightarrow{\text{a.s.}} 0$ with $\mathbb{E}[X_n] = 1$ for every n, so there is no $\mathcal{L}^1$ convergence. And if $X \sim N(0,1)$ and $X_n := -X$ for every n, then $X_n \xrightarrow{d} X$ by symmetry while $|X_n - X| = 2\,|X|$ does not converge to 0 in probability.

Convergence in distribution to a constant. If $X_n \xrightarrow{d} c$ then for $\varepsilon > 0$ the distribution function of X_n tends to 0 below $c - \varepsilon$ and to 1 above $c + \varepsilon$, both continuity points of the degenerate limit, so $\mathbb{P}(|X_n - c| > \varepsilon) \to 0$. ∎

> **Remark 14.8.1** (Where each mode is used). Consistency of an estimator is convergence in probability of $\hat\theta_n$ to θ_0; the notation $\operatorname{plim} \hat\theta_n = \theta_0$ is Definition 14.8.1(ii). Asymptotic normality is convergence in distribution of $\sqrt{n}(\hat\theta_n - \theta_0)$. Mean-square consistency is convergence in $\mathcal{L}^2$, and it is what the projection argument of Chapter 11 delivers directly, since $\mathbb{E}[|X_n - X|^2]$ is a squared distance in the Hilbert space $\mathcal{L}^2$. The hierarchy of Proposition 14.8.1 orders the first two: mean-square consistency implies consistency and the converse fails. Asymptotic normality does not belong to that chain, being a statement about a different sequence. Consistency concerns $\hat\theta_n - \theta_0$; asymptotic normality concerns the rescaled error $\sqrt{n}(\hat\theta_n - \theta_0)$. If the rescaled error converges in distribution at all, then $\hat\theta_n - \theta_0 = n^{-1/2}\big[\sqrt{n}(\hat\theta_n - \theta_0)\big] \xrightarrow{d} 0$ by Theorem 14.8.1(ii), hence $\xrightarrow{\mathbb{P}} 0$ by Proposition 14.8.1(iv). So asymptotic normality implies consistency, and supplies in addition the rate $n^{-1/2}$ and the limiting distribution. What mean-square consistency supplies beyond either is convergence of second moments, which a distributional limit alone does not give: $X_n = n\mathbb{1}_{[0,1/n]}$ converges to 0 in distribution and in probability with $\mathbb{E}[X_n] = 1$ throughout.

Theorem 14.8.1 (Continuous mapping, Slutsky, delta method). Let X_n, X take values in $\mathbb{R}^k$ and let $c \in \mathbb{R}^k$, $a \in \mathbb{R}$.

(i) *(Continuous mapping.)* If $X_n \xrightarrow{d} X$ and $h\colon \mathbb{R}^k \to \mathbb{R}^m$ is continuous on a set C with $\mathbb{P}(X \in C) = 1$, then $h(X_n) \xrightarrow{d} h(X)$. The same implication holds with $\xrightarrow{d}$ replaced throughout by $\xrightarrow{\mathbb{P}}$ or by almost sure convergence.

(ii) *(Slutsky.)* Let $X_n \xrightarrow{d} X$ in $\mathbb{R}^k$, $Y_n \xrightarrow{\mathbb{P}} c$ in $\mathbb{R}^k$, and $A_n \xrightarrow{\mathbb{P}} A$ for random $m \times k$ matrices A_n and a constant matrix A. Then $X_n + Y_n \xrightarrow{d} X + c$ and $A_n X_n \xrightarrow{d} AX$. If $k = 1$ and $a \neq 0$ with $Y_n \xrightarrow{\mathbb{P}} a$ real, then $X_n / Y_n \xrightarrow{d} X/a$.

(iii) *(Delta method.)* If $\sqrt{n}(X_n - a) \xrightarrow{d} X$ with $X_n, a \in \mathbb{R}^k$ and $h\colon \mathbb{R}^k \to \mathbb{R}^m$ is differentiable at a, then $\sqrt{n}\big(h(X_n) - h(a)\big) \xrightarrow{d} Dh(a)X$.

Proof. Throughout, convergence in distribution is used in the form $\mathbb{E}[g(X_n)] \to \mathbb{E}[g(X)]$ for every bounded continuous g, as in Definition 14.8.1(iv).

A tightness bound. If $X_n \xrightarrow{d} X$ then for every $\varepsilon > 0$ there is K with $\limsup_n \mathbb{P}(\|X_n\| > K + 1) \le \varepsilon$. Indeed $h_K(x) := \min\{1, \max\{0, \|x\| - K\}\}$ is bounded and continuous with $\mathbb{1}_{\{\|x\| > K+1\}} \le h_K(x) \le \mathbb{1}_{\{\|x\| > K\}}$, so $\limsup_n \mathbb{P}(\|X_n\| > K + 1) \le \lim_n \mathbb{E}[h_K(X_n)] = \mathbb{E}[h_K(X)] \le \mathbb{P}(\|X\| > K)$, and the right side tends to 0 as $K \to \infty$.

(ii), sums. Let g be continuous with $|g| \le M$ and fix $\varepsilon > 0$. Choose K as above. The closed ball of radius $K + 1 + \|c\| + 1$ is compact, so g is uniformly continuous on it by Theorem 5.2.2: there is $\delta \in (0, 1)$ with $|g(x) - g(y)| < \varepsilon$ for x, y in that ball with $\|x - y\| < \delta$. On the event $\{\|Y_n - c\| < \delta\} \cap \{\|X_n\| \le K + 1\}$ the two points $X_n + Y_n$ and $X_n + c$ lie in the ball and differ by less than δ, so

$$\big|\mathbb{E}[g(X_n + Y_n)] - \mathbb{E}[g(X_n + c)]\big| \le \varepsilon + 2M\big[\mathbb{P}(\|Y_n - c\| \ge \delta) + \mathbb{P}(\|X_n\| > K + 1)\big].$$

The first probability vanishes and the second has $\limsup \le \varepsilon$. Since $x \mapsto g(x + c)$ is bounded and continuous, $\mathbb{E}[g(X_n + c)] \to \mathbb{E}[g(X + c)]$, and therefore $\limsup_n |\mathbb{E}[g(X_n + Y_n)] - \mathbb{E}[g(X + c)]| \le \varepsilon(1 + 2M)$ for every $\varepsilon > 0$.

(ii), products and quotients. For the matrix product the same argument applies on the event $\{\|A_n - A\|_2 < \delta/(K + 1)\} \cap \{\|X_n\| \le K + 1\}$, on which $\|A_n X_n - AX_n\| \le \|A_n - A\|_2 \|X_n\| < \delta$, with the ball of radius $(\|A\|_2 + 1)(K + 1)$ and with $\mathbb{E}[g(AX_n)] \to \mathbb{E}[g(AX)]$ by (i). If $a \neq 0$ then $1/Y_n \xrightarrow{\mathbb{P}} 1/a$, since $t \mapsto 1/t$ is continuous at a and convergence in probability is preserved under a map continuous at the limit point; the product case then gives $X_n / Y_n \xrightarrow{d} X/a$. Degeneracy of the second limit is what makes these arguments work, and the failure recorded before Proposition 14.8.1 shows that it cannot be dropped.

(iii). Differentiability at a gives $h(x) - h(a) = Dh(a)(x-a) + \|x-a\|\,\psi(x)$ with $\psi(a) := 0$ and $\psi(x) \to 0$ as $x \to a$. Applying (ii) with the constant factor $n^{-1/2} \to 0$ gives $X_n - a \xrightarrow{d} 0$, hence $X_n - a \xrightarrow{\mathbb{P}} 0$ by Proposition 14.8.1(iv). For $\varepsilon > 0$ pick $\eta > 0$ with $\|\psi(x)\| < \varepsilon$ whenever $\|x-a\| < \eta$; then $\mathbb{P}(\|\psi(X_n)\| \geq \varepsilon) \leq \mathbb{P}(\|X_n - a\| \geq \eta) \to 0$, so $\psi(X_n) \xrightarrow{\mathbb{P}} 0$. The tightness bound applied to $\sqrt{n}(X_n - a)$ makes $\|\sqrt{n}(X_n - a)\|$ bounded in probability, so the product $\sqrt{n}\,\|X_n - a\|\,\psi(X_n) \xrightarrow{\mathbb{P}} 0$. Finally $\sqrt{n}\big(h(X_n) - h(a)\big) = Dh(a)\sqrt{n}(X_n - a) + \sqrt{n}\,\|X_n - a\|\,\psi(X_n)$, where the first term converges in distribution to $Dh(a)X$ by (i) applied to the continuous map $Dh(a)$, and (ii) absorbs the second.

Part (i) for convergence in distribution is Corbae et al. (2009, Thm. 9.4.3, p. 562); its proof runs through the portmanteau characterisation of weak convergence, which is not developed here. For convergence in probability and almost sure convergence the statement is elementary: almost sure convergence passes through h pointwise on the set where $X \in C$, and the probability version follows by the subsequence criterion. ■

14.8.1 Interchanging limits and expectations

Theorem 14.8.2 (Monotone convergence, Fatou, dominated convergence)**.** Let $X, X_1, X_2, \ldots$ be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

(i) *(Monotone convergence.)* If $0 \leq X_n \leq X_{n+1}$ for every n and $X_n \to X$ pointwise, then $\mathbb{E}[X_n] \to \mathbb{E}[X]$, both sides being allowed the value $+\infty$.

(ii) *(Fatou.)* If $X_n \geq 0$ for every n, then $\mathbb{E}[\liminf_n X_n] \leq \liminf_n \mathbb{E}[X_n]$.

(iii) *(Dominated convergence.)* If $X_n \to X$ almost surely and there is an integrable Y with $|X_n| \leq Y$ almost surely for every n, then X and every X_n are integrable and $\mathbb{E}[X_n] \to \mathbb{E}[X]$; moreover $\mathbb{E}[|X_n - X|] \to 0$.

These are Theorems 2.4 (p. 21), 2.8 (p. 25) and 2.11 (p. 30) of Le Gall (2022), stated there for a general measure space; the proofs are omitted here because they belong to the construction of the Lebesgue integral, which Chapter 10 deliberately does not carry out, and because nothing in this volume depends on their internal structure. Each is used below only as a licence to exchange a limit with an integral.

Remark 14.8.2 (Where each hypothesis is used)**.** Each of (i)–(iii) fails without its hypothesis, and the same example refutes all three. On $([0,1], \mathcal{B}, \mathrm{Leb})$ put $X_n = n\mathbb{1}_{[0,1/n]}$. Then $X_n \to 0$ pointwise, so $\mathbb{E}[\lim X_n] = 0$, while $\mathbb{E}[X_n] = 1$ for every n: monotonicity fails in (i), Fatou's inequality in (ii) is strict, and no integrable dominating Y exists in (iii), since $\sup_n X_n \geq 1/x$ on $(0,1]$, which is not integrable. In economic terms the example is a sequence of lotteries whose payoffs grow as fast as their probabilities shrink; expected utility is continuous along such sequences only under a domination or uniform-integrability condition, and this is what a bounded utility function buys.

Part (iii) is what lets expectations pass through limits in Theorem 14.4.2 and in Proposition 14.4.3, and Lemma 10.5.1 is the form of it needed to differentiate under an integral sign — the step behind every comparative static of an expected-utility problem.

14.8.2 Limit theorems

Theorem 14.8.3 (Laws of large numbers)**.** Let $(X_n)_{n\in\mathbb{N}}$ be independent and identically distributed with $\mu := \mathbb{E}[X_1]$, and let $\bar{X}_n := n^{-1}\sum_{i=1}^n X_i$.

(i) *(Weak law.)* If $\mathrm{Var}(X_1) = \sigma^2 < \infty$, then $\mathbb{E}[(\bar{X}_n - \mu)^2] = \sigma^2/n \to 0$, so $\bar{X}_n \to \mu$ in $\mathcal{L}^2$ and hence in probability.

(ii) *(Strong law.)* If $\mathbb{E}\,|X_1| < \infty$, then $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

Proof. (i) $\mathbb{E}[\bar{X}_n] = \mu$ by linearity, and independence gives $\mathrm{Var}(\bar{X}_n) = n^{-2}\sum_i \mathrm{Var}(X_i) = \sigma^2/n$ by Proposition 14.2.3. Convergence in probability follows from Proposition 14.8.1, or directly from (14.29) as in Remark 14.7.1.

(ii) is Theorem 10.8 (p. 206) of Le Gall (2022). The proof rests on Kolmogorov's zero-one law and a truncation argument, neither of which is developed here; what matters below is only the conclusion, and the hypothesis $\mathbb{E}\,|X_1| < \infty$ is not merely sufficient but necessary for the limit to be defined. ■

Theorem 14.8.4 (Central limit theorem)**.** Let $(X_n)_{n\in\mathbb{N}}$ be independent and identically distributed with $\mathbb{E}[X_1] = \mu$ and $\operatorname{Var}(X_1) = \sigma^2 \in (0, \infty)$. Then

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2). \tag{14.30}$$

This is Theorem 10.15 (p. 219) of Le Gall (2022); the proof uses characteristic functions and Lévy's continuity theorem, neither of which is introduced in this volume, and is not reproduced. Two features of the statement are what applied work relies on. The scaling is $\sqrt{n}$ and not n: by Theorem 14.8.3(i) the deviation $\bar{X}_n - \mu$ has standard deviation $\sigma/\sqrt{n}$, so $\sqrt{n}$ is the only normalisation under which the limit is non-degenerate. And the limit law does not depend on the distribution of X_1 beyond its first two moments, which is why a standard error and a normal quantile suffice to form a confidence interval without knowing the data-generating distribution.

Remark 14.8.3 (Asymptotic normality of least squares)**.** Return to the estimator $\hat{\beta} = (X^\top X)^{-1}X^\top y$ of Proposition 11.5.1 with $y = X\beta + \varepsilon$. Then

$$\sqrt{n}\,(\hat{\beta} - \beta) = \left(\tfrac{1}{n}X^\top X\right)^{-1} \cdot \tfrac{1}{\sqrt{n}}X^\top \varepsilon.$$

The hypotheses are that the pairs (x_i, ε_i) are i.i.d. and

$$\mathbb{E}[x_i\varepsilon_i] = 0, \qquad Q := \mathbb{E}[x_i x_i^\top] \text{ finite and positive definite}, \qquad \Omega := \mathbb{E}\big[\varepsilon_i^2 x_i x_i^\top\big] \text{ finite}, \tag{14.31}$$

the last of which is the statement $\mathbb{E}[\varepsilon_i^2\,\|x_i\|^2] < \infty$ up to a change of norm. Finiteness of Q makes the entries of $x_i x_i^\top$ integrable, so Theorem 14.8.3 gives $n^{-1}X^\top X \to Q$ almost surely; matrix inversion is continuous at every invertible matrix, so positive definiteness of Q and Theorem 14.8.1(i) give $(n^{-1}X^\top X)^{-1} \to Q^{-1}$ almost surely, hence in probability. Finiteness of Ω makes the i.i.d. vectors $x_i\varepsilon_i$ square-integrable with mean zero, so Theorem 14.8.4 gives $n^{-1/2}X^\top\varepsilon \xrightarrow{d} N(0, \Omega)$. Now Theorem 14.8.1(ii), in the form that a random matrix converging in probability to a constant matrix may be substituted into a product, yields $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Q^{-1}N(0, \Omega) = N(0, Q^{-1}\Omega Q^{-1})$, the sandwich formula. Dropping positive definiteness of Q breaks the argument at the inversion step, and dropping finiteness of Ω breaks it at the central limit theorem, leaving $\hat{\beta}$ consistent but with no $\sqrt{n}$ limit. Invertibility of Q is the population version of the full-rank condition, and Remark 2.6.1 says that when Q is nearly singular the asymptotic variance along the least-excited direction of the regressors is large: $\kappa(Q)$ measures how much information the design carries about the worst-identified linear combination of coefficients.

14.8.3 Martingales

Definition 14.8.2 (Filtration and martingale)**.** A *filtration* on $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing sequence $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$ of σ-algebras. A process $(M_n)_{n\geq 0}$ is *adapted* if each M_n is $\mathcal{F}_n$-measurable, and is a *martingale* if it is adapted, $\mathbb{E}\,|M_n| < \infty$ for every n, and

$$\mathbb{E}\big[M_{n+1} \mid \mathcal{F}_n\big] = M_n \qquad \text{for every } n \geq 0. \tag{14.32}$$

It is a *supermartingale* if $\leq$ holds in (14.32) and a *submartingale* if $\geq$ does. A *stopping time* is a random $T\colon \Omega \to \mathbb{N}_0 \cup \{+\infty\}$ with $\{T \leq n\} \in \mathcal{F}_n$ for every n.

The filtration is the formalisation of information arriving over time, and $\mathcal{F}_n$ is the σ-algebra of Definition 14.4.3 representing what is known at date n. A stopping time is a rule for quitting that uses only information available when it is exercised: the condition $\{T \leq n\} \in \mathcal{F}_n$ excludes rules such as "sell at the peak".

Proposition 14.8.2 (Two constructions). (i) *(Doob martingale.)* If $\mathbb{E}|Z| < \infty$ then $M_n := \mathbb{E}[Z \mid \mathcal{F}_n]$ is a martingale.

(ii) *(Random walk.)* If $(\xi_n)_{n\geq 1}$ are independent with $\mathbb{E}[\xi_n] = 0$ and $\mathbb{E}|\xi_n| < \infty$, and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$, then $M_n := \sum_{i\leq n} \xi_i$ is a martingale.

(iii) If (M_n) is a martingale and ϕ is convex with $\mathbb{E}|\phi(M_n)| < \infty$, then $(\phi(M_n))$ is a submartingale.

Proof. (i) Integrability is Theorem 14.4.2, and $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\big[\mathbb{E}[Z \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\big] = \mathbb{E}[Z \mid \mathcal{F}_n] = M_n$ by the tower property (14.16), which applies because $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$.

(ii) $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = M_n + \mathbb{E}[\xi_{n+1}] = M_n$, using independence of ξ_{n+1} from $\mathcal{F}_n$.

(iii) By conditional Jensen (Proposition 14.4.2), $\mathbb{E}[\phi(M_{n+1}) \mid \mathcal{F}_n] \geq \phi\big(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]\big) = \phi(M_n)$. ∎

Theorem 14.8.5 (Optional stopping). Let (M_n) be a martingale and T a stopping time. Then the stopped process $(M_{n\wedge T})_{n\geq 0}$ is a martingale. If in addition T is bounded, then M_T is integrable and $\mathbb{E}[M_T] = \mathbb{E}[M_0]$.

This is Theorem 12.12 (p. 265) of Le Gall (2022), where it is obtained by writing $M_{n\wedge T} = M_0 + (H \bullet M)_n$ for the predictable sequence $H_n = \mathbb{1}_{\{T\geq n\}}$ and using that a predictable transform of a martingale is a martingale; the argument is short and is not reproduced. The boundedness of T cannot be dropped: for the symmetric random walk with $M_0 = 0$ and $T := \inf\{n : M_n = 1\}$ one has $T < \infty$ almost surely and $\mathbb{E}[M_T] = 1 \neq 0 = \mathbb{E}[M_0]$. The gambler who plays until ahead is certain to succeed and has unbounded expected playing time; the failure of the conclusion at an unbounded stopping time is what rules out that strategy in a model with borrowing limits.

Remark 14.8.4 (Martingales in the models of this volume). Three results already proved acquire their standard names once (14.32) is available.

The tower property (14.16) says that the conditional expectation of a fixed random variable given a growing information set is a martingale, by Proposition 14.8.2(i). Applied to a forecast, this is the statement that revisions to a rational forecast are unpredictable from current information — the testable content of rational expectations, and the reason a regression of the forecast revision on any date-n variable should have zero coefficient.

In Theorem 9.7.2, absence of arbitrage in the one-period model produced a strictly positive state price vector ψ. Write $R_0 := 1/\sum_s \psi_s$ for the gross return on the riskless claim and $\mathbb{P}^*(s) := \psi_s R_0$, which is a probability measure because its weights are positive and sum to one. Then (9.11) becomes $q_j = \mathbb{E}^*[R_{\cdot j}]/R_0$: every asset's price is its expected payoff under $\mathbb{P}^*$, discounted at the riskless rate. Applying this at each date of a multi-period model, with $\mathcal{F}_n$ the information available at date n, gives $S_n = \mathbb{E}^*[S_{n+1} \mid \mathcal{F}_n]/R_0$, so the discounted price process $R_0^{-n} S_n$ is a martingale under $\mathbb{P}^*$. The multi-period statement is not proved here — it requires the one-period result at every node, together with a consistency condition tying the measures across dates — but the algebra at a single date is (9.11) rewritten, and $\mathbb{P}^*$ is the risk-neutral measure.

In the Lucas model of Section 15.7, the Euler equation $p_t u'(c_t) = \beta \mathbb{E}_t\big[(p_{t+1} + d_{t+1}) u'(c_{t+1})\big]$ says that $\beta^t u'(c_t) p_t$ plus accumulated discounted dividends is a martingale; the transversality condition is the requirement that this martingale not diverge, which is how Remark 15.2.2 rules out bubbles.

14.9 Chapter Summary

A probability space consists of a sample space, a σ-algebra of events closed under complementation and countable intersection, and a countably additive measure of total mass one; monotonicity, finite additivity and inclusion–exclusion follow from the two axioms. A random variable is a measurable real function on the sample space, its expectation the mass-weighted sum of its values, and the variance identity $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is the special case, at $c = 0$, of the decomposition $\mathbb{E}[(X - c)^2] = \mathrm{Var}(X) + (\mathbb{E}[X] - c)^2$ that identifies the mean as the best constant predictor. Jensen's inequality follows from the existence of a supporting line to a convex function, and yields the sign of the risk premium without differentiability.

Conditional probability makes $\mathbb{P}(\cdot \mid H)$ a probability measure in its own right; Bayes' rule reverses the direction of conditioning, and in odds form states that posterior odds are prior odds times the likelihood ratio. The denominator, computed by the law of total probability, is what the base-rate fallacy neglects. Conditional expectation given a random variable is itself a random variable, satisfies the pull-out property $\mathbb{E}[g(X)Y \mid X] = g(X)\mathbb{E}[Y \mid X]$ and the law of iterated expectations $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$, and, viewed in L^2, is the orthogonal projection of Y onto the functions of X. That identification supplies at once the best-predictor property, the law of total variance, and the fact that the best linear predictor is a strictly worse projection onto a strictly smaller subspace, the two coinciding exactly in the jointly normal case.

Gaussian updating is conjugate: posterior precision is the sum of prior and signal precisions, independent of the realisation, and the posterior mean is the corresponding precision-weighted average, linear in the signal. With many independent signals the same arithmetic applies to their sum, and the sequential and simultaneous updates agree. Applied to a CARA investor the linearity yields demand $(\mu - p)/(a\sigma^2)$ with no wealth effect, hence prices that are linear statistics of dispersed information; applied recursively it yields the Kalman recursion, whose variance obeys the Riccati equation shown in Chapter 6 to be a contraction. Markov's inequality bounds a tail by a mean with no distributional assumption, and through Chebyshev delivers the weak law of large numbers.

> **Exercise 14.9.1.** Prove the general Bayes formula (14.8) directly from Definition 14.3.1 without invoking (14.6), and verify it against Example 14.3.3. Then show that the odds form (14.10) extends to a partition of size n as the statement that $\mathbb{P}(H_i \mid D)/\mathbb{P}(H_j \mid D)$ equals the prior odds ratio times the likelihood ratio, so that the normalising constant never has to be computed to compare two hypotheses.

> **Exercise 14.9.2.** Prove the law of total variance (14.20) from Theorem 14.5.1 and Theorem 14.4.1. Then show that $\mathrm{Var}\big(\mathbb{E}[Y \mid X]\big) \le \mathrm{Var}(Y)$, with equality if and only if Y is almost surely a function of X, and interpret the ratio of the two as the population R^2 of the regression of Y on X.

> **Exercise 14.9.3.** In the setting of Proposition 14.3.1, compute $\mathbb{P}(V = H \mid B)$ after a buy, and show that the market maker's bid $\mathbb{E}[V \mid S]$ and ask $\mathbb{E}[V \mid B]$ satisfy $\mathbb{E}[V \mid S] < \mathbb{E}[V] < \mathbb{E}[V \mid B]$ with the spread strictly increasing in q and vanishing at $q = 0$. Then verify that the martingale property $\mathbb{E}\big[\mathbb{E}[V \mid \mathrm{trade}]\big] = \mathbb{E}[V]$ holds, and explain why this is Theorem 14.4.1 rather than an assumption about the market maker's behaviour.

Chapter 15

Discrete-Time Dynamic Programming

15.1 The Sequential Problem

Definition 15.1.1 (Sequential problem)**.** Let $r\colon \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}$ be the *return function*, $g\colon \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ the *transition function* and $\beta \in (0,1)$ the *discount factor*. The *sequential problem* from an initial state $x_0 \in \mathbb{R}^n$ is

$$\sup_{(u_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \qquad \text{subject to} \quad x_{t+1} = g(x_t, u_t) \;\; \text{for every } t \geq 0, \quad x_0 \text{ given.} \tag{15.1}$$

Its value is the *value function* $V(x_0)$. The variable $x_t \in \mathbb{R}^n$ is the *state*, $u_t \in \mathbb{R}^k$ the *control*, and a *policy function* is a map $h\colon \mathbb{R}^n \to \mathbb{R}^k$ prescribing $u = h(x)$.

Assumption 15.1.1 (Standing hypotheses)**.** Throughout this chapter:

(i) the state lies in a set $X \subseteq \mathbb{R}^n$ and the feasible controls at x form a set $\Gamma(x) \subseteq \mathbb{R}^k$, defining a correspondence $\Gamma\colon X \rightrightarrows \mathbb{R}^k$ that is non-empty- and compact-valued and continuous;

(ii) the transition $g\colon \operatorname{gr}\Gamma \to \mathbb{R}^n$ is continuous and satisfies $g(x,u) \in X$ for every $x \in X$ and $u \in \Gamma(x)$, so the problem is well defined;

(iii) r is continuous and bounded on $\operatorname{gr}\Gamma$;

(iv) $\beta \in (0,1)$.

Continuity of g in (ii) enters at one point, and that point carries the chapter: the Bellman operator (15.2) sends v to a supremum of $r(x,u) + \beta v(g(x,u))$, and the composite $(x,u) \mapsto v(g(x,u))$ is continuous on $\operatorname{gr}\Gamma$ only if g is. Without it T need not map bounded continuous functions to bounded continuous functions, Theorem 13.5.1 cannot be applied to the maximisation, and neither the continuity of V nor the upper hemicontinuity of the policy correspondence in Proposition 15.2.1 is available. The contraction argument itself would survive on the space of bounded measurable functions, but the identification of V as a continuous function would not.

Boundedness in (iii) is what makes the infinite sum in (15.1) converge absolutely: $|r| \leq \bar{r}$ gives $\left|\sum_t \beta^t r(x_t, u_t)\right| \leq \bar{r}/(1-\beta)$ for every feasible sequence, so the supremum is finite and no ordering or summability question arises. It is a genuine restriction — it excludes log utility on an unbounded state space, and the cake-eating problem of Section 15.5 violates it as $w \downarrow 0$ — and Remark 15.3.2 records what survives without it.

Remark 15.1.1 (Stationarity and the prime notation)**.** The problem is *stationary*: neither r, g nor Γ

depends on t, and the horizon is infinite at every date. Therefore

$$V(x_t) = \beta^{-t} \sup_{(u_\tau)_{\tau \geq t}} \sum_{\tau=t}^{\infty} \beta^\tau r(x_\tau, u_\tau)$$

subject to the same constraints: the problem faced at date t in state x_t is the same problem faced at date 0 in state $x_0 = x_t$. Only the state matters, not the date at which it is reached, and one may therefore drop time subscripts and write x for today's state and x' for tomorrow's (Fukuda 2026b, slide 8). Every statement about V and h below is a statement about functions of the state alone; the reduction from a sequence problem in infinitely many variables to a functional equation in one is the entire content of the method.

15.2 Bellman's Principle of Optimality

> "An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision." (Bellman 1957, p. 83)

Definition 15.2.1 (Bellman operator)**.** Under Assumption 15.1.1, write $\mathcal{C}_b(X) := \mathcal{C}(X, \mathbb{R}) \cap \mathcal{B}(X, \mathbb{R})$ for the space of bounded continuous functions $X \to \mathbb{R}$ with the supremum norm, and define T on it by

$$(TW)(x) := \max_{u \in \Gamma(x)} \big\{ r(x,u) + \beta W\big(g(x,u)\big) \big\}, \qquad x \in X. \tag{15.2}$$

A function W with $W = TW$ solves the *Bellman equation*.

Definition (15.2) is (6.3) of Chapter 6 with the supremum replaced by a maximum. The replacement is a theorem, not a convention: Assumption 15.1.1 makes the constraint correspondence compact-valued and the objective continuous, so Proposition 15.2.1 shows the supremum is attained. Where it is not, every statement below about the policy correspondence fails while the contraction argument of Proposition 6.6.1 survives unchanged.

Remark 15.2.1 (Which space T acts on)**.** Three spaces of functions on X appear in dynamic programming and they are not interchangeable. The largest is $\mathcal{B}(X, \mathbb{R})$, the bounded functions, complete in the supremum norm by Theorem 6.3.2; the smallest is $\mathcal{C}_b(X)$, the bounded continuous ones, a closed subspace and hence complete by Proposition 6.3.3; between them sits the space of bounded *measurable* functions, also closed under uniform limits. The Bellman operator is a contraction of modulus β on any of the three by the same two-line Blackwell argument, since neither monotonicity nor discounting sees the regularity of W. What differs is the object the fixed point turns out to be. On $\mathcal{B}(X, \mathbb{R})$ the fixed point is bounded and nothing more, and no policy need be selectable from H_V. On $\mathcal{C}_b(X)$ one needs T to preserve continuity, which is where Assumption 15.1.1 and Berge's theorem are spent, and the reward is that V is continuous and H_V is upper hemicontinuous with compact values. On the bounded measurable functions T preserves measurability under weaker hypotheses on Γ, and selecting a policy from H_V then requires a measurable selection theorem rather than Berge's; this is the route taken when the state evolves stochastically and continuity of g is unavailable, and it is developed in Stokey et al. (1989, Ch. 7). Everything in this chapter works on $\mathcal{C}_b(X)$.

Proposition 15.2.1 (T is well defined)**.** Under Assumption 15.1.1, T maps $\mathcal{C}_b(X)$ into itself, and the associated *policy correspondence*

$$H_W(x) := \operatorname*{arg\,max}_{u \in \Gamma(x)} \big\{ r(x,u) + \beta W\big(g(x,u)\big) \big\} \tag{15.3}$$

is non-empty- and compact-valued and upper hemicontinuous.

Proof. Fix $W \in \mathcal{C}_b(X)$. The objective $(x,u) \mapsto r(x,u) + \beta W(g(x,u))$ is continuous on $\operatorname{gr} \Gamma$, being a sum of a continuous function and the composition of the continuous W with the continuous g. Assumption 15.1.1(i) makes Γ continuous with non-empty compact values, so Theorem 13.5.1 applies: the maximum is attained, TW is continuous, and H_W is non-empty- and compact-valued and upper hemicontinuous. Boundedness is immediate from $|TW| \le \bar{r} + \beta \|W\|_\infty$. ■

Theorem 15.2.1 (Principle of optimality). Under Assumption 15.1.1:

(i) the value function V of (15.1) satisfies the Bellman equation $V = TV$;

(ii) conversely, if $W \in \mathcal{C}_b(X)$ satisfies $W = TW$, then $W = V$;

(iii) a feasible plan (u_t^*) from x_0 is optimal for (15.1) if and only if $u_t^* \in H_V(x_t^*)$ for every t, where $x_{t+1}^* = g(x_t^*, u_t^*)$.

Proof. (i) Fix $x \in X$ and write $S(x)$ for the value of (15.1). Any feasible plan from x decomposes into a first control $u \in \Gamma(x)$ and a continuation plan feasible from $g(x,u)$, and its objective is $r(x,u) + \beta$ times the objective of the continuation plan. Taking suprema over continuation plans first and then over u, and using that the supremum over a product decomposes,

$$S(x) = \sup_{u \in \Gamma(x)} \big\{ r(x,u) + \beta S(g(x,u)) \big\} = (TS)(x),$$

the interchange being legitimate because the inner supremum is finite, uniformly bounded by $\bar{r}/(1-\beta)$.

(ii) Let $W = TW$ with W bounded. By (i) $V = TV$, and by Theorem 15.3.1 below T has at most one fixed point in $\mathcal{C}_b(X)$, so $W = V$ provided $V \in \mathcal{C}_b(X)$; that V is bounded is the estimate $|V| \le \bar{r}/(1-\beta)$, and that it is continuous follows from (i) and Proposition 15.2.1 once V is known to be the limit of $T^m W_0$, which is Theorem 15.3.1.

(iii) ($\Leftarrow$) Suppose $u_t^* \in H_V(x_t^*)$ for all t. Then $V(x_t^*) = r(x_t^*, u_t^*) + \beta V(x_{t+1}^*)$ for every t; iterating m times,

$$V(x_0) = \sum_{t=0}^{m-1} \beta^t r(x_t^*, u_t^*) + \beta^m V(x_m^*). \tag{15.4}$$

Since $|V| \le \bar{r}/(1-\beta)$, the last term vanishes as $m \to \infty$, so the plan attains $V(x_0)$ and is optimal.

($\Rightarrow$) Suppose the plan is optimal but $u_{t_0}^* \notin H_V(x_{t_0}^*)$ for some t_0. Then $r(x_{t_0}^*, u_{t_0}^*) + \beta V(x_{t_0+1}^*) < V(x_{t_0}^*)$ by the definition of the argmax, so replacing the continuation from t_0 by an optimal one — which exists by Proposition 15.2.1 and the argument just given — raises the objective by β^{t_0} times a strictly positive amount, contradicting optimality. ■

Remark 15.2.2 (Where the transversality condition hides). The step $\beta^m V(x_m^*) \to 0$ in (15.4) is the transversality condition, and boundedness of V is what supplies it. Without Assumption 15.1.1(iii) it must be imposed separately: a plan satisfying the Bellman equation at every date but with $\limsup_m \beta^m V(x_m) > 0$ need not be optimal, and the standard example is a rational asset-price bubble, in which the price satisfies the pricing equation period by period while violating the terminal condition. The converse failure — a solution of the Bellman equation that is not the value function — is the same phenomenon read in the other direction.

15.3 The Contraction Argument

Theorem 15.3.1 (The Bellman operator is a contraction). Under Assumption 15.1.1, $T \colon \mathcal{C}_b(X) \to \mathcal{C}_b(X)$ is a contraction of modulus β in the supremum norm. Consequently:

(i) T has a unique fixed point $V \in \mathcal{C}_b(X)$, so the value function exists, is unique among bounded continuous functions, and is continuous;

(ii) for every $W_0 \in \mathcal{C}_b(X)$ the iterates $W_m := T^m W_0$ converge uniformly to V, with

$$\|W_m - V\|_\infty \le \frac{\beta^m}{1-\beta} \|W_1 - W_0\|_\infty \qquad \text{and} \qquad \|W_m - V\|_\infty \le \frac{\beta}{1-\beta} \|W_m - W_{m-1}\|_\infty ; \tag{15.5}$$

(iii) the policy correspondence H_V of (15.3) is non-empty- and compact-valued and upper hemicontinuous, and is a continuous function whenever the maximiser is unique.

Proof. $\mathcal{C}_b(X)$ is complete in the supremum norm by Proposition 6.3.3, being a closed subspace of the Banach space $\mathcal{B}(X, \mathbb{R})$ of Theorem 6.3.2, and T maps it into itself by Proposition 15.2.1. We verify Blackwell's two sufficient conditions of Theorem 6.5.1.

Monotonicity. If $W \leq W'$ pointwise then for every x and every $u \in \Gamma(x)$, $r(x,u) + \beta W(g(x,u)) \leq r(x,u) + \beta W'(g(x,u))$, and taking maxima over the same set $\Gamma(x)$ preserves the inequality: $TW \leq TW'$.

Discounting. For $\alpha \geq 0$ constant,

$$\big(T(W+\alpha)\big)(x) = \max_{u \in \Gamma(x)} \big\{ r(x,u) + \beta W(g(x,u)) + \beta\alpha \big\} = (TW)(x) + \beta\alpha,$$

the constant $\beta\alpha$ passing outside the maximum because it does not depend on u.

Theorem 6.5.1 therefore makes T a contraction of modulus β, and Theorem 6.4.1 gives (i) and the first bound in (ii); the second bound is (6.1). Continuity of V holds because $\mathcal{C}_b(X)$ consists of continuous functions and uniform limits of continuous functions are continuous. Claim (iii) is Proposition 15.2.1 applied to $W = V$, together with Proposition 13.3.1 for the single-valued case. ■

The proof separates two things that are easily conflated. Monotonicity is a property of maximisation — raising the continuation value cannot lower today's optimum — and holds regardless of discounting. Discounting is a property of $\beta < 1$ and holds regardless of the maximisation. Neither alone gives a contraction, and the failure of the second is what makes undiscounted infinite-horizon problems hard.

Remark 15.3.1 (Value function iteration). The two bounds in (15.5) play different roles. The first is *a priori*: before running anything, it says $m \geq \log\big(\varepsilon(1-\beta)/\|W_1 - W_0\|_\infty\big)/\log\beta$ iterations suffice for accuracy ε. The second is *a posteriori*: it converts the observed change $\|W_m - W_{m-1}\|_\infty$, which the algorithm computes anyway, into a certified bound on the distance to the unknown V. Stopping when $\|W_m - W_{m-1}\|_\infty < \varepsilon$, as on (Fukuda 2026b, slides 14–16), therefore guarantees an error of at most $\beta\varepsilon/(1-\beta)$, not ε — a distinction that matters when $\beta = 0.99$, where the inflation factor is 99.

Both bounds are uniform in the initial guess, so a bad W_0 costs iterations but never convergence. Both degrade as $\beta \to 1$, and this is the practical reason patient problems are computationally hard: the number of iterations needed for fixed accuracy grows like $1/(1-\beta)$ up to logarithms.

Implemented on a finite grid, W is a vector of values at nodes and the maximum in (15.2) is taken over grid points, which introduces a discretisation error that (15.5) does not control. The programs `DP_cake_eating.m` and `DP_cake_eating_CRRA.m` choose the state and control so that the transition maps grid points to grid points, avoiding interpolation entirely: consumption is read off as $c_{ij} = w_i - w_j$ from the matrix of all pairs of nodes, and the maximisation is a row-wise maximum. The Lucas program of Section 15.7 cannot do this, because the transition $y \mapsto y^\alpha r$ carries nodes off the grid, and it interpolates.

Remark 15.3.2 (Unbounded returns). Assumption 15.1.1(iii) fails for log utility near zero consumption and for CRRA with $\gamma \geq 1$. Two repairs are standard. The first bounds the *control* away from the singularity, not merely the state: compactness of the state space alone leaves $c = 0$ feasible and r unbounded below, so the restriction has to keep consumption in a compact subinterval of $(0, \infty)$, as Remark 15.6.1 does for the growth model. The second replaces the supremum norm by a weighted norm $\|W\|_\phi = \sup_x |W(x)|/\phi(x)$ for a suitable ϕ and re-establishes the contraction property in that norm; see Stokey et al. (1989, Ch. 4). The cake-eating problem of Section 15.5 is solved below by guess-and-verify because it falls outside Assumption 15.1.1, and its verification consists in checking (15.4) directly rather than invoking Theorem 15.3.1.

Theorem 15.3.2 (Weighted-norm contraction). Let S be a metric space and $\phi\colon S \to [1, \infty)$ continuous. Write

$$\mathcal{B}(S, \mathbb{R})_\phi := \Big\{ W\colon S \to \mathbb{R} \text{ continuous } : \|W\|_\phi := \sup_{x \in S} \frac{|W(x)|}{\phi(x)} < \infty \Big\}.$$

Then $(\mathcal{B}(S,\mathbb{R})_\phi, \|\cdot\|_\phi)$ is a Banach space, and an operator T on it satisfying, for some $\beta \in (0,1)$,

(i) *(monotonicity)* $W \le W'$ pointwise implies $TW \le TW'$ pointwise, and

(ii) *(weighted discounting)* $T(W + a\phi) \le TW + a\beta\phi$ pointwise for every $a \ge 0$,

is a contraction of modulus β on $\mathcal{B}(S, \mathbb{R})_\phi$ and has a unique fixed point there.

Proof. The map $W \mapsto W/\phi$ is a linear bijection from $\mathcal{B}(S, \mathbb{R})_\phi$ onto the space of bounded continuous functions on S carrying $\|\cdot\|_\phi$ to $\|\cdot\|_\infty$, so completeness is Proposition 6.3.3. For the contraction property, note that $W \le W' + \|W - W'\|_\phi\, \phi$ pointwise by the definition of $\|\cdot\|_\phi$; applying (i) and then (ii),

$$TW \le T\big(W' + \|W - W'\|_\phi\, \phi\big) \le TW' + \beta\, \|W - W'\|_\phi\, \phi.$$

Exchanging W and W' gives $|TW - TW'| \le \beta\, \|W - W'\|_\phi\, \phi$ pointwise, that is $\|TW - TW'\|_\phi \le \beta\, \|W - W'\|_\phi$. Apply Theorem 6.4.1. ∎

With $\phi \equiv 1$ this is Blackwell's criterion (Theorem 6.5.1), so the weighted version costs nothing when the return is bounded. For CRRA utility with $\gamma > 1$ on a state space $S = (0, \infty)$ the natural choice is $\phi(x) = 1 + \left|x^{1-\gamma}\right|$, which grows at the rate at which the return function diverges; verifying (ii) then amounts to a growth condition relating β to the maximal expansion rate of the state, and the details for the stochastic growth model are Stokey et al. (1989, Ch. 4). Nothing below uses the weighted norm; it is stated so that the boundedness in Assumption 15.1.1(iii) is visible as a convenience rather than a limitation of the method.

15.3.1 Policy iteration

Value function iteration converges geometrically at rate β and never terminates exactly. The alternative is to iterate on policies.

Call $h \colon X \to \mathbb{R}^k$ a *stationary policy* if it is continuous and satisfies $h(x) \in \Gamma(x)$ for every $x \in X$, and write

$$(T_h W)(x) := r\big(x, h(x)\big) + \beta\, W\big(g(x, h(x))\big), \qquad x \in X. \tag{15.6}$$

Lemma 15.3.1 (Evaluation is well posed)**.** Under Assumption 15.1.1, T_h maps $\mathcal{C}_b(X)$ into itself for every stationary policy h and is a contraction of modulus β there. Its unique fixed point W^h is the value of following h forever, and $W^h \le V$.

Proof. Continuity of h and of r, g makes $x \mapsto r(x, h(x))$ and $x \mapsto g(x, h(x))$ continuous, and r is bounded, so $T_h W \in \mathcal{C}_b(X)$ whenever $W \in \mathcal{C}_b(X)$. Monotonicity is immediate from (15.6) and $\beta > 0$, and $T_h(W + \alpha) = T_h W + \beta\alpha$ for constant $\alpha \ge 0$, so Theorem 6.5.1 applies. Unrolling (15.6) along the trajectory generated by h identifies the fixed point with $\sum_{t \ge 0} \beta^t r(x_t, h(x_t))$, one feasible plan, whence $W^h \le V$ by (15.4). ∎

Definition 15.3.1 (Policy iteration)**.** Under Assumption 15.1.1, fix a stationary policy h_0. Given h_m, the algorithm performs

(i) *policy evaluation*: set $W_m := W^{h_m}$, the unique element of $\mathcal{C}_b(X)$ with

$$W_m(x) = r\big(x, h_m(x)\big) + \beta\, W_m\big(g(x, h_m(x))\big) \qquad \text{for every } x \in X; \tag{15.7}$$

(ii) *policy improvement*: choose a stationary policy h_{m+1} with

$$h_{m+1}(x) \in H_{W_m}(x) = \operatorname*{arg\,max}_{u \in \Gamma(x)} \big\{ r(x, u) + \beta W_m\big(g(x, u)\big) \big\} \qquad \text{for every } x \in X, \tag{15.8}$$

retaining $h_{m+1}(x) = h_m(x)$ at any x where $h_m(x) \in H_{W_m}(x)$.

Step (i) is Lemma 15.3.1. In step (ii) the correspondence H_{W_m} is non-empty- and compact-valued and upper hemicontinuous by Proposition 15.2.1 applied to $W = W_m$, which is what Berge's theorem delivers. That is not the same as the existence of the continuous selection h_{m+1} the definition asks for, and the gap is real: an upper hemicontinuous compact-valued correspondence need not admit any continuous selection. Two situations close it. If $H_{W_m}(x)$ is a singleton at every x, the correspondence is a continuous function by Proposition 13.3.1, and step (ii) is automatic; this is the case in every application below, where strict concavity of r in u makes the maximiser unique. If X and each $\Gamma(x)$ are finite, continuity is vacuous and any selection will do.

Remark 15.3.3 (Selection when uniqueness fails)**.** Outside those two cases one weakens the requirement from a continuous to a measurable selection, and Berge's theorem does not supply it. What does is the measurable maximum theorem: if Γ is a weakly measurable correspondence with non-empty compact values from a measurable space into a Polish space and $f(x,\cdot)$ is continuous for each x with $f(\cdot,u)$ measurable for each u, then $x \mapsto \arg\max_{u\in\Gamma(x)} f(x,u)$ is measurable with non-empty compact values and admits a measurable selector. The statement and its proof — which rests on the Kuratowski–Ryll-Nardzewski selection theorem — are Aliprantis and Border (2006, Thm. 18.19). Bounded measurable functions are then the right space for T, as in Remark 15.2.1; nothing in this chapter needs that generality, and it is recorded so that the appeal to Berge in (15.8) is not mistaken for a selection theorem.

Theorem 15.3.3 (Policy iteration improves monotonically)**.** Under Assumption 15.1.1, suppose the selections in (15.8) exist, so that (h_m) is a sequence of stationary policies with values $W_m = W^{h_m}$. Then

(i) $W_0 \le W_1 \le \cdots \le V$ and $\|W_{m+1} - V\|_\infty \le \beta \|W_m - V\|_\infty$;

(ii) $W_m \ge T^m W_0$ for every m, so the iterates dominate those of value function iteration started at the same W_0;

(iii) if X and each $\Gamma(x)$ are finite, then $W_m = V$ for all m beyond some finite index, and the tie-breaking rule of Definition 15.3.1 makes (h_m) constant from that index on.

Proof. Improvement. By (15.8) and (15.7),

$$T_{h_{m+1}} W_m = T W_m \ge T_{h_m} W_m = W_m,$$

where T is the Bellman operator (15.2). Applying the monotone operator $T_{h_{m+1}}$ repeatedly preserves the inequality, so $T^k_{h_{m+1}} W_m$ is non-decreasing in k; letting $k \to \infty$ and using that $T_{h_{m+1}}$ is a contraction with fixed point W_{m+1} gives $W_{m+1} \ge W_m$.

Domination by V. This is Lemma 15.3.1.

Rate. The same iteration gives $W_{m+1} \ge T W_m$. With $TV = V$ and monotonicity of T,

$$0 \le V - W_{m+1} \le V - T W_m = TV - T W_m \le \beta \|V - W_m\|_\infty ,$$

which is (i).

(ii). Induction. At $m = 0$ the claim reads $W_0 \ge W_0$. If $W_m \ge T^m W_0$, then monotonicity of T and $W_{m+1} \ge T W_m$ give $W_{m+1} \ge T W_m \ge T^{m+1} W_0$.

(iii). With X and the constraint sets finite there are finitely many policies, hence finitely many possible values W^h. The sequence (W_m) is non-decreasing and takes values in that finite set, so it is eventually constant, say $W_{m+1} = W_m$ from some index on. Then W_m is the fixed point of $T_{h_{m+1}}$, so $T_{h_{m+1}} W_m = W_m$; but $T_{h_{m+1}} W_m = T W_m$ by (15.8), so $T W_m = W_m$ and $W_m = V$ by uniqueness of the fixed point. Once $W_m = V$ every $h_m(x)$ already lies in $H_V(x)$, because $T_{h_m} W_m = W_m = T W_m$ means $h_m(x)$ attains the maximum at each x; the tie-breaking rule then returns $h_{m+1} = h_m$. Without that rule the values still reach V in finitely many steps, but the policy sequence may keep moving among the maximisers of H_V, all of which have value V. ∎

Remark 15.3.4 (Which algorithm to use)**.** The rate bound in Theorem 15.3.3(i) carries the same β as (15.5), so on that evidence alone the two algorithms are only tied. What separates them is Theorem 15.3.3(ii): started from the value W_0 of a stationary policy, the policy iterates dominate the value iterates from the same W_0 at every step, and both lie below V, so policy iteration is at no step further from V. The comparison needs W_0 to be a policy value; it says nothing about value function iteration started above V. In practice policy iteration converges in a number of steps that grows very slowly with the size of the state space, at the cost of solving (15.7) exactly at each step. On a finite grid with N nodes that solve is the linear system $(I - \beta P_h)W = r_h$, where P_h is the

$N \times N$ transition matrix induced by h; it is non-singular because $\rho(\beta P_h) = \beta < 1$ by Corollary 2.7.1, and Corollary 2.7.2 gives $(I - \beta P_h)^{-1} = \sum_k \beta^k P_h^k \geq 0$, the matrix of discounted occupation times. Value function iteration costs $O(N^2)$ per sweep and no linear solve; policy iteration costs a solve of size N. The programs accompanying the course all use value function iteration, which is why their iteration counts run to the hundreds. Figure 15.1 shows the gap on a random problem with 40 states, 8 actions and $\beta = 0.95$: policy iteration reaches the optimum to machine precision at the second step, while value function iteration started from the same W_0 is still 0.78 away after 45 sweeps, tracking the envelope $\beta^m \|W_0 - V\|_\infty$ of (15.5) throughout.

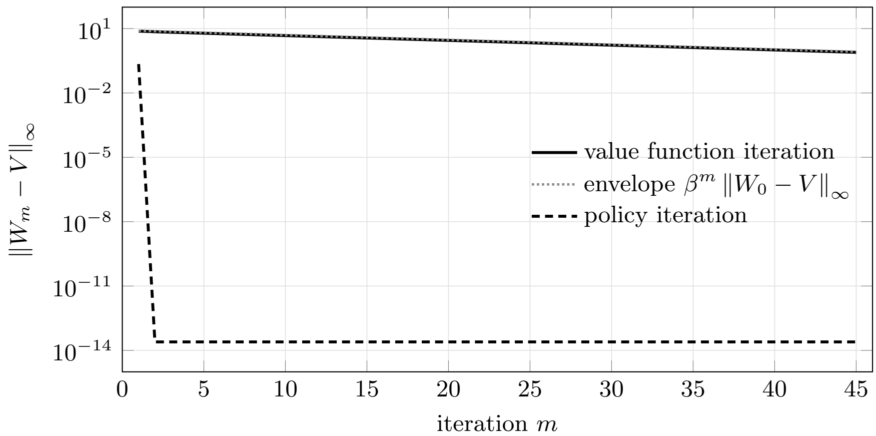


Figure 15.1: Distance to the optimal value function under the two algorithms, started from the value W_0 of the same stationary policy on a randomly generated Markov decision problem with 40 states, 8 actions and $\beta = 0.95$. The vertical axis is logarithmic. The dashed curve lies below the solid one at every step, which is Theorem 15.3.3(ii); it flattens at 2.5×10^{-14} because the linear solve in (15.7) returns the exact value of the optimal policy up to rounding, so the remaining error is the conditioning of $I - \beta P_h$ and not the contraction.

15.4 First-Order Conditions and the Envelope Formula

Contraction gives existence, uniqueness and an algorithm, but no characterisation. Characterisation comes from differentiating the Bellman equation, and that requires knowing that V is differentiable — which is a conclusion, not an assumption, since V is defined as a supremum.

Lemma 15.4.1 (Supergradients of a concave function)**.** Let $V : X \to \mathbb{R}$ be concave on a convex $X \subseteq \mathbb{R}^n$ and let $x_0 \in \operatorname{Int} X$. Then there exists $p \in \mathbb{R}^n$ with

$$V(x) \leq V(x_0) + \langle p, x - x_0 \rangle \qquad \text{for every } x \in X, \tag{15.9}$$

and V is differentiable at x_0 if and only if such a p is unique, in which case $p = \nabla V(x_0)$.

Proof. The hypograph $\operatorname{hyp} V = \{(x,t) \in X \times \mathbb{R} : t \leq V(x)\}$ is convex, by the concave analogue of Proposition 9.3.1, and $(x_0, V(x_0))$ is a boundary point of it. Theorem 9.4.2 supplies $(a, b) \neq 0$ with $\langle a, x \rangle + bt \leq \langle a, x_0 \rangle + bV(x_0)$ for all $(x,t) \in \operatorname{hyp} V$. Letting $t \to -\infty$ forces $b \geq 0$; and $b = 0$ would give $\langle a, x - x_0 \rangle \leq 0$ for all $x \in X$ with $a \neq 0$, impossible for x_0 interior. Dividing by $b > 0$ and setting $p := -a/b$ gives (15.9) on taking $t = V(x)$. The equivalence of uniqueness with differentiability is Rockafellar (1970, Thm. 25.1). ■

Theorem 15.4.1 (Benveniste–Scheinkman)**.** Let $X \subseteq \mathbb{R}^n$ be convex, $V : X \to \mathbb{R}$ concave, and $x_0 \in \operatorname{Int} X$. Suppose there is a neighbourhood $D \subseteq X$ of x_0 and a concave function $W : D \to \mathbb{R}$, differentiable at x_0, with

$$W(x_0) = V(x_0) \qquad \text{and} \qquad W(x) \leq V(x) \;\; \text{for all } x \in D. \tag{15.10}$$

Then V is differentiable at x_0 and $\nabla V(x_0) = \nabla W(x_0)$.

Proof. Let p satisfy (15.9) for V at x_0, which Lemma 15.4.1 provides. For $x \in D$,

$$W(x) \leq V(x) \leq V(x_0) + \langle p, x - x_0 \rangle = W(x_0) + \langle p, x - x_0 \rangle\,,$$

so p is a supergradient of W at x_0. But W is differentiable at x_0, so by Lemma 15.4.1 its supergradient there is unique and equals $\nabla W(x_0)$; hence $p = \nabla W(x_0)$. Since p was an arbitrary supergradient of V, the supergradient of V at x_0 is unique, and Lemma 15.4.1 makes V differentiable there with $\nabla V(x_0) = \nabla W(x_0)$. ∎

This is Benveniste and Scheinkman (1979). Applying it requires producing a W that is concave on a neighbourhood of x_0, differentiable at x_0, equal to V there and dominated by V nearby. In a dynamic programme the candidate is the value of freezing the control at its optimal value and letting the state vary, and whether that candidate is admissible depends on how the state enters the transition. The next theorem is therefore stated for problems written with the next state as the control, which is the form every application in this chapter takes.

Assumption 15.4.1 (Next state as control). $X \subseteq \mathbb{R}$ is an interval, the control is the next state, so that $g(x, u) = u$ and $\Gamma(x) \subseteq X$, and:

(i) r is continuously differentiable on $\operatorname{gr}\Gamma$ and $r(\cdot, u)$ is concave for each u;

(ii) the optimal policy h is single-valued and $h(x) \in \operatorname{Int}\Gamma(x)$ for $x \in \operatorname{Int} X$;

(iii) Γ expands locally at every $x \in \operatorname{Int} X$: there is a neighbourhood D of x with $h(x) \in \Gamma(z)$ for every $z \in D$.

Hypothesis (iii) is what makes the frozen control feasible at neighbouring states, and it holds in each application below: in the cake problem $\Gamma(w) = [0, w]$ and $h(w) = \beta w < w$; in the growth model $\Gamma(k) = [0, f(k) + (1 - \delta)k]$ and $h(k)$ is interior.

Theorem 15.4.2 (Euler equation and envelope formula). Let Assumption 15.1.1 and Assumption 15.4.1 hold and let V be concave. Then at every $x \in \operatorname{Int} X$ the optimal control $u^* = h(x)$ satisfies the *first-order condition*

$$\frac{\partial r}{\partial u}(x, u^*) + \beta V'(u^*) = 0, \tag{15.11}$$

and V is differentiable at x with the *envelope formula*

$$V'(x) = \frac{\partial r}{\partial x}(x, u^*). \tag{15.12}$$

Proof. Differentiability and (15.12). Fix $x \in \operatorname{Int} X$, write $u^* = h(x)$, and let D be the neighbourhood supplied by Assumption 15.4.1(iii). Define

$$W(z) := r(z, u^*) + \beta V(u^*), \qquad z \in D.$$

The second term is a constant, so W is concave on D by Assumption 15.4.1(i) and differentiable at x with $W'(x) = r_x(x, u^*)$; here is where writing the control as the next state pays, since the continuation value does not depend on z and no differentiability of V is presupposed. Feasibility of u^* at z gives $W(z) \leq V(z)$ on D, and optimality of u^* at x gives $W(x) = V(x)$. Theorem 15.4.1 now yields differentiability of V at x and $V'(x) = W'(x) = r_x(x, u^*)$.

The first-order condition. The preceding paragraph applies at every interior point, in particular at $u^* \in \operatorname{Int}\Gamma(x) \subseteq \operatorname{Int} X$, so V is differentiable at u^*. The map $u \mapsto r(x, u) + \beta V(u)$ is therefore differentiable at its interior maximiser u^*, and Theorem 7.2.1 gives (15.11). ∎

Remark 15.4.1 (The general transition). When the transition is a genuine function $g(x, u)$ of both arguments, the same construction gives $W(z) = r(z, u^*) + \beta V\big(g(z, u^*)\big)$, whose differentiability at x requires V to be differentiable at $g(x, u^*)$ — which is what was to be proved. The circularity is not an artefact: differentiability of V must then be obtained from the whole optimal plan rather than from one period, by taking

$$W(z) := \sum_{t\geq 0} \beta^t r(z_t, u_t^*), \qquad z_0 = z, \quad z_{t+1} = g(z_t, u_t^*),$$

with (u_t^*) the optimal control sequence from x. This W is concave whenever each $g(\cdot, u)$ is affine and each $r(\cdot, u)$ concave, and differentiable at x under a bound $\sup|g_x| \leq G$ with $\beta G < 1$ that makes the differentiated series converge. Every problem treated below is already in the form of Assumption 15.4.1, and any problem with $g(x, \cdot)$ invertible can be put in that form by taking the next state as the control, so the general construction is not needed here. It is set out in Stokey et al. (1989, Sec. 4.3).

Remark 15.4.2 (The derivation by differentiating the policy). The derivation on (Fukuda 2026b, slide 18) differentiates $V(x) = r(x, h(x)) + \beta V(h(x))$ directly, obtaining

$$V'(x) = \underbrace{\Big[r_u\big(x, h(x)\big) + \beta V'\big(h(x)\big)\Big]}_{=0 \text{ by } (15.11)} h'(x) + r_x\big(x, h(x)\big),$$

so that the term in h' drops out and (15.12) remains. The computation is right in content and presupposes what is at issue: that V and h are differentiable. Theorem 15.4.1 establishes the conclusion without differentiating h at all, and therefore without assuming the policy is smooth — which it need not be, since Theorem 15.3.1 delivers only upper hemicontinuity.

Corollary 15.4.1 (Euler equation for a depletable stock). Let Assumption 15.1.1 and Assumption 15.4.1 hold with $X = [0, \infty)$, $\Gamma(x) = [0, x]$ and $r(x, u) = \tilde{r}(x - u)$ for a continuously differentiable, strictly increasing and concave $\tilde{r}$, so that the state is a stock, the control is the stock carried forward and $c := x - u$ is the amount consumed. Let V be concave. Then along an interior optimal path

$$\tilde{r}'(c_t) = \beta\, \tilde{r}'(c_{t+1}) \qquad \text{for every } t. \tag{15.13}$$

Proof. Here $r_u(x, u) = -\tilde{r}'(x - u)$ and $r_x(x, u) = \tilde{r}'(x - u)$, so (15.11) reads $\tilde{r}'(c_t) = \beta V'(x_{t+1})$ and (15.12) reads $V'(x_t) = \tilde{r}'(c_t)$. Substituting the envelope formula at $t + 1$ into the first-order condition at t,

$$\tilde{r}'(c_t) = \beta V'(x_{t+1}) \overset{\text{envelope}}{=} \beta\, \tilde{r}'(c_{t+1}),$$

which is (15.13). ■

The chain in the proof — first-order condition, envelope, first-order condition — is the standard route from a Bellman equation to an Euler equation, and it eliminates the unknown V' entirely: the Euler equation involves only primitives (Fukuda 2026b, slide 32). Its economic reading is a no-arbitrage condition along the optimal path. Consuming Δ less today costs $\tilde{r}'(c_t)\Delta$ in current utility and returns $\beta\tilde{r}'(c_{t+1})\Delta$ in discounted utility tomorrow; at an optimum the two are equal, and the equality is precisely the first-order condition of the perturbed problem in Δ at $\Delta = 0$.

15.5 The Cake-Eating Problem

Problem. An agent endowed with a cake of size $w_0 > 0$ chooses consumption $(c_t)_{t\geq 0}$ to maximise $\sum_{t\geq 0} \beta^t \log c_t$ subject to $w_{t+1} = w_t - c_t$ and $c_t \geq 0$, $w_t \geq 0$.

There are two ways to assign the state and the control, and they are not computationally equivalent (Fukuda 2026b, slides 21–22). Taking the control to be consumption gives $V(w) = \max_{c\in[0,w]}\{\log c + \beta V(w - c)\}$, in which $w - c$ generally falls between grid points and must be interpolated. Taking the control to be *tomorrow's cake* gives

$$V(w) = \max_{w'\in[0,w]} \big\{\log(w - w') + \beta V(w')\big\}, \tag{15.14}$$

in which the transition is the identity on the grid. The second formulation is the one implemented.

Proposition 15.5.1 (Closed form). The cake-eating problem has value function and policy

$$V(w) = \frac{\log(1-\beta)}{1-\beta} + \frac{\beta\log\beta}{(1-\beta)^2} + \frac{\log w}{1-\beta}, \qquad h(w) = (1-\beta)w, \tag{15.15}$$

so that $c_t = (1-\beta)\beta^t w_0$ and $w_{t+1} = \beta w_t$.

Proof. The first two steps construct a candidate; the third proves it is the value function, and neither of the first two is used in that proof.

From the Euler equation. Corollary 15.4.1 with $\tilde{r} = \log$ gives $1/c_t = \beta/c_{t+1}$, that is $c_{t+1} = \beta c_t$, hence $c_\tau = \beta^{\tau-t}c_t$. Imposing exhaustion, $w_t = \sum_{\tau\ge t} c_\tau$ — provisionally, since $\lim_t w_t = 0$ is at this stage a conjecture about the optimal path rather than a consequence of anything proved — gives

$$w_t = \sum_{\tau=t}^{\infty} \beta^{\tau-t}c_t = \frac{c_t}{1-\beta}, \qquad \text{so} \qquad c_t = (1-\beta)w_t,$$

which is the stated policy, and then $w_{t+1} = w_t - c_t = \beta w_t$ and $c_t = (1-\beta)\beta^t w_0$.

Value. Substituting the optimal path into the objective,

$$\begin{aligned} V(w_0) &= \sum_{t=0}^{\infty} \beta^t \log\big((1-\beta)\beta^t w_0\big) = \log(1-\beta)\sum_{t\ge0}\beta^t + \log\beta\sum_{t\ge0} t\beta^t + \log w_0 \sum_{t\ge0}\beta^t \\ &= \frac{\log(1-\beta)}{1-\beta} + \frac{\beta\log\beta}{(1-\beta)^2} + \frac{\log w_0}{1-\beta}, \end{aligned}$$

using $\sum_{t\ge0}\beta^t = 1/(1-\beta)$ and

$$\sum_{t\ge0} t\beta^t = \beta\sum_{t\ge0} t\beta^{t-1} = \beta\frac{\mathrm{d}}{\mathrm{d}\beta}\sum_{t\ge0}\beta^t = \beta\frac{\mathrm{d}}{\mathrm{d}\beta}\frac{1}{1-\beta} = \frac{\beta}{(1-\beta)^2},$$

the term-by-term differentiation being legitimate on $|\beta| < 1$, where the power series converges uniformly on compact subsets.

Verification. Write $V(w) = A + B\log w$ with $B = 1/(1-\beta)$ and A as in (15.15). The objective in (15.14) is strictly concave in w' on $(0, w)$ and tends to $-\infty$ at both endpoints, so its unique maximiser is interior and solves $1/(w - w') = \beta B/w'$, that is $w' = \beta Bw/(1+\beta B) = \beta w$ since $\beta B = \beta/(1-\beta)$ and $1 + \beta B = 1/(1-\beta)$. Substituting back,

$$\log\big((1-\beta)w\big) + \beta\big(A + B\log(\beta w)\big) = \underbrace{\log(1-\beta) + \beta A + \beta B\log\beta}_{=A} + \underbrace{(1+\beta B)}_{=B}\log w,$$

the first bracket equalling A because $A(1-\beta) = \log(1-\beta) + \beta\log\beta/(1-\beta)$ by (15.15). Hence $TV = V$. Optimality of the associated plan follows from (15.4): along it $w_m = \beta^m w_0$, so $\beta^m V(w_m) = \beta^m(A + B\log w_0) + mB\beta^m\log\beta \to 0$, and the transversality condition of Remark 15.2.2 holds even though Assumption 15.1.1(iii) fails. ∎

The eaten fraction $1-\beta$ is independent of the cake size — a consequence of the homotheticity of log utility, under which the problem scales — and independent of time, which is stationarity. The value function inherits the logarithm from the return function, which is why the guess $A + B\log w$ works and why it would fail for a return function outside the CRRA class.

Remark 15.5.1 (What the numerical solution adds). `DP_cake_eating.m` solves (15.14) by value function iteration on a uniform grid of 2000 points from 0.001 to $w_0 = 1$ with $\beta = 0.85$, starting from $W_0 \equiv 0$ and stopping when successive iterates differ by less than 10^{-16} in the supremum norm (Fukuda 2026b, slides 23–30); `DP_cake_eating_CRRA.m` repeats it for CRRA utility and `DP_cake_eating_plot_iterations.m` displays the iterates W_m converging to V. Run as supplied on MATLAB R2025b, the first reports convergence after 279 iterations in 1.04 seconds, the second after 199, the third in 0.11 seconds. The lower endpoint is not zero because $\log 0 = -\infty$; the truncation discretises the state space rather than approximating the economics. Because (15.15) is available in closed form, the whole discrepancy is numerical error and can be located.

Measured against (15.15), the computed value function is off by 12.06 at $w = 0.001$, by at most 0.33 on $w \ge 0.1$ and by 0.033 at $w = 1$: the error lives at the bottom of the grid, where the truncation bites and log is unbounded. The computed policy differs from $(1-\beta)w$ by at most 0.00122 at every node above the first, which is under three grid spacings of 5.0×10^{-4}. Figure 15.2 plots both errors.

At the first node the reported consumption is -0.999, an artefact of the line

```
consumption(consumption<=0)=0.1^16.
```

At $w=0.001$ no grid choice of w' leaves positive consumption, so every entry of that row of the consumption matrix is the sentinel 10^{-16}; the entries of $\log c+\beta W$ then differ only through $\beta W(w')$, which is largest at $w'=1$, and the row maximum is decided by the continuation value alone. The plotted range $[0, 1.05(1-\beta)]$ keeps this out of the figure.

Replacing the sentinel by $-\infty$ removes that artefact and creates a worse one. The lowest node then has value $-\infty$; the second node's only feasible continuation is the lowest node, so it too becomes $-\infty$, and by induction every node does. The truncated state space $[0.001, 1]$ is not invariant under the optimal policy $w'=\beta w$, and a grid bounded away from zero cannot represent the tail of the optimal path — which is what the program's own comment about `num_period` records. The repair has to supply a boundary condition as well: set the infeasible cells to $-\infty$ and hold the lowest node at the analytic value $V(0.001)=-64.8388$ from (15.15). Run that way in MATLAB R2025b, the recursion converges in 51 iterations rather than 279, reports consumption 0 at the lowest node, and improves every error measure: the value error is 0.0238 rather than 0.325 on $w\geq0.1$ and 0.00236 rather than 0.0335 at $w=1$, and the policy error is at most 5.75×10^{-4} rather than 1.22×10^{-3}, that is one grid spacing rather than three. The dashed curves of Figure 15.2 are this variant.

The four sources of error in (15.14) separate cleanly, which is why the exercise is instructive:

- *iteration error*, bounded by (15.5), and measurable here because the analytic V is known — the observed ratio of successive errors is $\exp(-0.16252)=\beta$ to five decimals;
- *discretisation error*, from restricting w' to the grid, of order the spacing 5.0×10^{-4}, which is what the policy error attains;
- *truncation error*, from replacing the state space $(0,1]$ by $[0.001,1]$, which is what the boundary condition addresses and which is the largest term at the bottom of the grid;
- *interpolation and quadrature error*, both zero here, since the choice set is the grid itself and the problem is deterministic.

In `Lucas_Tree_DP.m` the last item is not zero: the conditional expectation is a 500-draw Monte Carlo average and the continuation value is a spline interpolant, and the residual 0.0103% against the closed-form price $\beta y/(1-\beta)$ is dominated by those two.

The stopping rule is also not the one analysed in Remark 15.3.1. Here $\|W_1-W_0\|_\infty=36.84$, so (15.5) gives $\|W_m-W_{m-1}\|_\infty<10^{-16}$ by $m=261$. The computed value function reaches -52.78 at the lowest node, where consecutive doubles are 7.1×10^{-15} apart — far above the tolerance. From about iteration 240 the computed increment is quantised at one unit in the last place, and the test can succeed only when the iteration lands on an exact floating-point fixed point. Reimplementing the same recursion with the continuation values added by broadcasting rather than by the matrix product `beta*ones(dimW,1)*V` terminates at 255 iterations instead of 279. A tolerance below the machine resolution of the value scale gives the termination decision to the rounding rather than to the contraction modulus; a tolerance of, say, 10^{-10} would be governed by (15.5) and would stop at a certified accuracy.

15.6 The Ramsey Growth Model

Problem. A planner chooses consumption to maximise $\sum_{t\geq0}\beta^t u(c_t)$ subject to

$$k_{t+1}=f(k_t)+(1-\delta)k_t-c_t,\qquad f(k)=Ak^\alpha,\ \alpha\in(0,1),\ A>0,\ \delta\in(0,1], \tag{15.16}$$

with $k_0>0$ given, $c_t\geq0$ and $k_t\geq0$; u is strictly increasing, strictly concave and continuously differentiable on $(0,\infty)$. In recursive form, writing $\varphi(k):=f(k)+(1-\delta)k$ for the resources available at k,

$$V(k)=\max_{k'\in[0,\varphi(k)]}\big\{u\big(\varphi(k)-k'\big)+\beta V(k')\big\}. \tag{15.17}$$

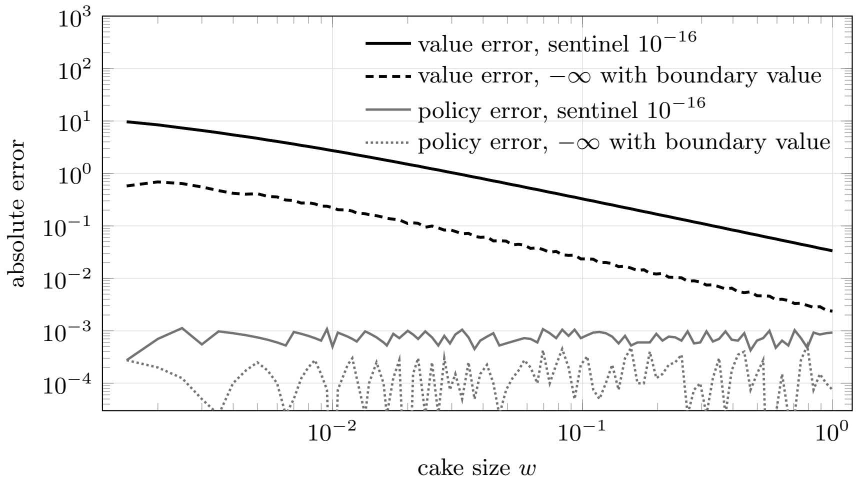


Figure 15.2: Absolute error of the converged value function iteration in `DP_cake_eating.m` against the closed form (15.15), and of the induced policy against $(1-\beta)w$, on the program's own grid with $\beta = 0.85$. Solid and light-solid curves use the sentinel 10^{-16} of the supplied code; dashed and dotted curves use $-\infty$ on infeasible pairs together with the analytic value at the lowest node. Both axes are logarithmic and the first node is omitted. The error in the value function is a boundary phenomenon under either encoding, falling by more than two orders of magnitude between $w = 0.0015$ and $w = 1$; the boundary condition lowers it by a further factor of about fourteen at the top of the grid, and halves the policy error to one grid spacing.

The theory of Section 15.3 does not apply to (15.17) as written: the state space $[0,\infty)$ is unbounded and u need not be bounded on the feasible consumption set, so Assumption 15.1.1 fails on both counts. Both are repaired before the Euler equation is derived.

Lemma 15.6.1 (Standing hypotheses for the growth model)**.** Put $k_g := (A/\delta)^{1/(1-\alpha)}$, the unique positive solution of $f(k) = \delta k$, and $\bar{k} := \max\{k_0, k_g\}$. On $X := [0,\bar{k}]$ with $\Gamma(k) := [0,\varphi(k)]$ and $g(k,k') := k'$:

(i) X is invariant, that is $\Gamma(k) \subseteq X$ for every $k \in X$;

(ii) Γ is non-empty- and compact-valued and continuous, and g is continuous;

(iii) if u extends to a real-valued continuous function on $[0,\infty)$, then $r(k,k') := u(\varphi(k) - k')$ is continuous and bounded on $\operatorname{gr}\Gamma$.

Under (iii), Assumption 15.1.1 holds on X for every $\beta \in (0,1)$.

Proof. (i) φ is strictly increasing, so $\max_{k\in X}\varphi(k) = \varphi(\bar{k})$, and $\varphi(\bar{k}) \le \bar{k}$ is equivalent to $A\bar{k}^\alpha \le \delta\bar{k}$, that is to $\bar{k} \ge k_g$, which holds by construction. Hence $\Gamma(k) \subseteq [0,\varphi(\bar{k})] \subseteq [0,\bar{k}] = X$.

(ii) $\Gamma(k)$ contains 0 and is a compact interval. For upper hemicontinuity, if $k_j \to k$ and $u_j \in \Gamma(k_j)$ with $u_j \to u$, then $0 \le u \le \lim\varphi(k_j) = \varphi(k)$ by continuity of φ. For lower hemicontinuity, given $u \in \Gamma(k)$ and $k_j \to k$, set $u_j := \min\{u, \varphi(k_j)\} \in \Gamma(k_j)$; then $u_j \to u$, again by continuity of φ. Continuity of g is immediate.

(iii) On $\operatorname{gr}\Gamma$ the consumption $c = \varphi(k) - k'$ ranges over the compact interval $[0,\varphi(\bar{k})]$, on which the continuous u is bounded by Corollary 5.4.1; and r is continuous as a composition of continuous maps. ■

Remark 15.6.1 (Utility unbounded at zero consumption)**.** Hypothesis (iii) is where $u = \log$ and CRRA with $\gamma \ge 1$ fail. Zero consumption is feasible at every k, and $u(0^+) = -\infty$, so r is unbounded below and Assumption 15.1.1(iii) holds on no state space that permits $c = 0$: the failure is in the control, not in the state, and confining k to a compact interval does not remove it. Two repairs are available.

A consumption floor. Fix $k_{\min} \in (0, k_g)$, so that $\varphi(k_{\min}) > k_{\min}$, and $\varepsilon \in (0, \varphi(k_{\min}) - k_{\min})$. On $X_\varepsilon := [k_{\min},\bar{k}]$ with $\Gamma_\varepsilon(k) := [k_{\min}, \varphi(k) - \varepsilon]$ the argument of Lemma 15.6.1 goes through unchanged, and consumption now ranges over $[\varepsilon, \varphi(\bar{k}) - k_{\min}]$, a compact subinterval of $(0, \infty)$ on which u is bounded. This is a different problem, and whether it has the same solution has to be checked rather than assumed. For $u = \log$ and $\delta = 1$ it can be: the closed-form policy $k' = \alpha\beta A k^\alpha$ leaves consumption $c = (1-\alpha\beta)Ak^\alpha \geq (1-\alpha\beta)Ak_{\min}^\alpha$, so any ε below that bound leaves the floor slack and the optimum unchanged.

A weighted norm. Alternatively keep Γ and replace the supremum norm by $\|\cdot\|_\phi$ of Theorem 15.3.2 with a ϕ growing at the rate at which u diverges. Verifying the weighted discounting condition (ii) of that theorem for the growth model is carried out in Stokey et al. (1989, Ch. 4) and is not reproduced here.

Proposition 15.6.1 (Euler equation and steady state). Assume Assumption 15.1.1 holds on the state space X, as it does under Lemma 15.6.1(iii) or under the consumption floor of Remark 15.6.1, and assume the optimal policy h is single-valued with $h(k) \in \operatorname{Int}\Gamma(k)$ for every $k \in \operatorname{Int} X$, so that consumption is strictly positive along the optimal path. Then

$$u'(c_t) = \beta\, u'(c_{t+1})\big(f'(k_{t+1}) + 1 - \delta\big), \tag{15.18}$$

and the unique interior steady state (k^*, c^*) with $c_t \equiv c^*$, $k_t \equiv k^*$ is

$$k^* = \left(\frac{\alpha A}{\beta^{-1} - (1-\delta)}\right)^{\frac{1}{1-\alpha}}, \qquad c^* = A(k^*)^\alpha - \delta k^*. \tag{15.19}$$

Proof. Apply Theorem 15.4.2 with state k, control k' and $r(k,k') = u(\varphi(k) - k')$, which is of the form required by Assumption 15.4.1: $r(\cdot, k')$ is concave as the composition of the concave increasing u with the concave $\varphi(\cdot) - k'$; part (ii) of that assumption is the interiority hypothesis; and Γ expands locally at every $k \in \operatorname{Int} X$ because φ is increasing and continuous, so $h(k) < \varphi(k)$ implies $h(k) < \varphi(z)$ for all z in a neighbourhood of k. Then $r_{k'} = -u'(c)$, so (15.11) gives

$$u'(c_t) = \beta V'(k_{t+1}); \tag{15.20}$$

and $r_k = u'(c)\big(f'(k) + 1 - \delta\big)$, so (15.12) gives

$$V'(k_t) = u'(c_t)\big(f'(k_t) + 1 - \delta\big). \tag{15.21}$$

Substituting (15.21) at $t+1$ into (15.20) yields (15.18). Concavity of V, required by Theorem 15.4.2, follows from concavity of u and of the constraint set: Proposition 13.5.1 applies to each iterate $T^m W_0$ started from a concave W_0, and Theorem 15.3.1 makes the uniform limit V concave, uniform limits preserving the inequality in Definition 9.3.1.

At a steady state, $c_t = c_{t+1}$ cancels u' from (15.18), leaving $1 = \beta\big(f'(k^*) + 1 - \delta\big)$, that is $\alpha A(k^*)^{\alpha-1} = \beta^{-1} - (1-\delta)$, which rearranges to the first half of (15.19); the exponent $1/(1-\alpha)$ is positive and the right-hand side is positive because $\beta^{-1} > 1 \geq 1 - \delta$, so k^* exists and is unique. Setting $k_{t+1} = k_t = k^*$ in (15.16) gives $c^* = f(k^*) - \delta k^*$. ∎

The steady state lies inside the state space of Lemma 15.6.1: $k^* < k_g$ is equivalent to $\alpha A/R < A/\delta$, that is to $\alpha\delta < \beta^{-1} - 1 + \delta$, which holds because $\beta < 1$ and $\alpha < 1$. Interiority of the policy, assumed in Proposition 15.6.1, is delivered by the Inada conditions $u'(0^+) = +\infty$ and $f'(0^+) = +\infty$ together with $k_0 > 0$: with infinite marginal utility at zero consumption no optimal plan sets $c_t = 0$, and with infinite marginal product at zero capital no optimal plan sets $k_{t+1} = 0$. The argument is in Stokey et al. (1989, Ch. 5) and is not reproduced here, since it requires the monotonicity and differentiability results for V that Theorem 15.4.2 presupposes.

Equation (15.18) is the intertemporal condition of the one-sector growth model: the marginal utility cost of investing a unit today equals its discounted marginal utility return, the return being the marginal product of capital plus the undepreciated remainder. The steady-state condition $\beta\big(f'(k^*) + 1 - \delta\big) = 1$ is the modified golden rule: the net marginal product $f'(k^*) - \delta$ equals the rate of time preference $\beta^{-1} - 1$. Capital is therefore below the golden-rule level that maximises steady-state consumption, at which $f'(k) = \delta$, and the wedge is impatience.

For comparative statics write $R := \beta^{-1} - (1-\delta) > 0$, so that $k^* = (\alpha A/R)^{1/(1-\alpha)}$ and

$$\log k^* = \frac{1}{1-\alpha}\big(\log\alpha + \log A - \log R\big).$$

Since R is decreasing in β and increasing in δ, and α is held fixed in both cases, k^* is increasing in β, increasing in A with elasticity $1/(1-\alpha)$, and decreasing in δ. The parameter α is different, because it enters both the level and the exponent:

$$\frac{\partial \log k^*}{\partial \alpha} = \frac{1}{(1-\alpha)^2} \log \frac{\alpha A}{R} + \frac{1}{\alpha(1-\alpha)}. \tag{15.22}$$

The second term is positive and the first has the sign of $\alpha A - R$, so k^* is increasing in α whenever $\alpha A \geq R$, which is the empirically relevant configuration — with $A = 1$, $\alpha = 0.3$, $\delta = 0.1$ and $\beta = 0.95$ one has $R \approx 0.153$ and $\alpha A/R \approx 1.97$, giving $\partial \log k^*/\partial\alpha \approx 6.14$. It is not a general property. Taking $A = 0.01$ with the same α, δ, β gives $\alpha A/R \approx 0.020$ and $\partial \log k^*/\partial\alpha \approx -3.26$, so raising the capital share *lowers* the steady-state stock. The reason is dimensional: k^* is measured in units in which A is a pure number, and when A is small the steady-state capital stock is below one, so a larger exponent $1/(1-\alpha)$ pushes it further down. Comparative statics in α are therefore statements about a particular normalisation of units, not about technology alone.

Unlike Section 15.5 the model has no closed form for general u and $\delta < 1$; the exception is $\delta = 1$ with $u = \log$, where the policy is $k' = \alpha\beta\big(Ak^\alpha\big)$. This is why the numerical treatment of (Fukuda 2026b, slides 37–41) restricts the grid to $[k_{\min}, k_{\max}]$ containing k^* with $k_{\min} > 0$, which is the consumption-floor construction of Remark 15.6.1 in numerical dress: the grid never offers a next-period capital above the largest node, so consumption is bounded away from zero at every node. On that compact set the return function is bounded, Assumption 15.1.1 holds, and Theorem 15.3.1 applies as stated. The program `deterministic_Ramsey_growth.m` computes k^* from (15.19) first and then lays 1000 nodes over $[0.25k^*, 1.75k^*]$, so the compact set is chosen by the theory rather than guessed; infeasible pairs (k, k') receive a large penalty in place of an undefined utility, which is the discrete analogue of the constraint $c_t \geq 0$. Run as supplied it converges in 211 iterations.

15.7 The Lucas Asset-Pricing Model

Environment. A representative agent in an endowment economy consumes a single non-storable good produced by a single tree. The endowment $(y_t)_{t\geq 0}$ is Markov, $y_{t+1} = G(y_t, \varepsilon_{t+1})$ with (ε_t) i.i.d. with distribution φ; in the numerical implementation $\log y_{t+1} = \alpha \log y_t + \sigma\varepsilon_{t+1}$ with ε_t standard normal. The agent ranks streams by $\mathbb{E}_0\big[\sum_{t\geq 0} \beta^t u(c_t)\big]$ with u of the CRRA form. A share $\pi_t \in [0, 1]$ of the tree, chosen at $t-1$, is the agent's only store of value, and an ex-dividend claim bought at price p_t entitles the holder to y_{t+1} and to the resale price p_{t+1}.

The agent's problem. With the share as state alongside the endowment,

$$v(\pi, y) = \max_{\pi'} \Big\{ u\big((y + p(y))\pi - p(y)\pi'\big) + \beta \int v\big(\pi', G(y, z)\big)\varphi(\mathrm{d}z) \Big\}, \tag{15.23}$$

the budget constraint $c + p(y)\pi' \leq (y + p(y))\pi$ binding because u is strictly increasing. The agent takes the pricing function $p(\cdot)$ as given; the guess that the price depends on the history only through the current endowment is verified by the construction below, and is legitimate because the endowment is Markov.

> **Theorem 15.7.1 (Consumption-based asset pricing).** In a competitive equilibrium of the Lucas economy the pricing function satisfies
>
> $$p(y) = \beta \int \frac{u'\big(G(y,z)\big)}{u'(y)} \Big(G(y,z) + p\big(G(y,z)\big)\Big)\varphi(\mathrm{d}z), \tag{15.24}$$
>
> equivalently, in sequence form,
>
> $$p_t = \mathbb{E}_t\Bigg[\underbrace{\beta\frac{u'(y_{t+1})}{u'(y_t)}}_{\text{stochastic discount factor}} \big(y_{t+1} + p_{t+1}\big)\Bigg]. \tag{15.25}$$

Proof. The first-order condition of (15.23) in π' is

$$u'(c)\,p(y) = \beta \int v_1\big(\pi', G(y,z)\big)\varphi(\mathrm{d}z), \qquad v_1 := \frac{\partial v}{\partial \pi}, \tag{15.26}$$

the cost in current marginal utility of an extra share equalling its discounted expected marginal value. Differentiating (15.23) in the state π with π' held at its optimum — which is legitimate by Theorem 15.4.1, v being concave in π as the value of a concave problem on a convex constraint set — gives the envelope formula

$$v_1(\pi, y) = u'(c)\big(y + p(y)\big), \tag{15.27}$$

an extra share being worth the marginal utility of the dividend plus the resale value it delivers.

Two market-clearing conditions pin down the equilibrium. The good is non-storable, so $c_t = y_t$; and with a representative agent there is no trade, so $\pi_t = 1$ for every t. Imposing $c = y$ and $\pi = \pi' = 1$ and substituting (15.27) into (15.26),

$$u'(y)p(y) = \beta \int u'\big(G(y,z)\big)\Big(G(y,z) + p\big(G(y,z)\big)\Big)\varphi(\mathrm{d}z),$$

which is (15.24) after division by $u'(y) > 0$. Rewriting the integral as a conditional expectation gives (15.25). ∎

Equation (15.25) is the central identity of asset pricing: the price of any claim is the conditional expectation of its payoff weighted by the *stochastic discount factor* $m_{t+1} = \beta u'(y_{t+1})/u'(y_t)$. Prices are high for claims that pay off when consumption is low, because marginal utility is then high; that is the entire economics, and the equity premium is the statement that the covariance between m_{t+1} and equity payoffs is empirically far too small to match observed returns for plausible γ.

Theorem 15.7.2 (Existence of the pricing function)**.** Suppose the map $y \mapsto \beta \int u'\big(G(y,z)\big)G(y,z)\,\varphi(\mathrm{d}z)$ defines a bounded continuous function h on $\mathbb{R}_+$. Then the operator

$$(\mathcal{T}f)(y) := h(y) + \beta \int f\big(G(y,z)\big)\varphi(\mathrm{d}z) \tag{15.28}$$

is a contraction of modulus β on $\mathcal{B}(\mathbb{R}_+, \mathbb{R})$, so it has a unique fixed point f^*, reached from any starting point, and the equilibrium price is recovered as

$$p^*(y) = \frac{f^*(y)}{u'(y)}. \tag{15.29}$$

Proof. Setting $f(y) := u'(y)p(y)$ turns (15.24) into $f = \mathcal{T}f$, because the integrand $u'(G(y,z))\big(G(y,z) + p(G(y,z))\big)$ splits as $u'(G(y,z))G(y,z)$, whose integral is $h(y)/\beta$, plus $f(G(y,z))$. This is Lucas' change of unknown, and its point is that the resulting operator is *affine* in f, whereas (15.24) in the unknown p is not.

For the contraction property, take $f_1, f_2 \in \mathcal{B}(\mathbb{R}_+, \mathbb{R})$. For each y,

$$\begin{aligned} |(\mathcal{T}f_1)(y) - (\mathcal{T}f_2)(y)| &= \beta \left| \int \big[f_1(G(y,z)) - f_2(G(y,z))\big]\varphi(\mathrm{d}z) \right| \\ &\le \beta \int |f_1 - f_2|\,\big(G(y,z)\big)\varphi(\mathrm{d}z) \le \beta\,\|f_1 - f_2\|_\infty, \end{aligned}$$

the first inequality by the triangle inequality for integrals and the second because φ is a probability measure. Taking the supremum over y gives $\|\mathcal{T}f_1 - \mathcal{T}f_2\|_\infty \le \beta\,\|f_1 - f_2\|_\infty$. Theorem 6.3.2 and Theorem 6.4.1 give the rest, and (15.29) inverts the substitution. ∎

Example 15.7.1 (Logarithmic utility)**.** Let $u = \log$, so $u'(c) = 1/c$. Then

$$h(y) = \beta \int \frac{1}{G(y,z)} G(y,z)\,\varphi(\mathrm{d}z) = \beta \int \varphi(\mathrm{d}z) = \beta,$$

a constant independent of y and of the law of motion G. The functional equation becomes $f = \beta + \beta f$ for a constant f, whose solution is $f^* = \beta/(1-\beta)$, and uniqueness in Theorem 15.7.2 rules out any other bounded solution. Hence

$$p^*(y) = \frac{f^*(y)}{u'(y)} = \frac{\beta}{1-\beta}\,y. \tag{15.30}$$

The price-dividend ratio is the constant $\beta/(1-\beta)$: with logarithmic utility the income and substitution effects of a change in expected future dividends cancel exactly, so no feature of the endowment process — persistence α, volatility σ, even the shape of φ — affects the ratio. This is the cleanest statement of why the log case is a knife-edge, and why matching observed variation in price-dividend ratios requires $\gamma \neq 1$.

> **Remark 15.7.1** (Numerical implementation). Iterating (15.28) requires evaluating f at $G(y,z) = y^\alpha r$, $r = e^{\sigma\varepsilon}$ log-normal, which is not a grid point; `Lucas_Tree_DP.m` and `Lucas_Tree_DP_Octave.m` interpolate with a spline (Fukuda 2026b, slide 61); run as supplied the first converges in 180 iterations, as does the second when translated into Matlab syntax, and the computed price differs from the closed form $\beta y/(1-\beta)$ of (15.30) by at most 0.0103%. Two approximations therefore compound with the contraction error (15.5): the integral is replaced by an average over 500 draws, contributing a Monte Carlo error of order $500^{-1/2}$ by Remark 14.7.1, and f off the grid is replaced by a spline. Neither is controlled by (15.5), which bounds only the distance between the exact iterates and the exact fixed point. The grid is centred using the stationary moments of the endowment: from $\log y_{t+1} = \alpha \log y_t + \sigma\varepsilon_{t+1}$ with $|\alpha| < 1$, $\mathbb{E}[\log y_t] = 0$ and $\mathrm{Var}(\log y_t) = \sigma^2/(1-\alpha^2)$ in the stationary distribution, and the grid spans four standard deviations either side. Checking the computed p^* against (15.30) in the log case is the available validation, and it is the reason that case is run first.

15.8 Chapter Summary

A stationary discounted sequential problem reduces to a functional equation in the state alone. Bellman's principle, made precise in Theorem 15.2.1, has two halves: the value function solves the Bellman equation, and — given boundedness, which supplies the transversality condition — any bounded solution of the Bellman equation is the value function and any plan attaining the maximum period by period is optimal. The Bellman operator satisfies Blackwell's monotonicity and discounting conditions, hence is a contraction of modulus β on the Banach space of bounded continuous functions; Theorem 6.4.1 then delivers existence, uniqueness, convergence of value function iteration from any starting point, and the two error bounds that make the algorithm usable. Berge's theorem supplies continuity of the value function and upper hemicontinuity of the policy correspondence at each iterate.

Characterisation requires differentiability of the value function, which the Benveniste–Scheinkman theorem obtains without assuming it: a concave function sandwiched above a differentiable concave function that touches it at an interior point is differentiable there, with the same derivative. Applied with the control fixed at its optimum, this yields the envelope formula, and combining the envelope formula with the first-order condition eliminates the value function and produces the Euler equation in primitives alone.

The three applications instantiate the same machinery. In the cake-eating problem the Euler equation $c_{t+1} = \beta c_t$ integrates to the policy $h(w) = (1-\beta)w$ and the value function $A + \log w/(1-\beta)$, verified directly because unbounded log utility falls outside the standing hypotheses. In the Ramsey model the Euler equation $u'(c_t) = \beta u'(c_{t+1})(f'(k_{t+1}) + 1 - \delta)$ has the modified golden rule $\beta(f'(k^*) + 1 - \delta) = 1$ as its steady state, placing capital below the golden-rule level by exactly the rate of time preference. In the Lucas economy the first-order and envelope conditions, combined with the market-clearing conditions $c = y$ and $\pi = 1$, give the consumption-based pricing equation and the stochastic discount factor $\beta u'(y_{t+1})/u'(y_t)$; substituting $f = u'p$ linearises the functional equation into a contraction, and under log utility the price-dividend ratio is the constant $\beta/(1-\beta)$ regardless of the endowment process.

> **Exercise 15.8.1.** Show that for any non-empty U, W and any $\Phi \colon U \times W \to [-\infty, +\infty]$,
>
> $$\sup_{(u,w)\in U\times W} \Phi(u,w) = \sup_{u\in U} \sup_{w\in W} \Phi(u,w),$$
>
> with no boundedness, measurability or finiteness hypothesis — so the interchange of suprema in the proof of Theorem 15.2.1(i) is free. Identify what boundedness *is* needed for in that proof, namely that $\sum_t \beta^t r(x_t, u_t)$ converge absolutely so that the objective is a well defined real number and the tail decomposition holds term by term. Then show that without Assumption 15.1.1(iii) the Bellman equation can have many solutions: with $X = \mathbb{R}_+$, $\Gamma(x) = \{x/\beta\}$ and $r \equiv 0$, every

$W(x) = cx$ satisfies $W = TW$, while the value of the sequential problem is $V \equiv 0$. Verify that $\beta^m W(x_m) = cx_0$ for every m, so that exactly one of these solutions satisfies the transversality condition of Remark 15.2.2.

Exercise 15.8.2. Solve the Ramsey model in closed form for $u = \log$ and $\delta = 1$ by guessing $V(k) = A + B \log k$. Show $B = \alpha/(1 - \alpha\beta)$ and that the policy is $k' = \alpha\beta A k^{\alpha}$, so the saving rate is the constant $\alpha\beta$. Verify that the steady state of this policy agrees with (15.19) at $\delta = 1$, and explain why the constant saving rate is a knife-edge property of the pairing of log utility with full depreciation.

Exercise 15.8.3. In the Lucas economy with $u(c) = c^{1-\gamma}/(1-\gamma)$ and $\log y_{t+1} = \alpha \log y_t + \sigma\varepsilon_{t+1}$, use Proposition 10.7.1 to compute $h(y)$ in closed form, and show that the price-dividend ratio depends on y unless $\gamma = 1$ or $\alpha = 0$. Determine the sign of $\partial(p^*(y)/y)/\partial y$ as a function of γ and interpret it in terms of the relative strength of income and substitution effects.

Chapter 16

Continuous-Time Dynamic Optimisation

16.1 From Sequences to Paths

The chapter preceding treated the choice of a sequence; this one treats the choice of a function of time. The change is more than notational. A sequence is a point in a space of countably many coordinates and can be perturbed one coordinate at a time; a path must be perturbed by another path, and the first-order condition is therefore an equation that must hold for *every* admissible perturbation. Converting such a family of scalar conditions into a pointwise differential equation is the office of the following lemma, and every result in this chapter rests on it.

> **Lemma 16.1.1** (Fundamental lemma of the calculus of variations)**.** Let $t_0 < t_1$ and let $g \in \mathcal{C}([t_0, t_1], \mathbb{R})$ satisfy
> $$\int_{t_0}^{t_1} g(t)h(t)\,\mathrm{d}t = 0 \qquad \text{for every } h \in \mathcal{C}([t_0, t_1], \mathbb{R}) \text{ with } h(t_0) = h(t_1) = 0. \tag{16.1}$$
> Then $g \equiv 0$ on $[t_0, t_1]$.

Proof. Suppose $g(s) \neq 0$ for some $s \in (t_0, t_1)$; without loss of generality $g(s) > 0$, otherwise replace g by $-g$. By continuity there are α, β with $t_0 < \alpha < s < \beta < t_1$ and $g(t) > \frac{1}{2}g(s)$ for every $t \in [\alpha, \beta]$. Define

$$h(t) := \begin{cases} (t-\alpha)(\beta - t), & t \in [\alpha, \beta], \\ 0, & \text{otherwise.} \end{cases}$$

This h is continuous on $[t_0, t_1]$ — it vanishes at α and β, where the two branches meet — non-negative, and satisfies $h(t_0) = h(t_1) = 0$. Then

$$\int_{t_0}^{t_1} g(t)h(t)\,\mathrm{d}t = \int_{\alpha}^{\beta} g(t)h(t)\,\mathrm{d}t > \frac{g(s)}{2}\int_{\alpha}^{\beta}(t-\alpha)(\beta-t)\,\mathrm{d}t = \frac{g(s)}{2}\cdot\frac{(\beta-\alpha)^3}{6} > 0,$$

using Proposition 10.2.3 for the strict monotonicity of the integral of a continuous function that is strictly positive on $[\alpha, \beta]$. This contradicts (16.1). Hence g vanishes on the open interval, and by continuity at the endpoints as well. ∎

This is Proposition 7.30 of Fukuda (2026d, p. 288). Read in the language of Chapter 11, it says that the only continuous function orthogonal, in the L^2 inner product $\langle g, h\rangle = \int gh$, to every continuous function vanishing at the endpoints is the zero function: the set of such h is dense in $L^2[t_0, t_1]$, so its orthogonal complement is trivial by Proposition 11.2.1. The elementary proof above avoids the density argument and is the one used here because it also delivers the conclusion for the piecewise-defined perturbations that appear in constrained problems.

16.2 The Calculus of Variations

Definition 16.2.1 (Variational problem). Given $f\colon [t_0, t_1] \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ continuously differentiable and endpoints $x_0, x_1 \in \mathbb{R}$, the *calculus-of-variations problem* is

$$\max_{x(\cdot)} \int_{t_0}^{t_1} f\big(t, x(t), \dot{x}(t)\big)\,\mathrm{d}t \qquad \text{subject to} \quad x(t_0) = x_0, \;\; x(t_1) = x_1, \tag{16.2}$$

the maximum being over continuously differentiable paths $x\colon [t_0, t_1] \to \mathbb{R}$.

Theorem 16.2.1 (Euler–Lagrange equation). Let x^* solve (16.2) and suppose $t \mapsto f_{\dot{x}}\big(t, x^*(t), \dot{x}^*(t)\big)$ is continuously differentiable. Then

$$f_x\big(t, x^*(t), \dot{x}^*(t)\big) - \frac{\mathrm{d}}{\mathrm{d}t}\Big[f_{\dot{x}}\big(t, x^*(t), \dot{x}^*(t)\big)\Big] = 0 \qquad \text{for every } t \in [t_0, t_1]. \tag{16.3}$$

Proof. Fix a continuously differentiable h with $h(t_0) = h(t_1) = 0$, so that $x^* + ah$ is admissible for every $a \in \mathbb{R}$, and set

$$J(a) := \int_{t_0}^{t_1} f\big(t, x^*(t) + ah(t), \dot{x}^*(t) + a\dot{h}(t)\big)\,\mathrm{d}t.$$

Optimality of x^* gives $J(0) \geq J(a)$ for all a, so $a = 0$ is an interior maximiser of J and Theorem 7.2.1 gives $J'(0) = 0$, provided J is differentiable. It is, by Lemma 10.5.1: the integrand is continuously differentiable in a and its a-derivative is continuous in (t, a), hence bounded on the compact set $[t_0, t_1] \times [-1, 1]$. Differentiating under the integral sign and using Theorem 8.3.1,

$$J'(a) = \int_{t_0}^{t_1} \Big\{ f_x\big(t, x, \dot{x}\big)h(t) + f_{\dot{x}}\big(t, x, \dot{x}\big)\dot{h}(t) \Big\}\,\mathrm{d}t, \qquad x = x^* + ah.$$

Integrating the second term by parts (Theorem 10.3.2),

$$\int_{t_0}^{t_1} f_{\dot{x}}\dot{h}\,\mathrm{d}t = f_{\dot{x}}h\big|_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{\mathrm{d}f_{\dot{x}}}{\mathrm{d}t}h\,\mathrm{d}t = -\int_{t_0}^{t_1} \frac{\mathrm{d}f_{\dot{x}}}{\mathrm{d}t}h\,\mathrm{d}t,$$

the boundary term vanishing because $h(t_0) = h(t_1) = 0$. Hence

$$0 = J'(0) = \int_{t_0}^{t_1} \Big[f_x\big(t, x^*, \dot{x}^*\big) - \frac{\mathrm{d}}{\mathrm{d}t}f_{\dot{x}}\big(t, x^*, \dot{x}^*\big)\Big]h(t)\,\mathrm{d}t$$

for every admissible h, and Lemma 16.1.1 gives (16.3). ■

The derivative $\frac{\mathrm{d}}{\mathrm{d}t}f_{\dot{x}}$ in (16.3) is a total derivative along the path, not a partial derivative: expanded by the chain rule it is $f_{\dot{x}t} + f_{\dot{x}x}\dot{x}^* + f_{\dot{x}\dot{x}}\ddot{x}^*$, so the Euler–Lagrange equation is a second-order ordinary differential equation in x^*, and the two endpoint conditions of (16.2) are exactly what a second-order equation requires. That count is the structural reason a variational problem with one free endpoint needs a substitute boundary condition, and the substitute is the transversality condition of Section 16.3.

Remark 16.2.1 (Necessary, not sufficient). Theorem 16.2.1 is a first-order necessary condition and nothing more: it is Theorem 7.2.1 applied along a one-parameter family, so it identifies stationary paths, among which are minima and saddles. Sufficiency requires concavity, exactly as in Theorem 7.2.3. If $(x, \dot{x}) \mapsto f(t, x, \dot{x})$ is concave for each t, then a path satisfying (16.3) is a global maximiser, because for any admissible x,

$$\int_{t_0}^{t_1} \big[f(t, x, \dot{x}) - f(t, x^*, \dot{x}^*)\big]\,\mathrm{d}t \leq \int_{t_0}^{t_1} \big[f_x \cdot (x - x^*) + f_{\dot{x}} \cdot (\dot{x} - \dot{x}^*)\big]\,\mathrm{d}t = 0,$$

the inequality by Theorem 9.3.1 and the final equality by the same integration by parts as in the proof, using (16.3) and $x(t_i) = x^*(t_i)$. The concavity hypothesis is satisfied in the economic problems below because U is concave and the constraint is linear in the control.

16.3 Optimal Control

Optimal control generalises Definition 16.2.1 by separating the control from the rate of change of the state, which permits constraints on the control and transitions that are not of the form $\dot{x} = u$.

Definition 16.3.1 (Optimal control problem). Fix a horizon $t_1 < \infty$, an open *control region* $U \subseteq \mathbb{R}$, and functions $f, g\colon [t_0, t_1] \times \mathbb{R} \times U \to \mathbb{R}$ that are continuously differentiable. An *admissible control* is a continuously differentiable $u\colon [t_0, t_1] \to U$ for which the initial value problem

$$\dot{x}(t) = g\big(t, x(t), u(t)\big), \qquad x(t_0) = x_0 \tag{16.4}$$

has a solution on the whole of $[t_0, t_1]$; that solution is then unique and is the *state trajectory* generated by u, by the Picard–Lindelöf theorem (Amann and Escher 2008, Thm. VII.8.14, p. 235) applied to the locally Lipschitz right-hand side. The problem is

$$\max_{u(\cdot)\ \text{admissible}} \int_{t_0}^{t_1} f\big(t, x(t), u(t)\big)\,\mathrm{d}t \qquad \text{subject to} \quad (16.4), \tag{16.5}$$

with $x(t_1)$ unrestricted. The function x is the *state*, u the *control*, and the *present-value Hamiltonian* is

$$\mathcal{H}(t, x, u, \lambda) := f(t, x, u) + \lambda(t)\, g(t, x, u), \tag{16.6}$$

with λ the *co-state variable*.

Openness of U is what makes the perturbation $u^* + ah$ admissible for small $|a|$, and it is why the theorem below produces a stationarity condition rather than a maximisation. The restriction to continuously differentiable controls is a genuine loss — optimal controls are frequently only piecewise continuous, and in bang-bang problems they are discontinuous — and Remark 16.3.1 records what the general theory delivers instead.

Theorem 16.3.1 (First-order necessary conditions, interior case). Let (u^*, x^*) solve (16.5), and assume that for every continuously differentiable $h\colon [t_0, t_1] \to \mathbb{R}$ the perturbed problem (16.4) with control $u^* + ah$ has a solution $y(\cdot, a)$ on $[t_0, t_1]$ for all $|a|$ small, with $(t, a) \mapsto y(t, a)$ continuous and $a \mapsto y(t, a)$ continuously differentiable, uniformly in t. Then there is a continuously differentiable $\lambda\colon [t_0, t_1] \to \mathbb{R}$ such that, for every t,

$$\frac{\partial \mathcal{H}}{\partial u} = 0 \iff f_u + \lambda g_u = 0, \tag{16.7}$$

$$-\frac{\partial \mathcal{H}}{\partial x} = \dot{\lambda} \iff \dot{\lambda}(t) = -\big(f_x + \lambda g_x\big), \tag{16.8}$$

$$\frac{\partial \mathcal{H}}{\partial \lambda} = \dot{x} \iff \dot{x} = g, \tag{16.9}$$

together with the initial condition $x(t_0) = x_0$ and the *transversality condition* $\lambda(t_1) = 0$.

Proof. Perturb the control, $u = u^* + ah$ for a fixed continuously differentiable h, and let $y(t, a)$ be the state path it induces through (16.4), so $y(t, 0) = x^*(t)$ and $y(t_0, a) = x_0$ for every a; the hypothesis of the theorem is that this family is well defined and differentiable in a. Let λ be any continuously differentiable function, to be chosen. Since the constraint holds identically along every admissible path, the objective is unchanged by adding $\lambda(g - \dot{x}) \equiv 0$ to the integrand, and integration by parts (Theorem 10.3.2) gives

$$\begin{aligned} J(a) &:= \int_{t_0}^{t_1} \Big\{ f\big(t, y, u^* + ah\big) + \lambda g\big(t, y, u^* + ah\big) - \lambda \dot{y} \Big\}\,\mathrm{d}t \\ &= \int_{t_0}^{t_1} \Big\{ f + \lambda g + \dot{\lambda}\, y \Big\}\,\mathrm{d}t - \lambda(t_1) y(t_1, a) + \lambda(t_0)\, \underbrace{y(t_0, a)}_{= x_0}. \end{aligned}$$

Differentiating at $a = 0$, legitimate by Lemma 10.5.1, and writing $y_a := \partial y / \partial a$ evaluated at $a = 0$,

$$J'(0) = \int_{t_0}^{t_1} \Big\{ \big(f_u + \lambda g_u\big) h + \big(f_x + \lambda g_x + \dot{\lambda}\big) y_a \Big\}\,\mathrm{d}t - \lambda(t_1)\, y_a(t_1), \tag{16.10}$$

the term in $\lambda(t_0)$ contributing nothing because $y(t_0, a) = x_0$ for all a.

The function λ is still free. Choose it to solve the linear differential equation (16.8) with terminal condition $\lambda(t_1) = 0$; a solution exists and is unique on $[t_0, t_1]$ because the equation is linear in λ with continuous coefficients, and the integrating-factor representation $\lambda(t) = \int_t^{t_1} f_x(s) \exp\big(\int_t^s g_x(\tau)\, \mathrm{d}\tau\big)\, \mathrm{d}s$ exhibits it. This choice annihilates the second and third terms of (16.10), and $J'(0) = 0$ by optimality, leaving

$$\int_{t_0}^{t_1} (f_u + \lambda g_u) h(t)\, \mathrm{d}t = 0 \qquad \text{for every admissible } h.$$

Lemma 16.1.1 then gives (16.7) pointwise. Equation (16.9) is the constraint, restated as a partial derivative of $\mathcal{H}$. ■

Remark 16.3.1 (What Pontryagin's maximum principle actually says). Theorem 16.3.1 is a stationarity result, not a maximum principle. It says $\partial\mathcal{H}/\partial u = 0$ along the optimal path, which locates a critical point of $\mathcal{H}$ in the control, and it obtains that conclusion from an interior perturbation in an open control region. It differs from the theorem of Pontryagin et al. (1962) in the following respects.

First, the conclusion. The maximum principle asserts

$$u^*(t) \in \arg\max_{u \in U} \ \mathcal{H}\big(t, x^*(t), u, \lambda(t)\big) \qquad \text{for almost every } t, \tag{16.11}$$

with U an arbitrary set, closed or finite included. Where U is open and $\mathcal{H}$ is differentiable in u, (16.11) implies (16.7); the converse fails, since a stationary point of $\mathcal{H}$ need be neither a maximum nor even a local maximum. In a problem linear in the control, $\partial\mathcal{H}/\partial u$ never vanishes except on singular arcs, (16.7) is empty, and (16.11) selects the bang-bang solution.

Second, the multipliers. The general statement produces a pair (λ_0, λ) with $\lambda_0 \in \{0, 1\}$, replaces $\mathcal{H}$ by $\lambda_0 f + \lambda g$, and asserts only $(\lambda_0, \lambda) \neq (0, 0)$. The *abnormal case* $\lambda_0 = 0$ deletes the objective from the Hamiltonian; it arises when the constraint structure alone determines the optimal path, and it cannot be excluded without a constraint qualification. Theorem 16.3.1 silently normalises $\lambda_0 = 1$, which the free terminal state and the open control region make legitimate here.

Third, the class of controls and the constraints. The general theorem admits measurable controls and yields (16.11) almost everywhere; it accommodates terminal constraints $x(t_1) \in S$, in which case $\lambda(t_1)$ is normal to S rather than zero, and pure state constraints $\psi(t, x) \geq 0$, in which case λ acquires jumps at the entry and exit times of the constrained arc and is only of bounded variation. None of these appears below, and each is treated in Kamien and Schwartz (1981).

Remark 16.3.2 (Existence of an optimal control). Theorem 16.3.1 and (16.11) are necessary conditions: they describe an optimal control if one exists, and every application below establishes optimality separately, by concavity or by a verification argument. Existence itself needs a compactness argument, and the compactness is supplied by Theorem 6.3.3.

Suppose the control values lie in a compact $U \subseteq \mathbb{R}^k$, that g is continuous and bounded by C on $[t_0, t_1] \times K \times U$ for a compact K containing every admissible trajectory, and that f is continuous. Every admissible trajectory then satisfies $|x(s) - x(t)| = \big|\int_t^s g\big| \leq C\,|s - t|$ and starts at x_0, so the admissible trajectories form a uniformly Lipschitz family with a common value at t_0 and, by Corollary 6.3.2, a relatively compact subset of $\mathcal{C}([t_0, t_1], \mathbb{R}^n)$. A maximising sequence of admissible pairs therefore has a subsequence whose trajectories converge uniformly.

The limit trajectory need not be admissible. What can fail is that the limit is generated by no single control, and the standard illustration is

$$\min_{u(\cdot)} \int_0^1 x(t)^2\, \mathrm{d}t \qquad \text{subject to} \quad \dot{x} = u, \quad u(t) \in \{-1, 1\}, \quad x(0) = 0.$$

Switching u between -1 and 1 every $1/n$ units of time keeps $|x| \leq 1/n$, so the infimum is 0; but a control attaining it would need $x \equiv 0$ and hence $\dot{x} = 0 \notin \{-1, 1\}$. The trajectories converge uniformly to $x \equiv 0$ and the limit is generated by no admissible control. Replacing $\{-1, 1\}$ by its convex hull $[-1, 1]$ restores existence, with $u \equiv 0$ optimal.

What the example isolates is convexity of the set of attainable velocities. The Filippov–Cesari theorem states that if, in addition to the compactness hypotheses above, the set

$$N(t,x) := \{(f(t,x,u)+\gamma,\ g(t,x,u)) : u \in U,\ \gamma \le 0\} \subseteq \mathbb{R}^{1+n}$$

is convex for every (t,x), and at least one admissible control exists, then an optimal control exists. The proof is the compactness argument above together with a closure theorem showing that the limit of admissible pairs is admissible, for which convexity of $N(t,x)$ is what is needed; see Cesari (1983) for the general theory of such existence and closure theorems and Seierstad and Sydsaeter (1987) for the version stated for economic problems. Convexity of $N(t,x)$ holds automatically when g is affine in u and f is concave in u with U convex, which covers the models of Section 16.5 and Section 16.6.

Remark 16.3.3 (What the three conditions say)**.** The co-state $\lambda(t)$ is the shadow price of the state at time t, the continuous-time counterpart of the multiplier of Theorem 12.3.1 and of the derivative $V'(x)$ of Theorem 15.4.2. The optimality condition (16.7) equates the marginal current return of the control to its marginal cost in terms of the state it consumes, valued at λ. The co-state equation (16.8) is an asset-pricing condition: rearranged as $-\dot{\lambda} = f_x + \lambda g_x$, it says the capital loss on a unit of the state equals its current dividend plus its contribution to future accumulation. The transversality condition $\lambda(t_1) = 0$ says a unit of the state has no value at the terminal date when the terminal state is free: anything left over is wasted, so nothing should be left over.

When $t_1 = \infty$ the terminal condition must be replaced, and the replacement is less automatic than the finite-horizon case suggests. The condition usually written is

$$\lim_{t\to\infty} \lambda(t)x(t) = 0, \qquad \text{equivalently} \qquad \lim_{t\to\infty} e^{-\rho t}\mu(t)x(t) = 0 \tag{16.12}$$

in current-value terms. It is *not* a necessary condition for optimality in general: Halkin (1974) exhibits an infinite-horizon control problem with a smooth, bounded, concave structure whose unique optimal path violates (16.12), and shows that no condition of that form can be appended to the maximum principle without further hypotheses. What is true, and is what the applications below use, is the converse direction: if f and g are concave in (x,u), if (u^*,x^*) together with a non-negative λ satisfies (16.7)–(16.9), and if $\liminf_{t\to\infty} \lambda(t)\big(x(t)-x^*(t)\big) \ge 0$ for every admissible x, then (u^*,x^*) is optimal. Condition (16.12) together with $\lambda \ge 0$ and $x \ge 0$ implies that hypothesis, so it functions as a sufficient condition and as a selection device among candidates. Section 16.5 is the illustration: the first-order conditions leave a free constant, and (16.12) is what pins it down — but the argument there proceeds by verification, exhibiting a solution of the Hamilton–Jacobi–Bellman equation and checking optimality directly, rather than by appeal to (16.12) as a necessary condition.

Proposition 16.3.1 (Current-value Hamiltonian)**.** Consider the discounted problem

$$\max_{u(\cdot)} \int_{t_0}^{t_1} e^{-\rho t} f\big(t,x(t),u(t)\big)\,\mathrm{d}t \qquad \text{subject to} \quad \dot{x} = g(t,x,u),\ \ x(t_0) = x_0, \tag{16.13}$$

and define the *current-value Hamiltonian* $H := e^{\rho t}\mathcal{H} = f(t,x,u) + \mu(t)g(t,x,u)$ with $\mu(t) := e^{\rho t}\lambda(t)$. Then (16.7)–(16.9) become

$$\frac{\partial H}{\partial u} = 0, \qquad \dot{\mu}(t) = \rho\mu(t) - \frac{\partial H}{\partial x}, \qquad \frac{\partial H}{\partial \mu} = \dot{x}. \tag{16.14}$$

Proof. The optimality condition is unaffected by multiplying $\mathcal{H}$ by the positive factor $e^{\rho t}$, which does not depend on u. For the co-state equation, (16.8) reads $-\dot{\lambda} = \partial\mathcal{H}/\partial x$, and multiplying by $e^{\rho t}$ gives $-e^{\rho t}\dot{\lambda} = \partial H/\partial x$. Since $\mu = e^{\rho t}\lambda$, the product rule (Proposition 7.3.2) gives

$$\dot{\mu} = \rho e^{\rho t}\lambda + e^{\rho t}\dot{\lambda} = \rho\mu - \frac{\partial H}{\partial x}.$$

The state equation is unchanged since $\partial H/\partial\mu = g$. ∎

The current-value formulation removes the explicit $e^{-\rho t}$ from every equation and leaves an autonomous system whenever f and g do not depend on t directly. That is the practical reason it is preferred in economics: the resulting pair of differential equations in (x, μ) has a phase diagram, and the steady state and saddle path of Section 16.6 are features of that diagram.

> **Proposition 16.3.2** (Calculus of variations as a special case)**.** The problem (16.2) is (16.5) with $u := \dot{x}$ and $g(t, x, u) = u$, and the maximum principle then reproduces the Euler–Lagrange equation (16.3).

Proof. With $g = u$ the Hamiltonian is $\mathcal{H} = f(t, x, u) + \lambda u$, so (16.7) reads $f_u + \lambda = 0$ and (16.8) reads $\dot{\lambda} = -f_x$. Differentiating the first along the path and using $u = \dot{x}$ from (16.9),

$$\dot{\lambda} = -\frac{\mathrm{d}}{\mathrm{d}t} f_u(t, x, u) = -\frac{\mathrm{d}}{\mathrm{d}t} f_{\dot{x}}(t, x, \dot{x}),$$

and eliminating $\dot{\lambda}$ between this and $\dot{\lambda} = -f_x$ gives $f_x = \frac{\mathrm{d}}{\mathrm{d}t} f_{\dot{x}}$. ∎

16.4 The Hamilton–Jacobi–Bellman Equation

The third route to the same conditions is dynamic programming, obtained as the continuous-time limit of Chapter 15.

Heuristic derivation. Divide time into periods of length $\delta > 0$ and discount one period by $1/(1+\delta\rho)$. Calendar date t is reached after t/δ periods, so the factor applied to a payoff at t is $(1 + \delta\rho)^{-t/\delta}$, and writing it as $\big[(1 + \delta\rho)^{1/(\delta\rho)}\big]^{-\rho t}$ shows

$$(1 + \delta\rho)^{-t/\delta} \longrightarrow e^{-\rho t} \qquad \text{as } \delta \downarrow 0,$$

since $(1 + h)^{1/h} \to e$ as $h \downarrow 0$. The problem is stationary, so its value depends on the state alone and the accounting can be done in current values, avoiding the factor entirely. Over one period the flow payoff is $r(x, u)\delta$ and the state moves to $x + g(x, u)\delta$, so

$$V(x) = \max_u \Big\{ r(x, u)\,\delta + \tfrac{1}{1+\rho\delta}\, V\big(x + g(x, u)\delta\big) \Big\}.$$

Assume V differentiable and expand both factors to first order, using Theorem 8.5.2 for the value and $(1 + \rho\delta)^{-1} = 1 - \rho\delta + o(\delta)$ for the discount:

$$\tfrac{1}{1+\rho\delta} V\big(x + g\delta\big) = V(x) + \big[V'(x)g(x, u) - \rho V(x)\big]\delta + o(\delta).$$

Substituting, cancelling $V(x)$, dividing by δ and letting $\delta \downarrow 0$ gives $0 = \max_u\{r(x, u) + V'(x)g(x, u) - \rho V(x)\}$, and $\rho V(x)$ does not depend on u.

> **Definition 16.4.1** (HJB equation)**.** The *Hamilton–Jacobi–Bellman equation* of the stationary infinite-horizon problem $V(x) = \max \int_0^\infty e^{-\rho t} r(x_t, u_t)\,\mathrm{d}t$ subject to $\dot{x} = g(x, u)$, $x_0 = x$, is
>
> $$\rho V(x) = \max_u \big\{ r(x, u) + V'(x)\, g(x, u) \big\}. \tag{16.15}$$

Equation (16.15) is the asset-pricing form of the Bellman equation and was met in (10.6): the required return ρV on the "asset" x equals the current dividend r plus the expected capital gain $V'(x)\dot{x}$. Comparing (16.15) with (16.6) identifies the maximand as the current-value Hamiltonian with $\mu = V'(x)$: the co-state is the derivative of the value function, exactly as λ was the shadow price in Remark 16.3.3 and as V' was in (15.21). The three methods of this chapter — variations, the maximum principle, dynamic programming — are therefore three presentations of one set of conditions, and which is most convenient depends on the problem rather than on the mathematics.

> **Remark 16.4.1** (The status of the derivation)**.** The passage to the limit above assumes what it should prove: that V is differentiable, that the limit and the maximum interchange, and that the discrete problems converge to the continuous one. None of these is automatic. The value function of a control problem is in general only continuous, and (16.15) then holds in the weaker *viscosity* sense rather than pointwise. Where a classical solution of (16.15) does exist and the associated policy

is admissible, a verification argument establishes optimality directly, without any limit: this is the content of Proposition 16.5.2 below, and it is why guess-and-verify is the honest way to use (16.15).

16.5 The Continuous-Time Cake-Eating Problem

Problem. Maximise $\int_0^\infty e^{-rt} \log c(t)\, \mathrm{d}t$ subject to $\dot{k}(t) = -c(t)$, $k(0) = k_0 > 0$, $c(t) > 0$, together with the transversality condition $\lim_{t\to\infty} k(t) = 0$.

Proposition 16.5.1 (Solution by three routes)**.** The problem has the unique solution

$$k^*(t) = k_0 e^{-rt}, \qquad c^*(t) = rk_0 e^{-rt}, \qquad V(k) = \frac{\log r - 1 + \log k}{r}, \tag{16.16}$$

and the optimal feedback rule is $c = rk$.

Proof. Variational route. Substituting $c = -\dot{k}$ turns the problem into (16.2) with $f(t, k, \dot{k}) = e^{-rt}\log(-\dot{k})$. Here $f_k = 0$ and $f_{\dot{k}} = -e^{-rt}/(-\dot{k}) \cdot (-1) = e^{-rt}/\dot{k}$, so (16.3) reduces to

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^{-rt}}{\dot{k}^*(t)}\right) = 0,$$

that is $e^{-rt}/\dot{k}^*(t)$ is constant in t, so $\dot{k}^*(t) = ae^{-rt}$ for some $a \in \mathbb{R}$. Integrating from 0 to t by Theorem 10.3.1,

$$k^*(t) = k_0 + \int_0^t ae^{-r\tau}\, \mathrm{d}\tau = k_0 + \frac{a}{r}\big(1 - e^{-rt}\big).$$

Letting $t \to \infty$ and imposing $\lim_t k^*(t) = 0$ gives $k_0 + a/r = 0$, that is $a = -rk_0$, whence $k^*(t) = k_0 e^{-rt}$ and $c^*(t) = -\dot{k}^*(t) = rk_0 e^{-rt}$.

Hamiltonian route. The current-value Hamiltonian is $H = \log c - \mu c$. Condition (16.14) gives $1/c = \mu$; and $\partial H/\partial k = 0$, so $\dot{\mu} = r\mu$. This separable equation integrates to $\mu(t) = \mu(0)e^{rt}$, hence $c(t) = 1/\mu(t) = ae^{-rt}$ with $a = 1/\mu(0)$. The state equation and the transversality condition then reproduce the previous paragraph with the sign of a reversed, giving $a = rk_0$ and the same paths. With the present-value Hamiltonian $\mathcal{H} = e^{-rt}\log c - \lambda c$ the co-state equation is $-\dot{\lambda} = \partial\mathcal{H}/\partial k = 0$, so λ is constant and $e^{-rt}/c(t) = \lambda$ delivers $c(t) = ae^{-rt}$ directly.

Dynamic programming route. See Proposition 16.5.2. ∎

Proposition 16.5.2 (Verification via HJB)**.** The function $V(k) = (\log r - 1 + \log k)/r$ solves (16.15) for the cake problem, the maximiser is $c = rk$, and the resulting path is optimal.

Proof. Here $r(k, c) = \log c$ and $g(k, c) = -c$, so (16.15) reads $rV(k) = \max_{c>0}\{\log c - V'(k)c\}$. Guess $V(k) = A + B\log k$, so $V'(k) = B/k > 0$ for $B > 0$; the objective is then strictly concave in c with $\lim_{c\downarrow 0}$ derivative $+\infty$, so by Theorem 7.2.3 the unique maximiser solves $1/c = B/k$, that is $c = k/B$. Substituting,

$$r(A + B\log k) = \log\frac{k}{B} - \frac{B}{k}\cdot\frac{k}{B} = \log k - \log B - 1.$$

Matching the coefficient of $\log k$ gives $rB = 1$, so $B = 1/r$; matching constants gives $rA = -\log B - 1 = \log r - 1$, so $A = (\log r - 1)/r$. The feedback rule is $c = k/B = rk$, so the state obeys $\dot{k} = -rk$, giving $k^*(t) = k_0 e^{-rt}$ and $c^*(t) = rk_0 e^{-rt}$, which satisfy the transversality condition.

Optimality. The usual verification argument integrates $\frac{\mathrm{d}}{\mathrm{d}t}[e^{-rt}V(k(t))] \le -e^{-rt}\log c(t)$ from 0 to T to obtain

$$\int_0^T e^{-rt}\log c(t)\, \mathrm{d}t \le V(k_0) - e^{-rT}V\big(k(T)\big)$$

and then discards the terminal term. That step is not available here: $V(k) \to -\infty$ as $k \downarrow 0$, so on a path whose stock is driven to zero quickly the term $-e^{-rT}V(k(T))$ diverges to $+\infty$ and the bound becomes vacuous rather than tight. Since $k(T) \to 0$ is precisely what the candidate path does, the terminal term must be avoided rather than estimated. The following argument does that, and uses nothing beyond concavity of the logarithm.

Let (c, k) be admissible: $c \geq 0$ measurable, $\dot{k} = -c$, $k(0) = k_0$ and $k(t) \geq 0$ for all t. If $c = 0$ on a set of positive measure, or if the negative part of $t \mapsto e^{-rt} \log c(t)$ is not integrable, the objective is $-\infty$ and there is nothing to prove; so assume $c > 0$ almost everywhere with the objective well defined in $[-\infty, \infty)$. Write $c^*(t) := rk_0 e^{-rt}$ for the candidate. Concavity of log gives the tangent-line bound (Theorem 9.3.1)

$$\log c(t) \leq \log c^*(t) + \frac{c(t) - c^*(t)}{c^*(t)} \qquad \text{for almost every } t.$$

Multiply by e^{-rt} and integrate. Using $\int_0^\infty e^{-rt}\mathrm{d}t = 1/r$ and $\int_0^\infty te^{-rt}\mathrm{d}t = 1/r^2$,

$$\int_0^\infty e^{-rt} \log c^*(t)\,\mathrm{d}t = \int_0^\infty e^{-rt}\big(\log(rk_0) - rt\big)\,\mathrm{d}t = \frac{\log(rk_0) - 1}{r} = V(k_0),$$
$$\int_0^\infty e^{-rt} \frac{c(t) - c^*(t)}{c^*(t)}\,\mathrm{d}t = \frac{1}{rk_0}\int_0^\infty c(t)\,\mathrm{d}t - \frac{1}{r},$$

the second line because $e^{-rt}/c^*(t) = 1/(rk_0)$ is constant. Feasibility gives $\int_0^T c(t)\,\mathrm{d}t = k_0 - k(T) \leq k_0$ for every T, hence $\int_0^\infty c \leq k_0$ and the second display is at most $1/r - 1/r = 0$. Therefore

$$\int_0^\infty e^{-rt} \log c(t)\,\mathrm{d}t \leq V(k_0),$$

and the candidate attains the bound, since the first display is an equality. Equality requires $c = c^*$ almost everywhere, because the tangent-line inequality is strict wherever $c \neq c^*$ and log is strictly concave; the optimal path is therefore unique up to a null set. ■

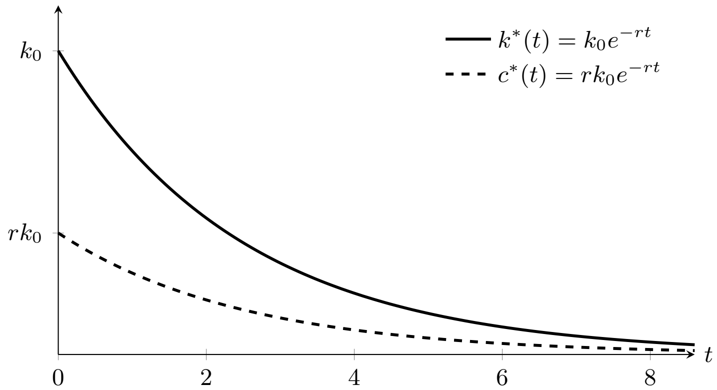


Figure 16.1: The optimal paths (16.16). Both decay at the common rate r, so the ratio $c^*(t)/k^*(t)$ is constant and equal to r: the feedback rule $c = rk$ is what makes the two curves scaled copies of each other. The cake is exhausted only in the limit, which is the content of the transversality condition $\lim_{t\to\infty} k(t) = 0$.

Remark 16.5.1 (The discrete-time limit). Compare (16.16), drawn in Figure 16.1, with (15.15). In discrete time with period length Δ and discount factor $\beta = e^{-r\Delta}$, the policy is $c = (1 - \beta)w$ per period, so the consumption *rate* is

$$\frac{(1 - e^{-r\Delta})w}{\Delta} \xrightarrow[\Delta\downarrow 0]{} rw,$$

by Theorem 7.4.1 applied to $\Delta \mapsto e^{-r\Delta}$, which is exactly the feedback rule $c = rk$. Correspondingly the cake decays at rate $\beta^{t/\Delta} = e^{-rt}$, matching $k^*(t) = k_0 e^{-rt}$. The rate of time preference r in continuous time and the discount factor β in discrete time are related by $\beta = 1/(1 + r\Delta) + o(\Delta)$, which is the substitution made in the derivation of (16.15). The agreement is not automatic — it requires the transversality conditions to correspond, which they do here because both take the form "the state is asymptotically exhausted".

16.6 The Ramsey Model in Continuous Time

Problem. A planner maximises $\int_0^\infty e^{-\rho t} u(c(t))\,\mathrm{d}t$ subject to

$$\dot{k}(t) = f\big(k(t)\big) - \delta k(t) - c(t), \qquad k(0) = k_0 > 0,\ c(t) \geq 0,\ k(t) \geq 0, \tag{16.17}$$

with f strictly increasing, strictly concave and satisfying the Inada conditions, u strictly increasing and twice differentiable with $u''(c) < 0$ at every $c > 0$, $\rho > 0$ and $\delta \in [0, 1]$. The pointwise condition $u'' < 0$ is strictly stronger than strict concavity, which for a C^2 function gives only $u'' \leq 0$ and permits isolated points where u'' vanishes. On $(0, 2)$ the function $u(c) = c - \frac{1}{4}(c-1)^4$ has $u'(c) = 1 - (c-1)^3 \in (0, 2)$ and $u''(c) = -3(c-1)^2$, so it is strictly increasing and strictly concave while $u''(1) = 0$ and $\gamma(1) = 0$. The pointwise condition is needed because (16.18) divides by $\gamma(c)$.

Theorem 16.6.1 (Keynes–Ramsey rule and the modified golden rule)**.** Along an interior optimal path of (16.17),

$$\frac{\dot{c}(t)}{c(t)} = \frac{f'\big(k(t)\big) - \delta - \rho}{\gamma\big(c(t)\big)}, \qquad \gamma(c) := -\frac{c\,u''(c)}{u'(c)}, \tag{16.18}$$

and the unique interior steady state (k^*, c^*) satisfies

$$f'(k^*) = \rho + \delta, \qquad c^* = f(k^*) - \delta k^*. \tag{16.19}$$

Proof. The current-value Hamiltonian is $H = u(c) + \mu\big(f(k) - \delta k - c\big)$. By Proposition 16.3.1,

$$\frac{\partial H}{\partial c} = u'(c) - \mu = 0, \qquad \dot{\mu} = \rho\mu - \frac{\partial H}{\partial k} = \rho\mu - \mu\big(f'(k) - \delta\big) = \mu\big(\rho + \delta - f'(k)\big). \tag{16.20}$$

Differentiating $u'(c) = \mu$ in t by Proposition 7.3.4 gives $u''(c)\dot{c} = \dot{\mu} = u'(c)\big(\rho + \delta - f'(k)\big)$, so

$$\frac{c\,u''(c)}{u'(c)} \cdot \frac{\dot{c}}{c} = \rho + \delta - f'(k), \qquad \text{that is} \qquad -\gamma(c)\frac{\dot{c}}{c} = \rho + \delta - f'(k),$$

which is (16.18). At a steady state $\dot{c} = 0$, and since $\gamma(c) > 0$ by the hypothesis $u'' < 0$, the numerator must vanish: $f'(k^*) = \rho + \delta$. Strict concavity of f makes f' strictly decreasing, and the Inada conditions make it range over $(0, \infty)$, so k^* exists and is unique. Setting $\dot{k} = 0$ in (16.17) gives $c^* = f(k^*) - \delta k^*$. ∎

Remark 16.6.1 (Reading the Keynes–Ramsey rule)**.** Equation (16.18) decomposes consumption growth into a rate of return and a willingness to substitute. Consumption rises when the net marginal product of capital $f'(k) - \delta$ exceeds the rate of time preference ρ, and the responsiveness is the elasticity of intertemporal substitution $1/\gamma$. For CRRA utility γ is constant and (16.18) is the continuous-time counterpart of (15.18); at $\rho = \beta^{-1} - 1$ the two steady-state conditions $f'(k^*) = \rho + \delta$ and $\beta\big(f'(k^*) + 1 - \delta\big) = 1$ coincide, as they must.

The modified golden rule places k^* strictly below the golden-rule stock k^g defined by $f'(k^g) = \delta$, which maximises steady-state consumption $f(k) - \delta k$. The gap is impatience: a planner with $\rho = 0$ would accumulate to k^g, and the discounted problem sacrifices steady-state consumption for consumption today. That the discounted planner is not dynamically inefficient — unlike a decentralised overlapping-generations economy, which can over-accumulate past k^g — follows from $f'(k^*) = \rho + \delta > \delta$.

Remark 16.6.2 (Saddle-path stability)**.** The pair (16.17) and (16.18) is an autonomous planar system in (k, c). Linearising about (k^*, c^*) by Theorem 8.5.2, the Jacobian is

$$J = \begin{pmatrix} f'(k^*) - \delta & -1 \\ \dfrac{c^* f''(k^*)}{\gamma(c^*)} & 0 \end{pmatrix} = \begin{pmatrix} \rho & -1 \\ \dfrac{c^* f''(k^*)}{\gamma(c^*)} & 0 \end{pmatrix},$$

the $(2, 2)$ entry vanishing because differentiating $c\big(f'(k) - \delta - \rho\big)/\gamma(c)$ in c leaves the factor $f'(k^*) - \delta - \rho = 0$. The determinant is $c^* f''(k^*)/\gamma(c^*) < 0$ because $f'' < 0$. By Definition 2.5.1 the eigenvalues of a 2×2 matrix are the roots of $p_J(\lambda) = \lambda^2 - (\operatorname{tr} J)\lambda + \det J$, so $\lambda_1\lambda_2 = \det J < 0$

and the discriminant $(\operatorname{tr} J)^2 - 4 \det J > 0$: the eigenvalues are real and of opposite signs. The steady state is therefore a saddle. Since $\operatorname{tr} J = \rho = \lambda_1 + \lambda_2$, the first row of $J - \lambda_s I$ gives the eigenvector v_s of the negative eigenvalue λ_s the slope $\rho - \lambda_s = \lambda_u$ in the (k, c) plane: the stable arm crosses the steady state rising at the rate of the *unstable* eigenvalue. The stable manifold theorem for a hyperbolic equilibrium then supplies a neighbourhood N of (k^*, c^*) and a one-dimensional C^1 manifold $W^s \subseteq N$, tangent at (k^*, c^*) to the eigenvector of the negative eigenvalue, consisting of the points of N whose forward orbit remains in N and converges to the steady state. That theorem is not proved in this volume; it belongs to the qualitative theory of ordinary differential equations rather than to the material developed here, and the statement used is the standard one for a hyperbolic equilibrium of a C^1 planar system. Two limitations of it bear on what follows.

First, W^s is local. For k_0 near k^* it determines $c(0)$ uniquely, and that is the sense in which the initial consumption level is pinned down. For k_0 far from k^* — the case drawn in Figure 16.2 — the relevant object is the *global* stable set, obtained by flowing W^s backwards in time. That this global set is again the graph of a single-valued function of k on all of $(0, \infty)$, so that $c(0)$ is unique for every k_0, is a separate result, proved from strict concavity of the planner's objective and of the feasible set — which make the optimal path unique — together with the transversality condition, which rules out the divergent paths. Neither ingredient is the eigenvalue calculation, and neither is established in this volume; the argument is Acemoglu (2009, Ch. 8).

Second, the eigenvalue calculation is a statement about the linearisation. It certifies that the non-linear system has a saddle structure near the steady state, and nothing about behaviour elsewhere in the phase plane. Paths starting off the stable set do exhibit the two failures the diagram suggests: with $c(0)$ too high, capital reaches zero in finite time and the path is infeasible; with $c(0)$ too low, capital rises without bound and consumption falls, and such a path is dominated by one that consumes the excess. Establishing that dichotomy requires the global argument just described, not the Jacobian.

The count of stable eigenvalues against the count of predetermined variables is the continuous-time form of the Blanchard–Kahn condition of Remark 2.8.1, with "outside the unit circle" replaced by "positive real part": here one predetermined variable k and one jump variable c face one negative and one positive eigenvalue, which is the configuration under which the linearised system has a locally unique non-explosive solution.

Figure 16.2 takes $f(k) = k^{0.3}$, $\delta = 0.1$, $\rho = 0.05$ and $u = \log$, so that $\gamma \equiv 1$. Then $k^* = 2.6918$, $c^* = 1.0767$, $f''(k^*) = -0.039007$, $\det J = -0.0420$, and the eigenvalues are $\lambda_s = -0.18146$ and $\lambda_u = 0.23146$; the arm therefore leaves the steady state with slope 0.23146 and the stable mode has half-life $\log 2 / |\lambda_s| = 3.82$. The arm drawn there is the global stable set, obtained by integrating the planar system backwards in time from $(k^*, c^*) \pm 10^{-9} v_s$: reversing time turns λ_s into $-\lambda_s > 0$, which carries the solution outwards along the manifold, and turns λ_u into $-\lambda_u < 0$, so a component off the manifold contracts. Backwards is thus the numerically stable direction, and forwards is not.

16.7 Chapter Summary

Continuous-time optimisation replaces the choice of a sequence by the choice of a path, and the first-order condition accordingly holds against every admissible perturbation. The fundamental lemma converts such a family of integral conditions into a pointwise differential equation, and it is the single analytic tool behind all three methods presented.

The calculus of variations perturbs the path directly and yields the Euler–Lagrange equation $f_x = \frac{\mathrm{d}}{\mathrm{d}t} f_{\dot{x}}$, a second-order differential equation matched to two endpoint conditions. It is necessary, not sufficient; concavity of f in $(x, \dot{x})$ upgrades it to sufficient. Optimal control separates the control from the state and perturbs the control, producing the Hamiltonian $\mathcal{H} = f + \lambda g$ and three conditions — optimality in the control, a differential equation for the co-state, and the state equation — together with $\lambda(t_1) = 0$ when the terminal state is free. The co-state is the shadow price of the state, and the co-state equation is a no-arbitrage condition on that price. The current-value Hamiltonian removes the discount factor from the equations and leaves an autonomous system whose phase diagram carries the dynamics. The calculus of variations is the case $g = u$. Dynamic programming gives the Hamilton–Jacobi–Bellman equation $\rho V(x) = \max_u \{r(x, u) + V'(x) g(x, u)\}$, in which the co-state is identified as $V'(x)$; the heuristic passage from discrete time assumes differentiability of the value function, and the rigorous route is verification against a candidate.

The continuous-time cake-eating problem yields $c = rk$, $k(t) = k_0 e^{-rt}$ by all three methods, and

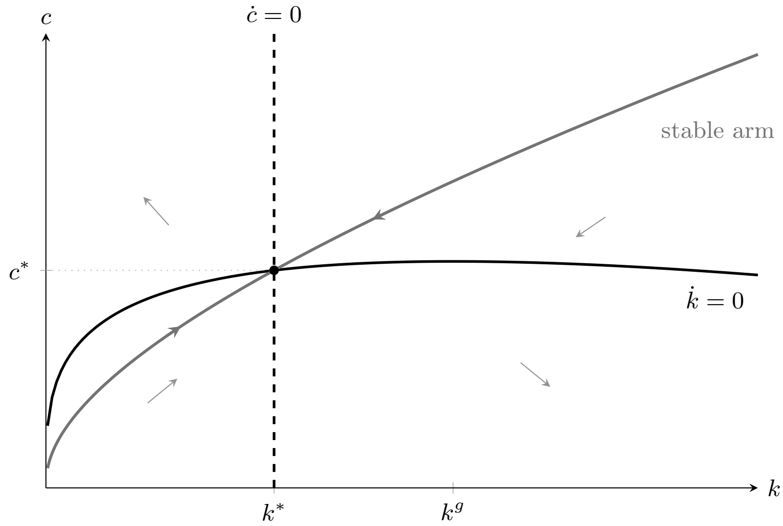


Figure 16.2: The two loci of Theorem 16.6.1 and the stable manifold of Remark 16.6.2. Consumption rises to the left of k^* and falls to its right; capital rises below the hump and falls above it. The grey arrows give the direction of motion in each of the four regions. The modified golden rule places k^* strictly below the golden-rule stock k^g at which $f'(k) = \delta$, the wedge being the rate of time preference ρ. The arm is the manifold itself, obtained by integrating the system backwards in time from the stable eigenvector as described in Remark 16.6.2, and not a sketch of it. To the right of k^* it runs well above the $\dot{k} = 0$ locus, so capital is drawn down; at the right-hand edge $k = 8.4$ it reaches $c = 2.15$ against output $f(k) = 1.89$, consumption exceeding output rather than merely net output.

matches the discrete-time policy $c = (1 - \beta)w$ in the limit of short periods. In each derivation the first-order conditions leave one free constant and only the transversality condition fixes it. The continuous-time Ramsey model yields the Keynes–Ramsey rule $\dot{c}/c = (f'(k) - \delta - \rho)/\gamma$ and the modified golden rule $f'(k^*) = \rho + \delta$, which places capital strictly below the golden-rule level by the amount of impatience. The linearisation shows the steady state is a saddle, so for k_0 near k^* initial consumption is the unique level placing the economy on the local stable manifold; the corresponding statement for arbitrary k_0 is a global result, and Remark 16.6.2 records what it needs beyond the eigenvalues.

Scope of the chapter. The results of this chapter carrying full proofs are the fundamental lemma (Lemma 16.1.1), the Euler–Lagrange equation and its concavity-based sufficiency, the maximum principle in the fixed-endpoint and free-endpoint forms of Theorem 16.3.1 under the differentiability hypotheses stated there, the current-value transformation, the verification theorem Proposition 16.5.2, and the Keynes–Ramsey rule with the modified golden rule. Four things are stated and used but not established. The maximum principle in the generality of Pontryagin et al. (1962), with control sets that are merely compact and no differentiability in u, needs the needle-variation argument. The existence of an optimal control needs the compactness and convexity conditions of Remark 16.3.2. The stable manifold theorem, and with it the global saddle-path property, belongs to the qualitative theory of differential equations. And the value function of a control problem is in general not differentiable, so (16.15) holds only in the viscosity sense; the heuristic derivation above assumes away the difficulty, and the verification theorem is what makes the chapter's use of (16.15) legitimate without it. Each of the four is a standard topic of a graduate course in dynamic optimisation, and Seierstad and Sydsaeter (1987) and Cesari (1983) are the references used above.

> **Exercise 16.7.1.** Prove Lemma 16.1.1 under the weaker hypothesis that (16.1) holds only for infinitely differentiable h supported in (t_0, t_1), by replacing the quadratic bump with $h(t) = \exp\bigl(-1/((t-\alpha)(\beta-t))\bigr)$ on $[\alpha, \beta]$ and 0 elsewhere, and verifying that this h is smooth. Explain why the strengthening matters for problems in which the admissible perturbations are constrained.

Exercise 16.7.2. Solve $\max \int_0^T e^{-\rho t} u(c)\,\mathrm{d}t$ subject to $\dot{a} = ra - c$, $a(0) = a_0$ and $a(T) = 0$, with $u(c) = c^{1-\gamma}/(1-\gamma)$. Show that $\dot{c}/c = (r-\rho)/\gamma$, solve for $c(0)$ from the terminal condition, and verify that the solution converges as $T \to \infty$ to the infinite-horizon solution under the transversality condition $\lim_{t\to\infty} e^{-\rho t} u'(c(t))a(t) = 0$. Identify precisely where the terminal condition $a(T) = 0$ replaces $\lambda(T) = 0$ of Theorem 16.3.1 and why both cannot be imposed.

Exercise 16.7.3. Consider the Hotelling problem $\max \int_0^\infty e^{-\rho t} p(t)q(t)\,\mathrm{d}t$ subject to $\dot{S} = -q$, $S(0) = S_0$, where $p(t)$ is a given price path and $q \geq 0$ is extraction. Show that the maximum principle gives $\dot{\mu} = \rho\mu$, so the shadow price of the resource grows at the rate of interest, and deduce Hotelling's rule that the price of the extracted resource net of extraction cost must grow at rate ρ along any interior path. State what fails when extraction is constrained to a capacity $q \leq \bar{q}$, and relate the answer to Remark 12.1.1.

Appendix

Appendix A

The Implicit Function Theorem

Theorem 8.6.1 was stated in Chapter 8 and its proof deferred. It is given here in full, because the theorem carries the comparative statics of every model in this volume and because the argument is the most substantial application of Theorem 6.4.1 outside dynamic programming.

A.1 Two Lemmas

Lemma A.1.1 (Neumann invertibility). Let $M \in M_{m\times m}(\mathbb{R})$ satisfy $\|I - M\| < 1$ in the operator norm. Then M is invertible, $M^{-1} = \sum_{j=0}^{\infty}(I-M)^j$, and $\left\|M^{-1}\right\| \le \left(1 - \|I-M\|\right)^{-1}$.

Proof. Write $N := I - M$ and $q := \|N\| < 1$. Submultiplicativity of the operator norm gives $\left\|N^j\right\| \le q^j$, so the partial sums $S_k := \sum_{j=0}^{k} N^j$ satisfy $\|S_k - S_l\| \le \sum_{j=l+1}^{k} q^j \le q^{l+1}/(1-q)$ for $k > l$ and are therefore Cauchy. The space $M_{m\times m}(\mathbb{R})$, identified with $\mathbb{R}^{m^2}$, is complete by Theorem 6.3.1, so $S_k \to S$ for some matrix S with $\|S\| \le \sum_{j\ge 0} q^j = (1-q)^{-1}$. Since $S_k(I-N) = (I-N)S_k = I - N^{k+1}$ and $\left\|N^{k+1}\right\| \le q^{k+1} \to 0$, passing to the limit gives $SM = MS = I$. ∎

Lemma A.1.2 (Mean value inequality). Let $U \subseteq \mathbb{R}^m$ be open and convex and let $\Phi\colon U \to \mathbb{R}^m$ be continuously differentiable with $\|D\Phi(z)\| \le \kappa$ for every $z \in U$. Then $\|\Phi(z_1) - \Phi(z_2)\| \le \kappa\,\|z_1 - z_2\|$ for all $z_1, z_2 \in U$.

Proof. Convexity places the segment $s \mapsto z_2 + s(z_1 - z_2)$, $s \in [0,1]$, inside U. The map $s \mapsto \Phi\big(z_2 + s(z_1 - z_2)\big)$ is continuously differentiable with derivative $D\Phi\big(z_2 + s(z_1-z_2)\big)(z_1 - z_2)$ by Theorem 8.3.1, so applying Theorem 10.3.1 to each coordinate,

$$\Phi(z_1) - \Phi(z_2) = \int_0^1 D\Phi\big(z_2 + s(z_1 - z_2)\big)(z_1 - z_2)\,\mathrm{d}s.$$

Taking norms and using the triangle inequality for integrals gives the bound. The scalar mean value theorem (Theorem 7.2.5) cannot be applied coordinate by coordinate here, since each coordinate would produce its own intermediate point and no single point serves for all of them; the integral form avoids the difficulty, and that is why it is stated separately. ∎

A.2 Proof of the Theorem

Proof of Theorem 8.6.1. Write $A := \nabla_y f(\bar{x}, \bar{y})$, invertible by hypothesis, and define

$$\Phi(x,y) := y - A^{-1} f(x,y), \qquad (x,y) \in \Lambda. \tag{A.1}$$

Because A^{-1} is injective, $f(x,y) = 0$ if and only if $\Phi(x,y) = y$: the solutions of the system are the fixed points of $\Phi(x,\cdot)$.

Step 1: $\Phi(x,\cdot)$ is a contraction. Differentiating (A.1) in y gives $\nabla_y \Phi(x,y) = I - A^{-1}\nabla_y f(x,y)$, which vanishes at $(\bar{x},\bar{y})$. Since f is continuously differentiable, $\nabla_y f$ is continuous, so there are $r > 0$ and $\delta > 0$

with $\overline{B_\delta(\bar{x})} \times \overline{B_r(\bar{y})} \subseteq \Lambda$ and

$$\left\|I - A^{-1}\nabla_y f(x,y)\right\| \le \tfrac{1}{2} \qquad \text{whenever } \|x-\bar{x}\| \le \delta,\ \|y-\bar{y}\| \le r. \tag{A.2}$$

Lemma A.1.2, applied on the convex set $B_r(\bar{y})$ and extended to its closure by continuity, gives

$$\|\Phi(x,y_1) - \Phi(x,y_2)\| \le \tfrac{1}{2}\|y_1 - y_2\|, \qquad y_1, y_2 \in \overline{B_r(\bar{y})},\ \|x-\bar{x}\| \le \delta. \tag{A.3}$$

Step 2: the ball is invariant. Since $f(\bar{x},\bar{y}) = 0$ we have $\Phi(\bar{x},\bar{y}) = \bar{y}$, and $x \mapsto \Phi(x,\bar{y})$ is continuous, so after shrinking δ we may assume $\|\Phi(x,\bar{y}) - \bar{y}\| \le r/2$ for $\|x-\bar{x}\| \le \delta$. For such x and any $y \in \overline{B_r(\bar{y})}$,

$$\|\Phi(x,y) - \bar{y}\| \le \|\Phi(x,y) - \Phi(x,\bar{y})\| + \|\Phi(x,\bar{y}) - \bar{y}\| \le \tfrac{1}{2}\|y-\bar{y}\| + \tfrac{r}{2} \le r,$$

so $\Phi(x,\cdot)$ maps $\overline{B_r(\bar{y})}$ into itself.

Step 3: existence and uniqueness of e. The set $\overline{B_r(\bar{y})}$ is closed in the complete space $\mathbb{R}^m$, hence complete by Proposition 6.3.2. By (A.3) and Step 2, $\Phi(x,\cdot)$ is a contraction of modulus $\frac{1}{2}$ of that set into itself, so Theorem 6.4.1 supplies a unique fixed point, called $e(x)$. Setting $\Gamma := B_\delta(\bar{x})$ gives $e\colon \Gamma \to \overline{B_r(\bar{y})}$ with $f(x,e(x)) = 0$, which is (i); uniqueness of the fixed point at $x = \bar{x}$ forces $e(\bar{x}) = \bar{y}$, which is (ii). Uniqueness also yields the final claim of the theorem: inside $\Gamma \times B_r(\bar{y})$ the solution set of $f = 0$ is exactly the graph of e.

Step 4: (iii). Estimate (A.2) says $\left\|I - A^{-1}\nabla_y f(x,y)\right\| \le \frac{1}{2} < 1$, so Lemma A.1.1 makes $A^{-1}\nabla_y f(x,y)$ invertible and hence $\nabla_y f(x,y)$ invertible throughout the box, in particular at $(x,e(x))$.

Step 5: e is Lipschitz. For $x_1, x_2 \in \Gamma$, using $e(x_i) = \Phi(x_i, e(x_i))$ and then (A.3),

$$\|e(x_1) - e(x_2)\| \le \underbrace{\|\Phi(x_1,e(x_1)) - \Phi(x_1,e(x_2))\|}_{\le \frac{1}{2}\|e(x_1)-e(x_2)\|} + \|\Phi(x_1,e(x_2)) - \Phi(x_2,e(x_2))\|,$$

and the second term equals $\left\|A^{-1}\left[f(x_2,e(x_2)) - f(x_1,e(x_2))\right]\right\| \le \left\|A^{-1}\right\| M \|x_1 - x_2\|$, where M is the supremum of $\|\nabla_x f\|$ over the closed box — finite by Corollary 5.4.1 — and Lemma A.1.2 is applied in the x variable. Rearranging,

$$\|e(x_1) - e(x_2)\| \le L\|x_1 - x_2\|, \qquad L := 2\left\|A^{-1}\right\| M; \tag{A.4}$$

in particular e is continuous.

Step 6: differentiability and (iv). Fix $x \in \Gamma$ and write $y := e(x)$, $B := \nabla_y f(x,y)$ — invertible by Step 4 — and $C := \nabla_x f(x,y)$. For small h put $k := e(x+h) - e(x)$, so $\|k\| \le L\|h\|$ by (A.4). Differentiability of f at (x,y) gives

$$0 = f(x+h, y+k) - f(x,y) = Ch + Bk + o\big(\|h\| + \|k\|\big),$$

and $\|h\| + \|k\| \le (1+L)\|h\|$ converts the remainder into $o(\|h\|)$. Multiplying by B^{-1},

$$k = -B^{-1}Ch + o(\|h\|),$$

which is precisely the statement that e is differentiable at x with $De(x) = -B^{-1}C$, that is (8.9). Continuity of De follows because $x \mapsto \nabla_y f(x,e(x))$ and $x \mapsto \nabla_x f(x,e(x))$ are continuous, by continuity of Df and of e, and because matrix inversion is continuous on the invertible matrices — itself a consequence of Lemma A.1.1. Hence e is continuously differentiable. ∎

> **Remark A.2.1** (Deriving the inverse function theorem). Corollary 8.6.1 follows by applying Theorem 8.6.1 to $F(z,x) := g(x) - z$ on $\mathbb{R}^n \times A$ at the point $\big(g(\bar{x}), \bar{x}\big)$: here the derivative in the second block is $\nabla_x F = Dg(\bar{x})$, nonsingular by hypothesis, so there is a continuously differentiable e with $g(e(z)) = z$ near $g(\bar{x})$, and (8.9) gives $De(z) = -\big[Dg(e(z))\big]^{-1}(-I) = \big[Dg(e(z))\big]^{-1}$.

Appendix B

Matrix Differentiation

The identities below are used repeatedly in Chapters 8, 11 and 14. All vectors are columns, gradients of scalar functions are columns, and Df denotes the Jacobian, so that $Df(x) = \nabla f(x)^{\mathsf{T}}$ for scalar f. This convention is the one fixed in the front matter, and it is the only source of transposition errors in what follows.

B.1 Linear and Quadratic Forms

Proposition B.1.1. Let $a \in \mathbb{R}^n$, $A \in M_{n\times n}(\mathbb{R})$, $B \in M_{m\times n}(\mathbb{R})$, $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^n$.

(i) $\nabla\big(\langle a, x\rangle\big) = a$ and $D(Bx) = B$.

(ii) $\nabla\big(x^{\mathsf{T}}Ax\big) = (A+A^{\mathsf{T}})x$; in particular $\nabla\big(x^{\mathsf{T}}Ax\big) = 2Ax$ when A is symmetric, and $\nabla \|x\|^2 = 2x$.

(iii) $\mathrm{D}^2\big(x^{\mathsf{T}}Ax\big) = A + A^{\mathsf{T}}$, a constant matrix.

(iv) If $\varphi(x) := f(Bx)$ with f differentiable, then $\nabla\varphi(x) = B^{\mathsf{T}}\nabla f(Bx)$.

(v) $\nabla_x\big(x^{\mathsf{T}}Ay\big) = Ay$ and $\nabla_y\big(x^{\mathsf{T}}Ay\big) = A^{\mathsf{T}}x$.

Proof. (i) $\langle a, x\rangle = \sum_i a_i x_i$ has $\partial/\partial x_j = a_j$; the i-th coordinate of Bx is $\sum_j B_{ij}x_j$, whose partial derivative in x_j is B_{ij}.

(ii) Writing $q(x) = \sum_{i,j} A_{ij}x_i x_j$,

$$\frac{\partial q}{\partial x_k} = \sum_j A_{kj}x_j + \sum_i A_{ik}x_i = \big(Ax\big)_k + \big(A^{\mathsf{T}}x\big)_k,$$

the two sums arising from the occurrences of x_k in the first and in the second factor. Taking $A = I$ gives $\nabla \|x\|^2 = 2x$.

(iii) Differentiate (ii) once more: $x \mapsto (A + A^{\mathsf{T}})x$ is linear with Jacobian $A + A^{\mathsf{T}}$ by (i).

(iv) Theorem 8.3.1 gives $D\varphi(x) = Df(Bx)B$, and transposing, $\nabla\varphi(x) = B^{\mathsf{T}}\nabla f(Bx)$. The transpose is where the convention matters: Df is a row and ∇f a column.

(v) $x^{\mathsf{T}}Ay = \sum_{i,j} x_i A_{ij} y_j$ is linear in x for fixed y with coefficient vector Ay, and linear in y for fixed x with coefficient vector $A^{\mathsf{T}}x$; apply (i). ■

Example B.1.1 (Where each identity is used). Identity (ii) with symmetric A produced (8.3) and the second-order conditions of Theorem 8.5.3. Identity (iv) with $f = \|\cdot\|^2$ gives the least-squares normal equations: minimising $\|y - X\beta\|^2$ over β has gradient $-2X^{\mathsf{T}}(y - X\beta)$, whose vanishing is $X^{\mathsf{T}}X\beta = X^{\mathsf{T}}y$, the calculus route to Proposition 8.8.1 and the algebraic counterpart of the projection argument of Proposition 11.5.1. Identity (iii) shows that the Hessian of a quadratic form is constant, so a quadratic objective is concave globally or not at all — which is why Theorem 8.5.4 needs to be checked only once for such objectives rather than at every point.

B.2 Inverse, Trace and Determinant

Proposition B.2.1. Let $X \in M_{n\times n}(\mathbb{R})$ be invertible and $A \in M_{n\times n}(\mathbb{R})$.

(i) If $t \mapsto X(t)$ is differentiable with $X(t)$ invertible, then

$$\frac{\mathrm{d}}{\mathrm{d}t} X(t)^{-1} = -X(t)^{-1}\Big(\frac{\mathrm{d}}{\mathrm{d}t} X(t)\Big) X(t)^{-1}. \tag{B.1}$$

(ii) $\partial \operatorname{tr}(AX)/\partial X = A^\mathsf{T}$, entrywise.

(iii) $\partial \log\det X/\partial X = (X^{-1})^\mathsf{T}$.

Proof. (i) Differentiate $X(t)X(t)^{-1} = I$ by the product rule (Proposition 7.3.2, applied entrywise), obtaining $\dot{X}X^{-1} + X\frac{\mathrm{d}}{\mathrm{d}t}(X^{-1}) = 0$, and multiply on the left by X^{-1}.

(ii) $\operatorname{tr}(AX) = \sum_{i,j} A_{ij}X_{ji}$, whose partial derivative in X_{ji} is A_{ij}; assembling these into the matrix indexed by (j, i) gives A^T.

(iii) Jacobi's formula $\mathrm{d}(\det X) = \det X \cdot \operatorname{tr}\big(X^{-1}\,\mathrm{d}X\big)$, a consequence of the cofactor expansion of Definition 2.4.1, gives $\mathrm{d}\log\det X = \operatorname{tr}\big(X^{-1}\,\mathrm{d}X\big)$; comparing with (ii) at $A = X^{-1}$ gives the stated gradient. A derivation of Jacobi's formula is in Axler (2024, Ch. 10). ■

Identity (B.1) is the matrix analogue of $\frac{\mathrm{d}}{\mathrm{d}t}(1/x) = -\dot{x}/x^2$, and it is what the Riccati recursion (14.27) and the comparative statics formula (8.9) reduce to in the scalar case. Identity (iii) is the first-order condition of Gaussian maximum likelihood with respect to a covariance matrix, and it is the reason the log-likelihood is concave in the precision matrix rather than in the covariance matrix.

Appendix C

The Multivariate Normal Distribution

Theorem 14.6.1 and Proposition 14.5.1 are the scalar and bivariate cases of one statement, recorded here in general form because econometric and macroeconomic applications use it with several signals and several states at once.

C.1 Definition and Affine Images

Definition C.1.1 (Multivariate normal). A random vector $X = (X_1, \dots, X_n)^\top$ is *multivariate normal* with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \succ 0$, written $X \sim \mathcal{N}(\mu, \Sigma)$, if it has density

$$f(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big), \qquad x \in \mathbb{R}^n. \tag{C.1}$$

That (C.1) integrates to one is Proposition 10.7.2 combined with the substitution $z = \Sigma^{-1/2}(x - \mu)$, whose Jacobian determinant is $(\det \Sigma)^{-1/2}$ by Theorem 10.4.2; the symmetric square root $\Sigma^{1/2} = Q\Lambda^{1/2}Q^\top$ exists by Theorem 2.5.1 and has strictly positive diagonal $\Lambda^{1/2}$ by Proposition 2.5.1.

Proposition C.1.1 (Affine images and the moment generating function). Let $X \sim \mathcal{N}(\mu, \Sigma)$, $b \in \mathbb{R}^m$ and $B \in M_{m\times n}(\mathbb{R})$ of rank m. Then

$$b + BX \sim \mathcal{N}\big(b + B\mu,\ B\Sigma B^\top\big), \qquad \mathbb{E}\big[e^{\langle t, X\rangle}\big] = \exp\Big(\langle t, \mu\rangle + \tfrac{1}{2}t^\top \Sigma t\Big) \quad \text{for } t \in \mathbb{R}^n. \tag{C.2}$$

In particular every linear combination $\langle a, X\rangle$ is univariate normal with mean $\langle a, \mu\rangle$ and variance $a^\top \Sigma a$.

Proof. For the moment generating function, complete the square in the exponent of (C.1): with $\nu := \mu + \Sigma t$,

$$\langle t, x\rangle - \tfrac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu) = \langle t, \mu\rangle + \tfrac{1}{2}t^\top\Sigma t - \tfrac{1}{2}(x-\nu)^\top\Sigma^{-1}(x-\nu),$$

as multiplying out both sides and using the symmetry of Σ^{-1} verifies. Integrating the remaining Gaussian kernel gives 1 and leaves the stated formula. For the affine image, the moment generating function of $b + BX$ at $s \in \mathbb{R}^m$ is

$$\mathbb{E}\big[e^{\langle s, b+BX\rangle}\big] = e^{\langle s, b\rangle}\,\mathbb{E}\big[e^{\langle B^\top s, X\rangle}\big] = \exp\Big(\langle s, b + B\mu\rangle + \tfrac{1}{2}s^\top B\Sigma B^\top s\Big),$$

which is the moment generating function of $\mathcal{N}(b + B\mu, B\Sigma B^\top)$; a distribution on $\mathbb{R}^m$ whose moment generating function is finite on a neighbourhood of the origin is determined by it (Ash and Doléans-Dade 1999, Ch. 5). The rank condition keeps $B\Sigma B^\top$ positive definite. ■

Corollary C.1.1 (Uncorrelated implies independent). Let $(X_1^\top, X_2^\top)^\top \sim \mathcal{N}(\mu, \Sigma)$, with Σ partitioned into blocks $\Sigma_{11}, \Sigma_{12}, \Sigma_{21}, \Sigma_{22}$ conformably. Then X_1 and X_2 are independent if and only if $\Sigma_{12} = 0$.

Proof. Independence implies zero covariance for any square-integrable pair, by Proposition 14.2.1(iv). Conversely, if $\Sigma_{12} = 0$ then Σ and Σ^{-1} are block diagonal and $\det \Sigma = \det \Sigma_{11} \det \Sigma_{22}$, so (C.1) factors into the product of the two marginal densities. ■

The corollary is false without joint normality. Let Z be standard normal and let S be independent of Z with $\mathbb{P}(S = 1) = \mathbb{P}(S = -1) = \frac{1}{2}$. Then SZ is standard normal and $\operatorname{Cov}(Z, SZ) = \mathbb{E}[S]\mathbb{E}[Z^2] = 0$, yet $|Z| = |SZ|$, so the two are far from independent. The pair (Z, SZ) is not jointly normal, since $Z + SZ$ equals 0 with probability $\frac{1}{2}$ and no non-degenerate normal variable has an atom.

C.2 The Conditional Law

Theorem C.2.1 (Conditional distribution)**.** Let $(X_1^\top, X_2^\top)^\top$ be multivariate normal with mean $(\mu_1^\top, \mu_2^\top)^\top$ and covariance blocks $\Sigma_{11}, \Sigma_{12}, \Sigma_{21}, \Sigma_{22}$, with Σ_{22} invertible. Then

$$X_1 \mid X_2 = x_2 \ \sim\ \mathcal{N}\Big(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2),\ \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big). \tag{C.3}$$

The conditional mean is affine in x_2 and the conditional covariance does not depend on x_2.

Proof. Set $Z := X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2$. By Proposition C.1.1 the pair $(Z^\top, X_2^\top)^\top$ is an affine image of $(X_1^\top, X_2^\top)^\top$ and is therefore jointly normal, and

$$\operatorname{Cov}(Z, X_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0.$$

Corollary C.1.1 makes Z independent of X_2. Hence, conditionally on $X_2 = x_2$, the vector $X_1 = Z + \Sigma_{12}\Sigma_{22}^{-1}x_2$ is Z shifted by a constant, so its conditional law is normal with mean

$$\mathbb{E}[Z] + \Sigma_{12}\Sigma_{22}^{-1}x_2 = \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2 + \Sigma_{12}\Sigma_{22}^{-1}x_2$$

and covariance

$$\operatorname{Var}(Z) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21},$$

which together give (C.3). ■

Remark C.2.1 (Recovering the results of Chapter 14). Take the blocks to be scalars with $X_1 = Y$ and $X_2 = X$. Then $\Sigma_{12}\Sigma_{22}^{-1} = \operatorname{Cov}(X, Y)/\operatorname{Var}(X)$ and (C.3) is (14.22), so Proposition 14.5.1 is the bivariate case.

Take instead $X_1 = X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $X_2 = Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent of X. The pair is jointly normal by Proposition C.1.1, with $\Sigma_{11} = \Sigma_{12} = \sigma_x^2$ and $\Sigma_{22} = \sigma_x^2 + \sigma_\varepsilon^2$, so (C.3) gives conditional mean $\mu_x + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\varepsilon^2}(y - \mu_x)$ and conditional variance $\sigma_x^2 - \frac{\sigma_x^4}{\sigma_x^2 + \sigma_\varepsilon^2} = \frac{\sigma_x^2\sigma_\varepsilon^2}{\sigma_x^2 + \sigma_\varepsilon^2}$, which is (14.23). The completing-the-square derivation given there is therefore an instance of the orthogonalisation argument used above, and the statement that precisions add is the identity $\big(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\big)^{-1} = \sigma_x^{-2} + \sigma_\varepsilon^{-2}$ in this parametrisation.

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